MA-105 (Survey of Math)

Worcester State University

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Contents

Chapter 1

(Module 1.) Voting and Apportionment

Beyond just marking a ballot, these concepts shape how our voices are heard and how resources are distributed. This chapter breaks down the mechanics behind decision-making, offering clear examples of how mathematical fairness directly impacts your community and representation.

1.1 Voting methods

How to determine a Winner?

> **Preference ballots** are ballots in which a voter is asked to rank all candidates in order of preference. A **preference table** shows how often each particular outcomes occurred.

Example 1.1 Four candidates are running for student body president: Alice (A), Bonnie (B), Carlos (C) and Daniel (D). The students were asked to rank all the candidates in order of preference.

Table 1.2 Preference Table For The Election Of Student Body President

Number of Votes	130	120	100	150
First choice	A	D	D	C
Second choice	B	B	B	B
Third choice	C	C	A	A
Fourth choice	D	A	C	D

1. How many students voted in the election?
2. How many students selected the candidates in this order : D, B, A, C?
3. How many students selected Daniel (D) as their first choice for student body president?

Solution.

1. To find the total number of students, sum the votes in the top row: $130 + 120 + 100 + 150 = 500$ students.

2. Looking at the columns, the order ***D, B, A, C*** appears in the third column. There were 100 students who voted in this order.

3. Daniel (D) is listed as the first choice in the second and third columns. The total number of students is $120 + 100 = 220$ students.

▲

Definition 1.3 The Plurality Method. The candidate (or candidates, if there is more than one) with the most first-place votes is the winner. ◆

Example 1.4 Table 1.2 shows the preference table for the four candidates running for student body president. Who is declared the winner using the Plurality method?

Solution. A received 130 first-place votes, D received $120 + 100 = 220$ first-place votes and C received 150 first-place votes. Thus Daniel is the winner. ▲

Majority Rule Decision is when more than 50% of the voters rank a candidate in first-place. The majority rule will almost always have a winner.

Definition 1.5 The Plurality-with-Elimination Method. The candidate with the majority (over 50% votes) of first-place votes is the winner. If no candidate receives a majority, eliminate the candidate with the fewest first-place votes. Move the candidates below them up one place and recount. Repeat until a candidate has a majority. ◆

Example 1.6 Elimination Method Example. Determine the winner for the student president election from Table 1.2, using Plurality-with-Elimination.

Solution. Total votes = 500. Majority needed = 251.

Round 1: A=130, D=220, C=150, B=0. No majority. Bonnie (B) is eliminated.

Table 1.7 Preference Table: Round 2 (Bonnie Eliminated)

Number of Votes	130	120	100	150
First choice	A	D	D	C
Second choice	C	C	A	A
Third choice	D	A	C	D

Round 2: After shifting votes up, the first-place votes remain: A=130, D=220, C=150. Still no majority. Alice (A) has the fewest (130) and is eliminated.

Table 1.8 Preference Table: Final Round (Alice and Bonnie Eliminated)

Number of Votes	130	120	100	150
First choice	C	D	D	C
Second choice	D	C	C	D

Round 3: Alice's 130 votes go to her next preferred candidate, Carlos (C). New totals: Daniel = 220, Carlos = $150 + 130 = 280$.

Carlos now has 280 votes (a majority). Carlos is the winner. ▲

Definition 1.9 The Borda Count Method. Each voter ranks the candidates from the most favorable to the least favorable. Each last-place vote is given 1 point, each next-to-last-place vote is given 2 points, and so on. The points are totaled for each candidate separately. The candidate with the most points is the winner. ◆

Example 1.10 Borda Count Example. Using the preference data from Table 1.2, who is declared the winner using the Borda Count Method?

Solution. With 4 candidates, 1st place gets 4 pts, 2nd gets 3 pts, 3rd gets 2 pts, and 4th gets 1 pt.

Table 1.11 Borda Count Calculation Table

Number of Votes	130	120	100	150
1st choice: 4 pts	A: $130 \times 4 = 520$	D: $120 \times 4 = 480$	D: $100 \times 4 = 400$	C: $150 \times 4 = 600$
2nd choice: 3 pts	B: $130 \times 3 = 390$	B: $120 \times 3 = 360$	B: $100 \times 3 = 300$	B: $150 \times 3 = 450$
3rd choice: 2 pts	C: $130 \times 2 = 260$	C: $120 \times 2 = 240$	A: $100 \times 2 = 200$	A: $150 \times 2 = 300$
4th choice: 1 pt	D: $130 \times 1 = 130$	A: $120 \times 1 = 120$	C: $100 \times 1 = 100$	D: $150 \times 1 = 150$

- ***Alice (A):*** $(130 \times 4) + (120 \times 1) + (100 \times 2) + (150 \times 2) = 520 + 120 + 200 + 300 = 1140$ pts
- ***Bonnie (B):*** $(130 \times 3) + (120 \times 3) + (100 \times 3) + (150 \times 3) = 390 + 360 + 300 + 450 = 1500$ pts
- ***Carlos (C):*** $(130 \times 2) + (120 \times 2) + (100 \times 1) + (150 \times 4) = 260 + 240 + 100 + 600 = 1200$ pts
- ***Daniel (D):*** $(130 \times 1) + (120 \times 4) + (100 \times 4) + (150 \times 1) = 130 + 480 + 400 + 150 = 1160$ pts

Bonnie is the winner with 1500 points. ▲

Definition 1.12 The Pairwise Comparison Method. Every candidate is matched on a one-to-one basis with every other candidate. If candidate A beats candidate B in one-to-one competition, then A receives 1 point and B receives 0. If they tie, each receives 0.5 points. The candidate with the most total points is the winner.

In an election with n candidates, the number of comparisons, C, is calculated as:

$$C = \frac{n(n-1)}{2}$$

◆

Example 1.13 Pairwise Comparison Example. Using the same preference Table 1.2, for the four candidates (Alice, Bonnie, Carlos, and Daniel), who is the winner using the Pairwise Comparison Method?

Solution. With $n = 4$ candidates, there are $\frac{4(3)}{2} = 6$ total comparisons (A vs.B, A vs.C, A vs.D, B vs.C, B vs.D, C vs.D) and determine the winner using the pairwise comparison method.

130	120	100	150
A	B	B	B
B	A	A	A

vs B

A

130	120	100	150
A	C	A	C
C	A	C	A

vs C

A

130	120	100	150
A	D	D	A
D	A	A	D

vs D

A

130	120	100	150
B	B	B	C
C	C	C	B

vs C

B

130	120	100	150
B	D	D	B
D	B	B	D

vs D

B

130	120	100	150
C	D	D	C
D	C	C	D

vs D

C

1. ***A vs B:*** Alice is preferred over Bonnie in 130 ballots. Bonnie is preferred

over Alice in $120 + 100 + 150 = 370$ ballots. **Winner: Bonnie (B gets 1 pt)**

2. ***A vs C:*** Alice is preferred over Carlos in $130 + 120 + 100 = 350$ ballots. Carlos is preferred over Alice in 150. **Winner: Alice (A gets 1 pt)**

3. ***A vs D:*** Alice is preferred over Daniel in 130 ballots. Daniel is preferred over Alice in $120 + 100 + 150 = 370$ ballots. **Winner: Daniel (D gets 1 pt)**

4. ***B vs C:*** Bonnie is preferred over Carlos in $130 + 120 + 100 = 350$ ballots. Carlos is preferred over Bonnie in 150. **Winner: Bonnie (B gets 1 pt)**

5. ***B vs D:*** Bonnie is preferred over Daniel in $130 + 150 = 280$ ballots. Daniel is preferred over Bonnie in $120 + 100 = 220$. **Winner: Bonnie (B gets 1 pt)**

6. ***C vs D:*** Carlos is preferred over Daniel in $130+150 = 280$ ballots. Daniel is preferred over Carlos in $120 + 100 = 220$. **Winner: Carlos (C gets 1 pt)**

Final Point Totals:

- Bonnie (B): 3 points
- Alice (A): 1 point
- Daniel (D): 1 point
- Carlos (C): 1 point

Bonnie is the winner using Pairwise Comparison. ▲

If a candidate wins all pair comparisons with other candidates, that candidate is called the **Condorcet winner**. A Condorcet winner automatically wins by the pairwise comparisons method.

Example 1.14 Condorcet Winner. Is there a Condorcet winner in the following election?

Table 1.15 Preference Table for Condorcet Example

Number of Votes	30	30	30	20
First choice	B	B	A	A
Second choice	A	A	C	D
Third choice	C	D	D	C
Fourth choice	D	C	B	B

Solution. To find a Condorcet winner, we must see if one candidate wins every head-to-head matchup. Let's check candidate ***B*** and candidate ***A***:

- ***B vs A:*** B is ranked higher than A in $30 + 30 = 60$ ballots. A is ranked higher than B in $30 + 20 = 50$ ballots. **Winner: B**.
- ***B vs C:*** B is ranked higher than C in $30 + 30 + 30 + 20 = 110$ ballots (every ballot). **Winner: B**.
- ***B vs D:*** B is ranked higher than D in $30 + 30 + 30 + 20 = 110$ ballots (every ballot). **Winner: B**.

Since ***B*** beats every other candidate in a one-on-one comparison, ***B is the Condorcet winner***. ▲

Arrow's Impossibility Theorem states that a ranked-voting electoral system cannot reach a community-wide ranked preference by converting individuals' preferences while meeting all the conditions of a fair voting system.

Consider the following situation with 3 candidates in one-to-one competition (pairwise comparison method):

- Suppose candidate A beats candidate B
- Suppose candidate B beats candidate C
- Suppose candidate C beats candidate A

When this occurs, it is because the conflicting majorities are each made up of different groups of individuals. No winner can be determined.

This is called the **Condorcet Paradox** (also known as the voting paradox).

Example 1.16 Condorcet Paradox. Who would win this election through the pairwise comparison method?

Table 1.17 Preference Table: Three-Candidate Election

Number of Votes	8	9	10
First choice	C	B	A
Second choice	A	C	B
Third choice	B	A	C

Solution. We perform the three possible head-to-head comparisons:

- ***A vs B:*** A is preferred on 8 (col 1) + 10 (col 3) = 18 ballots. B is preferred on 9 (col 2) ballots. **Winner: A** (A gets 1 pt).
- ***B vs C:*** B is preferred on 9 (col 2) + 10 (col 3) = 19 ballots. C is preferred on 8 (col 1) ballots. **Winner: B** (B gets 1 pt).
- ***C vs A:*** C is preferred on 8 (col 1) + 9 (col 2) = 17 ballots. A is preferred on 10 (col 3) ballots. **Winner: C** (C gets 1 pt).

Each candidate (A, B, and C) has exactly 1 point. This is a **Condorcet Paradox** because there is a "rock-paper-scissors" loop (A beats B, B beats C, and C beats A), resulting in a three-way tie. ▲

Checkpoint 1.18 Mayor of a Small Town. Three candidates A, B, and C are running for mayor of a small town. The results of the election are shown in the following preference table.

Table 1.19 Mayor Election Preference Table

Number of Votes	1200	900	900	600
First choice	A	C	B	B
Second choice	B	A	C	A
Third choice	C	B	A	C

1. Determine the winner using the Plurality method.

2. Determine the winner using the Borda count method.
3. Determine the winner using the Plurality-with-elimination method.
4. Is there a Condorcet Winner?

Answer.

1. ***Plurality:*** A = 1200, B = $900 + 600 = 1500$, C = 900. **Winner: B**.
2. ***Borda Count (3-2-1):***
   - A: $(1200 \times 3) + (900 \times 2) + (900 \times 1) + (600 \times 2) = 3600 + 1800 + 900 + 1200 = 7500$
   - B: $(1200 \times 2) + (900 \times 1) + (900 \times 3) + (600 \times 3) = 2400 + 900 + 2700 + 1800 = 7800$
   - C: $(1200 \times 1) + (900 \times 3) + (900 \times 2) + (600 \times 1) = 1200 + 2700 + 1800 + 600 = 6300$

   Winner: B.
3. ***Plurality-with-Elimination:*** Total votes = 3600; Majority = 1801. Round 1: A=1200, B=1500, C=900. No majority. Eliminate C. C's 900 votes go to A (second choice). Final: A = $1200 + 900 = 2100$, B = 1500. **Winner: A**.
4. ***Condorcet Winner Check:***
   - A vs B: A is preferred on $1200 + 900 = 2100$. B on 1500. (A wins)
   - A vs C: A is preferred on $1200 + 600 = 1800$. C on $900 + 900 = 1800$. (Tie)

   Since A does not win all matches, **No Condorcet Winner exists**.

Solution.

1. *Plurality Method.*

   We count only the first-place votes for each candidate:
   - ***A:*** 1200 votes
   - ***B:*** $900 + 600 = 1500$ votes
   - ***C:*** 900 votes

   Winner: B (1500 votes).
2. *Borda Count Method (3-2-1 Points).*

   Table 1.20 Borda Count Point Distribution

Candidate	1st (3 pts)	2nd (2 pts)	3rd (1 pt)	Total Poin
A	$1200 \times 3 = 3600$	$(900 + 600) \times 2 = 3000$	$900 \times 1 = 900$	7500
B	$(900 + 600) \times 3 = 4500$	$1200 \times 2 = 2400$	$900 \times 1 = 900$	7800
C	$900 \times 3 = 2700$	$900 \times 2 = 1800$	$(1200 + 600) \times 1 = 1800$	6300

   Winner: B (7800 points).

3. *Plurality-with-Elimination.*

   Total Votes: 3600. Majority needed: 1801.

   Round 1: A=1200, B=1500, C=900. No candidate has 1801. Eliminate C.

   Table 1.21 Final Round (C Eliminated)

Votes	1200	900	900	600
1st	A	A	B	B
2nd	B	B	A	A

   New totals: $\boldsymbol{A} = 1200 + 900 = 2100$. $\boldsymbol{B} = 900 + 600 = 1500$. **Winner: A** (2100 votes).

Checkpoint 1.22 Math Club Presidential Election. Four candidates are running for president of the Math Club: Paula (P), Sylvia (S), Craig (C), and Brian (B). The results of the election are shown in the following preference table:

Table 1.23 Election Preference Schedule

Number of Votes	15	19	23	10	18	15
First choice	B	C	P	P	S	B
Second choice	S	P	S	B	C	S
Third choice	P	S	B	C	P	C
Fourth choice	C	B	C	S	B	P

1 How many students voted in the election?

2 How many students voted Brian as their first choice?

3 Who would win the presidency using the Plurality method?

4 Is there a Majority Winner?

5 Who would win the presidency using the Plurality with Elimination method?

6 Who would win the presidency using the Borda count method?

7 Who would win the presidency using the pairwise comparison method? If a tie, use the Plurality method between the winners to determine the tie-breaker.

Answer.

1 100 students

2 30 students

3 Paula (33 votes)

4 No (Majority requires 51 votes)

5 Craig

6 Sylvia (279 points)

7 Paula (3 points)

Solution.

1 Total Votes: 100.

2 Brian 1st choice: 30.

3 Plurality Winner: Paula (33).

4 Majority: No (Requires 51).

5 Plurality with Elimination:

Round 1: Sylvia has the fewest 1st-place votes (18) and is eliminated.

Table 1.24 Round 2 (Sylvia Eliminated)

Votes	15	19	23	10	18	15
1st	B	C	P	P	C	B
2nd	P	P	B	B	P	C
3rd	C	B	C	C	B	P

New totals: P=33, B=30, C=37 (19 + 18). Brian is eliminated.

Table 1.25 Round 3 (Brian Eliminated)

Votes	15	19	23	10	18	15
1st	P	C	P	P	C	C
2nd	C	P	C	C	P	P

Round 3: Paula vs. Craig. Craig wins the final matchup against Paula 52 to 48.

Winner: Craig.

6 Borda Count Method:

Using 4 pts for 1st, 3 for 2nd, 2 for 3rd, and 1 for 4th:

- Paula: $33(4) + 19(3) + 33(2) + 15(1) = 270$
- Sylvia: $18(4) + 53(3) + 19(2) + 10(1) = 279$
- Craig: $19(4) + 18(3) + 25(2) + 38(1) = 218$
- Brian: $30(4) + 10(3) + 23(2) + 37(1) = 233$

Winner: Sylvia.

7 Pairwise Comparison:

For the **Pairwise Comparison Method**, we compare every possible pair. There are 100 total votes, so 51 are needed to win a matchup:

- P vs. S: P has 52; S has 48. (P gets 1 pt)
- P vs. C: P has 63; C has 37. (P gets 1 pt)
- P vs. B: P has 70; B has 30. (P gets 1 pt)
- S vs. C: S has 81; C has 19. (S gets 1 pt)
- S vs. B: S has 60; B has 40. (S gets 1 pt)
- C vs. B: C has 60; B has 40. (C gets 1 pt)

Total Points: Paula = 3, Sylvia = 2, Craig = 1, Brian = 0. Paula is the winner; no tie-breaker is required.

1.2 Apportionment

Apportionment is the method of distributing power in political systems based on population density. Apportionment may reflect the availability of buses for city routes or doctors for patients.

Definition 1.26 The Standard Divisor. The standard divisor is found by dividing the total population under consideration by the number of items to be allocated.

$$\text{Standard divisor} = \frac{\text{total population in a group}}{\text{total number of items to be apportioned}}$$

◆

Definition 1.27 The Standard Quota. The standard quota for a particular group is found by dividing that group's population by the standard divisor.

$$\text{Standard quota} = \frac{\text{population of a particular group}}{\text{standard divisor}}$$

◆

Example 1.28 Palm Springs is composed of four states: A, B, C, and D. According to the country's constitution, the congress will have 30 seats, divided among the four states according to their respective populations. Find the standard quotas for states A, B, C, and D and complete the table below.

Table 1.29 Population of Palm Springs by State

State	**A**	**B**	**C**	**D**	**Total**
Population (in thousands)	275	383	465	767	1890
Standard Quota					

Solution. First, we find the **Standard Divisor** (SD):

$$SD = \frac{1890}{30} = 63$$

Now, divide each state's population by the divisor to find the **Standard Quota** (SQ):

- State A: $275/63 \approx 4.3651$
- State B: $383/63 \approx 6.0794$
- State C: $465/63 \approx 7.3810$
- State D: $767/63 \approx 12.1746$

Table 1.30

State	**A**	**B**	**C**	**D**	**Total**
Population (in thousands)	275	383	465	767	1890
Standard Quota	4.365	6.079	7.381	12.175	

▲

The **lower quota** is the standard quota rounded down to the nearest whole number. The **upper quota** is the standard quota rounded up to the nearest whole number.

A group's apportionment should be either its upper quota or its lower quota. An apportionment method that guarantees that this will always occurs is said to **satisfy the quota rule**.

Algorithm 1.31 Hamilton's Method.

1. *Calculate each group's standard quota.*
2. *Find the lower quota for each group.*
3. *Give surplus items, one at a time, to groups with the largest decimal parts until no surplus remains.*

Example 1.32 From Table 1.29, use Hamilton's method to apportion the 30 congressmen for Palm Springs.

Table 1.33 Hamilton's Method Table

State	**A**	**B**	**C**	**D**	**Total**
Population	275	383	465	767	1890
Standard Quota	4.365	6.079	7.381	12.175	30.00
Lower Quota	4	6	7	12	29
Surplus	+0	+0	+1	+0	1
Apportionment	4	6	8	12	30

1. The Standard Divisor is 63.
2. Lower quotas sum to 29, leaving 1 surplus seat.
3. State C has the largest decimal part (.381), so it receives the surplus seat.

Solution. Following the steps of **Hamilton's Method**:

1. The sum of the **Lower Quotas** is $4 + 6 + 7 + 12 = 29$. Since we must allocate 30 seats, there is $30 - 29 = 1$ surplus seat.
2. We look at the decimal parts of the Standard Quotas:
   - A: 0.365
   - B: 0.079
   - C: 0.381
   - D: 0.175
3. State C has the largest decimal part (0.381), so it receives the 1 surplus seat.
4. Final counts: A=4, B=6, C=8, D=12.

▲

Modified Divisor (MD).

A **Modified divisor** is a divisor, chosen by guess-and-check, that is used to make the sum of the modified quotas exactly equal to the number of items to be apportioned.

Algorithm 1.34 Jefferson's Method. *Find a* ***modified divisor*** *such that when each group's modified quota is rounded down (****modified lower quota****), the sum equals the number of items to be apportioned. Apportion to each group its modified lower quota.*

1. *Calculate the* ***Standard Divisor*** *(SD = Total Pop/Seats).*
2. *Calculate* ***Standard Quotas*** *and round down to find* ***Lower Quotas****.*
3. *If the sum of Lower Quotas is less than the total seats, choose a* ***Modified Divisor*** *(MD) that is* smaller *than the SD.*
4. *Recalculate quotas using MD and round down until the sum exactly matches the total seats.*

Example 1.35 From Table 1.29, apportion 30 seats to Palm Springs using Jefferson's Method.

Solution. The standard divisor (63) produced a sum of 29. For Jefferson's method, we must *decrease* the divisor to increase the quotas. Using a **Modified Divisor** of 61:

- A: $275/61 \approx 4.50 \to 4$
- B: $383/61 \approx 6.27 \to 6$
- C: $465/61 \approx 7.62 \to 7$
- D: $767/61 \approx 12.57 \to 13$

Sum: $4 + 6 + 7 + 13 = 30$. State D receives the extra seat.

Table 1.36 Jefferson's Method: Palm Springs

State	A	B	C	D	Total
Population	275	383	465	767	1890
Standard Quota (SD=63)	4.365	6.079	7.381	12.175	30
Lower Quota	4	6	7	12	29
Modified Quota (MD=61)	4.508	6.279	7.623	12.574	30.98
Modified LQ (Apportionment)	4	6	7	13	30

▲

Checkpoint 1.37 Apportioning 11 buses. Apportion 11 buses to the three towns using Jefferson's method.

Table 1.38 Bus Apportionment Table

Town	Westboro	Shrewbury	Worcester	Total
Population	89	120	168	377
Final Apportionment				

Solution. First, find the Standard Divisor: $SD = 377/11 \approx 34.27$.

Table 1.39 Trial 1: Using Standard Divisor (34.27)

Town	Westboro	Shrewbury	Worcester	Total
Population	89	120	168	377
Standard Quota	2.59	3.50	4.90	10.99
Lower Quota	2	3	4	9

The sum (9) is too low. We need 11 buses. We must *decrease* the divisor.

Let's try a smaller divisor, $MD = 31$.

Table 1.40 Trial 2: Modified Divisor (31)

Town	Westboro	Shrewbury	Worcester	Total
Modified Quota	2.87	3.87	5.41	12.15
Lower Quota	2	3	5	10

The sum (10) is still too low. We must decrease the divisor further. Trying $MD = 30$.

Table 1.41 Final Apportionment: Modified Divisor (30)

Town	Westboro	Shrewbury	Worcester	Total
Modified Quota	2.96	4.00	5.60	12.56
Modified LQ	2	4	5	11

The sum is exactly 11. Final Apportionment: Westboro: 2, Shrewbury: 4, Worcester: 5.

Table 1.42 Bus Apportionment Work Table

Town	Westboro	Shrewbury	Worcester	Total
Population	89	120	168	377
Standard Quota	2.59	3.50	4.90	10.99
Lower Quota	2	3	4	9
Modified Quota (MD=31)	2.87	3.87	5.41	12.15
Modified LQ	2	3	5	10
Modified Quota (MD=30)	2.96	4.00	5.60	12.56
Final Apportionment	2	4	5	11

Algorithm 1.43 Adam's Method.

1. *Find a **modified divisor**, d, so that when each group's **modified quota** is the upper quota, the sum of the whole numbers for all the groups is the number of items to be apportioned. The modified quotients that are rounded up are called **modified upper quota**.*

2. *Apportion to each group its modified upper quota.*

When using Adam's method, begin with a modified divisor that is slightly greater than the standard divisor.

Example 1.44 The Palm Springs is composed of four states: A, B, C, and D. According to the country's constitution, the congress will have 30 seats, divided among the four states according to their respective populations.

Find the standard quotas for states A, B, C, and D in Palm Springs and complete the table below. Use Adam's method with a modified divisor to apportion the congressmen.

Table 1.45 Population Data and Apportionment

State	A	B	C	D	Total
Population (1000s)	275	383	465	767	1890
Standard Quota					
Upper Quota					
Modified UQ					
Apportionment					30

Solution. 1. First, find the Standard Divisor (SD):

$$SD = \frac{1890}{30} = 63$$

2. Calculate Standard Quotas ($Pop/63$):

- A: 4.365 (Upper Quota: 5)
- B: 6.079 (Upper Quota: 7)
- C: 7.381 (Upper Quota: 8)
- D: 12.175 (Upper Quota: 13)

Sum of Upper Quotas $= 5 + 7 + 8 + 13 = 33$ (Too many seats).

3. Since the sum is too high, we increase the divisor. Let's try $md = 70.5$:

- A: $275/70.5 \approx 3.90 \to 4$
- B: $383/70.5 \approx 5.43 \to 6$
- C: $465/70.5 \approx 6.60 \to 7$
- D: $767/70.5 \approx 10.88 \to 11$

Sum $= 4 + 6 + 7 + 11 = 28$ (Too low).

4. Refined $md = 67.5$: Sums will result in A:5, B:6, C:7, D:12 = 30.
Using $md = 66.5$:

State	A	B	C	D	Total
Standard Quota ($d = 63$)	4.37	6.08	7.38	12.18	30
Upper Quota	5	7	8	13	33
Mod. Quota ($d = 66.5$)	4.14	5.76	6.99	11.53	—
Modified UQ	5	6	7	12	30

▲

Finding the Modified Divisor for Adam's Method:.

1. Compute the standard divisor

$$D = \frac{\text{Total population}}{\text{Number of seats}}.$$

2. Increase D by an amount so that when allocations (State population / Modified D) are rounded upward, they add up to the exact number of seats.

Checkpoint 1.46 Apportioning 7 Doctors. Apportion 7 doctors to 4 clinics using Adam's method.

Clinic	West	North	South	East	Total
Population	120	180	160	240	700
Appointment					

Solution. Standard Divisor (SD) = 700/7 = 100.

The final apportionment using Adam's method with a modified divisor of md=120 is: West: 1, North: 2, South: 2, and East: 2.

Clinic	**West**	**North**	**South**	**East**	**Total**
Population	120	180	160	240	700
Standard Quota ($d = 100$)	1.2	1.8	1.6	2.4	7
Upper Quota (Initial)	2	2	2	3	9 (Too High)
Mod. Quota ($d = 120$)	1.0	1.5	1.33	2.0	—
Modified UQ (Final)	1	2	2	2	7

Checkpoint 1.47 Apportioning 40 HMO Doctors. An HMO has 40 doctors to be apportioned among four clinics: A, B, C, and D.

Clinic	**A**	**B**	**C**	**D**
Patient Load	275	392	611	724
Apportionment				

1 Use Hamilton's method.

2 Use Jefferson's method with $d = 48$.

3 Use Adam's method.

Solution 1 (Hamilton's Method Solution). Calculate Standard Divisor (SD) and Quotas

- Total Patient Load: $275 + 392 + 611 + 724 = 2002$
- $SD = 2002/40 = 50.05$

Clinic	**A**	**B**	**C**	**D**	**Total**
Load	275	392	611	724	2002
Standard Quota	5.4945	7.8322	12.2078	14.4655	40
Lower Quota	5	7	12	14	38
Fractional Part	0.4945	0.8322	0.2078	0.4655	—
Surplus Seat	+1	+1	0	0	+2
Apportionment	6	8	12	14	40

Solution 2 (Jefferson's Method Solution (d=48)).

Clinic	**A**	**B**	**C**	**D**	**Total**
Modified Quota	5.73	8.17	12.73	15.08	—
Apportionment	5	8	12	15	40

Solution 3 (Adam's Method Solution (md=52)).

Clinic	**A**	**B**	**C**	**D**	**Total**
Modified Quota	5.29	7.54	11.75	13.92	—
Apportionment	6	8	12	14	40

Checkpoint 1.48 Apportioning Math Course Sections. Thirty sections of math courses are to be offered in introductory algebra, intermediate algebra, college algebra, and liberal arts math. The preregistration figures for the number of students planning to enroll in each section are given in the following table.

Course	**Intro**	**Intermed**	**College**	**Lib Arts**	**Total**
Enrollment	388	405	213	345	1351

1 Use Hamilton's method to determine how many sections of each course

should be offered.

2 Use Jefferson's method with Modified divisor md = _____ to determine how many sections of each course should be offered.

3 Use Adam's method with Modified divisor md = _____ to determine how many sections of each course should be offered.

Solution. Thirty sections of math courses are to be offered based on preregistration figures.

1 Hamilton's Method (SD = 45.0333):

Course	Intro	Intermed	College	Lib Arts	Total
Enrollment	388	405	213	345	1351
Standard Quota	8.6158	8.9933	4.7298	7.6610	30
Lower Quota	8	8	4	7	27
Fractional Part	0.6158	0.9933	0.7298	0.6610	—
Surplus Row	0	+1	+1	+1	+3
Apportionment	8	9	5	8	30

2 Jefferson's Method (using $d = 42.5$):

Course	Intro	Intermed	College	Lib Arts	Total
Mod. Quota	9.129	9.529	5.011	8.117	—
Apportionment	9	9	5	8	31 (Try Again)
Mod. Quota ($d = 43$)	9.023	9.418	4.953	8.023	—
Apportionment	9	9	4	8	30

3 Adam's Method (using $d = 48$):

Course	Intro	Intermed	College	Lib Arts	Total
Mod. Quota	8.083	8.437	4.437	7.187	—
Apportionment	9	9	5	8	31 (Try Again)
Mod. Quota ($d = 49$)	7.918	8.265	4.346	7.040	—
Apportionment	8	9	5	8	30

Example 1.49 Hamilton's Method Apportionment of 100 seats. Suppose there were four states, A, B, C ,and D with populations as shown in the table below. Suppose these four states decide to create a legislature with 100 seats. Apply Hamilton's method to determine the apportionment.

Table 1.50

Category	State A	State B	State C	State D	Total
Population	936	2,726	2,603	3,735	10,000
Standard Quota					
Lower Quota					
Surplus					
Apportioned Seats					100

The following table demonstrates the apportionment using Hamilton's Method with a standard divisor of $SD = 10,000/100 = 100$.

Table 1.51 Hamilton's Method Apportionment

Category	**State A**	**State B**	**State C**	**State D**	**Total**
Population	936	2,726	2,603	3,735	10,000
Standard Quota	9.36	27.26	26.03	37.35	100.00
Lower Quota	9	27	26	37	99
Fractional Part	0.36	0.26	0.03	0.35	-
Surplus	+1				
Final Seats	10	27	26	37	100

▲

Theorem 1.52 The Alabama Paradox. *In a fair apportionment system, adding extra seats must not result in fewer seats for any state. The* ***Alabama Paradox*** *occurs when the total number of available seats increases, yet one state (or more) loses seats as a result. This paradox can occur when using* HAMILTON'S METHOD.

Example 1.53 Hamilton's Method Apportionment of 101 seats. Using Hamilton's method, recompute the apportionment from Example 1.49, if there are 101 seats instead of 100. Does the Alabama paradox occur?

Table 1.54

States	**State A**	**State B**	**State C**	**State D**	**Total**
Population	936	2,726	2,603	3,735	10,000
Standard Quota					
Lower Quota					
Surplus					
Apportioned Seats					101

The table below shows the Hamilton's Method apportionment with 101 seats.

Note that State A loses a seat compared to the 100-seat scenario, demonstrating the Alabama Paradox.

Table 1.55 Hamilton's Method (101 Seats)

States	**State A**	**State B**	**State C**	**State D**	**Total**
Population	936	2,726	2,603	3,735	10,000
Standard Quota	9.4536	27.5326	26.2903	37.7235	101.0000
Lower Quota	9	27	26	37	99
Fractional Part	0.4536	0.5326	0.2903	0.7235	-
Surplus		+1		+1	
Final Seats	9	28	26	38	101

▲

1.3 Chapter Exercises

Voting and Apportionment Practice Problems

1. The city council gave a questionnaire to its citizens, asking them their priorities for next year's budget: (P)olice, (R)oads, (S)chools, and (T)rash removal.

Preference	**15**	**30**	**18**	**17**	**2**
1st	P	T	P	R	R
2nd	R	R	S	T	S
3rd	S	S	T	S	P
4th	T	P	R	P	T

(a) Which option is selected using the plurality method?

(b) Which option is selected using the plurality-with-elimination method?

(c) Which option is selected using the Borda count method?

(d) Which option is selected using the pairwise comparison method?

Solution.

(a) Plurality Method: Count only first-place votes.

- P: $15 + 18 = 33$
- T: 30
- R: $17 + 2 = 19$
- S: 0

Winner: Police (P) with 33 votes.

(b) Plurality-with-Elimination:

- Round 1: S has 0 votes. Eliminate S.

Preference	**15**	**30**	**18**	**17**	**2**
1st	P	T	P	R	R
2nd	R	R	T	T	P
3rd	T	P	R	P	T

- Round 2: R has the fewest 1st place votes (19). Eliminate R. The 17 votes from (R, T, S, P) move to T. The 2 votes from (R, S, P, T) move to P.

Preference	**15**	**30**	**18**	**17**	**2**
1st	P	T	P	T	P
2nd	T	P	T	P	T

New totals: P: $15 + 18 + 2 = 35$; T: $30 + 17 = 47$.

Winner: Trash Removal (T).

(c) Borda Count: 1st = 4pts, 2nd = 3pts, 3rd = 2pts, 4th = 1pt.

- P: $15(4) + 30(1) + 18(4) + 17(1) + 2(2) = 183$
- R: $15(3) + 30(3) + 18(1) + 17(4) + 2(4) = 229$
- S: $15(2) + 30(2) + 18(3) + 17(2) + 2(3) = 184$
- T: $15(1) + 30(4) + 18(2) + 17(3) + 2(1) = 224$

Winner: Roads (R) with 229 points.

(d) Pairwise Comparison:

- P vs R: P=33, R=49. (R wins)

- P vs S: P=63, S=19. (P wins)
- P vs T: P=35, T=47. (T wins)
- R vs S: R=64, S=18. (R wins)
- R vs T: R=52, T=30. (R wins)
- S vs T: S=35, T=47. (T wins)

R has 3 wins, T has 2 wins, P has 1 win. Winner: Roads (R).

2. The preference table shows the results of an election among three candidates, A, B, and C.

Number of votes	**7**	**6**	**3**
1st choice	A	B	C
2nd choice	B	C	B
3rd choice	C	A	A

(a) Using the plurality method, who is the winner?

(b) Is the Majority satisfied?

(c) Is the Condorcet satisfied?

Solution. Total votes: $7 + 6 + 3 = 16$. Votes needed for a majority: $16/2 = 8$, so 9 votes are required.

(a) Plurality Method: Count only 1st choice votes:

- A: 7 votes
- B: 6 votes
- C: 3 votes

The winner is Candidate A.

(b) Majority Criterion: A candidate must have more than 50% of the first-place votes to be a majority winner. Since the leader (A) has only 7 votes and 9 were needed for a majority, there is no majority winner. The criterion is satisfied (vacuously) because the criterion only says "if a majority winner exists, they must win." Since none exists, the rule isn't broken.

(c) Condorcet Criterion: We check if the plurality winner (A) is also the Condorcet winner by doing pairwise comparisons:

- A vs B: A is preferred by 7 voters; B is preferred by $6 + 3 = 9$ voters. B wins.
- B vs C: B is preferred by $7 + 3 = 10$ voters; C is preferred by 6 voters. B wins.
- A vs C: A is preferred by 7 voters; C is preferred by $6 + 3 = 9$ voters. C wins.

Candidate B wins all head-to-head matches and is the Condorcet Winner. Since the plurality method chose A instead of B, the Condorcet Criterion is not satisfied in this election.

3. The preference table shows the results of an election among three candidates, A, B, and C.

Number of votes	10	4	2
1st choice	A	B	C
2nd choice	B	C	B
3rd choice	C	A	A

(a) Using the plurality-with-elimination method, who is the winner?

(b) Is the Majority satisfied?

(c) In the actual election, the 2 voters in the last column who voted C, B, and A, change their vote to A, B, C. Using the plurality-with-elimination method, which candidate wins the actual election?

Solution. Total votes: $10 + 4 + 2 = 16$. Majority needed: 9 votes.

(a) Plurality-with-Elimination:

- Round 1: A has 10 votes, B has 4, and C has 2. Since A has 10 votes (more than the majority of 9), A is the winner immediately.

Winner: Candidate A.

(b) Majority: Yes. Candidate A has a majority of first-place votes (10 out of 16) and won the election. Therefore, the Majority is satisfied.

(c) Changed Election: If the 2 voters for C change to A, the new totals are:

- A: $10 + 2 = 12$ votes
- B: 4 votes
- C: 0 votes

Winner: Candidate A.

4. If a candidate wins all pairwise comparisons with other candidates, that candidate is called the ***Condorcet winner***.

The mathematics department is hiring a new faculty member and the five-person hiring committee has interviewed four candidates: Adam(A), Beth(B), Carol(C), and Dan(D). They have decided to use Condorcet's method on their five ballots. Who gets the offer?

Choice	Voter 1	Voter 2	Voter 3	Voter 4	Voter 5
First	B	D	C	D	B
Second	A	B	A	A	A
Third	C	C	B	C	D
Fourth	D	A	D	B	C

Solution. To find the Condorcet winner, we must compare each candidate head-to-head. There are 5 voters, so 3 votes are needed to win a comparison.

- A vs. B:

  Voter 1: B, Voter 2: B, Voter 3: A, Voter 4: A, Voter 5: B.

  Result: B wins 3 to 2.

- A vs. C:

  Voter 1: A, Voter 2: C, Voter 3: C, Voter 4: A, Voter 5: A.

Result: A wins 3 to 2.

- A vs. D:
  Voter 1: A, Voter 2: D, Voter 3: A, Voter 4: D, Voter 5: A.
  Result: A wins 3 to 2.
- B vs. C:
  Voter 1: B, Voter 2: B, Voter 3: C, Voter 4: C, Voter 5: B.
  Result: B wins 3 to 2.
- B vs. D:
  Voter 1: B, Voter 2: D, Voter 3: B, Voter 4: D, Voter 5: B.
  Result: B wins 3 to 2.
- C vs. D:
  Voter 1: C, Voter 2: D, Voter 3: C, Voter 4: D, Voter 5: D.
  Result: D wins 3 to 2.

Summary of Wins:

- Beth (B) won 3 comparisons (vs. A, C, and D).
- Adam (A) won 2 comparisons (vs. C and D).
- Dan (D) won 1 comparison (vs. C).
- Carol (C) won 0 comparisons.

Since Beth (B) won every head-to-head match, she is the Condorcet winner and gets the offer.

5. A small city has 30 police officers to be apportioned among 4 precincts based on the population of each precinct. The populations are given in the following table.

Precinct	**1**	**2**	**3**	**4**	**Total**
Population	3117	1883	2776	3174	10,950

(a) Find the standard divisor.

(b) Find the standard quota for each precinct.

(c) Find the modified quota for the fourth precinct using Jefferson's method with a modified divisor of 350.

(d) Find the apportionment for the second precinct using Adams's method with a modified divisor of 380.

Solution. Part (a) & (b): Standard Divisor and Standard Quotas
The Standard Divisor (SD) is $10,950/30 = 365$.

Table 1.56 Hamilton's Method (Standard Quotas)

Category	Precinct 1	Precinct 2	Precinct 3	Precinct 4	Total
Population	3117	1883	2776	3174	10,950
Std. Quota	8.54	5.16	7.61	8.70	30.00
Lower Quota	8	5	7	8	28
Fraction	0.54	0.16	0.61	0.70	-
Surplus			+1	+1	
Hamilton Final	8	5	8	9	30

Part (c): Jefferson's Method (Modified Divisor $d = 350$)

Table 1.57 Jefferson's Method Calculation

Category	Precinct 1	Precinct 2	Precinct 3	Precinct 4	Total
Mod. Quota	8.91	5.38	7.93	9.07	-
Jefferson Apport.	8	5	7	9	29

The modified quota for the fourth precinct is 9.07.
Part (d): Adams's Method (Modified Divisor $d = 380$)

Table 1.58 Adams's Method Calculation

Category	Precinct 1	Precinct 2	Precinct 3	Precinct 4	Total
Mod. Quota	8.20	4.96	7.31	8.35	-
Adams Apport.	9	5	8	9	31

The apportionment for the second precinct using Adams's method is 5 officers.

6. A mother has an incentive program to get her five children to read more. She has 30 pieces of candy to divide among her children at the end of the week based on the number of minutes each of them spends reading.

Child	Abby	Bobby	Charli	Dave	Eddie	Total
Reading Times	138	142	188	218	64	750

(a) Use Hamilton's method to apportion 30 pieces of candy.

(b) At the last minute, the mother finds another piece of candy and does the apportionment again. Now, she has 31 pieces of candy to divide among her children at the end of the week based on the number of minutes each of them spends reading. Use Hamilton's method to apportion the candy among the children.

(c) Look at your answers for parts (a) and (b). What changed and why? Why is the change interesting?This is an example of which paradox?

Solution. Part (a): 30 Pieces (Standard Divisor = 25)

Table 1.59 Apportionment for 30 Pieces

Category	Abby	Bobby	Charli	Dave	Eddie	Total
Reading Min	138	142	188	218	64	750
Std. Quota	5.52	5.68	7.52	8.72	2.56	30.00
Lower Quota	5	5	7	8	2	27
Fraction	0.52	0.68	0.52	0.72	0.56	-
Surplus		+1		+1	+1	
Final Seats	5	6	7	9	3	30

Part (b): 31 Pieces (Standard Divisor = 24.1935)

Table 1.60 Apportionment for 31 Pieces

Category	**Abby**	**Bobby**	**Charli**	**Dave**	**Eddie**	**Total**
Reading Min	138	142	188	218	64	750
Std. Quota	5.704	5.870	7.771	9.011	2.645	31.00
Lower Quota	5	5	7	9	2	28
Fraction	0.704	0.870	0.771	0.011	0.645	-
Surplus	+1	+1	+1			
Final Seats	6	6	8	9	2	31

Part (c): Increasing the total number of items from 30 to 31 creates the **Alabama Paradox**. In this specific case, Eddie loses a candy even though the total amount of candy increased.

Chapter 2

(Module 2.) Personal Finance

In an era of digital banking and shifting economies, understanding your personal finances is a superpower. This resource moves beyond the classroom to show how smart financial habits today can create long-term stability and open doors for the opportunities you care about most.

2.1 Understanding Interest: Simple Interest

Interest is the dollar amount that we get paid for lending money or pay for borrowing money. When we deposit money in a financial institution, the institution pays us interest for its use. When we borrow money, interest is the price we pay for the privilege of using the money until we repay it.

Definition 2.1 Simple Interest. The amount of money that we deposit or borrow is called the ***principal***. The amount of interest depends on the principal, the interest ***rate***, which is given as a percent and various from bank to bank, and the length of time for which the money is deposited. ***Simple interest*** involves interest calculated ***only on the principal***.

$$Interest = Principal \times rate \times time$$

$$I = Prt$$

The rate, r, is expressed as a decimal when you calculating simple interest. ◆

Example 2.2 Calculating Simple interest for a year. You deposit \$2,000 in a savings account, which has a rate 6.0%. Find the interest at the end of the first year.We can use the formula for simple interest to find the interest at the end of the first year. $Interest = Principal \times rate \times time$ $Interest = 2000 \times 0.06 \times 1 = 120$ ▲

Example 2.3 Calculating Simple interest for more than a year. A student took out a simple interest loan for \$1,300 for two years at a rate of 7.5%to purchase a used car. What is the interest on the loan?We can use the formula for simple interest to find the interest on the loan. $Interest = Principal \times rate \times time$ $Interest = 1300 \times 0.075 \times 2 = 195$ ▲

When a loan is repaid, the interest is added to the original principal to find the total amount due. In Example 2.3, at the end of 2 years the student will have to repay.

$Principal + Interest = 1300 + (1300 \times 0.075 \times 2) = 1300 + 195 = 1495$

In general, after t years the amount due, A, can be determined as follows:

$$A = P + \underbrace{I}_{I=Prt} = P + Prt = P(1 + rt)$$

Definition 2.4 Future Value. ***Future value = Principal + Interest***: The amount due, A, is called the ***future value*** of the loan. The principal borrowed P is also known as the loan's ***present value***.

$$A = P(1 + rt)$$

◆

Example 2.5 Calculating Future Value. What is the future value of a \$1,000 loan at 5% for 3 years?We can use the formula for future value to find the amount due at the end of 3 years. $A = P(1 + rt)$ $A = 1000(1 + 0.05 \times 3) = 1000(1 + 0.15) = 1000 \times 1.15 = 1150$ ▲

Example 2.6 Determining a simple interest rate. You borrow \$2,500 from a friend and promise to pay back \$2,655 in six months. What simple interest rate will you pay?We can use the formula for future value to find the interest rate.

$$\begin{aligned} A &= P(1 + rt) \\ 2655 &= 2500(1 + r \times 0.5) \\ 2655 &= 2500 + 1250r \\ 1250r &= 155 \\ r &= \frac{155}{1250} \\ r &= 0.124 = 12.4\% \end{aligned}$$

▲

Example 2.7 Determining a present value. How much should you put an investment paying a simple interest rate of 8.0% if you need \$3000 in six months?We can use the formula for future value to find the present value.

$$\begin{aligned} A &= P(1 + rt) \\ 3000 &= P(1 + 0.08 \times 0.5) \\ 3000 &= P(1 + 0.04) & &= P(1.04) \\ P &= \frac{3000}{1.04} \\ P &= 2884.62 \end{aligned}$$

▲

Exercises for Practice

1. **Calculating a simple interest rate for a T-bill.** Treasury bills (T-bills) can be purchased from the U.S. Treasury Department. You buy a T-bill for \$981.60 that pays \$1,000 in 13 weeks. What simple interest rate, to the nearest tenth of a percent, does this T-bill earn?

   Answer. We can use the formula for future value to find the interest rate.

$$A = P(1 + rt)$$

$$\begin{aligned}
1000 &= 981.60(1 + r \times \frac{13}{52}) \\
1000 &= 981.60(1 + 0.25r) \\
1000 &= 981.60 + 245.40r \\
245.40r &= 18.40 \\
r &= \frac{18.40}{245.40} = 0.0749 = 7.5\%
\end{aligned}$$

2. **Calculating a present value for a CD.** A bank offers a CD that pays a simple interest rate of 4.5%. How much must you put in this CD now in order to have \$8,000 for a kitchen remodeling project in two years? Round up to the nearest cent.

   Answer. We can use the formula for future value to find the present value.

$$\begin{aligned}
A &= P(1 + rt) \\
8000 &= P(1 + 0.045 \times 2) \\
8000 &= P(1 + 0.09) = P(1.09) \\
P &= \frac{8000}{1.09} = 7339.45
\end{aligned}$$

3. **Calculating a future value for a savings account.** You deposit \$1,000 in a savings account at a bank that has a rate of 1.95%. Find a future value at the end of 5 years.

   Answer. We can use the formula for future value to find the amount due at the end of 5 years. $A = P(1 + rt)$ $A = 1000(1 + 0.0195 \times 5) = 1000(1 + 0.0975) = 1000 \times 1.0975 = 1097.50$

4. **Determining whether a statement makes sense or not, explain your reasoning..**

   (a) After depositing \$1,500 in an account at a rate of 4%, my balance at the end of the first year was \$(1500)(0.04).

   (b) I planned to save \$5,000 in four years, computed the present value to be \$3,846.153, so I round the principal to \$3,846.15.

   Answer.

   (a) This statement does not make sense because the balance at the end of the first year should be the future value, which is the sum of the principal and interest. The correct balance should be $\$1500 + (1500)(0.04) = \$1560$.

   (b) This makes sense. In fact, rounding down is the safer move here. Rounding Down (\$3,846.15): By starting with slightly less than the exact "perfect" number, you technically might end up a fraction of a cent short of your \$5,000 goal after four years. However, in practice, this is unlikely to be a significant issue, and it provides a small buffer to ensure you meet your target.

2.2 Saving Money: Compound Interest

In 1626, Peter Minuit convinced the wrappinger Indians to sell him Manhattan Island for \$24. If the native American had ***put the \$24 into a bank account***

at a 5% interest rate compound annually, by the year 2021 there would be well over \$5.6 billion in the account!

So, how did the present value of Manhattan in 1626 attain a future value of over \$5.6 billion in 2021, 395 years at 5% simple interest is

$$A = P(1+rt) = 24(1+(0.05)(395)) = 498.00$$

or \$498.00 compared to over \$5.6 billion.

Compound Interest vs. Simple Interest.

The difference between simple interest and compound interest is that simple interest is calculated only on the original principal, while compound interest is calculated on the original principal and any accumulated interest.

Compound interest is interest computed on the original principal as well as on any accumulated interest.

For example, suppose you deposit \$1,000 in a savings account at a rate of 5%. Following table shows how the investment grows.

Calculating Compound interest for a year.

Table 2.8 Compound Interest Growth of \$1000 in a savings account at a rate of 5%

Year	Beginning Balance (\$)	Interest Earned: $I = Prt$	New Balance(\$)
1	\$1,000.00	$I = \$1,000.00 \times 0.05 \times 1 = 50$	\$1,050.00
2	\$1,050.00	$I = \$1,050.00 \times 0.05 \times 1 = 52.50$	\$1,102.50
3	\$1,102.50	$I = \$1,102.50 \times 0.05 \times 1 = 55.13$	\$1,157.63
4	\$1,157.63	$I = \$1,157.63 \times 0.05 \times 1 = 57.88$	\$1,215.51
5	\$1,215.51		
6			

The fast way to determine the amount, A, in the account subject to compound interest is the following formula. If P dollars are deposited at rate r, in decimal form, subject to compound interest, then the amount, A, of money in the account after t years is given by

Definition 2.9 Compound Interest Paid Once a Year. Calculating the amount in an account ***for compound interest paid once year:***

$$A = P(1+r)^t$$

where the amount A is the account's ***future value*** and the principal P is its ***present value***. ◆

Example 2.10 Calculating Compound interest for a year. You deposit \$2,000 in a savings account, which has a rate 6%. Find the amount, A, of money in the account after 3 years subject to compound interest and find the interest.To find the amount A after 3 years, we use the compound interest formula

$$A = P(1+r)^t$$

where $P = 2000$, $r = 0.06$, and $t = 3$.

Substituting the values:

$$A = 2000(1+0.06)^3 = 2000(1.06)^3 = \$2382.03$$

The interest earned is calculated as:

$$I = A - P = 2382.03 - 2000 = \$382.03$$

▲

Calculating the amount in an account ***for compound interest paid more than once a year.*** The period of time between two interest payments is called the ***compounding period***.

If compound interest is paid once a year, the compounding period is one year. We say the interest is ***compounded annually***.

If compound interest is paid twice a year, the compounding period is six months. We say the interest is ***compounded semiannually***.

If compound interest is paid four times a year, the compounding period is three months. We say the interest is ***compounded quarterly***.

If compound interest is paid every month of a year, the compounding period is twelve months. We say the interest is ***compounded monthly***.

In general, if compound interest is paid $\boldsymbol{n}$ times per year, we say the interest is ***n compounding periods per year***.

Definition 2.11 Compound interest: Future Value. If P dollars are deposited at rate r, in decimal form, subject to compound interest paid n times per year, then the amount, A, of money in the account after t years is given by

$$A = P\left(1 + \frac{r}{n}\right)^{nt} \tag{2.1}$$

where the amount A is the account's ***future value*** and the principal P is its ***present value***. ◆

Example 2.12 Calculating Compound Interest for More Than a Year. You deposit \$7,500 in a savings account that has a rate of 6%. The interest is compounded monthly. How much money will you have after five years? Find the interest after five years.

Solution. To find the future value of the account, we use the compound interest formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Given $P = 7500$, $r = 0.06$, $n = 12$, and $t = 5$:

$$\begin{aligned} A &= 7500\left(1 + \frac{0.06}{12}\right)^{12\times 5} \\ &= 7500(1.005)^{60} \\ &\approx 10116.38 \end{aligned}$$

The account balance after five years is approximately \$10,116.38. The interest earned over five years is:

$$I = 10116.38 - 7500 = 2616.38$$

The interest earned is \$2,616.38. ▲

Definition 2.13 Compound Interest: Present Value. If A dollars are to be accumulated in t years in an account that pays rate r compounded n times per year, then the present value, P, that needs to be invested now is given by

$$P = \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}}$$

◆

Example 2.14 Calculating present value. How much money should be deposited in an account today that earns 8% compounded monthly so that it will accumulate to $20,000 in five years?To find the required deposit (Present Value), we use the formula:

$$P = \frac{A}{(1 + \frac{r}{n})^{nt}}$$

where:

- $A = 20,000$ (Future Value)
- $r = 0.08$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 5$ (Time in years)

Substituting these values into the formula gives:

$$P = \frac{20,000}{(1 + \frac{0.08}{12})^{12\times 5}}$$
$$P = \frac{20,000}{(1.0066667)^{60}}$$
$$P \approx 13,424.21$$

Therefore, a deposit of $\$13,424.21$ is required today. ▲

The ***annual percentage yield (APY),*** or the ***effective interest rate***, is the simple interest rate that produces the same amount of money in an account at the end of one year as when the account is subject to compound interest at a stated rate.

Definition 2.15 Annual percentage yield (APY). Suppose that an investment has a nominal interest rate, r, in decimal form, and pays compound interest n times per year. The investment's ***effective annual yields***, APY, is given by

$$Y = \left(1 + \frac{r}{n}\right)^{n} - 1 \tag{2.2}$$

◆

Example 2.16 Understanding effective annual yield. You deposit $4000 in an account that pays 8% interest compounded monthly. Use the future value formula for simple interest to determine the effective annual yield.By the formula for future value

$$A = P(1 + \frac{r}{n})^{nt} = 4000(1 + \frac{0.08}{12})^{12\cdot 1} \approx \$4,332.00$$

Rounded to nearest cent, the amount in the savings account after one year in $4332.00.

The effective annual yield is a simple interest rate. We use the future value formula for simple interest to determine the simple interest rate that produces a future value of $4332 for $4000 deposit after one year.

$$A = P(1 + rt)$$
$$4332 = 4000(1 + r(1))$$
$$4332 = 4000 + 4000r$$

$$332 = 4000r$$
$$r = 0.083 = 8.3\%$$

You will pay a simple interest rate of 8.3%. This means that money invested at 8.3% simple interest earns the same amount in one year as money invested at 8%interest compounded monthly. ▲

Example 2.17 Calculating Effective Annual Yield. A passbook savings account has a nominal rate of 5%. The interest is compounded daily. Find the account's effective annual yield, assuming a 365-day year.

Solution. The Effective Yield (APY) represents the actual interest earned in one year. It is calculated using the formula:

$$Y = \left(1 + \frac{r}{n}\right)^n - 1$$

where:

- $r = 0.05$ (Nominal annual interest rate)
- $n = 365$ (Compounding periods per year)

Substituting the given values:

$$Y = \left(1 + \frac{0.05}{365}\right)^{365} - 1$$
$$Y \approx (1.000136986)^{365} - 1$$
$$Y \approx 1.051267 - 1$$
$$Y \approx 0.051267$$

Converted to a percentage, the effective annual yield is approximately 5.13%. ▲

Example 2.18 Quarterly Compounding and Interest Calculation. You deposit $\$1,000$ in a savings account at a bank that has a rate of 2.95%. The interest is compounded quarterly.

a How much money will you have after ten years? Round up to the nearest cent.

b Find the interest.

Solution. First, we identify the variables for the compound interest formula:

- $P = 1,000$ (Principal)
- $r = 0.0295$ (Annual rate)
- $n = 4$ (Quarterly compounding)
- $t = 10$ (Years)

For part (a), we calculate the future value (A):

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$
$$A = 1,000\left(1 + \frac{0.0295}{4}\right)^{4\times10}$$
$$A = 1,000(1.007375)^{40}$$

$$A \approx 1,341.517$$

Rounding up, the account balance will be $\$1,341.52$.

For part (b), we find the interest earned (I) by subtracting the principal from the future value:

$$\begin{aligned} I &= A - P \\ I &= 1,341.52 - 1,000 \\ I &= 341.52 \end{aligned}$$

The interest earned is $341.52. ▲

Checkpoint 2.19 Comparing Investment Options. Suppose that you have $\$10,000$ to invest. Which investment yields the greater return over 5 years: 7.55% compounded quarterly or 7.6% compounded semiannually?

Solution. To determine which investment is superior, we calculate the future value (A) for both options using the formula $A = P(1+\frac{r}{n})^{nt}$, where $P = 10,000$ and $t = 5$.

Option 1: 7.55% **compounded quarterly** Here, $r = 0.0755$ and $n = 4$.

$$\begin{aligned} A &= 10,000\left(1 + \frac{0.0755}{4}\right)^{4\times 5} \\ A &= 10,000(1.018875)^{20} \\ A &\approx 14,530.01 \end{aligned}$$

Option 2: 7.6% **compounded semiannually** Here, $r = 0.076$ and $n = 2$.

$$\begin{aligned} A &= 10,000\left(1 + \frac{0.076}{2}\right)^{2\times 5} \\ A &= 10,000(1.038)^{10} \\ A &\approx 14,520.23 \end{aligned}$$

Comparing the two totals, Option 1 results in $\$14,530.01$ while Option 2 results in $\$14,520.23$.

Therefore, the 7.55% investment compounded quarterly yields the greater return by $9.78.

Checkpoint 2.20 Present Value with Quarterly Compounding. How much money should be deposited today in an account that earns 5% compounded quarterly so that it will accumulate to $\$15,000$ in ten years?

Solution. To find the required initial deposit, we use the Present Value formula:

$$P = \frac{A}{(1+\frac{r}{n})^{nt}}$$

where:

- $A = 15,000$ (The target future value)
- $r = 0.05$ (The annual interest rate)
- $n = 4$ (Compounded quarterly)
- $t = 10$ (The time in years)

Substituting these values into the formula:

$$
\begin{aligned}
P &= \frac{15,000}{(1+\frac{0.05}{4})^{4\times 10}} \\
P &= \frac{15,000}{(1.0125)^{40}} \\
P &\approx \frac{15,000}{1.643619} \\
P &\approx 9,126.20
\end{aligned}
$$

Therefore, a deposit of $9,126.20 is required today to reach the goal of $15,000 in ten years.

Exercises for Practice

1. **Calculating Effective Annual Yield.** You deposit $7,500 in an account that pays 4.75% interest compounded monthly. Determine the effective annual yield.

   Solution. The Effective Annual Yield (APY) is the actual annual rate of return and is independent of the principal amount deposited. We use the formula:
   $$Y = \left(1+\frac{r}{n}\right)^n - 1$$
   where:

   - $r = 0.0475$ (Nominal annual interest rate)
   - $n = 12$ (Compounding periods per year)

   Substituting these values:

   $$
   \begin{aligned}
   Y &= \left(1+\frac{0.0475}{12}\right)^{12} - 1 \\
   Y &\approx (1.0039583)^{12} - 1 \\
   Y &\approx 1.048545 - 1 \\
   Y &\approx 0.048545
   \end{aligned}
   $$

   Converted to a percentage and rounded to two decimal places, the effective annual yield is 4.85%.

2. **Future Value for a Long-Term Investment.** At the time of her grandson's birth, a grandmother deposits $5,000 in an account that pays 6% compounded monthly. What will be the value of the account at the child's twenty-first birthday, assuming that no other deposits or withdrawals are made during this period?

   Solution. To find the future value of the account, we use the compound interest formula:
   $$A = P\left(1+\frac{r}{n}\right)^{nt}$$
   where:

   - $P = 5,000$ (Initial deposit)
   - $r = 0.06$ (Annual interest rate)
   - $n = 12$ (Compounding periods per year)

- $t = 21$ (Time in years until the 21st birthday)

Substituting these values into the formula:

$$A = 5,000\left(1 + \frac{0.06}{12}\right)^{12\times 21}$$
$$A = 5,000(1.005)^{252}$$
$$A \approx 5,000(3.514418)$$
$$A \approx \$17,572.09$$

On the child's twenty-first birthday, the account will be worth $\$17,572.09$.

3. Interest Rate Concepts: True or False. Determine whether each statement is True or False and explain your reasoning.

1 My bank provides simple interest at 3.25% per year, but I cannot determine if this is a better deal than a competing bank offering 3.25% compound interest without knowing the compounding period.

2 My bank advertises a compound interest rate of 2.4%, although, without making deposits or withdrawals, the balance in my account increased by 2.43% in one year.

Solution.

1 False. Any amount of compounding is better than simple interest when the nominal rates are identical (and greater than zero). While the compounding period determines *how much* better the deal is, we already know compound interest will yield more than simple interest regardless of whether it is compounded annually, monthly, or daily.

2 True. The advertised rate is the nominal rate (2.4%), but the 2.43% increase represents the *Effective Annual Yield*. Because interest is added to the balance throughout the year, the actual growth is slightly higher than the nominal rate. For example, 2.4% compounded monthly results in an effective yield of approximately 2.427%.

4. Quarterly Compounding vs. Simple Interest. A sum of $\$10,000$ is invested at an annual rate of 3%. Find the balance in the account after 5 years subject to:

1 quarterly compounding

2 simple interest

Solution. We are given $P = 10,000$, $r = 0.03$, and $t = 5$.

1 *Quarterly Compounding.*
Using the formula $A = P(1 + \frac{r}{n})^{nt}$ with $n = 4$:

$$A = 10,000\left(1 + \frac{0.03}{4}\right)^{4\times 5}$$
$$A = 10,000(1.0075)^{20}$$
$$A \approx 10,000(1.161184)$$
$$A \approx 11,611.84$$

The balance with quarterly compounding is $\$11,611.84$.

2 *Simple Interest.*

Using the formula $A = P(1 + rt)$:

$$\begin{aligned} A &= 10,000(1 + 0.03 \times 5) \\ A &= 10,000(1 + 0.15) \\ A &= 10,000(1.15) \\ A &= 11,500.00 \end{aligned}$$

The balance with simple interest is $\$11,500.00$.

Quarterly compounding results in an additional \$111.84 compared to simple interest.

5. Future Value and Effective Yield. You deposit $\$3,000$ in an account that pays 4.5% interest compounded monthly.

1 Find the future value after one year.

2 Determine the effective annual yield.

Solution. Given: $P = 3,000$, $r = 0.045$, $n = 12$, and $t = 1$.

1 To find the future value (A) after one year:

$$\begin{aligned} A &= P\left(1 + \frac{r}{n}\right)^{nt} \\ A &= 3,000\left(1 + \frac{0.045}{12}\right)^{12\times 1} \\ A &= 3,000(1.00375)^{12} \\ A &\approx 3,000(1.0459398) \\ A &\approx 3,137.82 \end{aligned}$$

The future value after one year is $\$3,137.82$.

2 To find the effective annual yield (EAY):

$$\begin{aligned} Y &= \left(1 + \frac{r}{n}\right)^{n} - 1 \\ Y &= \left(1 + \frac{0.045}{12}\right)^{12} - 1 \\ Y &\approx 1.0459398 - 1 \\ Y &\approx 0.04594 \end{aligned}$$

The effective annual yield is approximately 4.59%.

6. Planning for Education: Present Value. Parents wish to have $\$120,000$ available for a child's education. If the child is now 1 year old, how much money must be set aside at 8% compounded monthly to meet their financial goal when the child is 18?

Solution. To find the required initial investment, we use the Present Value formula:

$$P = \frac{A}{(1 + \frac{r}{n})^{nt}}$$

where:

- $A = 120,000$ (The target goal)

- $r = 0.08$ (Annual interest rate)
- $n = 12$ (Monthly compounding)
- $t = 17$ (Years remaining until the child is 18: $18 - 1 = 17$)

Substituting these values into the formula:

$$P = \frac{120,000}{(1 + \frac{0.08}{12})^{12\times 17}}$$
$$P = \frac{120,000}{(1.0066667)^{204}}$$
$$P \approx \frac{120,000}{3.876778}$$
$$P \approx 30,953.54$$

The parents must set aside $30,953.54 today to meet the $120,000 goal in 17 years.

7. **The Historical DeHaven Debt Calculation.** In 1777, Jacob DeHaven loaned George Washington's army $450,000 in gold and supplies. Due to a disagreement over repayment, he was never repaid. In 1989, his descendants sued the U.S. government over the 212-year-old debt. If the DeHavens used an interest rate of 6% and daily compounding, how much money did the DeHaven family demand in their lawsuit?

Solution. To find the total value of the debt after 212 years, we use the compound interest formula:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

where:

- $P = 450,000$ (Initial loan)
- $r = 0.06$ (Annual interest rate)
- $n = 365$ (Daily compounding)
- $t = 212$ (Years from 1777 to 1989)

Substituting these values:

$$A = 450,000\left(1 + \frac{0.06}{365}\right)^{365\times 212}$$
$$A = 450,000(1.00016438)^{77,380}$$
$$A \approx 450,000(336,543.15)$$
$$A \approx 151,444,417,500$$

The DeHaven family's demand in the lawsuit was approximately $151.4 billion. (Depending on rounding, specific historical claims often cited amounts ranging from $141 to $151 billion).

2.3 Saving Money: Annuities

An **annuity** is a sequence of equal (regular) payments made into an account over time periods. An IRA is an example of an annuity. The value of an annuity is the sum of all deposits plus all interest paid.

Example 2.21 Annuity Basics. Suppose you deposit \$1,000 into a savings plan at the end of each year for three years. The interest rate is 8% per year compounded annually.

1 Find the value of the annuity after three years.

2 Find the interest earned.

Solution. We calculate the balance step-by-step:

- Value at the end of year 1: \$1,000
- Value at the end of year 2: $1,000(1 + 0.08) + 1,000 = \$2,080$
- Value at the end of year 3: $2,080(1 + 0.08) + 1,000 = \$3,246.40$

The value of the annuity at the end of three years is \$3,246.40. Since you made three payments of \$1,000, the total principal deposited is \$3,000.

The interest is:

$$I = 3,246.40 - 3,000 = \$246.40$$

▲

Definition 2.22 Value of an Annuity: Interest Compounded n Times per Year. If P is the deposit made at the end of each compounding period for an annuity that pays an annual interest rate r compounded n times per year, the future value, FV or A, of the annuity after t years is:

$$FV = A = \frac{P\left[\left(1 + \frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)} \tag{2.3}$$

◆

Example 2.23 Retirement Savings: IRA Annuity. To save for retirement, you decide to deposit \$1,000 into an IRA at the end of each year for the next 30 years. If you can count on an interest rate of 10% per year compounded annually:

1 How much will you have from the IRA after 30 years?

2 Find the interest.

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1 + \frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)}$$

where $P = 1,000$, $r = 0.10$, $n = 1$, and $t = 30$.

1 Substituting the values into the formula:

$$A = \frac{1,000\left[(1 + 0.10)^{30} - 1\right]}{0.10}$$

$$A = \frac{1,000\left[(1.1)^{30} - 1\right]}{0.10}$$
$$A \approx \frac{1,000(17.4494 - 1)}{0.10}$$
$$A \approx 164,494.02$$

After 30 years, the IRA will have $164,494.02.

2 To find the interest, we subtract the total deposits from the future value. Total deposits $= 1,000 \times 30 = \$30,000$.

$$I = 164,494.02 - 30,000$$
$$I = 134,494.02$$

The total interest earned is $134,494.02.

▲

Example 2.24 Retirement Savings with Monthly Compounding. At age 25, to save for retirement, you decide to deposit $200 at the end of each month into an IRA that pays 7.5% compounded monthly.

1 How much will you have from the IRA when you retire at age 65?

2 Find the interest.

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1 + \frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 200$ (Monthly deposit)
- $r = 0.075$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 40$ (Years of investing: $65 - 25 = 40$)

1 Substituting the values into the formula:

$$A = \frac{200\left[\left(1 + \frac{0.075}{12}\right)^{12\times 40} - 1\right]}{\left(\frac{0.075}{12}\right)}$$
$$A = \frac{200\left[(1.00625)^{480} - 1\right]}{0.00625}$$
$$A \approx \frac{200(19.92053 - 1)}{0.00625}$$
$$A \approx 605,457.09$$

At age 65, the IRA will be worth $605,457.09.

2 To find the interest, subtract the total deposits from the future value. Total deposits $= 200 \times 12 \times 40 = \$96,000$.

$$I = 605,457.09 - 96,000$$
$$I = 509,457.09$$

The total interest earned is $509,457.09.

▲

Checkpoint 2.25 Calculating Future Value of an Annuity. To save for a down payment on a home, you deposit \$500 at the end of each quarter into an account that pays 4% compounded quarterly. How much will be in the account after 6 years?

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 500$ (Quarterly deposit)
- $r = 0.04$ (Annual interest rate)
- $n = 4$ (Compounded quarterly)
- $t = 6$ (Number of years)

Substituting these values into the formula:

$$A = \frac{500\left[\left(1+\frac{0.04}{4}\right)^{4\times 6} - 1\right]}{\left(\frac{0.04}{4}\right)}$$
$$A = \frac{500\left[(1.01)^{24} - 1\right]}{0.01}$$
$$A \approx \frac{500(1.26973 - 1)}{0.01}$$
$$A \approx 13,486.50$$

After 6 years, the account balance will be $\$13,486.50$.

Example 2.26 Logic Check: Annuities vs. Lump Sums. Determine whether each statement is True or False and explain your reasoning.

1 With the same interest rate, compounding period, and time period, an annuity will generate more interest than a lump sum deposit.

2 By putting \$20 at the end of each month into an IRA that pays 3.5% compounded monthly, I'll be able to retire comfortably in just 30 years.

Solution.

1 False. A lump sum deposit of the same total amount invested earns more interest because the entire amount is present at the beginning and earns interest for the full duration. In an annuity, the money is deposited gradually, so later payments have less time to grow.

2 False. We can calculate the total using the annuity formula where $P = 20$, $r = 0.035$, $n = 12$, and $t = 30$:

$$A = \frac{20\left[(1+\frac{0.035}{12})^{12\times 30} - 1\right]}{\frac{0.035}{12}}$$
$$A \approx \$12,705.54$$

A total of $\$12,705.54$ is not nearly enough to sustain a comfortable retirement.

▲

Example 2.27 Sinking Fund: Monthly Savings for College. Parents of a baby girl are in a financial position to begin saving for her college education. They plan to have $100,000 in a college fund in 18 years by making regular, end-of-month deposits in an annuity that pays 9% compounded monthly.

1 How much should they deposit each month?

2 How much of the goal comes from interest?

Solution. To find the required monthly payment (P), we use the sinking fund formula, which is the annuity formula solved for P:

$$P = \frac{A\left(\frac{r}{n}\right)}{\left[\left(1+\frac{r}{n}\right)^{nt} - 1\right]}$$

where $A = 100,000$, $r = 0.09$, $n = 12$, and $t = 18$.

1 Substituting the values:

$$P = \frac{100,000\left(\frac{0.09}{12}\right)}{\left[\left(1+\frac{0.09}{12}\right)^{12\times 18} - 1\right]}$$
$$P = \frac{750}{(1.0075)^{216} - 1}$$
$$P \approx \frac{750}{5.022575 - 1}$$
$$P \approx 186.45$$

The parents should deposit $186.45 each month.

2 To find the interest, we first calculate the total principal deposited:

$$186.45 \times 12 \times 18 = \$40,273.20$$

Subtracting this from the future value:

$$I = 100,000 - 40,273.20 = \$59,726.80$$

The amount from interest is $59,726.80.

▲

Checkpoint 2.28 Calculating the Value of an Annuity. You deposit $2,000 into a savings plan at the end of each year for three years. The interest rate is 10% per year compounded annually.

1 Find the value of the annuity after three years.

2 Find the interest earned.

Solution. We use the future value formula for an annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)}$$

where $P = 2000$, $r = 0.10$, $n = 1$, and $t = 3$.

1 Substituting the values into the formula:

$$\begin{aligned} A &= \frac{2000\left[(1+0.10)^3-1\right]}{0.10} \\ &= \frac{2000\left[(1.1)^3-1\right]}{0.10} \\ &= \frac{2000(1.331-1)}{0.10} \\ &= 6620 \end{aligned}$$

The value of the annuity after three years is \$6,620.

2 The total amount of deposits made is $3 \times 2000 = 6000$. The interest earned is the difference between the future value and the total deposits:

$$I = 6620 - 6000 = 620$$

The interest earned is \$620.

Checkpoint 2.29 Retirement Savings: Growth After Contributions Cease. To save for retirement, you decide to deposit \$3,000 into an IRA at the end of each year for the ***next 10*** years. If you can count on an interest rate of 9% per year compounded annually:

1 How much will you have from the IRA ***after 40*** years, assuming no additional deposits or withdrawals after the 10 years of contribution?

2 Find the interest earned.

Solution. This problem requires two steps. First, we find the value of the annuity after the 10-year contribution period. Second, we treat that amount as a lump sum that grows for the remaining 30 years.

Step 1: Future Value of the Annuity (Years 1-10)

$$\begin{aligned} A_{10} &= \frac{3000\left[(1+0.09)^{10}-1\right]}{0.09} \\ &= \frac{3000(2.36736-1)}{0.09} \\ &\approx 45578.80 \end{aligned}$$

Step 2: Compound Interest on the Lump Sum (Years 11-40) The balance grows for $40 - 10 = 30$ more years using the formula $A = P(1+r)^t$:

$$\begin{aligned} A_{40} &= 45578.80(1+0.09)^{30} \\ &= 45578.80(13.26768) \\ &\approx 604725.10 \end{aligned}$$

1 After 40 years, the total value will be \$604,725.10.

2 The total deposits were $3000 \times 10 = 30000$.

$$I = 604725.10 - 30000 = 574725.10$$

The interest earned is \$574,725.10.

Exercises for Practice

1. **Retirement Savings Analysis.** At age 30 to save for retirement, you decide to deposit \$200 at the end of each month into an IRA that pays 13.8% compounded monthly.

   1 How much will you have from the IRA when you retire at age 65?

   2 Find the interest earned.

   Solution. We use the future value formula for an ordinary annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

   where:

   - $P = 200$ (Monthly deposit)
   - $r = 0.138$ (Annual interest rate)
   - $n = 12$ (Compounding periods per year)
   - $t = 35$ (Years of saving: $65 - 30 = 35$)

   1 Substituting the values into the formula:

$$A = \frac{200\left[\left(1+\frac{0.138}{12}\right)^{12\times 35}-1\right]}{\left(\frac{0.138}{12}\right)}$$
$$A = \frac{200\left[(1.0115)^{420}-1\right]}{0.0115}$$
$$A \approx 2,100,992.95$$

   You will have approximately $\$2,100,992.95$ in the IRA at age 65.

   2 First, find the total amount of your deposits over 35 years:

$$200 \times 12 \times 35 = \$84,000$$

   The interest earned is the future value minus these deposits:

$$2,100,992.95 - 84,000 = \$2,016,992.95$$

   The total interest earned is $\$2,016,992.95$.

2. **Calculating Monthly Deposits to Reach a Goal.** How much should you deposit at the end of each month into an IRA that pays 6.5% compounded monthly to have \$2 million when you retire in 45 years? How much of the \$2 million comes from interest?

   Solution. To find the required monthly payment (P), we use the sinking fund formula:

$$P = \frac{A\left(\frac{r}{n}\right)}{\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}$$

   where:

   - $A = 2,000,000$ (The target goal)

- $r = 0.065$ (Annual interest rate)
- $n = 12$ (Monthly compounding)
- $t = 45$ (Time in years)

Substituting these values into the formula:

$$P = \frac{2,000,000\left(\frac{0.065}{12}\right)}{\left[\left(1+\frac{0.065}{12}\right)^{12\times 45} - 1\right]}$$
$$P = \frac{10,833.333}{(1.0054167)^{540} - 1}$$
$$P \approx \frac{10,833.333}{21.21736 - 1}$$
$$P \approx 510.59$$

The required monthly deposit is \$510.59.

To find the interest, we first calculate the total of all deposits:

$$510.59 \times 12 \times 45 = \$275,718.60$$

Subtracting this from the $\$2,000,000$ final value:

$$I = 2,000,000 - 275,718.60 = \$1,724,281.40$$

Over the 45 years, $\$1,724,281.40$ of the final balance comes from interest.

3. **Long-Term Retirement Savings Analysis.** At age 25, to save for retirement, you decide to deposit \$100 at the end of each month in an IRA that pays 6.25% compounded monthly. How much will you have from the IRA when you retire at age 70? What is the interest you earned?

Solution. We use the future value formula for an ordinary annuity:

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt} - 1\right]}{\left(\frac{r}{n}\right)}$$

where:

- $P = 100$ (Monthly deposit)
- $r = 0.0625$ (Annual interest rate)
- $n = 12$ (Compounding periods per year)
- $t = 45$ (Years of saving: $70 - 25 = 45$)

Substituting the values into the formula:

$$A = \frac{100\left[\left(1+\frac{0.0625}{12}\right)^{12\times 45} - 1\right]}{\left(\frac{0.0625}{12}\right)}$$
$$A = \frac{100\left[(1.0052083)^{540} - 1\right]}{0.0052083}$$
$$A \approx \frac{100(16.28236 - 1)}{0.0052083}$$
$$A \approx 293,421.36$$

At age 70, you will have approximately $\$293,421.36$ in the IRA.

To find the interest earned, first calculate the total of your deposits:

$$100 \times 12 \times 45 = \$54,000$$

Subtracting the deposits from the future value:

$$I = 293,421.36 - 54,000 = \$239,421.36$$

The total interest earned is $\$239,421.36$.

4. **Quarterly Deposits to Reach a Retirement Goal.** How much should you deposit at the end of every 3 months into an IRA that pays 6% compounded quarterly to have \$1.5 million when you retire in 40 years? How much of the \$1.5 million comes from interest?

Solution. To find the required quarterly payment (P), we use the sinking fund formula:

$$P = \frac{A\left(\frac{r}{n}\right)}{\left[\left(1+\frac{r}{n}\right)^{nt} - 1\right]}$$

where:

- $A = 1,500,000$ (The target goal)
- $r = 0.06$ (Annual interest rate)
- $n = 4$ (Quarterly compounding)
- $t = 40$ (Time in years)

Substituting these values into the formula:

$$P = \frac{1,500,000\left(\frac{0.06}{4}\right)}{\left[\left(1+\frac{0.06}{4}\right)^{4\times 40} - 1\right]}$$

$$P = \frac{22,500}{(1.015)^{160} - 1}$$

$$P \approx \frac{22,500}{10.85055 - 1.85055}$$

$$P \approx 2,284.14$$

The required quarterly deposit is $\$2,284.14$.

To find the interest, we first calculate the total of all deposits over 40 years:

$$2,284.14 \times 4 \times 40 = \$365,462.40$$

Subtracting this from the $\$1,500,000$ final value:

$$I = 1,500,000 - 365,462.40 = \$1,134,537.60$$

Over the 40 years, $\$1,134,537.60$ of the final balance comes from interest.

2.4 Borrowing Money: Mortgage

This section introduces fixed-installment loans featuring constant interest rates. The standard amortization formula explored here applies broadly to any fixed-rate financing, including auto loans, student loans, and home improvement loans. Mortgages follow this exact same mathematical structure, differing only by their extended repayment timelines, which typically span 15, 30, or 40 years.

.

A ***mortgage*** is a long-term loan (perhaps up to 15 or 30 years) for the purpose of buying a home, and for which the property is pledged as security for payment.

The ***down payment*** is the portion of the sale price of the home that the buyer initially pays to the seller.

The minimum required down payment is computes as a percentage of the sale price.

The ***amount of the mortgage*** is the difference between the sale price and the down payment.

The ***points*** are charged for your mortgage. One point equals one percent of the loan amount.

For example, you make a 10% down payment of \$120,000 home, the amount of the mortgage is $\$120,000 - \$12,000 = \$108,000$. So, if your mortgage is \$108,000, one point is \$1,080. You pay these points up front when you begin the mortgage.

Many financial institutions or mortgage companies provide mortgage loans. Some companies, called ***mortgage broker*** offer to find you a mortgage lender willing to make you a loan. Mortgages can have a fixed interest rate or a variable interest rate. ***Fixed-rate mortgages*** have same monthly payment during the entire time of the loan.

Definition 2.30 Loan Payment Formula for Fixed Installment Loans. The regular payment amount, PMT, required to repay a loan of P dollars paid n times per year over t years at an annual rate r is given by:

$$PMT = \frac{P\left(\frac{r}{n}\right)}{\left[1 - \left(1 + \frac{r}{n}\right)^{-nt}\right]} \tag{2.4}$$

where:

PMT	is regular Payment Amount (usually a monthly payment)
P	is the principal amount (loan balance)
r	is the annual interest rate (as a decimal)
n	is the number of payments per year
t	is the total number of years

◆

Example 2.31 . The price of a home is \$195,000. The bank requires a 10% down payment and two points at the time of closing. The cost of the home is financed with 30-year fixed rate mortgage at 7.5%.

1. Find the required down payment.
2. Find the amount of the mortgage.
3. How much must be paid for the two points at closing?
4. Find the monthly payment
5. Find the total interest paid over 30 years.

Solution.

1. The required down payment is 10% of \$195,000 or

$$0.10 \times \$195,000 = \$19,500.$$

2. The amount of the mortgage is the difference between the price of the home and the down payment:

$$\text{Amount of mortgage} = \$195,000 - \$19,500 = \$175,500.$$

3. To find the cost of two points on a mortgage of \$175,500:

$$2 \text{ points} = 2\% \text{ of mortgage} = 0.02 \times \$175,500 = \$3,510.$$

   The down payment (\$19,500) is paid to the seller and the cost of two points (\$3,510) is paid to the lending institution.

   Computing Monthly Mortgage Payments. The monthly mortgage payment, PMT, required to repay a loan of P dollars paid n times per year over t years at an annual rate r is:

$$PMT = \frac{P\left(\frac{r}{n}\right)}{\left[1 - \left(1 + \frac{r}{n}\right)^{-nt}\right]}$$

4. The monthly payments are:

$$PMT = \frac{175500\left(\frac{0.075}{12}\right)}{\left[1 - \left(1 + \frac{0.075}{12}\right)^{-12\cdot 30}\right]} \approx \$1,227.12$$

5. The total payments are calculated by multiplying the monthly payment by the total number of periods (360 months):

$$\begin{aligned} \text{Total Payments} &= \text{monthly payments} \times \text{month} \times \text{years} \\ &= \$1,227.12 \times 360 = \$441,763.20. \end{aligned}$$

   The interest paid over 30 years is the total payments minus the original principal:

$$\begin{aligned} \text{Interest} &= \text{total payments} - \text{amount of mortgage} \\ &= \$441,763.20 - \$175,500 = \$266,263.20. \end{aligned}$$

▲

Example 2.32 Comparing 30-Year and 15-Year Mortgages. In Example 2.31,a mortgage of \$175,500 was financed with a 30-year fixed rate at 7.5%.

1. Find the monthly payment for the 30-year mortgage.
2. Find the total interest paid over 30 years.
3. Find the total interest paid if the loan was instead for 15 years at the same rate.
4. How much interest is saved by reducing the mortgage from 30 to 15 years?

Solution.

1. We use the formula

$$PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$$

with $P = 175500$, $r = 0.075$, $n = 12$, and $t = 30$:

$$PMT = \frac{175500(0.075/12)}{1-(1+0.075/12)^{-12\cdot 30}} \approx \$1,227.12$$

2. The total interest for 30 years is the total payments minus the principal:

$$\text{Total Payments} = \$1,227.12 \times 360 = \$441,763.20$$

$$\text{Interest}_{30} = \$441,763.20 - \$175,500 = \$266,263.20$$

3. First, find the 15-year monthly payment ($t = 15$):

$$PMT = \frac{175500(0.075/12)}{1-(1+0.075/12)^{-12\cdot 15}} \approx \$1,626.91$$

Now calculate the total 15-year interest:

$$\text{Total Payments} = \$1,626.91 \times 180 = \$292,843.80$$

$$\text{Interest}_{15} = \$292,843.80 - \$175,500 = \$117,343.80$$

4. The interest saved is the difference between the two interest totals:

$$\text{Savings} = \$266,263.20 - \$117,343.80 = \$148,919.40$$

▲

Example 2.33 Condominium Mortgage Analysis. The price of a condominium is \$280,000. The bank requires a 5% down payment and one point at the time of closing. The cost of the condominium is financed with a 30-year fixed-rate mortgage at 8%.

1. Find the required down payment.
2. Find the amount of the mortgage.
3. How much must be paid for the one point at closing?
4. Find the monthly payment by using the monthly payment formula.
5. Find the total cost of interest over 30 years.

Solution.

1. The down payment is 5% of the purchase price:

$$0.05 \times \$280,000 = \$14,000$$

2. The mortgage amount is the price minus the down payment:

$$\$280,000 - \$14,000 = \$266,000$$

3. One point is 1% of the mortgage amount:

$$0.01 \times \$266,000 = \$2,660$$

4. Using the formula
$$PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$$
with $P = 266,000$, $r = 0.08$, $n = 12$, and $t = 30$:

$$PMT = \frac{266000(0.08/12)}{1-(1+0.08/12)^{-360}} \approx \$1,951.81$$

5. Total payments over 30 years are

$$\$1,951.81 \times 360 = \$702,651.60.$$

The total interest is:

$$\text{Total Interest} = \$702,651.60 - \$266,000 = \$436,651.60$$

▲

Checkpoint 2.34 Student Loan Term Comparison. A student graduates from college with a loan of \$36,000 at an 8% annual interest rate.

1. Find the monthly payments and the total interest for a 10-year loan term.
2. If the interest rate remains at 8% and the loan term is reduced to 5 years, how much more must the student pay each month and how much less will be paid in total interest?

Answer.

1. Monthly payment: \$436.78; Total interest: \$16,413.60
2. Monthly increase: \$293.07; Interest saved: \$8,414.40

Solution.

1. For $t = 10$:

$$PMT = \frac{36000(0.08/12)}{1-(1+0.08/12)^{-120}} \approx \$436.78$$

$$\text{Total Interest} = (\$436.78 \times 120) - \$36,000 = \$16,413.60$$

2. For $t = 5$:
$$PMT = \frac{36000(0.08/12)}{1-(1+0.08/12)^{-60}} \approx \$729.85$$
$$\text{Total Interest} = (\$729.85 \times 60) - \$36,000 = \$7,791.00$$

The monthly increase is $\$729.85 - \$436.78 = \$293.07$. The interest saved is $\$16,413.60 - \$7,791.00 = \$8,422.60$. (Note: Small differences may occur due to rounding).

Checkpoint 2.35 Mortgage Comparison with Points and Closing Costs. Which mortgage loan has the greater total cost and by how much?

Consider a \$210,000 mortgage with these two options:

- **Mortgage A**: 20-year fixed at 7% with closing costs of \$3,000 and 2 points.
- **Mortgage B**: 20-year fixed at 6.75% with closing costs of \$1,750 and 4 points.

Answer. Mortgage A has the greater total cost by \$4,577.20.

Solution. The total cost for each mortgage includes the sum of all monthly payments, the cost of points, and the fixed closing costs.

1. *Mortgage A Calculations.*

   First, calculate the monthly payment (PMT):

   $$PMT = \frac{210000(0.07/12)}{1-(1+0.07/12)^{-240}} \approx \$1,628.13$$

   Next, find the total cost of payments, points, and closing:

   $$\begin{aligned} \text{Total Payments} &= \$1,628.13 \times 240 = \$390,751.20 \\ \text{Points Cost} &= 0.02 \times \$210,000 = \$4,200.00 \\ \text{Total Cost}_A &= \$390,751.20 + \$4,200.00 + \$3,000.00 = \$397,951.20 \end{aligned}$$

2. *Mortgage B Calculations.*

   First, calculate the monthly payment (PMT):

   $$PMT = \frac{210000(0.0675/12)}{1-(1+0.0675/12)^{-240}} \approx \$1,596.76$$

   Next, find the total cost:

   $$\begin{aligned} \text{Total Payments} &= \$1,596.76 \times 240 = \$383,222.40 \\ \text{Points Cost} &= 0.04 \times \$210,000 = \$8,400.00 \\ \text{Total Cost}_B &= \$383,222.40 + \$8,400.00 + \$1,750.00 = \$393,372.40 \end{aligned}$$

3. *Comparison.*

   Mortgage A is more expensive. The difference is:

   $$\$397,951.20 - \$393,372.40 = \$4,578.80$$

   (Note: Differences in final cents may vary based on rounding the monthly payment).

Definition 2.36 Maximum Purchase Price Formula. The ***maximum purchase price*** for a house is given by

$$\text{maximum purchase price} = \frac{PMT \cdot \left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$$

where:

PMT	is the monthly mortgage payment
n	is the number of payments per year
t	is the total number of years (loan term)

r is the annual interest rate (as a decimal)

◆

Example 2.37 Calculating Maximum Mortgage Amount. Amanda and Fred are buying a house on a 30-year mortgage. They can afford a monthly payment of \$800. If the annual interest rate (APR) is 3.75%, what is the maximum mortgage amount they can take out?

Solution. We use the maximum purchase price formula with $PMT = 800$, $r = 0.0375$, $n = 12$, and $t = 30$:

$$\text{Max Price} = \frac{800 \cdot \left[1 - \left(1 + \frac{0.0375}{12}\right)^{-12 \cdot 30}\right]}{\left(\frac{0.0375}{12}\right)}$$

First, calculate the periodic interest rate: $\frac{0.0375}{12} = 0.003125$.
Now, substitute into the formula:

$$\begin{aligned} \text{Max Price} &= \frac{800 \cdot \left[1 - (1.003125)^{-360}\right]}{0.003125} \\ \text{Max Price} &= \frac{800 \cdot [1 - 0.325016]}{0.003125} \\ \text{Max Price} &= \frac{800 \cdot 0.674984}{0.003125} \\ \text{Max Price} &\approx \$172,795.90 \end{aligned}$$

Amanda and Fred can take out a mortgage of approximately ***\$172,795.90***.

▲

Checkpoint 2.38 Home Affordability and Loan Terms. Suppose you want to purchase a house and your take-home pay is \$3,220 per month. You wish to stay within recommended guidelines by spending 1/4 of your take-home pay on a house payment. You have \$15,300 saved for a down payment. With your excellent credit, you can get an APR of 3.28% compounded monthly.

1. What is the total cost of a house you could afford with a 15-year mortgage?
2. What is the most that you could afford with a traditional 30-year mortgage?

Answer.

1. 15-year mortgage: \$130,175.76
2. 30-year mortgage: \$199,444.60

Solution. First, calculate the allowed monthly payment (PMT):

$$PMT = \frac{1}{4} \times \$3,220 = \$805$$

The monthly interest rate is $\frac{r}{n} = \frac{0.0328}{12} \approx 0.0027333$.

1. *15-Year Mortgage Affordability.*

   Using the Maximum Purchase Price formula with $n = 12$ and $t = 15$ ($nt = 180$):

$$P = \frac{805 \cdot [1 - (1 + 0.0027333)^{-180}]}{0.0027333} \approx \$114,875.76$$

The total house cost includes the down payment:

$$\text{Total Cost} = \$114,875.76 + \$15,300 = \$130,175.76$$

2. *30-Year Mortgage Affordability.*
Using the formula with $t = 30$ ($nt = 360$):

$$P = \frac{805 \cdot [1 - (1 + 0.0027333)^{-360}]}{0.0027333} \approx \$184,144.60$$

The total house cost includes the down payment:

$$\text{Total Cost} = \$184,144.60 + \$15,300 = \$199,444.60$$

2.5 Identify Formulas to use: Simple, Compound, Annuity, Loans

Exercises

1. Lonnie needs extra money to buy a truck to start up a delivery service. He takes out a simple interest loan for $7,000 for 6 months at a rate of 7.25%. How much interest must he pay, and what is the future value of the loan?

Answer. $I = Prt$; Interest: $253.75; Future Value: $7,253.75

2. If you placed $1 into an account that paid interest at a rate of 5% and compounded the interest daily, how much would that account is worth in 300 years?

Answer. $A = P(1 + r/n)^{nt}$; $3,268,864.51

3. You are purchasing a house for $155,000. The bank requires a down payment of 15% and 2.25 points at closing. The bank will give you a 20 year mortgage at 3.8%. What is the monthly payment you will make? What is the total interest paid?

Answer. $PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$; Monthly Payment: $783.50; Total Interest: $56,289.44

4. You deposit $8,500 for 9 years in an account that earns 2.5% interest and compounds weekly. How much interest will you earn?

Answer. $A = P(1 + r/n)^{nt}$; $2,143.91

5. You lend 650 dollars to your younger brother for tuition. He agrees to pay you back $720 in 4 months. What is the interest rate he paid you?

Answer. $r = \frac{I}{Pt}$; 32.31%

6. A mother invests $ 2000 in a bank account at the time of her daughter's birth. The interest is compounded quarterly at a rate of 8%. What will be the value of the daughter's account on her twentieth birthday?

Answer. $A = P(1 + r/n)^{nt}$; $9,750.88

7. Your grandmother lent you $2000 for 4 years to buy a car. You agreed to pay her 5% interest. How much do you have to pay her in total at the end of 4 years?

Answer. $A = P(1 + rt)$; $2,400.00

8. Which is the better choice: $1000 deposited for a year at a rate of 5.5% compounded semiannually or at a rate of 5.4% compounded quarterly?

Answer. $APY = (1 + r/n)^n - 1$; 5.5% compounded semiannually ($1,055.76 vs $1,055.10)

9. Ellen puts $100 each month into an IRA that pays 7.4% annually. How much money will she have in 30 years for her retirement? How much will come from interest?

Answer. $A = \frac{PMT[(1+r/n)^{nt}-1]}{r/n}$; Total Value: $132,410.87; Interest: $96,410.87

10. You want to save for your three year olds college education. You decide that you need $200,000 in 12 years. How much do you need to deposit monthly into an account that pays 8.7% interest and compounds monthly.

Answer. $PMT = \frac{A(r/n)}{(1+r/n)^{nt}-1}$; $796.86

11. Joe and Susan are purchasing a condo for $350,000. The bank requires a down payment of 30% and 4.75 points at closing. You are considering a 15 year mortgage at 4.75% and a 30 year mortgage at 3.2%. Which mortgage will result in a lower monthly payment?

Answer. $PMT = \frac{P(r/n)}{1-(1+r/n)^{-nt}}$; Option B: 30-year at 3.2% ($1,061.27 vs Option A: $1,906.18)

12. Pete wants to buy a boat in 4 years. The boat costs $100,000. How much money does he need to deposit weekly into an account that pays 3.5% interest compounded weekly?

Answer. $PMT = \frac{A(r/n)}{(1+r/n)^{nt}-1}$; $447.93

13. How much money should be deposited today in an account that earns 10% compounded semiannually so that it will accumulate to $ 1000 in 12 years?

Answer. $P = \frac{A}{(1+r/n)^{nt}}$; $310.07

14. Juan can save $50 every week for his honeymoon. If Juan is getting married in two years how much money will he have spent on the trip if he puts it into an annuity that pays 6%?

Answer. $A = \frac{PMT[(1+r/n)^{nt}-1]}{r/n}$; $5,516.48

15. How much money should be deposited today in an account that earns 8% compounded quarterly so that it will accumulate to $ 3800 in 12 years?

Answer. $P = \frac{A}{(1+r/n)^{nt}}$; $1,468.32

2.6 Chapter Exercises

Personal Finance Practice Problem

1. Suppose a local sales tax rate is 4.5% and you purchase a car for $\$28,500$.

a How much sales tax is paid?

b How much is the total cost of the car?

Answer.

a $\$1,282.50$

b $\$29,782.50$

Solution.

a Tax $= 28500 \times 0.045 = 1282.50$.

b Total $= 28500 + 1282.50 = 29782.50$.

2. A bank offers a CD that pays a simple interest rate of 5.25%. How much must you put in this CD now in order to have \$5,000 for college tuition in five years?

Answer. \$3,960.40

Solution. Use $A = P(1+rt)$. $5000 = P(1+0.0525\times 5)$ $5000 = P(1.2625)$ $P = 5000/1.2625 \approx 3960.40$.

3. How much money should be deposited today in an account that earns 5% compounded quarterly so that it will accumulate to \$15,000 in ten years?

Answer. \$9,126.10

Solution. Use $P = A(1 + r/n)^{-nt}$. $P = 15000(1 + 0.05/4)^{-40}$ $P = 15000(1.0125)^{-40} \approx 9126.10$.

4. The cost of washer-dryer is \$1,100. We can finance this by paying \$100 down and \$110 per month for 12 months. Determine the finance charge.

Answer. \$320

Solution. Total paid = Down payment + (Monthly payment $\times$ months) Total $= 100 + (110 \times 12) = 100 + 1320 = 1420$. Finance Charge $= 1420 - 1100 = 320$.

5. At the time of her grandson's birth, a grandmother deposits \$5,000 in an account that pays 6% compounded monthly. What will be the value of the account at the child's twenty-first birthday?

Answer. \$17,560.10

Solution. $A = 5000(1 + 0.06/12)^{(12\times 21)}$ $A = 5000(1.005)^{252} \approx 17560.10$.

6. You deposit \$7,500 in an account that pays 4.75% interest compounded monthly. Find the future value after one year and the effective annual yield.

Answer. FV: \$7,864.08; Yield: 4.85%

Solution. FV $= 7500(1 + 0.0475/12)^{12} \approx 7864.08$. Yield $= (1 + 0.0475/12)^{12} - 1 \approx 0.0485$ or 4.85%.

7. Find the value of an annuity of \$1,000 at the end of every 6 months at 4.5% compounded semiannually for 10 years.

Answer. Value: \$24,911.52; Interest: \$4,911.52

Solution. Use $A = P\frac{(1+r/n)^{nt}-1}{r/n}$. $A = 1000\frac{(1+0.045/2)^{20}-1}{0.045/2} \approx 24911.52$. Total deposits $= 1000\times 20 = 20000$. Interest $= 24911.52 - 20000 = 4911.52$.

8. Financial goal of \$150,000 at 4.75% compounded quarterly for 35 years. Find the periodic deposit.

Answer. Deposit: \$168.32

Solution. $P = \frac{A(r/n)}{(1+r/n)^{nt}-1}$ $P = \frac{150000(0.0475/4)}{(1+0.0475/4)^{140}-1} \approx 168.32$.

9. Loan of \$36,000 at 8% for 10 years.

a Find monthly payments and total interest.

b Compare to a 5-year term.

Answer.

a Payment: \$436.78; Interest: \$16,413.60

b Pay \$293.10 more per month; Save \$8,620.80 interest.

Solution. Use $PMT = \frac{Pr/n}{1-(1+r/n)^{-nt}}$.

10-yr: $PMT \approx 436.78$. Total paid = 52413.60. Interest = 16413.60.

5-yr: $PMT \approx 729.88$. Total paid = 43792.80. Interest = 7792.80.

Chapter 3

(Module 3.) Probability

Probability is the mathematical study of uncertainty. Whether we are tossing a coin or predicting the weather, we use probability to quantify the likelihood of different outcomes.

In this chapter, we will define the fundamental building blocks of the theory, starting with the concept of a **sample space**.

3.1 Fundamentals of Probability

Definition 3.1 Probability Formula. The ***theoretical probability*** of an event is a measure of how likely that event is to occur based on the assumption that all outcomes in the sample space are equally likely.

If an event E has $n(E)$ equally likely outcomes and its sample space S has $n(S)$ equally likely outcomes, the ***probability*** of event, $P(E)$, is

$$P(E) = \frac{\text{number of outcomes in event } E}{\text{total number of outcomes in sample space}} = \frac{n(E)}{n(S)}$$

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Example 3.2 Probability of a Single Die Roll. A die is rolled. The set of equally likely outcomes is $\{1, 2, 3, 4, 5, 6\}$. Find the probability of rolling:

1. a 2
2. a number less than 4
3. a number greater than 7
4. a number less than 7

Solution.

1. $P(2) = \dfrac{1}{6}$
2. The outcomes are $\{1, 2, 3\}$, so $P(\text{less than } 4) = \frac{3}{6} = \frac{1}{2}$
3. There are no such outcomes, so $P(\text{greater than } 7) = \frac{0}{6} = 0$
4. All outcomes are less than 7, so $P(\text{less than } 7) = \frac{6}{6} = 1$

▲

Example 3.3 Probability of a Pair of Dice. A single die is rolled twice. Find the probability of rolling:

1. two even numbers
2. two numbers whose sum is 6
3. two numbers whose sum exceeds 12
4. two prime numbers
5. two numbers whose sum is less than 7

Solution.

Table 3.4 Sample Space for Two Six-Sided Dice

Die 1 \ 2	**1**	**2**	**3**	**4**	**5**	**6**
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

1. The even numbers on a die are $\{2, 4, 6\}$. There are $3 \times 3 = 9$ outcomes where both numbers are even. The probability is $\frac{9}{36} = \frac{1}{4}$,
2. The outcomes with a sum of 6 are $\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$. There are 5 such outcomes, so the probability is $\frac{5}{36}$.
3. The maximum possible sum is $6 + 6 = 12$. It is impossible for the sum to exceed 12. The probability is $\frac{0}{36} = 0$.
4. The prime numbers on a die are $\{2, 3, 5\}$. There are $3 \times 3 = 9$ outcomes where both numbers are prime. The probability is $\frac{9}{36} = \frac{1}{4}$.
5. The outcomes with a sum less than 7 are:
   - Sum 2: (1,1) [1 outcome]
   - Sum 3: (1,2), (2,1) [2 outcomes]
   - Sum 4: (1,3), (2,2), (3,1) [3 outcomes]
   - Sum 5: (1,4), (2,3), (3,2), (4,1) [4 outcomes]
   - Sum 6: (1,5), (2,4), (3,3), (4,2), (5,1) [5 outcomes]

   Total outcomes $= 1 + 2 + 3 + 4 + 5 = 15$. The probability is $\frac{15}{36} = \frac{5}{12}$.

▲

Example 3.5 Probability of Drawing a Single deck of Cards. You are dealt one card from a standard 52-card deck. Find the probability of being dealt:

1. an ace
2. a red card
3. a red king
4. a spade

5. an even number heart
6. an odd number diamond
7. a face card

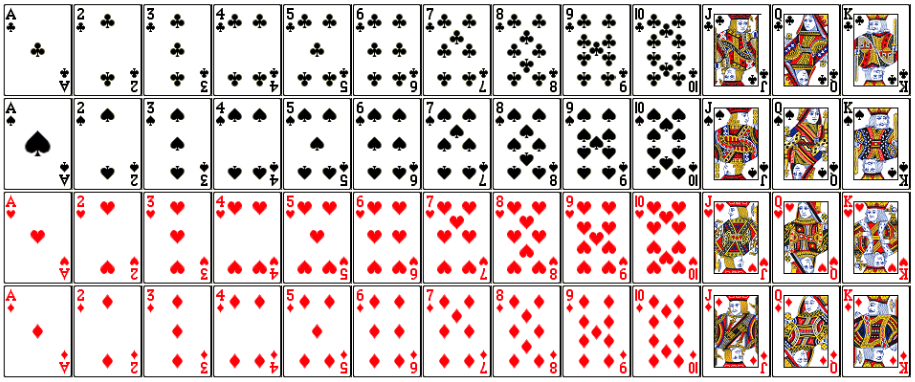


Figure 3.6 A standard deck of 52 playing cards.

Solution. There are 52 total outcomes in a standard deck.

1. There are 4 aces in a deck. $P(\text{ace}) = \frac{4}{52} = \frac{1}{13}$
2. There are 26 red cards (hearts and diamonds). $P(\text{red}) = \frac{26}{52} = \frac{1}{2}$
3. There are 2 red kings (king of hearts and king of diamonds). $P(\text{red king}) = \frac{2}{52} = \frac{1}{26}$
4. There are 13 spades in a deck. $P(\text{spade}) = \frac{13}{52} = \frac{1}{4}$
5. The even numbered hearts are $\{2, 4, 6, 8, 10\}$. There are 5 such cards. $P(\text{even heart}) = \frac{5}{52}$
6. The odd numbered diamonds (excluding face cards) are $\{3, 5, 7, 9\}$. (Note: Ace is usually not considered an odd number in this context, but if included, the count would be 5). Using $\{3, 5, 7, 9\}$, there are 4 such cards. $P(\text{odd diamond}) = \frac{4}{52} = \frac{1}{13}$
7. There are 12 face cards (Jack, Queen, King of each suit). $P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$

▲

Example 3.7 Probability of Three Coin Tosses. A fair coin is tossed three times in succession. The set of equally likely outcomes is $\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. Find the probability of getting:

1. Exactly one tail.
2. Exactly 2 heads.
3. At least one head.
4. Five tails.

Solution.

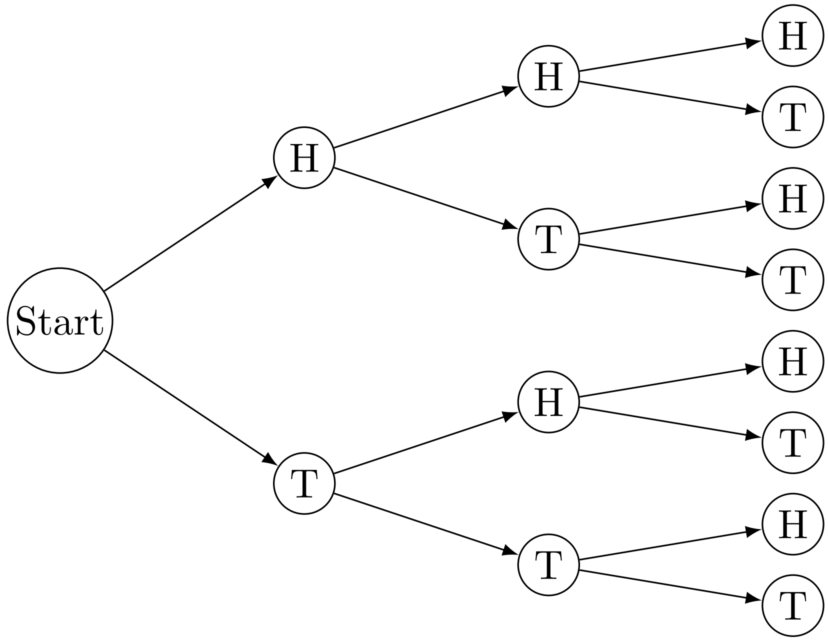


Figure 3.8 Tree diagram for three coin tosses

The total number of outcomes is 8.

1. The outcomes with exactly one tail are $\{HHT, HTH, THH\}$. There are 3 such outcomes, so $P(\text{exactly one tail}) = \frac{3}{8}$.
2. The outcomes with exactly two heads are $\{HHT, HTH, THH\}$. There are 3 such outcomes, so $P(\text{exactly 2 heads}) = \frac{3}{8}$.
3. The only outcome with no heads is $\{TTT\}$. Therefore, there are $8 - 1 = 7$ outcomes with at least one head. $P(\text{at least one head}) = \frac{7}{8}$.
4. It is impossible to get five tails in only three tosses. $P(\text{five tails}) = \frac{0}{8} = 0$.

▲

Checkpoint 3.9 Probability of a Coin Toss and Die Roll. A coin is tossed and a die is rolled. The tree diagram below illustrates the sample space of 12 equally likely outcomes.

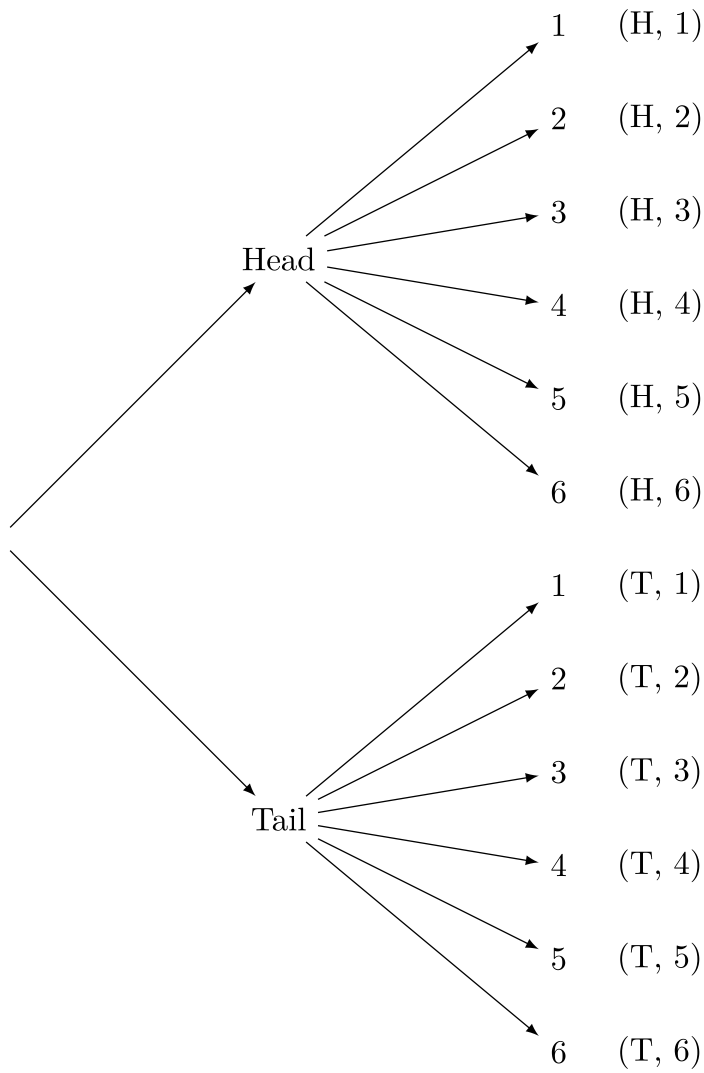


Figure 3.10 Tree diagram for tossing a coin and rolling a die.

Find the probability of getting:

1. a head and an even number.
2. a tail and a number less than 3.
3. a head and a number greater than 6.

Solution. The total number of equally likely outcomes is $2 \times 6 = 12$.

1. The outcomes for a head and an even number are $\{(H, 2), (H, 4), (H, 6)\}$. There are 3 such outcomes, so $P(\text{head and even}) = \frac{3}{12} = \frac{1}{4}$.
2. The outcomes for a tail and a number less than 3 are $\{(T, 1), (T, 2)\}$. There are 2 such outcomes, so $P(\text{tail and less than 3}) = \frac{2}{12} = \frac{1}{6}$.
3. Since a standard die only goes up to 6, it is impossible to roll a number greater than 6. The probability is $\frac{0}{12} = 0$.

3.2 Events Involving NOT and OR

A survey asked 500 Americans to rate their health. Of the surveyed, 270 rated their health as good/excellent. This means that $500 - 270 = 230$ people surveyed did not rate their health as good/excellent.

$$\begin{aligned} P(\text{good/excellent}) + P(\text{not good/excellent}) &= \frac{270}{500} + \frac{230}{500} \\ &= \frac{500}{500} = 1 \end{aligned}$$

In general, since the sum of probabilities of all possible outcomes in any situation is 1:

$$P(E) + P(\text{not } E) = 1$$

Definition 3.11 The Probability of an Event Not Occurring. The probability that an event E will not occur is equal to 1 minus the probability that it will occur:

$$P(\text{not } E) = 1 - P(E)$$

◆

Example 3.12 If you are dealt one card from a standard 52-card deck (Figure 3.6), find the probability that you are not dealt a queen.

Solution. $P(\text{not a queen}) = 1 - P(\text{queen}) = 1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}$. ▲

Checkpoint 3.13 If you are dealt one card from a standard 52-card deck, find the probability that you are not dealt a diamond.

Solution. $P(\text{not a diamond}) = 1 - P(\text{diamond}) = 1 - \frac{13}{52} = \frac{39}{52} = \frac{3}{4}$.

Definition 3.14 Mutually Exclusive Events. Two events are ***mutually exclusive*** if they cannot occur at the same time. In other words, if one event occurs, the other cannot occur.

Or Probabilities with Mutually Exclusive Events: If A and B are mutually exclusive events, then:

$$P(A \text{ or } B) = P(A) + P(B)$$

◆

Example 3.15 If one card is randomly selected from a deck of cards (Figure 3.6), what is the probability of selecting a king or a queen?

Solution. These are mutually exclusive. $P(\text{king or queen}) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$. ▲

Checkpoint 3.16 If you roll a single, six-sided die, what is the probability of getting either a 4 or a 5?

Solution. $P(4 \text{ or } 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$.

If one card is randomly selected from a deck of cards (Figure 3.6), what is the probability of selecting a diamond or a picture card?

$$P(\text{diamond or picture card}) = P(\text{diamond}) + P(\text{picture card}) = \frac{13}{52} + \frac{12}{52}$$

However, there are three cards that are ***simultaneously diamonds and picture cards***. The events are not mutually exclusive.

To correct for the double-counting:

$$P(\text{diamond or picture card}) = P(\text{diamond}) + P(\text{picture card}) - P(\text{diamond and picture card})$$

$$= \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26}$$

Definition 3.17 Events that are NOT Mutually Exclusive. If it is possible for events A and B to occur simultaneously, the events are said to be ***not mutually exclusive***.

If A and B are not mutually exclusive events, then:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

◆

Checkpoint 3.18 In a group of 50 students, 23 take math, 11 take psychology, and 7 take both. Find the probability that a randomly selected student takes math or psychology.

Solution. Let M be the event that a student takes math, and Ps be the event that a student takes psychology. We are given the following:

- Total students: $n(S) = 50$
- Students in math: $n(M) = 23$
- Students in psychology: $n(Ps) = 11$
- Students in both: $n(M \text{ and } Ps) = 7$

To find the probability that a student takes math **or** psychology, we use the Addition Rule for Probability:

$$P(M \text{ or } Ps) = P(M) + P(Ps) - P(M \text{ and } Ps)$$

Substituting the known values as fractions of the total population:

$$P(M \text{ or } Ps) = \frac{23}{50} + \frac{11}{50} - \frac{7}{50}$$

Since the denominators are the same, we combine the numerators:

$$\frac{23 + 11 - 7}{50} = \frac{27}{50}$$

Thus, the probability that a randomly selected student takes math or psychology is $\frac{27}{50}$ (or 0.54).

The following table shows the distribution of active duty personnel in the U.S. military (in thousands).

Table 3.19 U.S. Military Personnel by Branch and Gender

	Air Force	**Army**	**Marine Corps**	**Navy**	**Total**
Male	290	400	160	320	1,170
Female	70	70	10	50	200
Total	360	470	170	370	1,370

Checkpoint 3.20 Using Table 3.19, find the probability that a person selected at random is in the Army or is a woman.

Solution. We use the Addition Rule for Probability:

$$P(\text{Army or Female}) = P(\text{Army}) + P(\text{Female}) - P(\text{Army and Female})$$

From the table, we identify the following values:

- Total Personnel: $1,370$

- Total in Army: 470
- Total Female: 200
- Females in the Army (the overlap): 70

Now, substitute these into the formula:

$$\frac{470}{1,370} + \frac{200}{1,370} - \frac{70}{1,370} = \frac{470 + 200 - 70}{1,370} = \frac{600}{1,370}$$

Simplifying the fraction, we get $\frac{60}{137} \approx 0.438$.

Checkpoint 3.21 Using Table 3.19, find the probability that a person selected at random is in the Navy or is a man.

Solution. To find the probability of a person being in the Navy or a man, we apply the Addition Rule:

$$P(\text{Navy or Male}) = P(\text{Navy}) + P(\text{Male}) - P(\text{Navy and Male})$$

Identify the relevant values from the table:

- Total Personnel: $1,370$
- Total in Navy: 370
- Total Male: $1,170$
- Males in the Navy (the overlap): 320

Substitute the values:

$$\frac{370}{1,370} + \frac{1,170}{1,370} - \frac{320}{1,370} = \frac{370 + 1,170 - 320}{1,370} = \frac{1,220}{1,370}$$

Simplifying the fraction, we get $\frac{122}{137} \approx 0.891$.

3.3 The Fundamental Counting Principle

The Fundamental Counting Principle is a powerful tool for counting the number of possible outcomes in a variety of situations. It allows us to determine the total number of combinations or arrangements by multiplying the number of choices for each independent event.

The probability of winning the top prize in the lottery is about the same as the probability of being struck by lightning. There are a million possible number combinations in the lottery games, and only one way of winning the grand prize.

We can generalize the idea to any two groups of items with the ***Fundamental Counting Principle***.

Definition 3.22 The Fundamental Counting Principle. If there are M ways to choose an item from one group and N ways to choose an item from another group, then there are $M \cdot N$ ways to choose one item from each group. ◆

Example 3.23 Restaurant two-course meal. The Greasy Spoon Restaurant offers 6 appetizers and 14 main courses. In how many ways can a person order a two-course meal?

Solution. Use the Fundamental Counting Principle: choose 1 appetizer and 1 main course.$6 \cdot 14 = 84$There are 84 different two-course meals. ▲

Checkpoint 3.24 Two-course meal. The Diner offers 10 appetizers and 15 main courses. In how many ways can a person order a two-course meal?

Solution. Use the Fundamental Counting Principle and multiply the number of appetizers by the number of main courses.$10 \cdot 15 = 150$There are 150 possible two-course meals.

Checkpoint 3.25 Psychology and social science scheduling. This semester, you need to enroll in your required psychology and social science courses. Because you are registering early, you have 15 psychology sections to choose from. Additionally, there are 9 social science sections available that do not conflict with the psychology schedule. How many different two-course schedules can you create to fulfill these requirements?

Solution. Multiply the number of psychology section options by the number of social science section options.$15 \cdot 9 = 135$There are 135 possible two-course schedules.

Definition 3.26 The Fundamental Counting Principle. If there are n independent events and the first event can occur in m_1 ways, the second event can occur in m_2 ways, the third event can occur in m_3 ways, and so on, then the total number of ways that the events can occur is given by

$$m_1 \cdot m_2 \cdot m_3 \cdots m_n$$

by ***multiplying*** the number of ways each individual event can happen. ◆

Example 3.27 One-topping pizza choices. A pizza can be ordered with two choices of size (medium or large), three choices of crust (thin, thick or regular), and five choices of toppings (ground beef, sausage, pepperoni, bacon and mushrooms). How many different one-topping pizzas can be ordered?

Solution. Multiply the number of size choices by the number of crust choices and the number of topping choices.$2 \cdot 3 \cdot 5 = 30$There are 30 different one-topping pizzas. ▲

Example 3.28 Three-wheel car options. Car manufacturers are now experimenting with lightweight three-wheel cars, designed for one person and considered ideal for city driving. Suppose you could order such a car with a choice of 9 possible colors, with or without air conditioning, electric or gas powered, and with or without an on-board computer. In how many ways can this car be ordered with regard to these options?

Solution. Multiply the number of choices for each independent feature.$9 \cdot 2 \cdot 2 \cdot 2 = 72$There are 72 possible car orders. ▲

Checkpoint 3.29 Ordering a customizable car. The car in Example Example 3.28 is now available in 10 possible colors. The options involving air conditioning, power, and on-board computer still apply. Furthermore, the car is available with or without a global positioning system. In how many ways can this car be ordered in terms of these options?

Solution. Multiply the number of choices for each independent option.$10 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 160$There are 160 possible car orders.

Example 3.30 Ten-question multiple-choice test. You are taking a multiple-choice test that has ten questions. Each of the questions has four answer choices, with one correct answer per question. If you select one of these four choices for each question and leave nothing blank, in how many ways can you answer the questions?

Solution. This situation involves making choices with ten questions.

Question 1 (4 choices)
Question 2
Question 3
...
Question 10

We use the Fundamental Counting Principle to find the number of ways that you can answer the questions on the test. Multiply the number of choices, 4, for each of the ten questions.$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 = 4^{10} = 1,048,576$ ▲

Exercises for Practice

1. **Six-question multiple-choice test.** You are taking a multiple-choice test that has six questions. Each of the questions has three answer choices, with one correct answer per question. If you select one of these three choices for each question and leave nothing blank, in how many ways can you answer the questions?

   Solution. Multiply the number of choices for each question.$3^6 = 729$There are 729 possible answer sheets.

2. **Telephone number possibilities.** Telephone numbers in the United States begin with three-digit area codes followed by seven-digit local telephone numbers. Area codes and local telephone numbers cannot begin with 0 or 1. How many different telephone numbers are possible?

   Solution. There are 8 choices for the first digit of the area code and 10 choices for each remaining area code digit. For the local number, the first digit also has 8 choices and each remaining digit has 10 choices.$8 \cdot 10 \cdot 10 \cdot 8 \cdot 10^6 = 6,400,000,000$There are 6,400,000,000 possible telephone numbers.

3. **Pen choices.** A popular type of pen comes in red, blue, or black ink. The writing tip varies from extra bold, bold, regular, fine, or micro. How many different choices of pens do you have with this type of pen?

   Solution. Multiply the ink color choices by the tip style choices.$3 \cdot 5 = 15$There are 15 different pen choices.

4. **Catered meal combinations.** A wedding caterer gives you three choices for the main course, six starter choices and five options for dessert. How many different meals (made up of starter, dinner and dessert) are there?

   Solution. Multiply the number of starter choices, main course choices, and dessert choices.$3 \cdot 6 \cdot 5 = 90$There are 90 different meals.

5. **Yes/no survey.** You take a survey with five "yes" or "no" answers. How many different ways could you complete the survey?

   Solution. Each question has 2 choices, and the choices are independent.$2^5 = 32$There are 32 different ways to complete the survey.

6. **Product code possibilities.** A company puts a code on each different product they sell. The code is made up of 3 numbers and 2 letters. How many different codes are possible?

   Solution. Each number can be one of 10 digits, and each letter can be one of 26 letters.$10^3 \cdot 26^2 = 676,000$There are 676,000 different product codes possible.

3.4 Factorials

In this section, we will explore the concept of factorials and how they are used in combinatorial counting problems. Factorials are a fundamental tool in counting arrangements and selections, and they play a crucial role in the formulas for permutations and combinations. We will define what a factorial is, how to compute it, and how to apply it in various counting scenarios.

Factorials.

Factorials are a fundamental concept in combinatorics and play a crucial role in counting problems. The factorial of a positive integer n, denoted as $n!$, is the product of all positive integers from n down to 1. For example,

$$\begin{aligned}
1! &= 1\\
2! &= 2 \cdot 1 = 2\\
3! &= 3 \cdot 2 \cdot 1 = 6\\
4! &= 4 \cdot 3 \cdot 2 \cdot 1 = 24\\
5! &= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120
\end{aligned}$$

By definition, 0! is equal to 1.

Factorials are used in various counting problems, such as permutations and combinations, where we need to count the number of ways to arrange or select items. In the following sections, we will explore how factorials are applied in these contexts and how they can help us solve complex counting problems.

If n is a positive integer, then $n!$ (n ***factorial***) is the product of all positive integers from n down through 1.By definition, 0! is 1.

Definition 3.31 Factorial formula. The factorial of n is

$$n! = n(n-1)(n-2)(n-3)\cdots(3)(2)(1)$$

And by definition, $0! = 1$. ◆

Example 3.32 Evaluate factorial expressions. Evaluate the following factorial expressions without using the factorial key on your calculator.

1. $\dfrac{8!}{5!}$
2. $\dfrac{21!}{24!}$
3. $\dfrac{500!}{499!}$
4. $\dfrac{11!}{7!}$
5. $\dfrac{13!}{16!}$
6. $\dfrac{99!}{100!}$
7. $\dfrac{7!}{(5-1)!}$
8. $\dfrac{7!}{3!(4-4)!}$

Solution.

1. Expand both factorials completely to show the common factors:

$$\frac{8!}{5!} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

After canceling the common factors $(5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)$ from the top and bottom:

$$8 \cdot 7 \cdot 6 = 336$$

2. Since the denominator is larger, the remaining factors will be in the denominator:

$$\frac{21!}{24 \cdot 23 \cdot 22 \cdot 21!} = \frac{1}{24 \cdot 23 \cdot 22} = \frac{1}{12,144}$$

3.
$$\frac{500 \cdot 499!}{499!} = 500$$

4. Expand both factorials completely:

$$\frac{11!}{7!} = \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$$

After canceling the common factors $(7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)$:

$$11 \cdot 10 \cdot 9 \cdot 8 = 7,920$$

5.
$$\frac{13!}{16 \cdot 15 \cdot 14 \cdot 13!} = \frac{1}{16 \cdot 15 \cdot 14} = \frac{1}{3,360}$$

6.
$$\frac{99!}{100 \cdot 99!} = \frac{1}{100}$$

7. First, simplify the expression inside the parentheses:

$$\frac{7!}{(5-1)!} = \frac{7!}{4!} = \frac{7 \cdot 6 \cdot 5 \cdot 4!}{4!} = 7 \cdot 6 \cdot 5 = 210$$

8. Recall that $0! = 1$:

$$\frac{7!}{3!(4-4)!} = \frac{7!}{3! \cdot 0!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3!}{3! \cdot 1} = 7 \cdot 6 \cdot 5 \cdot 4 = 840$$

▲

Definition 3.33 Permutations of duplicate items. The number of permutations of n items, where p items are identical, q items are identical, r items are identical, s items are identical, and so on, is given by

$$\frac{n!}{(p!\ q!\ r!\ s! \cdots)}$$

◆

Example 3.34 Arrange MISSISSIPPI. In how many distinct ways can the letters of the word "MISSISSIPPI" be arranged?

Solution. The word has 11 letters with 4 identical I's, 4 identical S's, and 2 identical P's. $\frac{11!}{(4!4!2!)} = \frac{39,916,800}{(24 \cdot 24 \cdot 2)} = 34,650$ ▲

Example 3.35 Arrange BANANA. In how many distinct ways can the letters of the word "BANANA" be arranged?

Solution. The word has 6 letters with 3 identical A's and 2 identical

N's. $\frac{6!}{(3!2!)} = \frac{720}{(6 \cdot 2)} = 60$ ▲

Checkpoint 3.36 Arrange MASSACHUSETTS. In how many distinct ways can the letters of the word "MASSACHUSETTS" be arranged?

Solution. The word has 13 letters with 4 identical S's, 2 identical A's, and 2 identical T's. $\frac{13!}{(4!2!2!)} = \frac{6{,}227{,}020{,}800}{(24 \cdot 2 \cdot 2)} = 64,864,800$

Checkpoint 3.37 Arrange balloons. There are seventeen balloons: 3 blue, 5 red, 2 green, 3 yellow, and 4 orange. In how many distinct ways can the balloons be arranged?

Solution. There are 17 total balloons with duplicates in several colors. $\frac{17!}{(3!5!2!3!4!)} = \frac{355{,}687{,}428{,}096{,}000}{(6 \cdot 120 \cdot 2 \cdot 6 \cdot 24)} = 1,715,313,600$

3.5 Permutations

Definition 3.38 Permutation. A **permutation** is an ordered arrangement of items that occurs when:

- No item is used more than once.
- The order of arrangement makes a difference.

◆

Example 3.39 The BTS Concert Permutation Problem. Suppose BTS, Blackstreet Boys, NSYNC and TLC are given a no-contact concert series. You decide that BTS should be the last group to perform at the four-group concert. Given this decision, in how many ways can you put together the concert?

Solution. You can choose any one of the three groups (Blackstreet Boys, NSYNC, or TLC) as the opening act. Once you have chosen the first group, you will then have two groups left to choose for the second performance. You will then have just one group left to choose for the third performance. There is also just one choice for the closing act: BTS.

Table 3.40 Concert Performance Choices

First Group to perform	Second Group to perform	Third Group to perform	Last Group to pe
3 choices	2 choices	1 choice	1 choice
Blackstreet Boys, NSYNC, TLC			BTS

We use the Fundamental Counting Principle to find the number of ways you can put together the concert. Multiply the choices:

$$3 \cdot 2 \cdot 1 \cdot 1 = 6$$

Thus, there are six different ways to arrange the concert if BTS is the final group to perform. ▲

Example 3.41 Arrange books on a shelf. You need to arrange five books along a small shelf. How many different ways can you arrange the books, assuming that the order of the books makes a difference to you?

Solution. There are 5 choices for the first position, 4 for the second, 3 for the third, 2 for the fourth, and 1 for the last. Using the Fundamental Counting Principle:

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$$

There are 120 different ways to arrange the books. ▲

Checkpoint 3.42 In how many different ways can a police department arrange eight suspects in a police lineup if each lineup contains all eight people?

Solution. Since all 8 people are used and order matters, this is a permutation of 8 items taken 8 at a time:

$$_8P_8 = 8! = 40,320$$

There are 40,320 different ways to arrange the lineup.

Checkpoint 3.43 If five digits 1, 2, 3, 4, 5 are being given and a three-digit code has to be made from it if the repetition of digits is allowed then how many such codes can be formed?

Solution. Because repetition is allowed, there are 5 choices for each of the three positions in the code:

$$5 \cdot 5 \cdot 5 = 5^3 = 125$$

There are 125 possible codes.

Example 3.44 Baseball Batting Lineup. You are the coach of a 13-player baseball team and must set a 9-player batting lineup. Because different positions in the lineup carry different responsibilities, ***the order makes a difference***. For example, a strong hitter like "Barry" in the clean-up spot (fourth) drives in more runs than a home run later in the order. How many different ways can you arrange your 9 starters?

Solution. You have 13 players to choose from for the first person at bat. This leaves 12 players for the second position, 11 for the third, and so on, until the 9th position. The total number of batting orders is:

$$13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 259,459,200$$

There are nearly 260 million possible batting orders for a team of 13 players.

We can derive a general formula for permutations by rewriting this calculation as a fraction of factorials:

$$\frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot (4 \cdot 3 \cdot 2 \cdot 1)}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{13!}{4!} = \frac{13!}{(13-9)!}$$

Using standard notation, we call this "the number of permutations of 13 things taken 9 at a time," written as:

$$_{13}P_9 = \frac{13!}{(13-9)!}$$

▲

Note 3.45 Calculator Note: Using the TI-30XIIS.

Figure 3.46 TI-30XIIS Calculator

To evaluate ${}_{13}P_9$ on the TI-30XIIS, follow these steps:

1. Enter the value of n: Type `13`.
2. Press the `PRB` key.
3. The menu will display `nPr nCr !`. Since `nPr` is already underlined, simply press `ENTER`.
4. Enter the value of r: Type `9`.
5. Press `ENTER` to get the result: `259,459,200`.

Definition 3.47 Permutation Formula. The notation ${}_nP_r$ means the **number of permutations of n things taken r at a time**:

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

◆

Example 3.48 Corporate Board Elections. A corporation has seven members on its board of directors. In how many different ways can the board elect a president, vice-president, secretary, and treasurer?

Solution. Since each officer holds a specific title, the order of selection matters. We are looking for the number of permutations of 7 items taken 4 at a time:

$$ {}_nP_r = {}_7P_4 = \frac{7!}{(7-4)!} = \frac{7!}{3!} $$

Expanding the factorials, we get:

$$ \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = 7 \cdot 6 \cdot 5 \cdot 4 = 840 $$

The board can elect the four officers in 840 different ways.

TI-30XIIS Keystrokes: To calculate this on your TI-30XIIS: Type `7` → Press `PRB` → Select `nPr` and press `ENTER` → Type `4` → Press `ENTER`. ▲

Checkpoint 3.49 How many different programming schedules can be arranged by choosing 5 situation comedies from a collection of 9 classic sitcoms?

Solution. We are choosing $r = 5$ items from a set of $n = 9$ where order matters:

$$_9P_5 = \frac{9!}{(9-5)!} = \frac{9!}{4!} = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15,120$$

There are 15,120 possible programming schedules.

TI-30XIIS Keystrokes: To calculate this on your TI-30XIIS: Type `9` → Press `PRB` → Select `nPr` and press `ENTER` → Type `5` → Press `ENTER`.

Checkpoint 3.50 In a race in which six automobiles are entered and there are no ties, in how many ways can the first three finishers come in?

Solution. We are looking for the number of permutations of 6 cars taken 3 at a time:

$$_6P_3 = \frac{6!}{(6-3)!} = \frac{6!}{3!} = 6 \cdot 5 \cdot 4 = 120$$

There are 120 ways the first three finishers can come in.

Checkpoint 3.51 Five singers are to perform on a weekend evening at a night club. How many different ways are there to schedule their appearances?

Solution. This is a permutation of all 5 singers:

$$5! = 120$$

There are 120 different scheduling arrangements.

Checkpoint 3.52 A stock can go up, go down, or stay unchanged. How many possibilities are there if you own seven stocks?

Solution. Each of the 7 stocks has 3 possible outcomes. Since the outcome of one stock is independent of the others and order (which stock does what) matters:

$$3^7 = 2,187$$

There are 2,187 different possibilities for the seven stocks.

Checkpoint 3.53 Seven seats are positioned in a row at a movie theater. Alice, Betty, Craig, Dan, Evelyn, Frank, and Gavin want to sit together.

1. How many different ways can they be arranged?
2. How many different ways can they be arranged if Betty sits in the second seat?
3. How many different ways can they be arranged if Craig and Gavin want to sit in the two aisle seats?

Solution.

1. Total arrangements of 7 people: $7! = 5,040$.
2. If Betty is fixed in the second seat, we only need to arrange the remaining 6 people in the remaining 6 seats: $6! = 720$.
3. There are two aisle seats (the first and the last). First, arrange Craig and Gavin in those 2 seats: $2! = 2$ ways (Craig-Gavin or Gavin-Craig). Then, arrange the remaining 5 people in the middle 5 seats: $5! = 120$. Total: $2 \cdot 120 = 240$.

3.6 Combinations

Definition 3.54 Combination. A **combination** of items occurs when:

- The items are selected from the same group.
- No item is used more than once.
- ***The order of items makes no difference***.

◆

Checkpoint 3.55 Identifying Permutations and Combinations. For each of the following, determine whether the problem involves permutations or combinations:

1. How many ways can you select 6 free videos from a list of 200 videos?
2. In a race with 50 runners and no ties, in how many ways can the first three finishers come in?
3. Baskin-Robbins offers 31 different flavors of ice cream. One item is a bowl consisting of three scoops, each a different flavor. How many such bowls are possible?
4. A three-person committee is needed to study the possibility of expanding the neighborhood park. How many different committees could be formed from six people?

Solution.

1. **Combination**: the order of the selected videos does not matter; only which 6 are chosen matters.
2. **Permutation**: first, second, and third place are distinct positions, so order matters.
3. **Combination**: the bowl is defined by the set of three flavors, and swapping flavors produces the same bowl.
4. **Combination**: the committee members are chosen without ordered roles.

Definition 3.56 The Combination Formula. The number of combinations of n things taken r at a time is denoted by ${}_nC_r$:

$${}_nC_r = \frac{n!}{(n-r)!\,r!}$$

◆

Example 3.57 Public Transportation Committee. A three-person committee is needed to study ways of improving public transportation. How many committees could be formed from the eight people on the board of supervisors?

Solution. This is a combination because the order of committee members does not matter.

Compute

$${}_8C_3 = \frac{8!}{5!3!}.$$

Simplify:

$$\frac{8 \cdot 7 \cdot 6 \cdot 5!}{5! \cdot (3 \cdot 2 \cdot 1)} = \frac{336}{6} = 56.$$

There are 56 possible committees.

TI-30XIIS Keystrokes: To calculate this on your TI-30XIIS: Type `8` → Press `PRB` , use the arrow keys to underline `nCr`, press `ENTER` → Type `3` → Press `ENTER` again to get the result of 56. ▲

Example 3.58 Pet Combinations. You are volunteering to pet-sit for a friend who has seven different animals. How many different pet combinations are possible if you take three of the seven pets?

Solution. Order does not matter when choosing pets for a group, so use combinations.

$${}_7C_3 = \frac{7!}{4!3!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

TI-30XIIS Keystrokes: To calculate this on your TI-30XIIS: Type `7` → Press `PRB` , use the arrow keys to underline `nCr`, press `ENTER` → Type `3` → Press `ENTER` again to get the result of 35. ▲

Example 3.59 Four-Card Hand. How many different 4-card hands can be dealt from a deck that has 16 different cards?

Solution. A hand is a combination because the order of the cards does not matter.

$${}_{16}C_4 = \frac{16!}{12!4!} = \frac{16 \cdot 15 \cdot 14 \cdot 13}{4 \cdot 3 \cdot 2 \cdot 1} = 1,820.$$

▲

Example 3.60 Poker Hands. In poker, a person is dealt 5 cards from a standard 52-card deck (Figure 3.6). The order in which the cards are dealt does not matter. How many different 5-card poker hands are possible?

Solution. Use combinations because only the set of cards matters.

$${}_{52}C_5 = \frac{52!}{47!5!} = 2,598,960.$$

TI-30XIIS Keystrokes: To calculate this on your TI-30XIIS: Type `52` → Press `PRB` , use the arrow keys to underline `nCr`, press `ENTER` → Type `5` → Press `ENTER` again to get the result of 2,598,960. ▲

Example 3.61 Senate Committee Selection. The U.S. Senate of the 107th Congress consisted of 50 Democrats, 49 Republicans, and one independent. How many committees can be formed if each committee must have 3 Democrats and 2 Republicans?

Solution. Use combinations because the order of committee members does not matter.

Choose 3 Democrats from 50:

$${}_{50}C_3 = 19,600.$$

Choose 2 Republicans from 49:

$${}_{49}C_2 = 1,176.$$

Total committees:

$$19,600 \times 1,176 = 23,049,600.$$

TI-30XIIS Keystrokes: To calculate ${}_{50}C_3$: Type 50 → Press PRB, use the arrow keys to underline `nCr`, press ENTER → Type 3 → Press ENTER. Repeat the process for ${}_{49}C_2$ and multiply the results. ▲

Checkpoint 3.62 Practice Problems. Answer the following problems. Identify whether each situation uses combinations, permutations, or the product rule, and then compute the requested count.

1. A popular brand of pen is available in three colors and four writing tips. How many different choices of pens do you have with this brand?
2. How many four-digit odd numbers are there?
3. An election ballot asks voters to select three city commissioners from a group of eight candidates. In how many ways can this be done?
4. Six singers are to perform on a weekend evening at a nightclub. How many different ways are there to schedule their appearances?
5. In how many ways can you arrange six books along a shelf, assuming that the order of the books makes a difference?
6. An ice cream store sells two drinks in four sizes and five flavors. In how many ways can a customer order a drink?
7. An electronic gate can be opened by entering five digits on a keypad containing the digits 0–9. How many different keypad sequences are possible if the digit 0 cannot be used as the first digit?
8. A math exam consists of 10 multiple-choice questions and 5 open-ended problems in which all work must be shown. If a student must answer 8 of the multiple-choice questions and 3 of the open-ended problems, in how many ways can the questions and problems be chosen?
9. For a temporary job, you are painting parking spaces for a new shopping mall with a letter of the alphabet and a single digit from 1 to 9. The first parking space is A1 and the last parking space is Z9. How many parking spaces can you paint with distinct labels?
10. In a race with 100 runners and no ties, in how many ways can the first three finishers come in?
11. Six people are on the board of supervisors for your neighborhood park. A three-person committee is needed to study the possibility of expanding the park. How many different committees could be formed from the six people?
12. Nine comedy acts will perform over two evenings. Five of the acts will perform on the first evening. How many ways can the schedule for the first evening be made?
13. To win Mega Millions, you must pick 5 numbers from a collection of 56 and one Megaball number from a collection of 46. The order of the first 5 does not matter. How many different selections are possible?
14. An exam consists of 20 multiple-choice questions and 10 open-ended problems. If a student must answer 15 of the multiple-choice and 5 of the open-ended questions, in how many ways can the questions and problems be chosen?

15. In a lucky draw, 10 names are placed in a box and three are drawn. Find the number of ways those three names can be selected.

Solution.

1. Since we are making two independent choices (brand and color), we use the **Product Rule**. Multiply the number of brand options by the number of color options:

$$3 \times 4 = 12 \text{ different pen choices.}$$

2. We use the **Fundamental Counting Principle** for each of the four digits:

   - First digit (1-9): 9 choices
   - Middle digits (0-9): 10 choices each
   - Last digit (odd): 5 choices (1, 3, 5, 7, 9)

$$9 \times 10 \times 10 \times 5 = 4,500 \text{ possible codes.}$$

3. This is a **combination** because the order of committee members does not matter.

$${}_8C_3 = \frac{8!}{3!(8-3)!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$$

4. This is a **permutation** of all 6 distinct subjects.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720 \text{ possible schedules.}$$

5. Arranging all items in a row is a **permutation**.

$$6! = 720 \text{ arrangements.}$$

6. By the **Product Rule**, multiply the choices for size, caffeine, and flavor:

$$2 \times 4 \times 5 = 40 \text{ possible drink orders.}$$

7. The first digit cannot be zero (9 choices), and the remaining four digits can be any number 0-9 (10 choices each):

$$9 \times 10 \times 10 \times 10 \times 10 = 9 \times 10^4 = 90{,}000 \text{ codes.}$$

8. Select the items separately then multiply:

$${}_{10}C_8 = 45 \text{ and } {}_5C_3 = 10$$

$$45 \times 10 = 450 \text{ total combinations.}$$

9. Multiply the number of letter options by the number of digit options:

$$26 \times 9 = 234 \text{ possible outcomes.}$$

10. Since the positions (President, VP, Secretary) are distinct, order matters; this is a **permutation**.

$${}_6P_3 = 6 \times 5 \times 4 = 120.$$

11. Choosing a subgroup where order is irrelevant is a **combination**.

$$_6C_3 = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20.$$

12. Select 5 players from 9 without regard to order:

$$_9C_5 = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126.$$

13. First, calculate the combinations for the 5 white balls, then multiply by the Megaball choice:

$$_{56}C_5 = 3{,}819{,}816$$

$$3{,}819{,}816 \times 46 = 175{,}711{,}536.$$

14. Calculate the ways to choose the two groups separately:

$$_{20}C_{15} = 15{,}504 \text{ and } {}_{10}C_5 = 252$$

$$15{,}504 \times 252 = 3{,}907{,}008.$$

15. Selecting a group of 3 from 10 is a **combination**:

$$_{10}C_3 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120.$$

3.7 Probability with the FCP, nPr and nCr

Probability with Fundamental Counting Principle and permutations, nPr

We return to our concert series with BTS, Blackstreet Boys, NSYNC and TLC. Now, **New Kids on the Block (NKOTB)** agrees to join the tour. The five groups determine the performance order by drawing names from a hat.

What is the probability of **TLC** performing fourth and **BTS** performing last?

Recall the probability formula (Definition 3.1), the probability is the fraction of the number of event outcomes over the total number of possible outcomes.

To solve this problem, we will use the **Fundamental Counting Principle (FCP)** to find the total number of possible arrangements of the five groups, and then determine how many of those arrangements have TLC performing fourth and BTS performing last.{}The probability is defined as:

$$P(\text{TLC 4th, BTS last}) = \frac{\text{number of permutations with TLC 4th, BTS last}}{\text{total number of possible permutations}}$$

First, we find the total number of ways to arrange the 5 groups using the **Fundamental Counting Principle**:

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

Next, we find the number of successful outcomes where TLC is 4th and BTS is last. By fixing these two positions, we only need to arrange the remaining 3 groups (Backstreet Boys, NSYNC, and NKOTB) in the first three slots:

- 1st spot: 3 choices

- 2nd spot: 2 choices
- 3rd spot: 1 choice
- 4th spot: ***Fixed (TLC)***
- 5th spot: ***Fixed (BTS)***

$$3 \cdot 2 \cdot 1 \cdot 1 \cdot 1 = 6$$

Therefore, calculate the probability:

$$P(\text{TLC 4th, BTS last}) = \frac{6}{120} = \frac{1}{20}$$

TI-30XIIS Keystrokes: To find the total permutations (5!), type 5 → PRB → arrow over to underline ! → ENTER → ENTER. The result is 120.

Checkpoint 3.63 Probability of Performers. Seven performers A, B, C, D, E, F, and G are to appear in a fundraiser. The order of performance is determined by random selection. Find the probability that:

a D will perform first.

b E will perform sixth and B will perform last.

c They will perform in the following order: C, D, B, A, G, F, E.

d F or G will perform first.

Solution. First, we find the **total number of possible outcomes** for the sample space. Since there are 7 positions to fill:

$$\underline{7} \times \underline{6} \times \underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1}$$

$$7! = 5{,}040$$

TI-30XIIS: Type 7 → Press PRB → Arrow right to underline ! → Press ENTER → Press ENTER.

a To find the number of ways ***D can perform first***, we fix D in the first spot:

$$\underline{1(D)} \times \underline{6} \times \underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 720$$

$$P(\text{D first}) = \frac{720}{5,040} = \frac{1}{7}$$

b To find the number of ways ***E performs 6th and B performs last***, we fix those two positions:

$$\underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} \times \underline{1(E)} \times \underline{1(B)} = 120$$

$$P(\text{E 6th, B last}) = \frac{120}{5,040} = \frac{1}{42}$$

c There is only ***one specific sequence*** out of the 5,040 possibilities:

$$\underline{1(C)} \times \underline{1(D)} \times \underline{1(B)} \times \underline{1(A)} \times \underline{1(G)} \times \underline{1(F)} \times \underline{1(E)} = 1$$

$$P(\text{Specific order C, D, B, A, G, F, E}) = \frac{1}{5,040}$$

d For ***F or G to perform first***, we have 2 choices for the first spot:

$$\underline{2(\text{F or G})} \times \underline{6} \times \underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 1{,}440$$

$$P(\text{F or G first}) = \frac{1,440}{5,040} = \frac{2}{7}$$

Probability with Combinations, nCr

Example 3.64 Probability of Winning LOTTO. Florida's lottery game LOTTO is set up so that each player chooses six different numbers from 1 to 53. If the six numbers chosen match the six numbers drawn randomly, the player wins the top cash prize. With one LOTTO ticket, what is the probability of winning this prize?Because the order of the six numbers does not matter, this is a situation involving combinations. We begin with the formula for probability:

$$P(\text{winning}) = \frac{\text{number of ways of winning}}{\text{total number of possible combinations}}$$

We are selecting $r = 6$ numbers from a collection of $n = 53$ numbers. Using the combination formula:

$${}_{53}C_6 = \frac{53!}{(53-6)! \times 6!} = \frac{53!}{47! \times 6!} = 22{,}957{,}480$$

TI-30XIIS Keystrokes: To calculate ${}_{53}C_6$: Type 53 → Press PRB → Underline `nCr` and press ENTER → Type 6 → Press ENTER.

There are nearly 23 million number combinations possible in LOTTO. If a person buys one LOTTO ticket, that person has selected only one combination of six numbers. With one LOTTO ticket, there is only one way of winning:

$$P(\text{winning}) = \frac{1}{22{,}957{,}480} \approx 0.0000000436$$

The probability of winning the top prize with one LOTTO ticket is $\frac{1}{22{,}957{,}480}$ or about 1 in 23 million. ▲

Checkpoint 3.65 A state lottery is designed so that a player chooses five numbers from 1 to 30 on one lottery ticket. What is the probability that a player with one lottery ticket will win? What is the probability of winning if 100 different lottery tickets are purchased?

Hint. To calculate ${}_{30}C_5$ on your **TI-30XIIS**:

1. Enter 30.
2. Press the PRB key.
3. Use the Right Arrow to underline **nCr**.
4. Press ENTER.
5. Enter 5 and press ENTER (Result: 142,506).
6. To find the probability, press 1 ÷ 2nd ANS ENTER.

Solution. The total number of possible outcomes is the number of ways to choose 5 numbers from 30:

$${}_{30}C_5 = 142,506$$

The probability of winning with one ticket is:

$$P(1 \text{ ticket}) = \frac{1}{142,506} \approx 0.000007$$

The probability of winning with 100 tickets is:

$$P(100 \text{ tickets}) = \frac{100}{142,506} \approx 0.0007$$

Example 3.66 Probability of Club Committee Selection. A club consists of five men and seven women. Three members are selected at random to attend a conference. Find the probability that the selected group consists of:

a three men.

b one man and two women.

Solution. First, we find the total number of ways to choose 3 members from the group of 12 (5 men + 7 women):

$$ {}_{12}C_3 = 220 $$

a To find the probability of selecting three men, we first find the number of ways to choose 3 men from the 5 available:

$$ {}_5C_3 = 10 $$

The probability is:

$$ P(3 \text{ men}) = \frac{{}_5C_3}{{}_{12}C_3} = \frac{10}{220} = \frac{1}{22} \approx 0.0455 $$

b To find the probability of selecting one man and two women, we calculate the ways to choose each and multiply them:

- Ways to choose 1 man from 5: ${}_5C_1 = 5$
- Ways to choose 2 women from 7: ${}_7C_2 = 21$

$$ P(1 \text{ man, } 2 \text{ women}) = \frac{{}_5C_1 \times {}_7C_2}{{}_{12}C_3} = \frac{5 \times 21}{220} = \frac{105}{220} = \frac{21}{44} \approx 0.4773 $$

TI-30XIIS Tip: To calculate part (b) in one string: 5 PRB (**nCr**) 1 × 7 PRB (**nCr**) 2 ÷ 220 ENTER. ▲

Checkpoint 3.67 Probability of Defective Transistors. A box contains 25 transistors, 6 of which are defective. If 6 are selected at random, find the probability that:

a all are defective.

b none are defective.

Solution. First, we determine the total number of ways to select 6 transistors from 25:

$$ {}_{25}C_6 = 177,100 $$

a To find the probability that all 6 are defective, we choose 6 from the 6 defective transistors:

$$ {}_6C_6 = 1 $$

$$ P(\text{all defective}) = \frac{1}{177,100} \approx 0.0000056 $$

b To find the probability that none are defective, all 6 must be chosen from the 19 non-defective transistors (25 total - 6 defective):

$$ {}_{19}C_6 = 27,132 $$

$$ P(\text{none defective}) = \frac{27,132}{177,100} = \frac{6,783}{44,275} \approx 0.1532 $$

TI-30XIIS Calculator Steps:

- For the denominator: 25 PRB (**nCr**) 6 ENTER (Result: 177,100).
- For part (b): 19 PRB (**nCr**) 6 ÷ 177,100 ENTER.

Checkpoint 3.68 Parent-Teacher Committee Selection. A parent-teacher committee consisting of four people is to be selected from fifteen parents and five teachers. Find the probability of selecting two parents and two teachers.

Solution. First, we find the total number of ways to choose 4 members from the group of 20 (15 parents + 5 teachers):

$$_{20}C_4 = 4,845$$

Next, we find the number of ways to select the specific group:

- Ways to select 2 parents from 15: $_{15}C_2 = 105$
- Ways to select 2 teachers from 5: $_5C_2 = 10$

Total successful outcomes: $105 \times 10 = 1,050$.

The probability is the number of successful outcomes divided by the total outcomes:

$$P(\text{2 parents, 2 teachers}) = \frac{_{15}C_2 \times {_5C_2}}{_{20}C_4} = \frac{1,050}{4,845}$$

Simplifying the fraction and providing the decimal:

$$P = \frac{70}{323} \approx 0.2167$$

TI-30XIIS Calculator Steps: 15 PRB (**nCr**) 2 × 5 PRB (**nCr**) 2 ÷ 20 PRB (**nCr**) 4 ENTER.

Checkpoint 3.69 Probability of Dealing Four Hearts. If you are dealt 4 cards from a shuffled deck of 52 cards (Figure 3.6), find the probability that all 4 are hearts.

Solution. First, we determine the total number of ways to choose 4 cards from a deck of 52:

$$_{52}C_4 = 270,725$$

Next, we determine the number of ways to choose 4 hearts. Since there are 13 hearts in a deck:

$$_{13}C_4 = 715$$

The probability that all 4 cards are hearts is the number of ways to choose 4 hearts divided by the total number of ways to choose 4 cards:

$$P(\text{4 hearts}) = \frac{_{13}C_4}{_{52}C_4} = \frac{715}{270,725}$$

Simplifying the fraction and providing the decimal:

$$P = \frac{11}{4,165} \approx 0.0026$$

TI-30XIIS Calculator Steps: 13 PRB (**nCr**) 4 ÷ 52 PRB (**nCr**) 4 ENTER.

Checkpoint 3.70 Probability of Full House: Kings and Aces. If you are dealt 5 cards from a shuffled deck of 52 cards (Figure 3.6), find the probability that you get 3 kings and two aces.

Solution. First, we find the total number of ways to choose 5 cards from a deck of 52:

$$_{52}C_5 = 2,598,960$$

Next, we calculate the number of ways to select the specific cards:

- Ways to select 3 kings from the 4 available: $_4C_3 = 4$
- Ways to select 2 aces from the 4 available: $_4C_2 = 6$

Total successful outcomes: $4 \times 6 = 24$.

The probability is the ratio of successful outcomes to the total possible outcomes:

$$P(3 \text{ kings and } 2 \text{ aces}) = \frac{_4C_3 \times {_4C_2}}{_{52}C_5} = \frac{24}{2,598,960}$$

Simplifying the fraction and providing the decimal:

$$P = \frac{1}{108,290} \approx 0.00000923$$

TI-30XIIS Calculator Steps: 4 PRB (**nCr**) 3 × 4 PRB (**nCr**) 2 ÷ 52 PRB (**nCr**) 5 ENTER.

3.8 Chapter Exercises

Probability Practice Problem

1. Probability of Disjoint or Overlapping Events. A fair die is rolled. What is the probability of rolling an odd number or a number less than 3?

Solution. The sample space for a fair die is $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$.

- Odd numbers: $\{1, 3, 5\}$
- Numbers less than 3: $\{1, 2\}$
- Overlap (Odd AND less than 3): $\{1\}$

Using the Addition Rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$P = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3} \approx 0.6667$$

2. Committee Selection. How many ways can a committee of 5 be selected from a club with 10 members?

Solution. Since the order of selection does not matter for a committee, we use combinations:

$$_{10}C_5 = 252$$

TI-30XIIS: 10 PRB (**nCr**) 5 ENTER.

3. Probability with Cards (Face or 3). A card is drawn at random from a shuffled deck of 52 cards (Figure 3.6). What is the probability of drawing a face card or a 3?

Solution. These are mutually exclusive events (a card cannot be both a face card and a 3).

- Face cards (J, Q, K): $3 \times 4 = 12$
- 3s: 4

$$P = \frac{12}{52} + \frac{4}{52} = \frac{16}{52} = \frac{4}{13} \approx 0.3077$$

4. Probability of a Card Range. One card is selected from a deck of cards. Find the probability that the card selected is greater than 3 and less than 8.

Solution. First, we identify the values in a single suit that are greater than 3 and less than 8:

- The numbers are: 4, 5, 6, and 7.
- There are 4 such values per suit.

Since there are 4 suits in a deck (hearts, diamonds, clubs, and spades), we multiply:

$$4 \text{ values} \times 4 \text{ suits} = 16 \text{ successful cards}$$

The probability is the number of successful cards divided by the total number of cards (52):

$$P = \frac{16}{52} = \frac{4}{13} \approx 0.3077$$

TI-30XIIS Calculator Steps: 16 ÷ 52 ENTER (Then press 2nd PRB if you need to convert back to a simplified fraction).

5. Probability of Marbles (Not Blue). A bag contains 8 red, 2 blue, and 1 green marble. What is the probability of choosing a marble that is not blue?

Solution. Total marbles: $8 + 2 + 1 = 11$. Marbles that are "not blue" are red or green: $8 + 1 = 9$.

$$P(\text{not blue}) = \frac{9}{11} \approx 0.8182$$

6. Probability of Even or Five. A lottery game has balls numbered 0 through 9. What is the probability of selecting an even numbered ball or a 5?

Solution. First, we identify the sample space $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. The total number of outcomes is $n(S) = 10$.

Next, we identify the successful outcomes:

- Even numbered balls: $\{0, 2, 4, 6, 8\}$ (Note: 0 is an even number). There are 5 even balls.
- The number 5: $\{5\}$. There is 1 such ball.

Since these events are mutually exclusive (a ball cannot be both even and 5), we add the probabilities:

$$P(\text{even or } 5) = \frac{5}{10} + \frac{1}{10} = \frac{6}{10}$$

Simplifying the fraction and providing the decimal:

$$P = \frac{3}{5} = 0.6$$

TI-30XIIS Calculator Steps: 6 ÷ 10 ENTER. To see the simplified fraction, press 2nd PRB (F<->D).

7. **Health Survey (Inclusion-Exclusion).** A survey shows 40% take BP meds, 47% take cholesterol meds, and 13% take both. What is the probability a senior takes either?

Solution. Using the Addition Rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$$P = 0.40 + 0.47 - 0.13 = 0.74$$

There is a 74% probability that a senior takes either BP meds or cholesterol meds.

8. **Probability of a Specific Birth Sequence.** A family has five children. The probability of having a girl is 1/2. What is the probability of having 2 girls followed by 3 boys?

Solution.

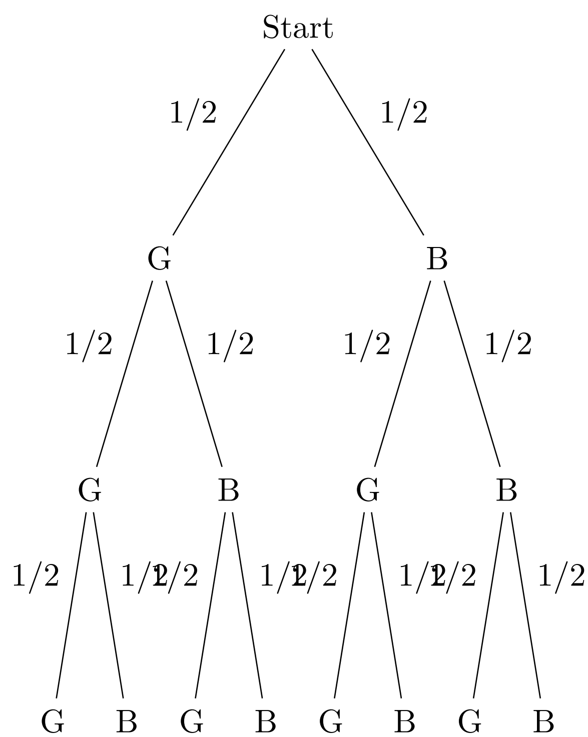


Figure 3.71 Probability Tree Diagram (First 3 Children)

This problem asks for the probability of one specific sequence: (G, G, B, B, B). Because the gender of each child is an independent event, we multiply the individual probabilities.

Given $P(G) = 1/2$ and $P(B) = 1/2$:

$$P(G \text{ then } G \text{ then } B \text{ then } B \text{ then } B) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Calculating the product:

$$P = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

In decimal form:

$$P = 0.03125$$

9. Probability of Marbles Without Replacement. Two marbles are drawn without replacement from a box with 3 white, 2 green, 2 red, and 1 blue marble.

a Find a total possible ways to choose two marbles without replacement.

b How many ways to choose two white marbles?

c Find the probability that both marbles are white.

Solution. First, we find the total number of marbles: $3+2+2+1=8$.

a The total number of ways to choose 2 marbles from 8 without regard to order:

$$_8C_2 = \frac{8\times 7}{2\times 1} = 28$$

b The number of ways to choose 2 white marbles from the 3 available:

$$_3C_2 = 3$$

c The probability that both marbles are white is the ratio of successful ways to total ways:

$$P(\text{both white}) = \frac{_3C_2}{_8C_2} = \frac{3}{28} \approx 0.1071$$

10. Counting Three-Digit Numbers. How many different three-digit numbers can be written using digits from the set $\{3,4,5,6,7\}$ without any repeating digits?

Solution. Since the order of the digits matters in a number (e.g., 345 is different from 543) and repetitions are not allowed, we use the permutation formula $_nP_r$.

There are $n=5$ digits available, and we are choosing $r=3$ of them:

$$_5P_3 = \frac{5!}{(5-3)!} = 5\times 4\times 3 = 60$$

There are 60 different three-digit numbers that can be formed.
TI-30XIIS Calculator Steps:

(a) Enter [5].

(b) Press the [PRB] button.

(c) Underline **nPr** and press [ENTER].

(d) Enter [3] and press [ENTER].

11. Distinct Arrangements (IMMUNOLOGY). In how many distinct ways can the letters in IMMUNOLOGY be arranged?

Solution. There are 10 letters total. We identify the repeats: M (2), O (2). The formula for distinguishable permutations is:

$$\frac{10!}{(2!\times 2!)} = \frac{3,628,800}{4} = 907,200$$

TI-30XIIS: [10] [PRB] (**!**) [÷] [(] [2] [PRB] [!] [×] [2] [PRB] [!] [)] [ENTER].

Chapter 4

(Module 4.) Introduction to Statistics

Statistics is the branch of mathematics that deals with collecting, analyzing, interpreting, presenting, and organizing data. It provides tools and techniques for making sense of data and drawing meaningful conclusions from it. In this handout, we will explore some fundamental concepts in statistics, including measures of central tendency, measures of variability, and basic probability.

In this chapter, we will cover the following topics:

- Data Graphs: Visual representations of data to identify patterns and trends.
- Center and Spread: Measures of central tendency (mean, median, mode) and measures of variability (range, variance, standard deviation).
- Empirical Rule: A rule that describes the distribution of data in a normal distribution.
- Z-scores: Standardized scores that indicate how many standard deviations a data point is from the mean.
- Regression: A statistical method for modeling the relationship between a dependent variable and one or more independent variables.

4.1 Visualizing Data Through Graphs

After collecting survey or experimental data, we must summarize and present it meaningfully for our audience. We will start with graphical data presentations before moving on to numerical summaries.

Raw information can often feel overwhelming and difficult to interpret. To reveal clear insights, researchers and data analysts rely heavily on visual graphics. The four primary tools used to display quantitative variables are bar graphs, line graphs, pie charts and histogram. The specific choice among them depends entirely on the characteristics of the data being evaluated.

Categorical (or qualitative) data helps us sort individual subjects into distinct groups. To make sense of this information, we usually begin by constructing a two-column ***frequency table***. The first column lists out the available categories, while the parallel column captures the corresponding frequency—meaning the absolute count of observations belonging to each classification.

Frequency Table.

A frequency table consists of two columns. The first column lists the data values and if the values are numeric, then they are listed from smallest to largest. The second column is the frequency, which indicates the number of times each value occurs.

Example 4.1 A questionnaire was given to students in an introductory statistics class during the first week of the course. One question asked, ``How stressed have you been in the last 2 1/2 weeks, on a scale of 0 to 10, with 0 being not at all stressed and 10 being as stressed as possible?" The students' responses are shown in the frequency table. Use this frequency table to answer the following questions.

Table 4.2 Stress Levels of Students

Stress Level	**Frequency**
0	2
1	3
2	5
3	8
4	12
5	10
6	13
7	31
8	26
9	11
10	9

1. Which stress rating describes the greatest number of students? How many students responded with this rating?
2. Which stress rating describes the least number of students? How many students responded with this rating?
3. How many students were involved in this study?
4. How many students had a stress rating of 8 or more?

Solution.

1. The most common stress level is 7, with 31 students responding with this rating.
2. The least common stress level is 0, with 2 students responding with this rating.
3. A total of 140 students were involved in this study.
$$2 + 3 + 5 + 8 + 12 + 10 + 13 + 31 + 26 + 11 + 9 = 140$$
4. The number of students who had a stress rating of 8 or more is:
$$26 + 11 + 9 = 46$$

▲

Example 4.3 The number of touchdown (TD) passes thrown by each of the 31 teams in the National Football League in the 2000 season are shown below:

Table 4.4 NFL Touchdown Passes (2000 Season)

37	33	33	32	29	28	28	23	22	22	22
21	21	21	20	20	19	19	18	18	18	18
16	15	14	14	14	12	12	9	6		

Construct a frequency table for the data of the number of touchdown passes thrown by each NFL team in the 2000 season.

Solution. To construct a frequency table, we first identify the unique values in the dataset and count how many times each value occurs. The frequency table for the NFL touchdown passes is as follows:

Table 4.5 Frequency Table for NFL Touchdown Passes (2000 Season)

Value	Frequency
6	1
9	1
12	2
14	3
15	1
16	1
18	4
19	2
20	2

▲

Definition 4.6 Bar graph. ***Bar graphs*** are best suited for comparing quantities or counts across distinct, independent categories. They utilize horizontal or vertical rectangular bars where the length or height of each bar directly corresponds to the numerical value it represents. ◆

For example, a political analyst would use a bar graph to compare the total number of votes received by different candidates in an election, allowing the reader to instantly identify the frontrunner.

Example 4.7 We return to the example Example 4.3 of the NFL touchdown passes data to demonstrate how a bar graph can be constructed.

Solution. Using the frequency table Table 4.5, we constructed earlier, we can create a bar graph where the x-axis represents the number of touchdown passes and the y-axis represents the frequency of each value. This will visually display how often each number of touchdown passes occurred across the 31 teams. Each bar's height will correspond to the frequency of that particular number of touchdown passes, allowing us to quickly see which values were most common and which were rare.

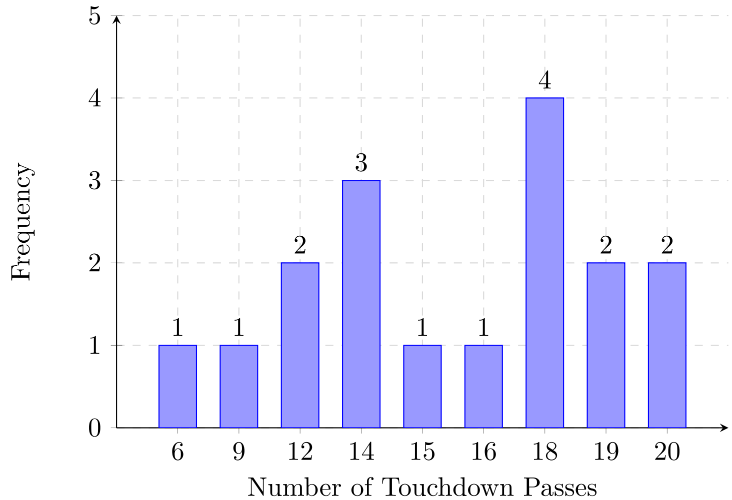


Figure 4.8 Touchdown Passes Frequency Bar Graph

▲

Example 4.9 A small business tracks the number of laptops sold over three months: January (15 units), February (22 units), and March (18 units). Determine which category goes on each axis to construct a vertical bar graph.

Solution. To construct a standard vertical bar graph, the independent nominal categories must be placed along the baseline axis, while the numerical quantitative metrics track along the vertical boundary:

- Horizontal Axis (x-axis): Contains the ***Months*** (January, February, and March).

- Vertical Axis (y-axis): Tracks the quantitative scale for ***Units Sold*** (ranging from 0 up to at least 22).

Each bar's height will correspond to the number of laptops sold in that month, allowing for an immediate visual comparison across the three categories.

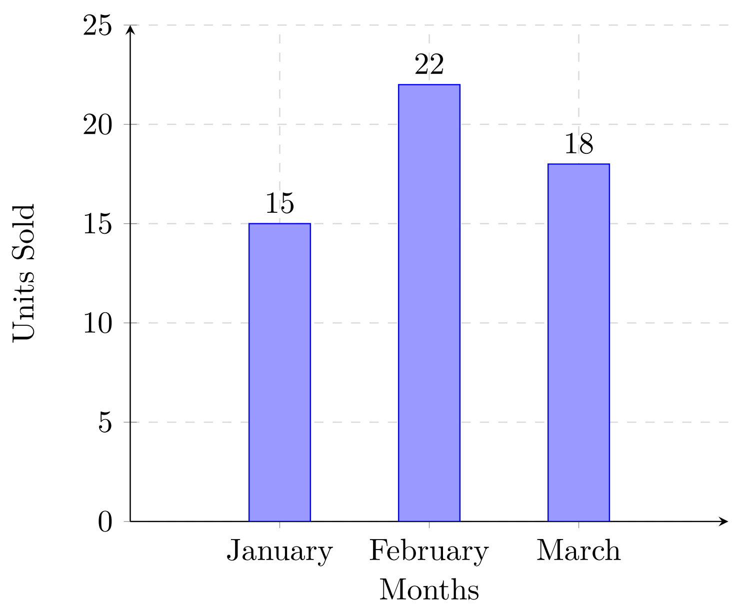


Figure 4.10 Sample structure for a vertical bar graph.

▲

An insurance provider calculates policy premiums by analyzing established risk variables. Drivers classified under higher-risk brackets are subject to increased premium rates. One variable often evaluated is vehicle color, under the assumption that certain car colors are linked to a higher probability of accidents.

To investigate this theory, the company reviewed local law enforcement records detailing recent total-loss traffic collisions. The gathered information is organized into the frequency distribution table below.

Table 4.11 Frequency of Total-Loss Collisions by Vehicle Color

Vehicle Color	**Frequency (Accidents)**
Blue	25
Green	52
Red	41
White	36
Black	39
Grey	23

To draw meaningful conclusions from this categorical data, we must first find the total sample size (n) by summing all frequencies:

$$n = 25 + 52 + 41 + 36 + 39 + 23 = 216$$

Next, we calculate the relative frequency (percentage) for each color category using the formula Relative Frequency $= \dfrac{\text{Frequency}}{n} \times 100\%$. Rounding to two decimal places, we find:

- ***Blue:*** $\frac{25}{216} \approx 11.57\%$
- ***Green:*** $\frac{52}{216} \approx 24.07\%$
- ***Red:*** $\frac{41}{216} \approx 18.98\%$
- ***White:*** $\frac{36}{216} \approx 16.67\%$
- ***Black:*** $\frac{39}{216} \approx 18.06\%$
- ***Grey:*** $\frac{23}{216} \approx 10.65\%$

The data suggests that ***Green*** vehicles accounted for the largest share of total-loss collisions (24.07%), while ***Grey*** vehicles accounted for the lowest share (10.65%).

Example 4.12 Using the total-loss collision dataset compiled by the insurance provider in Table 4.11, construct a bar graph to visually model the distribution of vehicle colors.

Solution. To construct the bar graph, we map each vehicle color category along the horizontal axis (x-axis) and place the frequency counts along the vertical axis (y-axis). The height of each rectangular column corresponds exactly to its recorded category frequency:

- ***Blue:*** 25 units high
- ***Green:*** 52 units high
- ***Red:*** 41 units high
- ***White:*** 36 units high

- ***Black:*** 39 units high
- ***Grey:*** 23 units high

This layout provides an immediate visual hierarchy, showing that green cars are involved in more total-loss accidents than any other color in this sample group.

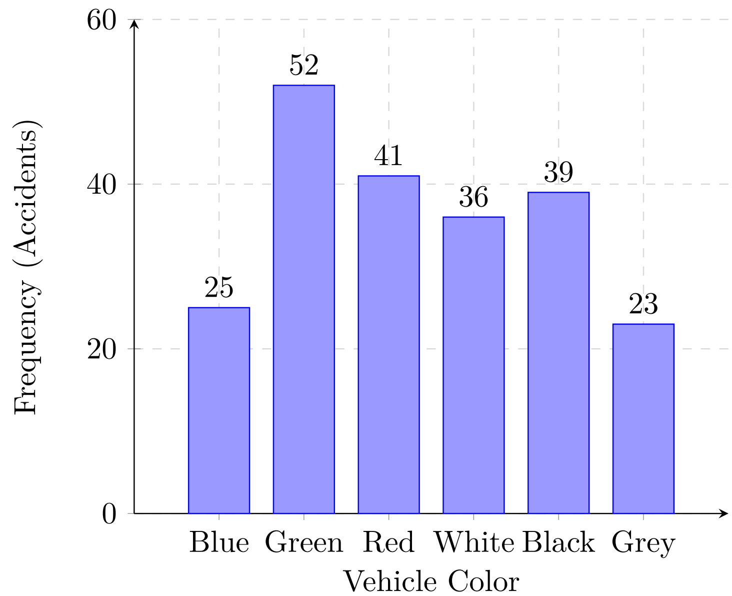


Figure 4.13 Bar Graph of Total-Loss Collisions by Vehicle Color

▲

Checkpoint 4.14 A school cafeteria counts the number of fruit items sold in a day: 40 apples, 55 bananas, and 25 oranges. If you create a vertical bar graph, which item will have the tallest bar, and what value will it represent?

Solution. The category with the highest count is bananas. Therefore, bananas will have the tallest bar, representing a value of 55.

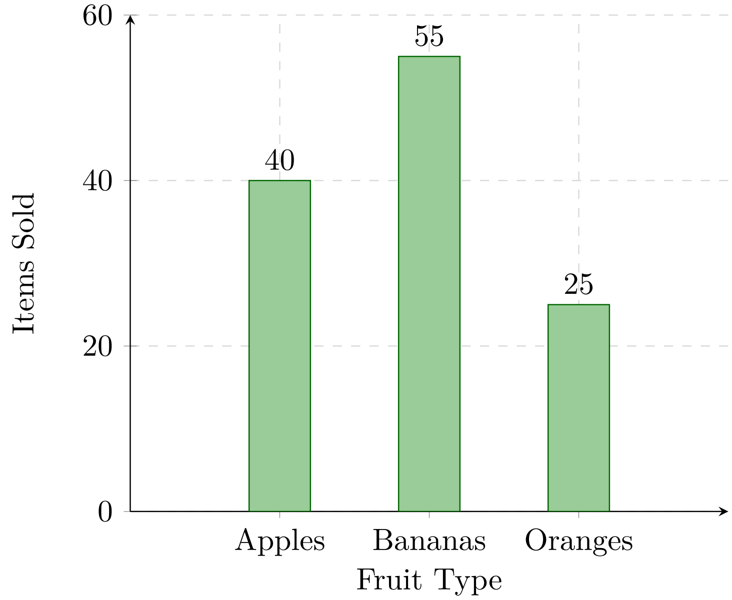


Figure 4.15 Vertical bar graph tracking daily cafeteria fruit sales.

After establishing class intervals and a corresponding frequency table for a quantitative dataset, we can construct a visual graph similar to a standard bar chart. However, because quantitative data is numerical and continuous, the horizontal axis must be treated as an unbroken number line with no spacing between adjacent bars. This specialized graphical representation is known as a ***histogram***.

Definition 4.16 Histogram. A ***histogram*** is a graphical tool used to display the frequency distribution of a single continuous numerical variable. Similar to a bar graph, it represents frequencies using the height of rectangular bars; however, its horizontal axis functions as a continuous number line. The data values are grouped into specific intervals, known as classes or bins. Unlike categorical bar charts, the vertical bars in a histogram must touch directly to accurately reflect the unbroken, sequential nature of the underlying scale. ◆

Example 4.17 Gym Workout Durations. A gym records the workout durations of its members in minutes. The data is grouped into three continuous intervals: 0–20 minutes (5 people), 20–40 minutes (15 people), and 40–60 minutes (10 people). Determine the total sample size tracked across this histogram layout.

Solution. To find the total sample size, sum the frequencies represented by the height of each adjacent bar:

$$5 + 15 + 10 = 30$$

The histogram tracks a total sample size of 30 gym members.

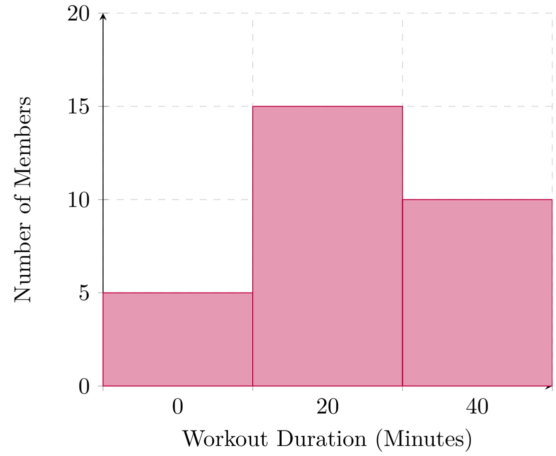


Figure 4.18 Frequency distribution across adjacent numerical bins.

▲

Example 4.19 Using the total-loss collision dataset compiled by the insurance provider in Table 4.11, construct a histogram-style bar chart to visually model the distribution of vehicle colors.

Solution. To present this categorical data in a touching-bar format, we plot the vehicle color categories along the horizontal axis and the collision frequencies on the vertical axis. By adjusting our bar spacing properties, we can configure the rectangles to touch side-by-side:

- ***Blue:*** Bar height is 25
- ***Green:*** Bar height is 52
- ***Red:*** Bar height is 41
- ***White:*** Bar height is 36
- ***Black:*** Bar height is 39
- ***Grey:*** Bar height is 23

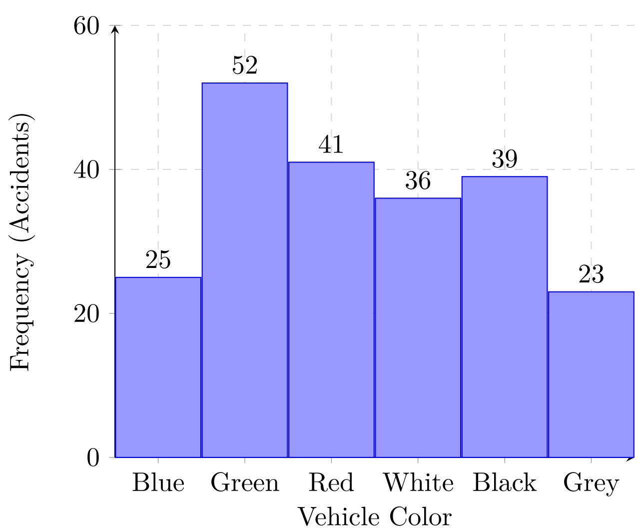


Figure 4.20 Histogram-Style Visual Distribution of Vehicle Colors

▲

Checkpoint 4.21 Nutrition Study Weight Distribution. Suppose that we have collected weights from 100 male subjects as part of a nutrition study. For our weight data, we have values ranging from a low of 121 pounds to a high of 263 pounds, giving a total span of:

$$263 - 121 = 142 \text{ pounds}$$

We can experiment with different bin allocations to summarize the data. Let us organize the dataset using a uniform interval width of 15, starting at a baseline value of 120.

Weight Interval (Pounds)	**Frequency (Count)**
120-134	4
135-149	14
150-164	16
165-179	28
180-194	12
195-209	8
210-224	7
225-239	6
240-254	2
255-269	3

Solution. Using the frequency chart, we construct a continuous frequency histogram. Each vertical bar spans an interval of 15 pounds on the horizontal axis, and the height matches the respective category count.

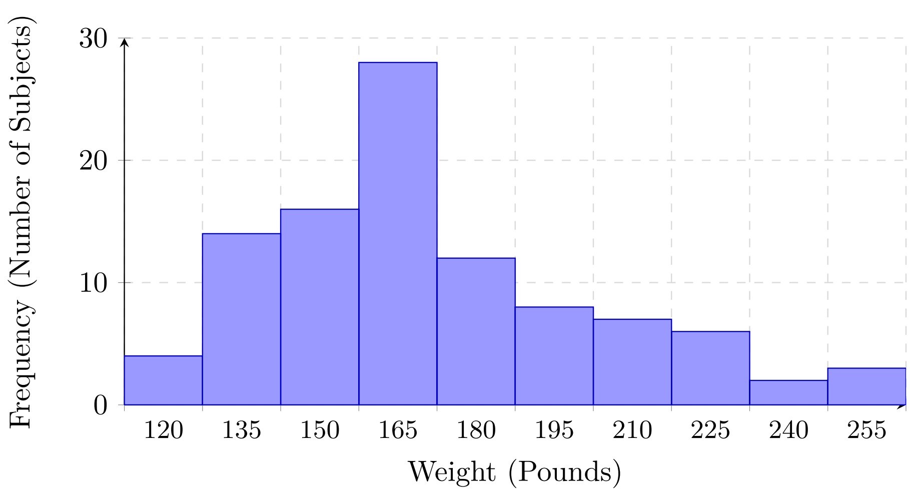


Figure 4.22 Frequency histogram of male subject weights with an interval width of 15.

Definition 4.23 Line graph. ***Line graphs*** are ideal for illustrating trends and demonstrating how numbers fluctuate over a continuous period. This format displays a sequential series of specific data points—frequently referred to as markers—that are connected by straight lines to show a clear trajectory. ◆

A common application includes tracking macroeconomic variables, such as a city's average monthly temperature or a company's fluctuating stock prices throughout a fiscal year.

Example 4.24 An analyst records the stock price of a company over four days: Day 1 ($10), Day 2 ($15), Day 3 ($12), and Day 4 ($20). Describe the overall trend shown by connecting these data points on a line graph.

Solution. The line segments will move upward from Day 1 to Day 2, slope downward from Day 2 to Day 3, and then rise sharply from Day 3 to Day 4. Despite the mid-period dip on Day 3, the overall trend over the four-day period is upward.

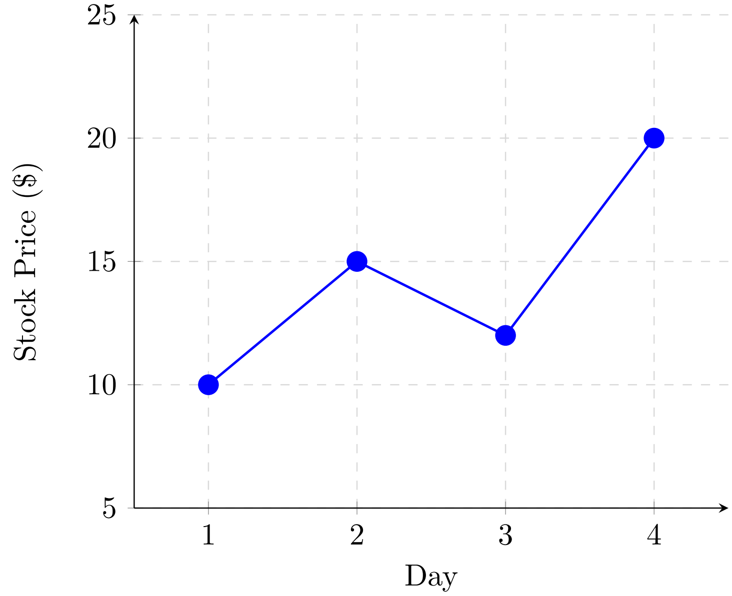


Figure 4.25 Sample trajectory for a continuous trend line.

▲

Example 4.26 Using the total-loss collision dataset compiled by the insurance provider in Table 4.11, construct a line graph to visually model the distribution of vehicle colors.

Solution. To construct a line graph for this dataset, we plot the vehicle color categories along the horizontal axis and the collision frequencies on the vertical axis. Instead of drawing columns, we plot a single coordinate point for each category and connect them sequentially with straight line segments:

- ***Blue:*** Point at 25
- ***Green:*** Point at 52
- ***Red:*** Point at 41
- ***White:*** Point at 36
- ***Black:*** Point at 39
- ***Grey:*** Point at 23

This visualization allows us to trace the ups and downs across categories, showing a sharp peak at Green and a drop at Grey.

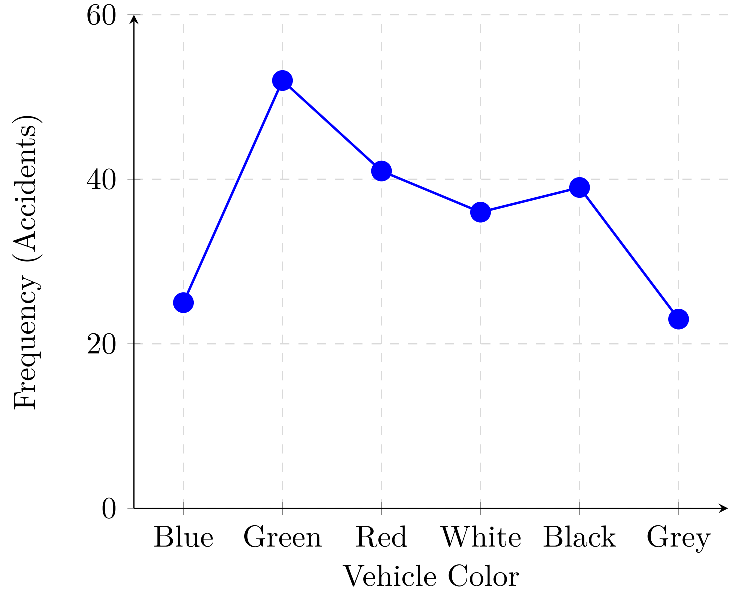


Figure 4.27 Line Graph of Total-Loss Collisions by Vehicle Color

▲

Checkpoint 4.28 A patient's body temperature is recorded every hour: Hour 1 (98.6°F), Hour 2 (100.2°F), Hour 3 (101.5°F), and Hour 4 (99.1°F). Between which two hours does the line graph show the sharpest increase?

Solution. To find the sharpest increase, we calculate the differences between consecutive hourly measurements:

The increase from Hour 1 to Hour 2 is:

$$100.2 - 98.6 = 1.6°\text{F}$$

The increase from Hour 2 to Hour 3 is:

$$101.5 - 100.2 = 1.3°\text{F}$$

Comparing these calculations, the graph shows the sharpest increase between Hour 1 and Hour 2.

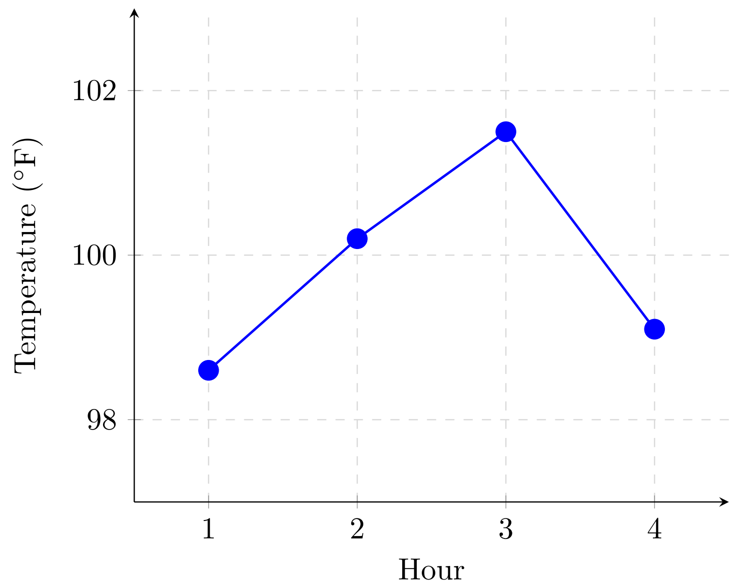


Figure 4.29 Hourly tracking of patient body temperature.

Definition 4.30 Pie Chart. ***Pie charts*** are designed for visualizing proportions and showing how a single, cohesive entity is divided into individual parts. The chart consists of a circle divided into sectors, or slices, where the arc length and central angle of each slice are strictly proportional to the percentage of the total dataset it represents. ◆

This format is particularly effective for illustrating the relative sizes of categories within a whole, such as market share distribution among companies or the breakdown of a budget into different expense categories.

Example 4.31 A budget consists of \$500 for Rent, \$250 for Food, and \$250 for Utilities. Calculate the percentage of the pie chart slice that represents Rent.

Solution. First, find the total budget by summing the individual expenses:

$$500 + 250 + 250 = 1000$$

Next, divide the Rent cost by the total budget to find its proportion:

$$\frac{500}{1000} = 0.50$$

Multiply by 100 to convert the decimal into a percentage:

$$0.50 \times 100 = 50\%$$

The Rent slice will take up exactly half (50%) of the pie chart.

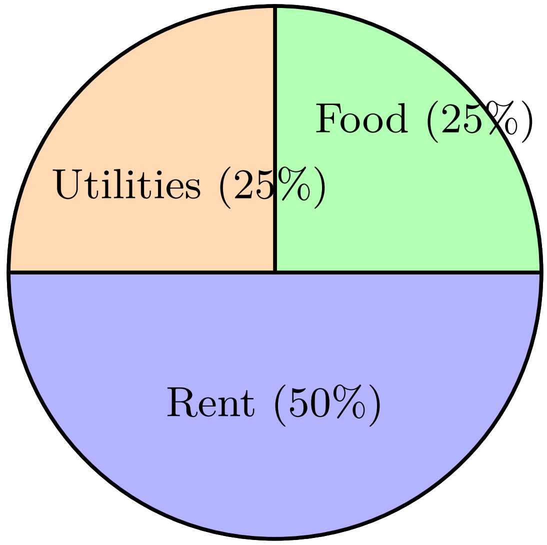


Figure 4.32 Proportional slice distribution in a circular layout.

▲

Example 4.33 Using the total-loss collision dataset compiled by the insurance provider in Table 4.11, construct a pie chart to visually model the distribution of vehicle colors.

Solution. To construct a pie chart, we convert each category's frequency into a proportional central angle of a circle ($360°$). Using the total sample size ($n = 216$), the angle for each slice is calculated using the formula Angle $= \frac{\text{Frequency}}{216} \times 360°$:

- ***Blue:*** $\frac{25}{216} \times 360° \approx 41.7°$ (11.57%)
- ***Green:*** $\frac{52}{216} \times 360° \approx 86.7°$ (24.07%)
- ***Red:*** $\frac{41}{216} \times 360° \approx 68.3°$ (18.98%)
- ***White:*** $\frac{36}{216} \times 360° \approx 60.0°$ (16.67%)
- ***Black:*** $\frac{39}{216} \times 360° \approx 65.0°$ (18.06%)
- ***Grey:*** $\frac{23}{216} \times 360° \approx 38.3°$ (10.65%)

Each sector is drawn sequentially around the center of the circle to construct the final visual proportion.

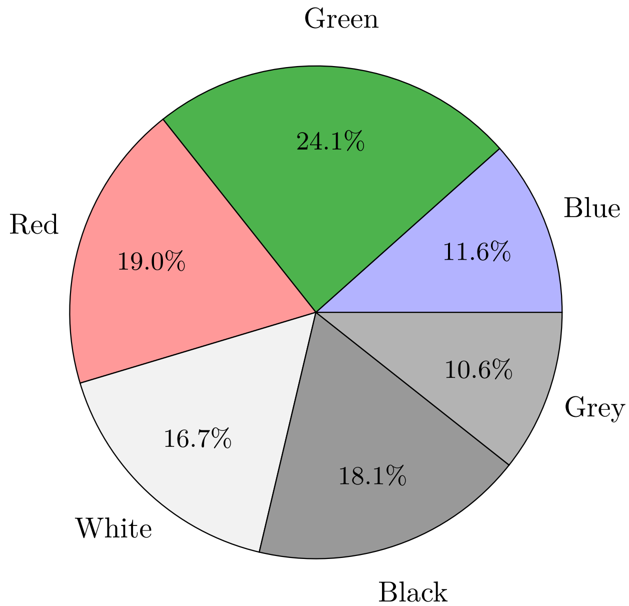


Figure 4.34 Pie Chart of Total-Loss Collisions by Vehicle Color

▲

Checkpoint 4.35 A survey of 200 people shows that 100 prefer vanilla ice cream, 60 prefer chocolate, and 40 prefer strawberry. What central angle degree should be used to draw the slice for chocolate?

Solution. The proportion for chocolate is:

$$\frac{60}{200} = 0.30 \quad (30\%)$$

Since a full circle has 360°, multiply the proportion by 360 to find the central angle:

$$0.30 \times 360 = 108$$

The chocolate slice requires a central angle of 108°.

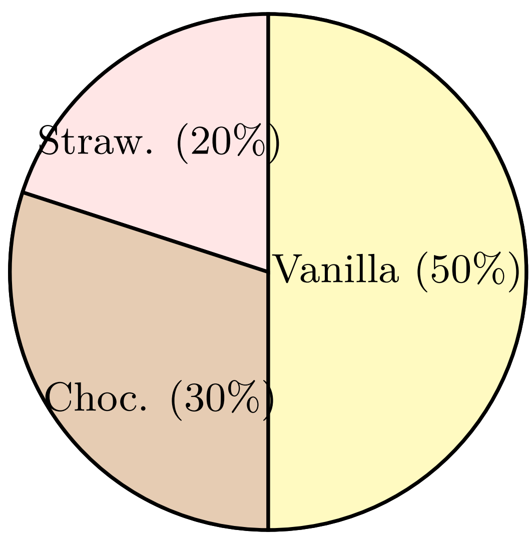


Figure 4.36 Pie chart representing ice cream flavor preferences.

Checkpoint 4.37 The following dataset represents the scores of 24 students on a recent history test:

80, 50, 50, 90, 70, 70, 100, 60, 70, 80, 70, 50, 90, 100, 80, 70, 30, 80, 80, 70, 100, 60, 60, 50

a Complete a frequency table for the test scores.

b Construct a bar graph of the data.

c Construct a line graph of the data.

d Construct a pie chart of the data.

e Construct a histogram of the data.

Solution. ***(a) Frequency Table:*** Sorting and counting each distinct score yields the following frequency distribution:

Table 4.38 Frequency Distribution of History Test Scores

Test Score	**Frequency (Students)**
30	1
50	4
60	3
70	6
80	5
90	2
100	3

(b) Bar Graph: We represent each distinct score on the horizontal axis and its frequency count on the vertical axis using separated vertical columns:

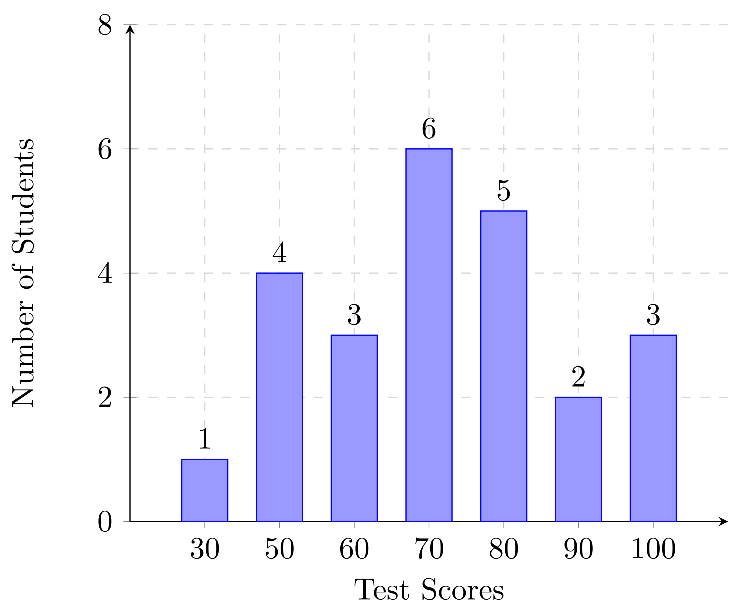


Figure 4.39 Bar Graph of Test Scores

(c) Line Graph: Using the same axes as the bar chart, coordinate markers are plotted at each data junction and linked sequentially using line tracks:

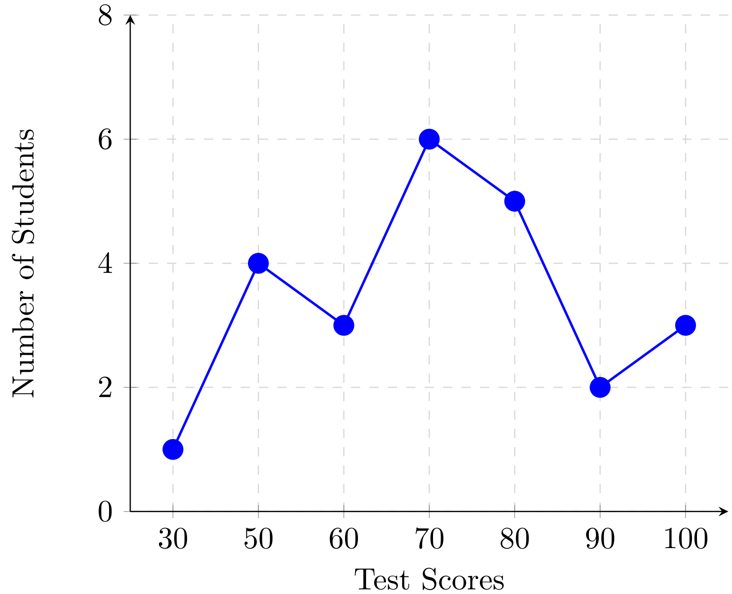


Figure 4.40 Line Graph of Test Scores

(d) Pie Chart: To map out global proportions, we compute percentage shares based on the total sample of $n = 24$ students (Angle $= \frac{\text{Freq}}{24} \times 360°$):

- ***Score 30:*** 4.17% (15°)
- ***Score 50:*** 16.67% (60°)
- ***Score 60:*** 12.50% (45°)
- ***Score 70:*** 25.00% (90°)
- ***Score 80:*** 20.83% (75°)
- ***Score 90:*** 8.33% (30°)
- ***Score 100:*** 12.50% (45°)

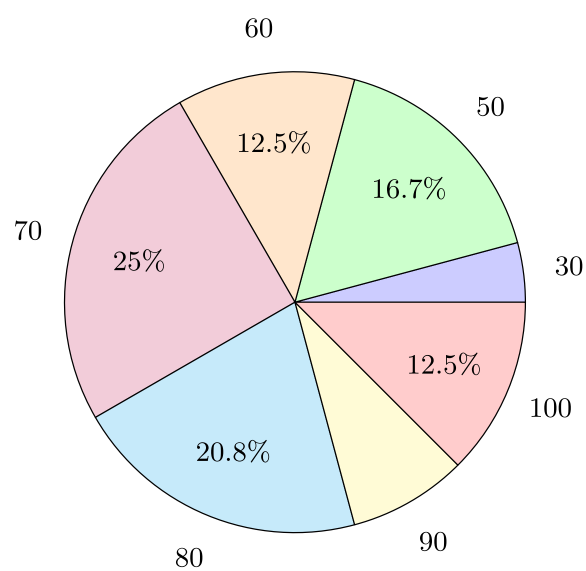


Figure 4.41 Pie Chart of Test Score Distributions

(e) Histogram: We group the scores into 10-point bins along a continuous horizontal axis causing the bars to lock tightly side-by-side:

- ***[25, 35):*** 1 student (score 30)
- ***[35, 45):*** 0 students
- ***[45, 55):*** 4 students (scores of 50)
- ***[55, 65):*** 3 students (scores of 60)
- ***[65, 75):*** 6 students (scores of 70)
- ***[75, 85):*** 5 students (scores of 80)
- ***[85, 95):*** 2 students (scores of 90)
- ***[95, 105]:*** 3 students (scores of 100)

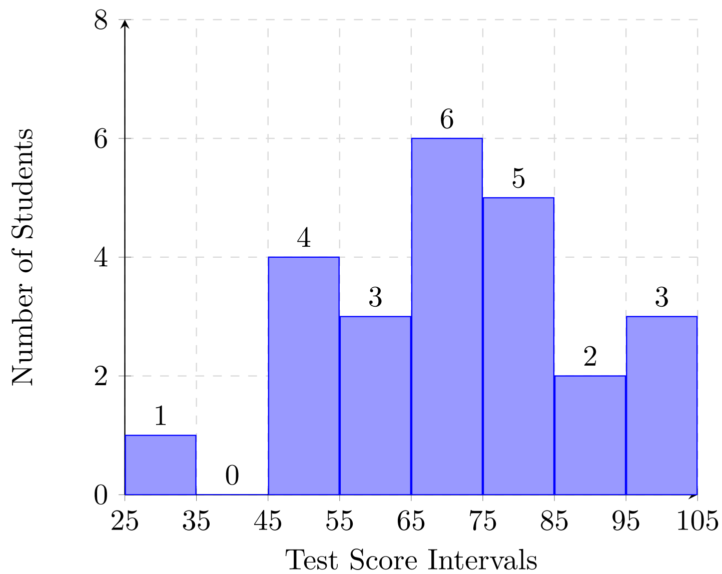


Figure 4.42 Histogram of Test Score Distributions

(e) Histogram: Unlike the categorical charts above, a true histogram tracks quantitative continuous intervals. We group the scores into 20-point bins along a continuous horizontal axis causing the bars to lock tightly side-by-side:

- ***[20, 40):*** 1 student (score 30)
- ***[40, 60):*** 4 students (scores of 50)
- ***[60, 80):*** 9 students (scores of 60 and 70)
- ***[80, 100):*** 7 students (scores of 80 and 90)
- ***[100, 120):*** 3 students (scores of 100)

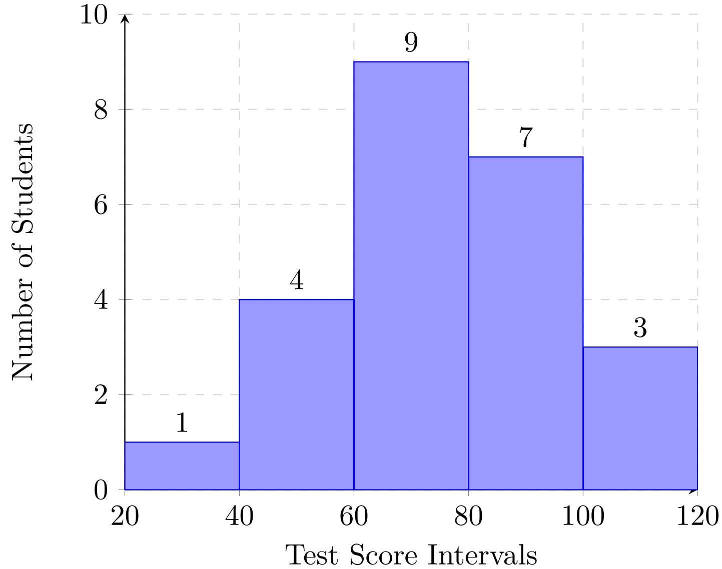


Figure 4.43 Histogram of Test Score Distributions

4.2 Measures of Center and Spread

Understanding distributions and their measures of center and spread is crucial for solving many statistics problems on the math practice. These concepts help summarize data sets concisely and allow for easier comparison and interpretation.

The center of a distribution describes a typical value of the data set and can be represented by the mean, median, or mode. The spread of a distribution indicates how much the data varies and can be measured using the range and standard deviation.

Definition 4.44 Mean. The mean, or average, is calculated by summing all values in a data set and dividing by the number of values. It represents a central point in the data. The mean is a measure of center that is sensitive to every value in the data set, making it particularly useful when the values are relatively close to each other.

The Greek letter sigma, $\sum$, called a symbol of summation, is used to indicate the sum of data items. The notation $\sum x$, read ``the sum of x,'' means to add all the data items in a given data set. We can use this symbol to give a formula for calculating the mean.

The **mean** of a set of data is the sum of the data values divided by the number of values;

$$\text{mean} = \bar{x} = \frac{\sum x}{n}$$

◆

Example 4.45 Find the mean of the data set $\{2, 4, 6, 8, 10\}$.

Solution. Sum the values ($\sum x = 2 + 4 + 6 + 8 + 10 = 30$) and divide by the number of values ($n = 5$). The mean is $\bar{x} = \dfrac{30}{5} = 6$. ▲

Example 4.46 Marci's exam scores for her last math class were: $79, 86, 82, 94$. Find the mean of these exam scores and round it appropriately.

Solution. To find the mean, sum the scores and divide by the total number of exams:

$$\bar{x} = \frac{\sum x}{n} = \frac{79 + 86 + 82 + 94}{4} = 85.25$$

Typically, we round means to one more decimal place than the original data had. Since the original scores are whole numbers, we round our final answer to one decimal place. Therefore, we round 85.25 to 85.3. ▲

Checkpoint 4.47 The number of touchdown (TD) passes thrown by each of the 31 teams in the National Football League in the 2000 season are shown below:

Table 4.48 NFL Touchdown Passes (2000 Season)

37	33	33	32	29	28	28	23	22	22	22
21	21	21	20	20	19	19	18	18	18	18
16	15	14	14	14	12	12	9	6		

Find the mean number of touchdown passes thrown during this season.

Solution. First, find the sum of all the values in the data set:

$$37 + 33 + 33 + 32 + 29 + 28 + 28 + 23 + 22 + 22 + 22 + 21 + 21+$$

$$21 + 20 + 20 + 19 + 19 + 18 + 18 + 18 + 18 + 16 + 15 + 14 + 14+$$

$$14 + 12 + 12 + 9 + 6 = 634 \text{ total TDs}$$

Next, divide by 31, which is the total number of data values:

$$\bar{x} = \frac{634}{31} \approx 20.4516$$

Following the standard rounding convention, we round the mean to one more decimal place than the original data. Since the raw touchdown counts are whole numbers, we round our final answer to one decimal place, giving 20.5.

It is most correct to report that "The mean number of touchdown passes thrown in the NFL in the 2000 season was 20.5 passes," though it is common to see the more casual term "average" used in place of "mean."

Checkpoint 4.49 The price of a jar of peanut butter at 5 stores was: \$3.29, \$3.59, \$3.79, \$3.75, and \$3.99. Find the mean price.

Solution. First, find the sum of the prices across all 5 stores:

$$3.29 + 3.59 + 3.79 + 3.75 + 3.99 = 18.41$$

Next, divide the total sum by the number of stores (5):

$$\text{Mean} = \bar{x} = \frac{18.41}{5} = 3.682$$

Following the standard rounding rule for the mean (rounding to one more decimal place than the original data), we round from two decimal places to three decimal places. The exact mean price is \$3.682 (or approximately \$3.68 when rounded to the nearest standard cent).

Definition 4.50 Median. The **median** is the middle value of a data set when the values are arranged in ascending order.

To find the median of a dataset:

1. Arrange the data in order from smallest to largest.
2. If the number of items in the dataset is ***odd***, the median is the exact

middle number in the ordered dataset.

3. If the number of items in the dataset is ***even***, the median is the mean of the two middle numbers in the ordered dataset.

◆

The median is a useful measure of center because it is not affected by extremely high or low values (outliers). This makes it a better representative of the data set when there are outliers present.

Example 4.51 Odd Number of Values. Find the median of $\{3, 5, 7, 9, 11\}$.

Solution. The data set is already in order. The median is 7. ▲

Example 4.52 Even Number of Values. Find the median of $\{5, 3, 1, 7, 11, 9\}$.

Solution. The middle values are 5 and 7. The median is $(5+7)/2 = 6$. ▲

Checkpoint 4.53 Returning to the football touchdown data from Checkpoint 4.47, find the median number of touchdown passes thrown in the NFL during the 2000 season.

Solution. To find the median, we start by listing the data in order. As seen in Table 4.48, the data is already in decreasing order, so we can work with it without needing to reorder it first.

Since there are 31 data values (an odd number), the median will be the middle number, which is the 16th data value. We can compute its position by evaluating $31/2 = 15.5$ and rounding up to 16, which leaves exactly 15 values below it and 15 values above it.

Counting to the 16th data value in the ordered list yields 20. Therefore, the median number of touchdown passes in the 2000 season was 20 passes. Notice that for this data, the median is fairly close to the mean of 20.5 that we calculated earlier in Checkpoint 4.47.

Checkpoint 4.54 Find the median of these quiz scores:

$$5, 10, 8, 6, 4, 8, 2, 5, 7, 7$$

Solution. We start by listing the data in ascending order:

$$2, 4, 5, 5, 6, 7, 7, 8, 8, 10$$

Since there are 10 data values (an even number), there is no single middle number. Instead, we find the mean of the two middle numbers, which are the 5th and 6th values: 6 and 7.

Calculating their average gives:

$$\frac{6+7}{2} = 6.5$$

Therefore, the median quiz score was 6.5.

Checkpoint 4.55 The price of a jar of peanut butter at 5 stores was: \$3.29, \$3.59, \$3.79, \$3.75, and \$3.99. Find the median price.

Solution. First, arrange the prices in ascending order:

$$3.29,\ 3.59,\ 3.75,\ 3.79,\ 3.99$$

Since there is an odd number of data values ($n = 5$), the median is the exact middle number, which is the 3rd value in our ordered list.

The middle value is 3.75. Therefore, the median price for a jar of peanut butter is \$3.75.

Definition 4.56 Mode. The **mode** is the value that appears most frequently (often) in a data set.

A data set can have no mode, one mode, or multiple modes.

The mode is useful for understanding which values are most common in the data set. The mode is particularly useful for categorical data, where we are interested in knowing the most frequent category. ◆

Example 4.57 Find the mode of the data set $\{4, 4, 6, 8, 8, 8, 9\}$.

Solution. The mode is 8 because it appears most frequently. ▲

Example 4.58 In a vehicle color survey, the following frequency data was collected:

Table 4.59 Vehicle Color Survey Frequencies

Color	**Frequency**
Blue	3
Green	5
Red	4
White	3
Black	2
Grey	3

Find the mode of this data set.

Solution. For this data, "Green" is the mode, since it is the data value that occurred most frequently with a count of 5.

It is possible for a data set to have more than one mode if several categories share the same highest frequency, or no modes if every single category occurs only once. ▲

Checkpoint 4.60 Reviewers were asked to rate a product on a scale of 1 to 5. The results are shown in the frequency table below:

Table 4.61 Product Rating Frequencies

Rating	**Frequency**
1	4
2	8
3	7
4	3
5	1

Find:

a. The mean rating

b. The median rating

c. The mode rating

Solution. First, determine the total number of reviewers (n) by summing the frequencies:

$$n = 4 + 8 + 7 + 3 + 1 = 23$$

a. To find the mean rating, calculate the weighted sum of the ratings and divide by the total number of reviewers:

$$\text{Sum} = (1 \times 4) + (2 \times 8) + (3 \times 7) + (4 \times 3) + (5 \times 1)$$

$$\text{Sum} = 4 + 16 + 21 + 12 + 5 = 58$$
$$\text{Mean} = \frac{58}{23} \approx 2.5217$$

Rounding to one more decimal place than the original data gives a mean rating of 2.5.

b. Since there are 23 data values (an odd number), the median is the middle value, located at position $(23+1)/2 = 12$. Accumulating frequencies from the lowest rating up to the 12th value:

- Ratings of 1 account for the first 4 values (positions 1–4).
- Ratings of 2 account for the next 8 values (positions 5–12).

The 12th value falls exactly at the end of the 2-rating group, so the median rating is 2.

c. The mode is the value with the highest frequency. A rating of 2 has the highest frequency with 8 reviews, so the mode rating is 2.

Definition 4.62 Measures of Spread. Measures of spread describe how much the data varies. Two common measures are ***range*** and ***standard deviation***. These measures help to understand the variability within the data set. ◆

Consider these three sets of quiz scores:

- Section A: $5, 5, 5, 5, 5, 5, 5, 5, 5, 5$
- Section B: $0, 0, 0, 0, 0, 10, 10, 10, 10, 10$
- Section C: $4, 4, 4, 5, 5, 5, 5, 6, 6, 6$

All three of these sets of data have a mean of 5 and a median of 5, yet the sets of scores are clearly quite different. In Section A, everyone had the same score. In Section B, half the class got no points and the other half got a perfect score, assuming this was a 10-point quiz. Section C was not as consistent as Section A, but not as widely varied as Section B.

In addition to the mean and median, which are measures of the "typical" or "middle" value, we also need a measure of how "spread out" or varied each data set is. There are several ways to measure this "spread" of the data. The first is the simplest and is called the **range**.

Note 4.63 Range. The range is the difference between the maximum and minimum values in a data set. It gives a quick sense of the spread of the data. A larger range indicates greater variability, while a smaller range indicates less variability.

Example 4.64 Find the range of the data set $\{1, 3, 5, 7, 9\}$.

Solution. The range is $9 - 1 = 8$. ▲

Example 4.65 Using the quiz scores from above, find the range for Section A, Section B, and Section C.

Solution. The range is calculated by subtracting the minimum score from the maximum score in each data set:

- For Section A, the maximum value is 5 and the minimum value is 5. The range is $5 - 5 = 0$.
- For Section B, the maximum value is 10 and the minimum value is 0. The range is $10 - 0 = 10$.

- For Section C, the maximum value is 6 and the minimum value is 4. The range is $6 - 4 = 2$.

▲

A second measure of spread, and one that is dependent on all of the data items, is called the ***standard deviation***.

Note 4.66 Standard Deviation. The **standard deviation** is a number that summarizes how far each data item differs from the mean. In comparing datasets, a dataset with a smaller standard deviation shows that the data are less spread out around the mean. A dataset with a larger standard deviation shows that the data are more spread out around the mean.

Example 4.67 Using the quiz scores from Example 4.65, Section A, Section B, and Section C all have the same mean of 5. However, they have very different spreads. Which sample do you think has the largest standard deviation? What does this measure tell us when comparing the spread of the three samples?

Solution. Section B has the largest standard deviation because its scores are the most spread out from the mean. The standard deviation is a more precise measure of spread than the range because it accounts for how every individual data point differs from the mean, rather than just looking at the extremes. ▲

Approximating Standard Deviation Without Computation.

To estimate the standard deviation of a small dataset, follow these steps:

1. Find the Mean: Calculate the average of your data points to establish the central point.
2. Find individual distances: Determine how far each data point sits from that mean value. Disregard negative signs.
3. Average the distances: Sum the distances and divide by the number of values to get a baseline average deviation.
4. Adjust upward: Select an estimate slightly higher than your average distance, as squaring the deviations shifts the geometric mean upward.

Checkpoint 4.68 Use the data 1, 2, 3, 5, 6, 7. Without actually computing the standard deviation, which of the following best approximates the standard deviation?

A. 2

B. 6

C. 10

D. 20

Answer. A.

Solution 1. Step 1: Find the Mean. Find the central balance point of the data:

$$\text{Mean} = \frac{1+2+3+5+6+7}{6} = 4$$

Step 2: Find the Absolute Distances. Calculate how far each value is from the mean of 4:

$$|1-4| = 3, \quad |2-4| = 2, \quad |3-4| = 1, \quad |5-4| = 1, \quad |6-4| = 2, \quad |7-4| = 3$$

Step 3: Average the Distances. Find the average of these absolute deviations:

$$\text{Average Distance} = \frac{3+2+1+1+2+3}{6} = \frac{12}{6} = 2$$

Step 4: Approximate. The true standard deviation is always slightly higher than the simple average distance. Therefore, we approximate the standard deviation to be a value **just above 2**.

Solution 2.

A. *Correct.*

Correct. The mean is 4 and the range is 6. The standard deviation represents the typical distance from the mean, so 2 is the best estimate.

B. *Incorrect.*

Incorrect. A standard deviation of 6 would imply the data points span much further from the mean than they actually do.

C. *Incorrect.*

Incorrect. The standard deviation cannot be larger than the total range of the dataset.

D. *Incorrect.*

Incorrect. This value is much too large for a small dataset spanning from 1 to 7.

Effect of Outliers

Outliers are values significantly different from other values in a data set. They can greatly affect summary statistics like the mean, median, mode, range, and standard deviation.

Consider the baseline data set $\{1, 2, 2, 3, 100\}$ where the outlier is 100. The sections below highlight how this outlier impacts each statistic:

Effect on Mean Outliers can significantly skew the mean of a data set. Including it, the mean is skewed higher. Removing it, the mean is more representative of the majority of the data.

Effect on Median The median is less affected by outliers because it is based on the middle values of the data set. In the data set $\{1, 2, 2, 3, 100\}$, the median remains 2 regardless of the outlier.

Effect on Mode Outliers have little to no effect on the mode since the mode is determined by the most frequently occurring values. In the data set $\{1, 2, 2, 3, 100\}$, the mode is still 2.

Effect on Range Outliers can drastically increase the range of a data set since the range is the difference between the maximum and minimum values. In the data set $\{1, 2, 2, 3, 100\}$, the range is $100 - 1 = 99$, which is significantly affected by the outlier.

Effect on Standard Deviation Outliers increase the standard deviation because they increase the average distance from the mean. The standard deviation is much larger when the outlier is included compared to when it is excluded.

Practice Questions

1. Find the mean of the data set $\{4, 6, 8, 10, 12\}$.

Solution. First, sum all the values in the data set:

$$4 + 6 + 8 + 10 + 12 = 40$$

Next, divide the sum by the total number of values ($n = 5$):

$$\text{Mean} = \frac{40}{5} = 8$$

2. Find the median of the data set $\{7, 3, 9, 1, 5\}$.

Solution. First, arrange the data set in ascending order: $\{1, 3, 5, 7, 9\}$. Since there is an odd number of values ($n = 5$), the median is the exact middle value. The middle value is 5.

3. Find the mode of the data set $\{2, 4, 4, 6, 6, 6, 8\}$.

Solution. Count the frequency of each value in the set:

- 2 appears 1 time
- 4 appears 2 times
- 6 appears 3 times
- 8 appears 1 time

The value 6 appears most frequently, so the mode is 6.

4. Find the range of the data set $\{14, 18, 21, 24, 29\}$.

Solution. Identify the maximum and minimum values in the data set:

$$\text{Maximum} = 29, \quad \text{Minimum} = 14$$

Subtract the minimum value from the maximum value:

$$\text{Range} = 29 - 14 = 15$$

5. If the mean of $\{2, 3, 5, 7, x\}$ is 5, what is x?

Solution. The formula for the mean of 5 values is:

$$\frac{2 + 3 + 5 + 7 + x}{5} = 5$$

Simplify the numerator by adding the known constants:

$$\frac{17 + x}{5} = 5$$

Multiply both sides by 5 to isolate the numerator:

$$17 + x = 25$$

Subtract 17 from both sides to find x:

$$x = 25 - 17 = 8$$

6. Compute the mean for each of the following data sets:

- Set A: $80, 80, 80, 80$
- Set B: $70, 70, 90, 90$

Using these means and without actually computing the standard deviation, can you tell which dataset has a lower standard deviation? Does a lower standard deviation imply that the dataset is less spread out or more spread out than the other dataset?

Solution. Both datasets have a mean of 80. We calculate this by summing the values and dividing by 4:

$$\text{Mean}_A = \frac{80 + 80 + 80 + 80}{4} = 80$$

$$\text{Mean}_B = \frac{70 + 70 + 90 + 90}{4} = 80$$

Without calculating the exact standard deviation, we can determine that Set A has the lower standard deviation. Every data point in Set A is identical to the mean, resulting in a standard deviation of exactly 0.

A lower standard deviation implies that the dataset is **less spread out** than the other dataset. It indicates that the individual data entries cluster tightly around the center point rather than deviating far from it.

Frequently Asked Questions

What is the difference between mean, median, and mode? The mean is the average (sum of values divided by count), the median is the middle value when data is ordered, and the mode is the most frequently occurring value. The median is less affected by outliers than the mean.

What is standard deviation? Standard deviation measures the typical spread from the mean—it is the average distance between each data point and the mean. A larger standard deviation indicates greater variability in the data set.

How do I find the median of an even number of values? When there is an even number of values, the median is the average of the two middle values. First arrange the data in order, then identify the two middle values and calculate their average.

4.3 The Empirical Rule

Definition 4.69 Bell-Curve. ***A bell curve*** is used in statistics to represent the continuous probability distribution of a naturally occurring variable. It is perfectly symmetrical, showing that data near the mean occurs more frequently than data far from the mean. ◆

For example, standardized test scores, human heights, and manufacturing errors typically follow this bell-shaped trajectory when plotted in mass quantities.

Example 4.70 An instructor plots final exam grades and notes they follow a normal distribution with a mean score of 75. Describe how the shape of this curve changes if you look at scores moving away from 75 in either direction.

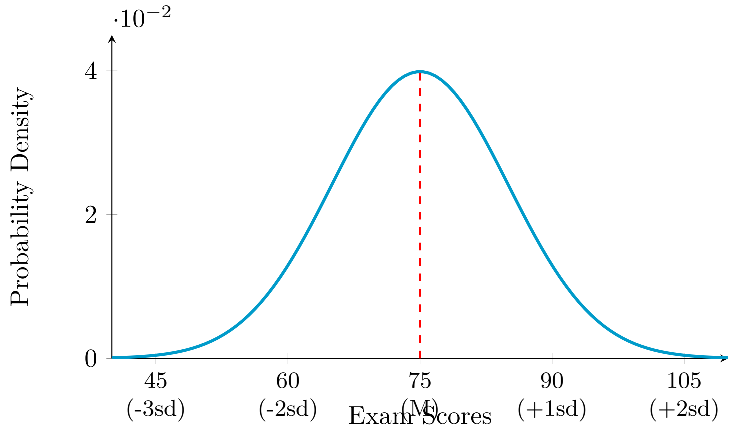


Figure 4.71 Symmetrical distribution curve centered around a population mean.

Solution. The highest point of the curve sits directly above the mean score of 75. As you move outward toward lower scores (left) or higher scores (right) standard deviations 15 away from the mean, the curve slopes downward symmetrically. The frequency of scores decreases as you move further from the mean, flattening out as it approaches the extremes of 45 and 105 where very few students scored. ▲

Checkpoint 4.72 If a dataset of adult heights perfectly matches a bell curve layout, what percentage of the population profile falls below the exact center peak value of the distribution?

Solution. Because a true normal distribution curve is perfectly symmetrical, the center peak represents both the mean and the median. Therefore, exactly 50% of the population data falls below this center peak line.

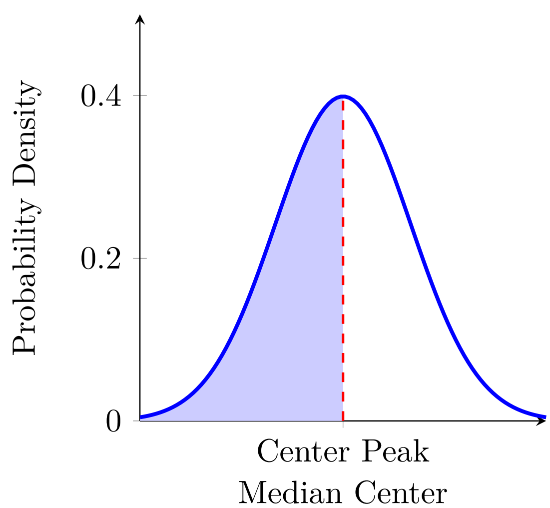


Figure 4.73 Symmetrical normal distribution curve with the lower half shaded.

Note that the standard deviation is a measure of how spread out data points are from their mean. To determine if a specific data value is far from the center, you calculate exactly how many standard deviations it sits away from that mean.

A data distribution shows how data points cluster or spread across their range of values. ***A normal distribution*** is a specific, symmetrical bell-shaped curve with predictable patterns. Exactly half (50%) of the data lies below the mean, and half lies above it, as illustrated in Figure 4.73.

Definition 4.74 Normal Distribution. A **normal distribution** is a symmetrical, bell-shaped probability distribution where the mean, median, and mode are all equal. ◆

Example 4.75 Normal Distribution to IQ Scores. IQ scores follow a normal distribution with a mean of 100 and a standard deviation of 16. Sketch the normal curve for this example and be sure to denote one, two, and three standard deviations from the mean.

Solution. The normal curve for IQ scores would be centered at 100, with the highest point of the bell curve directly above this mean. One standard deviation from the mean would be at 84 (100 - 16) and 116 (100 + 16). Two standard deviations from the mean would be at 68 (100 - 32) and 132 (100 + 32). Three standard deviations from the mean would be at 52 (100 - 48) and 148 (100 + 48). The curve would slope downward symmetrically as it moves away from the center, tapering off near the horizontal axis at the extremes.

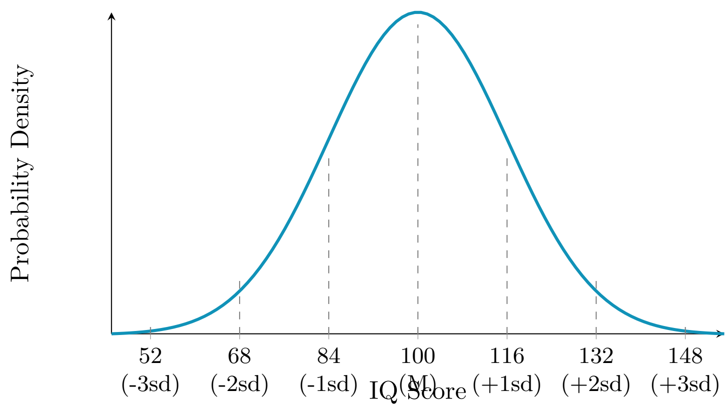


Figure 4.76 Normal distribution curve for IQ scores ($Mean = 100$, $sd = 16$).

▲

Definition 4.77 The empirical rule. The **empirical rule** (or 68-95-99.7 rule) states that for any data set following a symmetrical, bell-shaped normal distribution, almost all data points lie within three standard deviations of the mean:

- Approximately 68% of the data points fall within one standard deviation of the mean.
- Approximately 95% of the data points fall within two standard deviations of the mean.
- Approximately 99.7% of the data points fall within three standard deviations of the mean.

◆

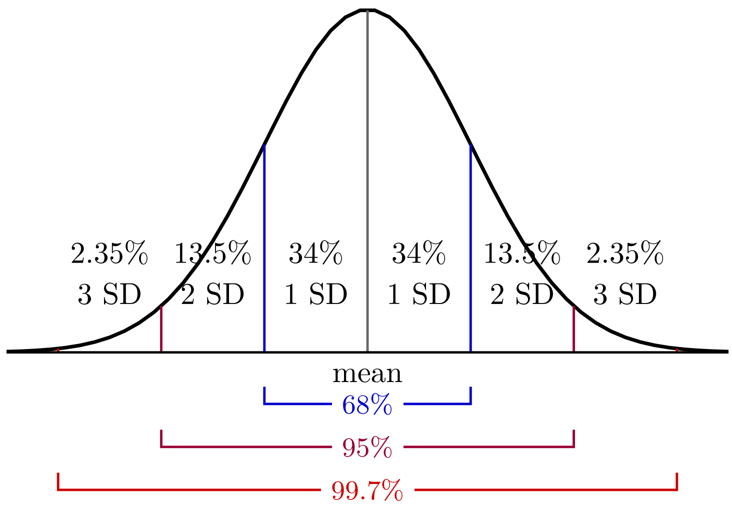


Figure 4.78 The 68-95-99.7 empirical rule for a normal distribution.

This graphic provides a concise summary of the vital statistics of a normal distribution. Because the graph resembles a bell, the normal distribution is also widely known as a "bell curve." It possesses unique, predictable probability properties:

- Exactly 50% of the data lies above the mean, and 50% lies below it.
- Approximately 68% of the data occurs within 1 standard deviation of the mean.
- Approximately 95% of the data occurs within 2 standard deviations of the mean.
- Approximately 99.7% of the data occurs within 3 standard deviations of the mean.

Because of these distinct probability ranges, the empirical rule is also frequently called the 68-95-99.7 rule.

Example 4.79 Applying the Empirical Rule to IQ Scores. Return to the distribution of IQ scores from Example 4.75. Recall that IQ scores follow a normal distribution with a mean of 100 and a standard deviation of 16 (see Figure 4.76).

1 According to the Empirical Rule, approximately what percentage of people have IQ scores between 68 and 132?

2 According to the Empirical Rule, approximately 99.7% of people have IQ scores between what two values?

3 According to the Empirical Rule, approximately what percentage of people have an IQ score greater than 116?

4 According to the Empirical Rule, approximately what percentage of people have an IQ score less than 68?

Solution. The Empirical Rule (or 68-95-99.7 Rule) states that for a normally distributed dataset, approximately 68% of the data falls within 1 standard deviation of the mean, 95% falls within 2 standard deviations, and 99.7% falls within 3 standard deviations. Given $Mean = 100$ and $sd = 16$:

1 The scores 68 and 132 represent exactly two standard deviations below and above the mean, respectively:

$$100 - 2(16) = 68 \quad \text{and} \quad 100 + 2(16) = 132$$

According to the Empirical Rule, approximately 95% of people have IQ scores within this range.

2 The rule states that 99.7% of data falls within 3 standard deviations of the mean ($\pm 3s$):

$$100 - 3(16) = 52 \quad \text{and} \quad 100 + 3(16) = 148$$

Therefore, 99.7% of people have IQ scores between 52 and 148.

3 An IQ score of 116 is exactly 1 standard deviation above the mean ($100 + 16 = 116$). Since 68% of people fall within 1 standard deviation (84 to 116), the remaining 32% are split equally between both outer tails.

$$\frac{100\% - 68\%}{2} = 16\%$$

Thus, approximately 16% of people have an IQ score greater than 116.

4 A score of 68 is exactly 2 standard deviations below the mean. Since 95% of people fall within 2 standard deviations (68 to 132), the remaining 5% are distributed evenly across the outer left and right tails.

$$\frac{100\% - 95\%}{2} = 2.5\%$$

Thus, approximately 2.5% of people have an IQ score less than 68.

▲

Example 4.80 Human Height Distribution. Human height is commonly considered an approximately normally distributed measure. If the mean height of adult males in the United States is 5' 10", with a standard deviation of 1.5", how common are men with heights greater than 6' 1"?

Solution. Since the mean is 5' 10" and the standard deviation is 1.5", a height of 6' 1" is exactly 3 inches above the mean. This places it exactly 2 standard deviations above the average.

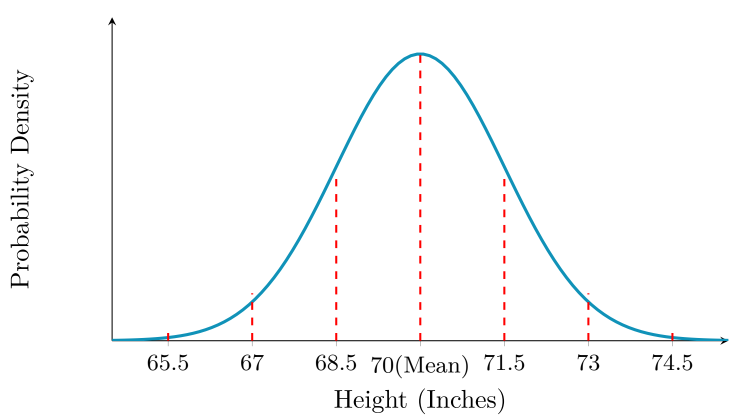


Figure 4.81 Normal distribution curve for U.S. adult male height ($Mean = 70$, $sd = 1.5$).

Since the mean is 5' 10" (70") and the standard deviation is 1.5", a height of 6' 1" (73") is exactly 3 inches above the mean. This places it exactly 2 standard deviations above the average.

According to the empirical rule, about 95% of data falls within 2 standard deviations of the mean, meaning 5% falls outside this range. Because the distribution is symmetrical, half of that remaining amount (2.5%) sits in the upper tail. Therefore, only about 2.5% of men are taller than 6' 1", making them quite rare in this population. ▲

Checkpoint 4.82 High School High Jumpers. If the maximum jumping height of US high school high jumpers is normally distributed with a mean of 5' 11.5" and a standard deviation of 2.2", how unusual is it to see a high school jumper clear 6' 3.9"?

Solution. If the mean is 5' 11.5", then 1 standard deviation above the mean is 6' 1.7" and 2 standard deviations above the mean is 6' 3.9".

According to the empirical rule, 95% of all jumpers max out within 2 standard deviations of the mean. This leaves 5% split equally between the extremely low and extremely high ends. This means that less than 2.5% of jumpers clear 6' 3.9", making it very unusual to see a competitor exceed this height.

Checkpoint 4.83 Yard Snow Depth Distribution. Suppose the depth of the snow in a yard is normally distributed with a mean of 2.5 inches and a standard deviation of 0.25 inches. What is the probability that a randomly chosen location in the yard has a snow depth between 2.25 and 2.75 inches?

Solution. First, find how many standard deviations the given depths sit away from the center. A depth of 2.25 inches is exactly one standard deviation below the mean $(2.5 - 0.25)$. A depth of 2.75 inches is exactly one standard deviation above the mean $(2.5 + 0.25)$.

According to the empirical rule, approximately 68% of the data in a normal distribution falls within one standard deviation of the mean. Therefore, the probability that a randomly chosen location has a snow depth between 2.25 and 2.75 inches is approximately 68% (or 0.68).

4.4 Z-Scores

We will learn how z-scores can be used to evaluate how extreme a given value is within a particular set or population. By connecting these standard values to probability density distributions, you will develop the skills necessary to calculate the probability of a data point falling between two specified z-scores, as well as the ability to work backward to determine an exact z-score when given a target probability threshold.

Using the Empirical Rule can give you a good idea of the probability of occurrence of a value that happens to be exactly one, two, or three standard deviations to either side of the mean. But how do you compare the probabilities of values that are in between those standard deviations?

Z-scores are closely related to the Empirical Rule; both offer methods for evaluating how extreme a particular value is within a given dataset. You can think of a z-score as the exact number of standard deviations there are between a given value and the mean of the set. While the Empirical Rule only allows you to associate the first three integer standard deviations with fixed percentages (68%, 95%, and 99.7%), the z-score allows you to state, with arbitrary precision, exactly how many standard deviations a value lies above or below the mean.

Conceptually, the calculation is straightforward. To find the number of standard deviations between a value and the mean, first calculate the difference between your specific data point and the population mean. Then, divide that difference by the standard deviation of the dataset.

Definition 4.84 Z-Score Formula. The mathematical formula to calculate a standard z-score is:

$$z = \frac{\text{item value} - \text{mean}}{\text{standard deviation}}$$

◆

For example, with mean= 65 and standard deviation= 3, a score of 59 gives:

$$z = \frac{59 - 65}{3} = -2$$

This means the value is 2 standard deviations *below* the mean.

Example 4.85 Positive Z-score. Let mean= 24 and standard deviation= 2. Find the z-score for $x = 27$.

$$z = \frac{27 - 24}{2} = 1.5$$

The value is 1.5 standard deviations *above* the mean. ▲

Example 4.86 Negative Z-score. Let mean= 125 and standard deviation= 6.2. Find the z-score for $x = 104.5$.

$$z = \frac{104.5 - 125}{6.2} \approx -3.31$$

The resulting value is approximately 3.31 standard deviations *below* the mean. ▲

Example 4.87 Finding a Value Given a Z-score. Find the original raw data value x if the mean is 63, the standard deviation is 4.25, and the given z-score is 2.403.

$$2.403 = \frac{x - 63}{4.25}$$

Multiplying both sides by 4.25 and adding 63 yields:

$$x = 63 + (2.403)(4.25) \approx 73.21$$

▲

Example 4.88 Which monthly utility bill is better with respect to relative standing?

- A monthly utility bill of \$120 with a mean of \$130 and a standard deviation of \$10.
- A monthly utility bill of \$150 with a mean of \$170 and a standard deviation of \$5.

Both distributions are approximately bell-shaped and symmetric.

Solution. To evaluate relative standing for financial costs, we calculate the **Z-score** for each scenario. Because utility bills represent an expense, a more negative Z-score is preferable as it indicates a cost further below the group average.

For the first utility bill:

$$Z_1 = \frac{120 - 130}{10} = \frac{-10}{10} = -1.0$$

This indicates the cost is exactly 1 standard deviation below its baseline mean.

For the second utility bill:

$$Z_2 = \frac{150 - 170}{5} = \frac{-20}{5} = -4.0$$

This indicates the cost is an exceptional 4 standard deviations below its baseline mean.

Comparing the two relative standing positions, $-4.0 < -1.0$. Therefore, the \$150 utility bill is significantly better with respect to relative standing. ▲

Checkpoint 4.89 A student scores 70 on an arithmetic test and 66 on a vocabulary test. The scores for both tests are normally distributed.

- The arithmetic test has a mean of 60 and a standard deviation of 20.
- The vocabulary test has a mean of 60 and a standard deviation of 2.

On which test did the student have the better score relative to the class?

Solution. To compare academic performances across distributions with different spreads, we calculate the **Z-score** for each exam. A larger positive value indicates a score further above the classroom average.

For the arithmetic test:

$$Z_{\text{arithmetic}} = \frac{70 - 60}{20} = \frac{10}{20} = +0.5$$

This indicates the student scored 0.5 standard deviations above the arithmetic class mean.

For the vocabulary test:

$$Z_{\text{vocabulary}} = \frac{66 - 60}{2} = \frac{6}{2} = +3.0$$

This indicates the student scored an exceptional 3.0 standard deviations above the vocabulary class mean.

Comparing the two positions of relative standing, $+3.0 > +0.5$. Therefore, the student performed significantly better on the vocabulary test relative to the class.

Practice Exercises

1. A retail store calculates that the mean price for a specific model of skis is mean= \$279 with a standard deviation of = \$16. Find the z-score for a pair of skis priced at \$247.

Solution.

$$z = \frac{247 - 279}{16} = -2$$

The price is exactly 2 standard deviations below the mean.

2. The mean number of ice cream scoops served per order at a shop is 3 scoops with a standard deviation of 1 scoop. Find the z-score for an order containing 5 scoops.

Solution.

$$z = \frac{5 - 3}{1} = 2$$

The order size is 2 standard deviations above the mean.

3. The mean weight of a specific cattle breed population is mean= 1150 lbs with a standard deviation of = 77 lbs. Calculate the z-score for a cow that weighs 825 lbs.

Solution.

$$z = \frac{825 - 1150}{77} \approx -4.22$$

The cow's weight is approximately 4.22 standard deviations below the mean.

4. Compute the z-score for a sample measurement of $x = 0.0034$ taken from a population where mean= 0.0041 and standard deviation= 0.0008.

Solution.

$$z = \frac{0.0034 - 0.0041}{0.0008} = -0.875$$

4.5 Scatter Plots and Regression Lines

When two data items are collected for every person or object in a sample, the data items can be visually displayed using a scatter plot.

Definition 4.90 Scatter Plot. A scatter plot is a type of plot that displays the relationship between two numerical variables. Each point on the plot represents a single observation, with its position determined by the values of the two variables. ◆

Example 4.91 Consider the relationship between the number of hours studied and the test scores of students. A scatter plot would display each student as a point, with the x-axis representing hours studied and the y-axis representing test scores.

Table 4.92 Student Study Hours versus Test Scores

Student	**Hours Studied (x)**	**Test Score (y)**
A	1.5	65
B	2.0	72
C	3.5	80
D	4.0	78
E	5.5	88
F	6.0	95
G	6.5	90
H	7.0	98

Solution.

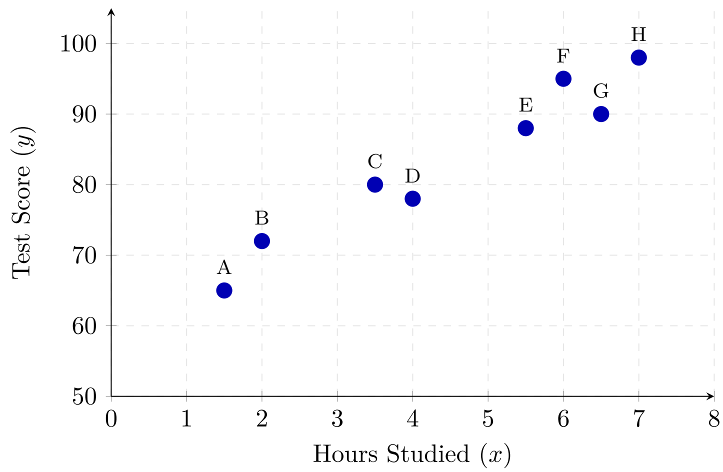


Figure 4.93 Scatter Plot of Student Study Hours versus Test Scores

By plotting the data points on a scatter plot, we can visually inspect the relationship between the two variables. The resulting chart displays a clear upward trend from left to right, showing a positive linear association. ▲

Checkpoint 4.94 Make a scatter plot for the given data provided in Table 4.95. Use the scatter plot to describe whether or not the variables appear to be related.

Table 4.95 Dataset of Variables x and y

Observation	**Variable x**	**Variable y**
1	1	12
2	2	18
3	3	22
4	4	35
5	5	31
6	6	48
7	7	52

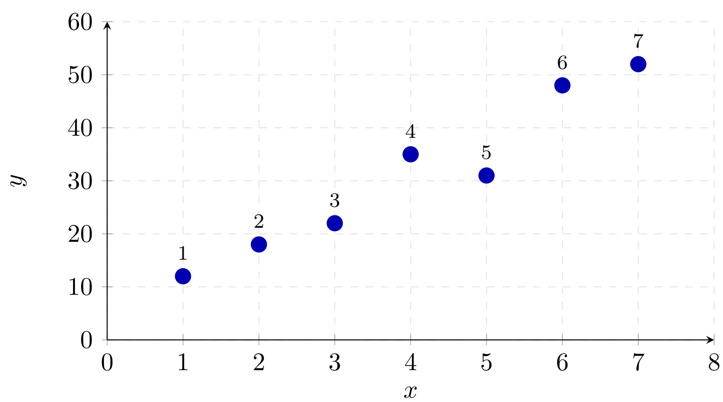


Figure 4.96 Scatter plot modeling the relationship between variables x and y.

Solution. By examining the scatter plot in Figure 4.96, we can visually interpret the relationship between the two variables.

As we look from left to right along the horizontal axis, the data points show a clear upward pattern. This tells us that as variable x increases, variable y

generally increases as well. Because the points follow a relatively straight, rising path, the variables appear to have a strong, positive linear correlation.

Note 4.97 A scatter plot can be used to determine whether two quantities are related. If there is a clear relationship, the quantities are said to be **correlated**. **Correlations** can often be seen when data items are displayed on a scatter plot. Although the scatter plot in Example 4.91 indicates a **correlation** between education and prejudice, we cannot conclude that increased education causes a person's level of prejudice to decrease.

Definition 4.98 Correlation Coefficient. correlation coefficient: A measure used to describe the strength and direction of a relationship between variables whose data points lie on or near a line is called the correlation coefficient, designated by r. A correlation coefficient, r, is a number between -1 and 1, inclusive. ◆

Example 4.99 Classifying Correlation Types and Strengths. The linear correlation coefficient (r) details both the direction (positive or negative) and strength (perfect, strong, weak, or zero) of the linear relationship between two variables. Match each of the following real-world study results with its correct statistical description:

1 A study on employee training hours and production errors yields $r = -0.85$.

2 An analysis of weekly grocery spending and family hair length yields $r = 0.02$.

3 A dataset comparing daily steps taken and body fat percentage yields $r = -0.25$.

4 An observation of local temperature and outdoor swimming pool attendance yields $r = 0.88$.

5 A report correlating a person's age with their vocabulary size yields $r = 0.31$.

Solution. Here is the correct classification for each scenario using standard interpretation boundaries for the correlation coefficient:

- ***Scenario 1 (**$r = -0.85$***): Strong Negative Correlation.*** The value is negative, meaning errors drop as training increases. Because $|-0.85|$ is close to 1.0, the data points form a tightly clustered downward trend line.
- ***Scenario 2 (**$r = 0.02$***): No Correlation.*** The value is approximately 0. Changes in grocery spending show a completely random distribution when plotted against hair length, indicating no linear link.
- ***Scenario 3 (**$r = -0.25$***): Weak Negative Correlation.*** The negative sign shows an inverse direction (more steps link to lower body fat). However, being close to 0 means the data forms a highly dispersed, scattered downward cloud.
- ***Scenario 4 (**$r = 0.88$***): Strong Positive Correlation.*** The positive sign indicates both variables rise together. The high value shows the data pairs cluster tightly along an upward-sloping trend line.
- ***Scenario 5 (**$r = 0.31$***): Weak Positive Correlation.*** The relationship is positive (older age links to larger vocabulary), but the low decimal value indicates a loosely scattered upward trend with significant variance.

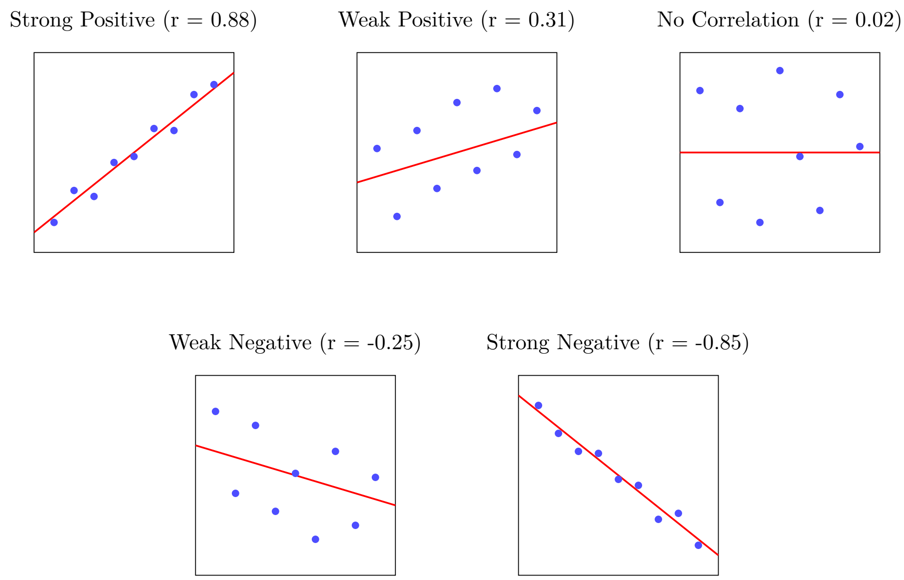


Figure 4.100

▲

Example 4.101 Stress and Illness Analysis. In a 1971 study involving 232 subjects, researchers found a relationship between the subjects' level of stress and how often they became ill. The correlation coefficient in this study was $r = 0.22$. What does this indicate about the relationship between stress and illness?

Stress and illness have a ________ and ________ relationship.

Solution. The value $r = 0.22$ provides two key pieces of information about the linear trend:

- ***Direction:*** The value is positive ($r > 0$). This means the two variables move in the same direction: as a person's recorded stress levels increase, their frequency of falling ill also tends to increase.
- ***Strength:*** The absolute value is low ($|0.22| < 0.30$). This means that while a pattern exists, the data points are highly scattered around a line of best fit, making it a weak predictable link.

Therefore, stress and illness have a ***weak*** and ***positive*** relationship.

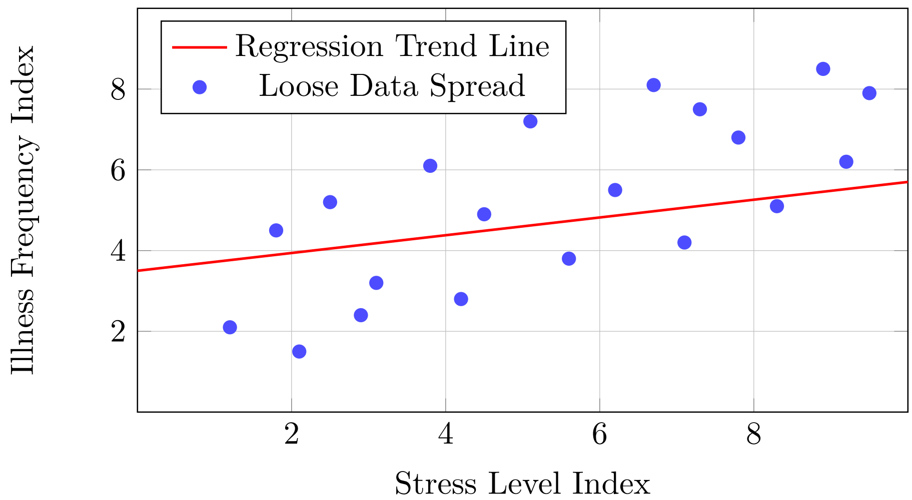


Figure 4.102

▲

Example 4.103 Real-World Analysis: Literacy and Child Mortality. The scatter plot below shows the relationship between the percentage of adult females in a country who are literate (X) and the mortality rate of children under five per thousand (Y), alongside its linear regression line.

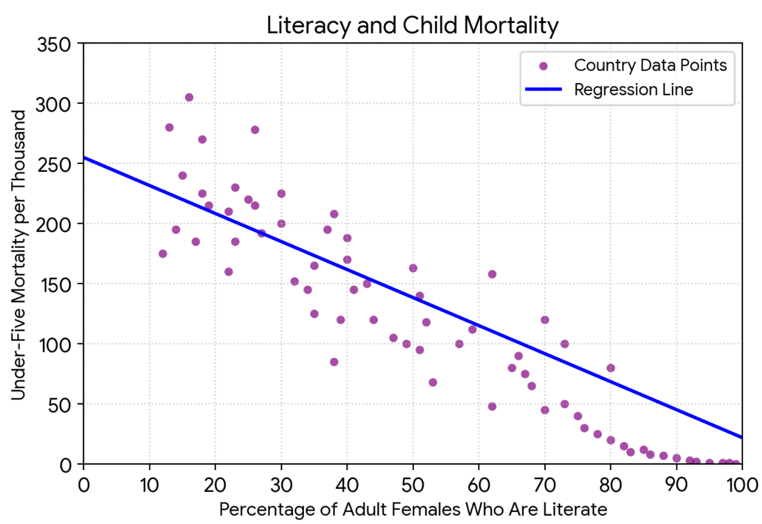


Figure 4.104 Scatter Plot of Female Literacy vs. Under-Five Mortality

Use the provided chart to determine whether each of the following statements is ***True*** or ***False***:

1 There is a perfect negative correlation between the percentage of adult females who are literate and under-five mortality.

2 As the percentage of adult females who are literate increases, under-five mortality tends to decrease.

3 There is no correlation between the percentage of adult females who are literate and under-five mortality.

Solution. By evaluating the direction and spatial dispersion of the data clusters on the graph, we find the following solutions:

- ***Statement 1 is False.*** A **perfect negative correlation** ($r = -1.0$) requires every observed country node to land exactly on the blue trajectory pathway line. Because data nodes are spread out in a loose cloud formation above and below the trendline, the pairing represents a strong, imperfect negative correlation.

- ***Statement 2 is True.*** The simple linear regression model features a distinct downward slope tracking from left to right. This confirms that as female literacy coordinates increase along the horizontal axis, the corresponding under-five mortality index shows a consistent, predictable decline.

- ***Statement 3 is False.*** A condition of **no correlation** ($r = 0$) generates a completely unstructured cloud of circular nodes accompanied by a flat, zero-slope horizontal regression bar. This graph shows a clear linear pattern, proving a strong statistical relationship between the two metrics.

▲

Simple Linear Regression.

Simple linear regression is a statistical method that models the relationship between a single independent variable and a continuous dependent variable by fitting a linear equation to observed data. It plots a straight line of best fit. The equation of the line is typically expressed as a linear function of the form

$$Y = mx + b,$$

where m represents the slope of the line and b represents the y-intercept.

Example 4.105 A linear model calculated from the dataset below yields a slope of 0.25 and a y-intercept of 8.25. The table details the relationship between the number of widgets in a package and the total length of the package, measured in inches.

Table 4.106

Number of Widgets (x)	**Length of Package in.** (y)
3	9.00
4	9.25
5	9.50
6	9.75

a) Form the equation of the regression line and interpret the slope and y-intercept in the context of the problem.

b) Using the regression line, predict the length of a package that contains 7 widgets.

Solution.

a Using the slope ($m = 0.25$) and the y_-intercept ($b = 8.25$), the simple linear regression equation is:

$$Y = 8.25 + 0.25X$$

Interpretation of the Slope: For each additional widget added to the package ($\Delta X = 1$), the total length of the package increases by exactly ***0.25 inches***.

Interpretation of the y-Intercept: When there are zero widgets in the package ($X = 0$), the baseline length of the packaging material itself is ***8.25 inches***.

b To predict the total packaging length for a shipment containing 7 widgets, substitute $X = 7$ into our linear regression model:

$$Y = 8.25 + 0.25(7)$$
$$Y = 8.25 + 1.75$$
$$Y = 10.00 \text{ inches}$$

A package holding 7 widgets is predicted to be exactly ***10.00 inches*** long.

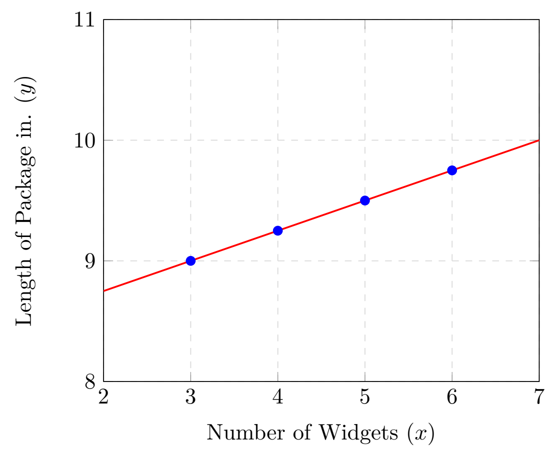


Figure 4.107 Linear Regression Trend for Widget Packaging

▲

Example 4.108 A linear model calculated from a dataset yields a slope of 13.3 and a y-intercept of 142.6 when modeling the relationship between total fat grams (x) and total calories (y) in a selection of fast food sandwiches.

a Form the linear regression equation that models this data, rounding all baseline parameters to the nearest integer, and interpret the slope and y-intercept in the context of the problem.

b Using the regression model, predict the total number of calories in a menu item that contains 25 grams of fat. Round your final prediction to the nearest whole integer.

Table 4.109

Total Fat g (x)	**Total Calories (y)**
9	260
13	320
21	420
30	530
31	560
32	580
34	590

Solution.

a Rounding the y-intercept ($142.6 \approx 143$) and the slope ($13.3 \approx 13$), the linear regression equation is:

$$Y = 143 + 13X$$

Slope Interpretation: For each additional 1 gram of fat in a sandwich, the energy content is predicted to increase by approximately ***13 calories***.

y-Intercept Interpretation: If a sandwich contains 0 grams of fat, its predicted baseline energy value from non-fat ingredients is ***143 calories***.

b Substitute $X = 25$ into the unrounded regression line equation to find the precise calorie projection:

$$Y = 142.6 + 13.3(25)$$
$$Y = 475.1 \approx 475 \text{ calories}$$

An item with 25 grams of fat is estimated to contain ***475 calories***.

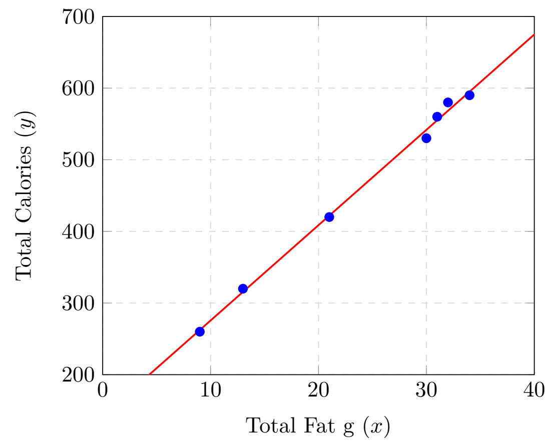


Figure 4.110 Linear Fit of Fat Content to Total Calories

▲

Checkpoint 4.111 During a solar eclipse, the temperature begins to drop as the Moon comes between the Earth and the Sun. The cooling temperatures, up through totality, of the eclipse are shown in the table below. Using a linear regression equation with the slope -4.4 and y-intercept 75.1, predict the temperature at the 4 minute mark during the cooling, to the nearest degree.

Table 4.112

Time minutes (x)	**Temp. °F (y)**
0	75
1	71
2	66
3	62
4	?

Select the predicted temperature at the 4 minute mark.

Solution. **Step 1: Calculate the Regression Equation** $Y = -4.4x + 75.1$.

Step 2: Evaluate at 4 Minutes Substitute $x = 4$ directly into our regression model to find the expected cooling value:

$$Y = -4.4(4) + 75.1$$

$$Y = -17.6 + 75.1 = 57.5$$

Rounding 57.5°F to the nearest whole integer gives our target choice of **58°**.

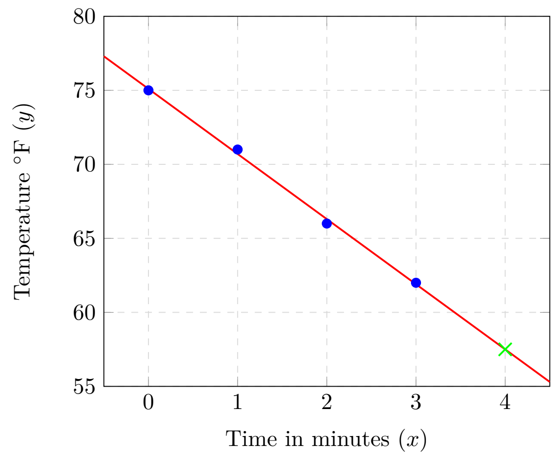


Figure 4.113 Linear Regression Line for Eclipse Cooling Pattern

Checkpoint 4.114 A morning radio talk show is running a "Winter Coat Campaign" to collect and deliver coats to local shelters. They are storing the coats until they reach their goal of 2100 coats. The table below shows the number of coats in storage at the end of each day of the campaign.

Table 4.115

Day of Campaign (x)	**Number of Coats Stored (y)**
1	860
2	930
3	1000
4	1150
5	1200
6	1360

Find the linear regression equation with the slope 98.86 and y-intercept 737.33 rounded to the nearest hundredths, and predict the day the campaign will meet its goal.

Solution. **Part a) Linear Regression Equation:** Thus, the linear modeling line of best fit is

$$Y = 98.86x + 737.33$$

Part b) Goal Prediction: To compute the target milestone, substitute $Y = 2100$ into the verified equation:

$$\begin{aligned} 2100 &= 98.86x + 737.33 \\ 1362.67 &= 98.86x \\ x &\approx 13.78 \end{aligned}$$

Because the data tracks totals at the end of each day, reaching 13.78 means the campaign will successfully complete its goal on **Day 14**.

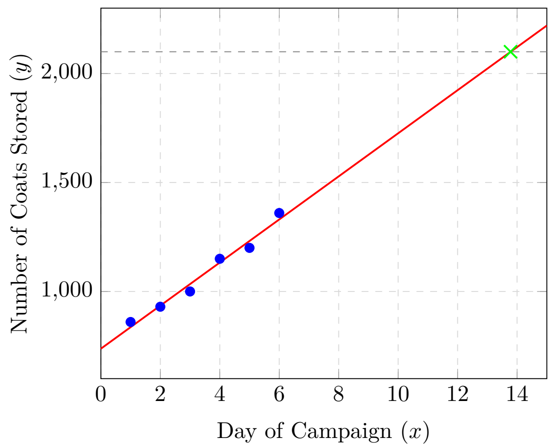


Figure 4.116 Linear Path Tracking Coat Collection Goal Delivery

4.6 Chapter Exercises

Statistics Practice Problem

1. Find the mean, median, and mode for the following dataset:

$$\{29, 24, 31, 40, 30, 35, 25, 28\}$$

Solution. To simplify the statistical calculations, we first arrange the dataset of $n = 8$ values in ascending order:

$$\{24, 25, 28, 29, 30, 31, 35, 40\}$$

1. Mean Calculation: The mean is calculated by summing all the values and dividing by the total number of data points ($n = 8$):

$$\text{Mean} = \frac{24 + 25 + 28 + 29 + 30 + 31 + 35 + 40}{8}$$
$$\text{Mean} = \frac{242}{8} = 30.25$$

Thus, the mean of the dataset is **30.25**.

2. Median Calculation: Since there is an even number of data points (8), the median is found by taking the average of the two middle values (the 4th and 5th numbers in our ordered list, which are 29 and 30):

$$\text{Median} = \frac{29 + 30}{2} = 29.5$$

Thus, the median of the dataset is **29.5**.

3. Mode Calculation: The mode is the value that appears most frequently in a dataset. Because every unique number in this list occurs exactly once, there is **no mode**.

2. Use the data 20, 24, 34, 38. Without actually computing the standard deviation, which of the following best approximates the standard deviation?

A. 2

B. 4

C. 8

D. 10

Answer. C.

Solution.

A. *Incorrect.*

Incorrect. The data points span a total range of 18 units (from 20 to 38), so they sit much further than 2 units away from the center.

B. *Incorrect.*

Incorrect. While 4 is a factor, the data points are spread out further from the mean of 29 than this value suggests.

C. *Correct.*

Correct. The mean is 29. The data points sit 5 and 9 units away from this mean. An average distance of roughly 7-8 units is the most accurate estimate.

D. *Incorrect.*

Incorrect. This value overestimates the standard deviation, as most points are closer to the mean than 10 units.

3. The rainfall in a small town is recorded daily for two weeks, in inches:

$$\{1.2, 2.3, 0.8, 2.3, 1.0, 3.1, 4.2, 1.4, 2.1, 0.4, 1.3, 2.0, 1.8, 3.2\}$$

What are the mean and the approximate sample standard deviation for this data set?

Solution. Step 1: Calculate the Mean. Sum all 14 data entries and divide by the count:

$$Mean = \frac{27.1}{14} \approx 1.94 \text{ inches}$$

Step 2: Find individual distances. Find the absolute difference between each entry and the mean of 1.94. Summing these 14 positive distances gives 11.3.

Step 3: Average the distances. Divide the sum of distances by the number of values:

$$\text{Average Distance} = \frac{11.3}{14} \approx 0.81$$

Step 4: Approximate Standard Deviation. Because the geometric process of squaring numbers shifts standard deviation upwards, the true standard deviation value will sit slightly higher than our baseline average distance. We can approximate the sample standard deviation to be **just above 0.81** (approximately 0.9 to 1.0).

4. The heights of adult men in a certain city are normally distributed with a mean of 70 inches and a standard deviation (SD) of 3 inches. Use the Empirical Rule to answer the following questions:

a) What percentage of men are between 67 and 73 inches tall?

b) What percentage of men are between 61 and 79 inches tall?

c) What percentage of men are taller than 76 inches?

Solution. The **Empirical Rule** (also known as the 68-95-99.7 rule) states that for a perfectly normal distribution:

- Approximately 68% of the data falls within 1 standard deviation of the mean (±1SD).
- Approximately 95% of the data falls within 2 standard deviations of the mean (±2SD).
- Approximately 99.7% of the data falls within 3 standard deviations of the mean (±3SD).

Given parameters: $Mean = 70$ inches and Standard Deviation SD = 3 inches. Let us map out the standard deviation markers:

- $\text{mean}\pm 1\text{SD} = 70 \pm 3 = [67, 73]$
- $\text{mean}\pm 2\text{SD} = 70 \pm 6 = [64, 76]$
- $\text{mean}\pm 3\text{SD} = 70 \pm 9 = [61, 79]$

Part a) Interval [67, 73]: These values correspond exactly to mean−1SD and mean+1SD. According to the rule, approximately **68%** of the men fall within this height range.

Part b) Interval [61, 79]: These values correspond exactly to mean −3SD and mean +3SD. According to the rule, approximately **99.7%** of the men fall within this height range.

Part c) Heights above 76 inches: A height of 76 inches corresponds exactly to mean+2SD. Since a normal curve is perfectly symmetrical, 50% of the data falls above the mean of 70. The region between the mean (70) and 2 standard deviations above the mean (76) contains exactly half of the 95% range, which is 47.5%. To find the tail value remaining above 76:

$$50\% - 47.5\% = 2.5\%$$

Thus, approximately **2.5%** of men in the city are taller than 76 inches.

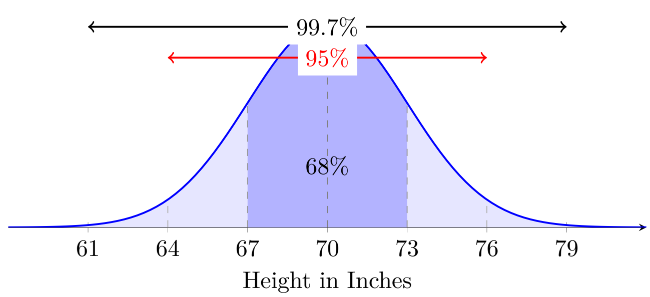


Figure 4.117 Normal Distribution of Heights with Empirical Rule Breakdowns

5. What is the z-score of the price of a pair of skis that cost \$247, if the mean ski price is \$279, with a standard deviation of \$16?

Solution. Identify the given parameters: value $x = 247$, mean $= 279$, and standard deviation $sd = 16$.

Substitute these values into the z-score formula:

$$z = \frac{247 - 279}{16} = \frac{-32}{16} = -2.0$$

The price of the skis is exactly **2.0 standard deviations below** the mean.

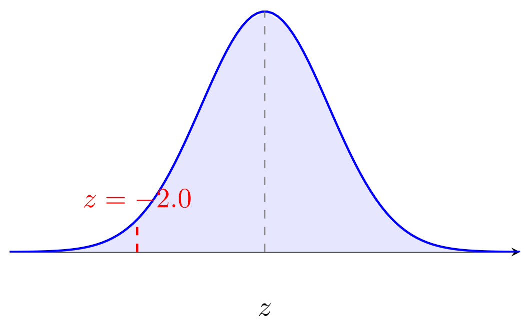


Figure 4.118 Normal distribution curve highlighting a z-score of -2.0.

6. What is the z-score of a 5-scoop ice cream cone if the mean number of scoops is 3, with a standard deviation of 1 scoop?

Solution. Identify the given parameters: value $x = 5$, mean $= 3$, and standard deviation $sd = 1$.

Substitute these values into the z-score formula:

$$z = \frac{5 - 3}{1} = \frac{2}{1} = 2.0$$

The 5-scoop cone is exactly **2.0 standard deviations above** the mean.

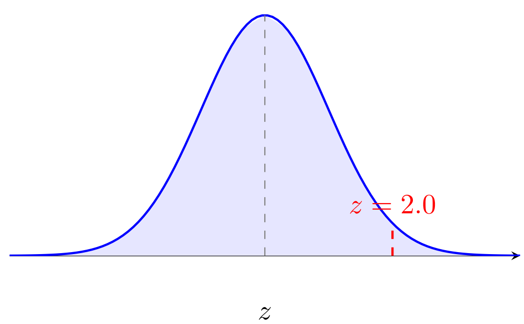


Figure 4.119 Normal distribution curve highlighting a z-score of 2.0.

7. What is the z-score of the weight of a cow that tips the scales at 825 lbs, if the mean weight for cows of her type is 1150 lbs, with a standard deviation of 77 lbs?

Solution. Identify the given parameters: value $x = 825$, mean $= 1150$, and standard deviation $sd = 77$.

Substitute these values into the z-score formula and round to two decimal places:

$$z = \frac{825 - 1150}{77} = \frac{-325}{77} \approx -4.22$$

The cow's weight is approximately **4.22 standard deviations below** the mean.

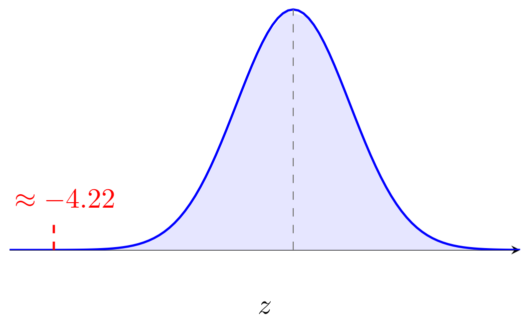


Figure 4.120 Normal distribution curve highlighting an extreme z-score of -4.22.

8. What is the z-score of a measured value of 0.0034, given mean = 0.0041 and $sd = 0.0008$?

Solution. Identify the given parameters: value $x = 0.0034$, mean = 0.0041, and standard deviation $sd = 0.0008$.

Substitute these values into the standard z-score formula:

$$z = \frac{x - mean}{sd} = \frac{0.0034 - 0.0041}{0.0008} = \frac{-0.0007}{0.0008} = -0.875$$

The measured value is exactly **0.875 standard deviations below** the population mean.

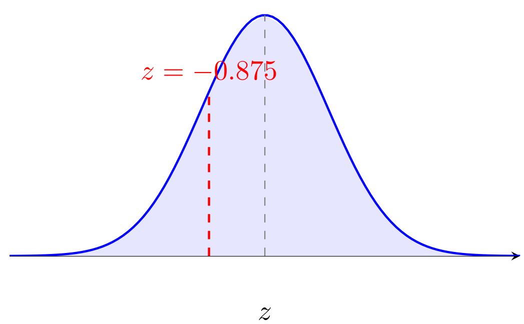


Figure 4.121 Normal distribution curve highlighting a z-score of -0.875.

9. What is the z-score of a student's test score of 89, if the class average was 80 with a standard deviation of 4 points?

Solution. Identify the given parameters: raw score $x = 89$, class mean = 80, and standard deviation $sd = 4$.

Substitute these values into the z-score expression:

$$z = \frac{x - mean}{sd} = \frac{89 - 80}{4} = \frac{9}{4} = 2.25$$

The student's test performance translates to a z-score of **2.25**, meaning it is **2.25 standard deviations above** the classroom average.

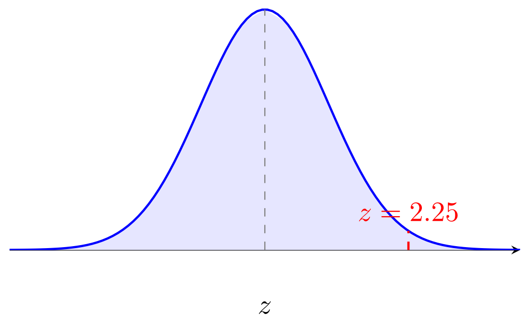


Figure 4.122 Normal distribution curve highlighting a z-score of 2.25.

10. Kori categorized her spending for this month into four categories: food, rent, fun, and other. The percentage she spent in each category is displayed in the chart below.

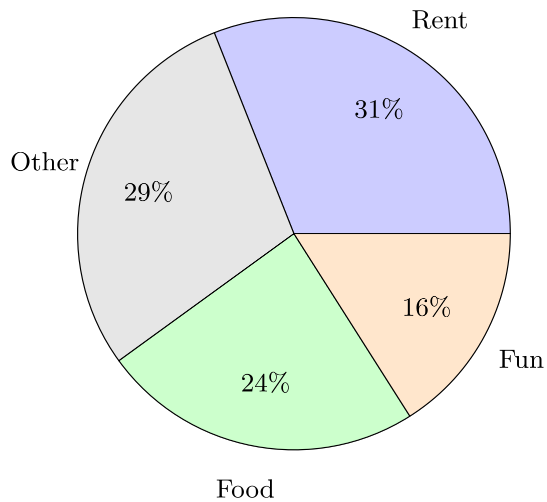


Figure 4.123 Pie Chart of Kori's Monthly Spending Shares

If she spent a total of \$2600 this month, how much money did she allocate to rent?

Solution. To find the exact dollar amount spent on rent, we convert the percentage share from the pie chart into a decimal and multiply it by the total monthly expenditures:

$$\text{Rent Share} = 31\% = 0.31$$

$$\text{Rent Expenditures} = 0.31 \times \$2600 = \$806$$

Therefore, Kori spent exactly \$806 on rent this month.

11. Suppose the height of adult women in the United States is normally distributed with a mean of 5' 8" and a standard deviation of 1.5". What is the probability that a randomly chosen woman in the United States is shorter than 5' 5"?

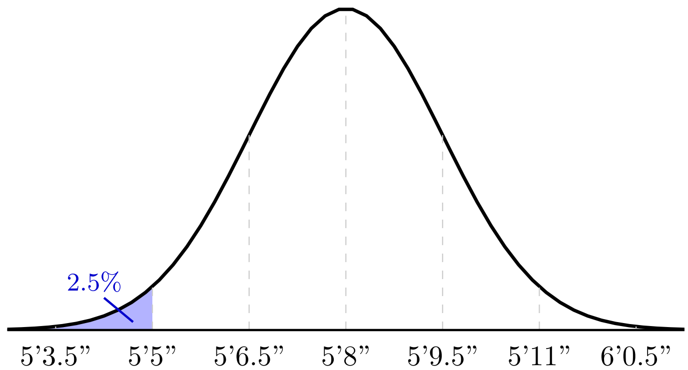


Figure 4.124 Normal distribution curve of women’s heights with the lower tail below 5’ 5" shaded.

Solution. First, determine how many standard deviations 5’ 5" is from the mean of 5’8". Since each standard deviation is 1.5", a height of 5’ 6.5" is 1 standard deviation below the mean, and 5’ 5" is exactly 2 standard deviations below the mean.

We can find the total percentage of women who are 5’ 5" or taller by adding up the known regions of the normal distribution:

- 50% of women are taller than the mean of 5’ 8".
- 34% of women are between 5’ 6.5" and 5’ 8" (within 1 standard deviation below the mean).
- 13.5% of women are between 5’ 5" and 5’ 6.5" (between 1 and 2 standard deviations below the mean).

Adding these regions together gives: $50\% + 34\% + 13.5\% = 97.5\%$. Since 97.5% of women are 5’ 5" or taller, the remaining portion is $100\% - 97.5\% = 2.5\%$.

Alternative Method: The empirical rule states that 95% of the data falls within 2 standard deviations of the mean. This leaves 5% split equally between the two extreme outer tails ($5\% \div 2 = 2.5\%$). Therefore, the probability that a randomly chosen woman is shorter than 5’ 5" is 2.5% (or 0.025).

12. Positive Correlation: Summer Heatwaves and Ice Cream Sales. A beachside shop owner wants to examine the relationship between the daily afternoon temperature (X, measured in degrees Fahrenheit) and the number of ice cream cones sold daily (Y). The data collected over a five-day period is presented below.

Table 4.125 Daily Temperature vs. Ice Cream Sales

Day	**Temperature (X in °F)**	**Ice Cream Cones Sold (Y)**
1	70	100
2	75	150
3	80	190
4	85	260
5	90	300

Using the simple linear regression model calculated from the observed data table above, predict the total number of ice cream cones sold if the afternoon temperature reaches 100°F.

A. 340 cones

B. 404 cones

C. 450 cones

D. 512 cones

Answer. B.

Solution 1. To predict the number of ice cream cones sold at 100°F, we first need to graph the data points from the table and calculate the line of best fit using linear regression.

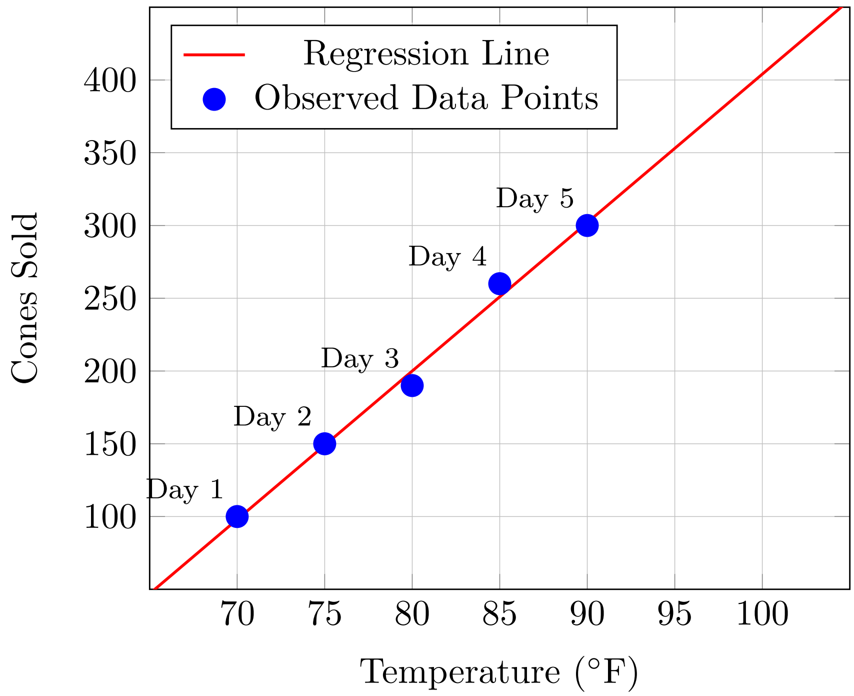


Figure 4.126 Scatter Plot of Temperature vs. Ice Cream Cones Sold

Solution 2.

A. *Incorrect.*

Incorrect. This value underestimates the strong upward trend observed as the temperature steps up to 100 degrees.

B. *Correct.*

Correct! Observe the data points and plug directly into the line: 404.

C. *Incorrect.*

Incorrect. While sales are rising fast, this estimate is slightly higher than what the mathematical trend line predicts.

D. *Incorrect.*

Incorrect. This value assumes exponential growth rather than a steady simple linear progression rate.

13. A store is doing inventory for one week to determine the popularity of a new product. The table below shows the number of items sold each day. Find the linear regression equation that models this data with the slope 8.66 and y-intercept 14.57, rounding coefficients to the nearest integer. Then, predict the number of items sold on the 10th day if the pattern of growth continues.

Table 4.127

Day Number (x)	**Number Sold (y)**
1	24
2	32
3	41
4	50
5	60
6	68
7	77

Solution. **Step 1: Calculate the Regression Equation** The linear regression equation is given by $y = mx + b$, where m is the slope and b is the y-intercept. Substituting the provided values, we have:

$$Y = 8.66x + 14.57$$

Rounding the coefficients to the nearest integer gives us the simplified regression equation:

$$Y = 9x + 15$$

Step 2: Predict for Day 10 To estimate item performance on the tenth day, we evaluate our regression equation at $x = 10$:

$$Y = 9(10) + 15$$
$$Y = 90 + 15 = 105$$

Rounding to the nearest whole item integer yields a final estimate of **105** items sold.

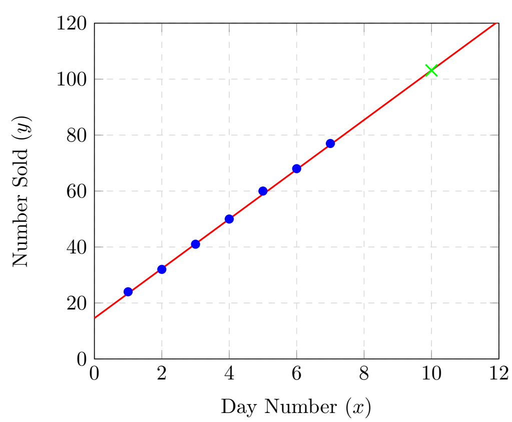


Figure 4.128 Linear Regression Mapping Weekly Product Growth

Chapter 5

MyOpenMath Interactive Activities

These files contain copies of the interactive MyOpenMath exercises that guided our learning for each topic. Special thanks to Professor Jason Hardin for sharing his MyOpenMath question bank.

01. Activity Voting Methods

In many decision making situations, it is necessary to gather the group consensus using some sort of vote. While the basic idea of voting is fairly universal, the method by which those votes are used to determine a winner can vary. In deciding upon a winner, there is always one main goal: to reflect the preferences of the people in the most fair way possible. We will discuss four different methods for determining the winner of an election using preference ballots, and will see that we can sometimes have different winners under different methods of voting.

See Section for the voting notes.

1. Four candidates are running for president of the Worcester State Math Club: Alan (A), Brenda (B), Craig (C), and Diane (D). The club members are asked to submit preference ballots and the results are shown in the preference table below.

	19	15	11	7	2
First	B	C	D	A	C
Second	A	A	C	D	D
Third	C	D	A	C	A
Fourth	D	B	B	B	B

Note: All parts of the questions must be correct to earn credit!

How many students are in the Math Club?

____

How many members have Craig as their first choice?

____

How many members have Brenda as their last choice?

____

Answer.

- 54
- 17
- 35

2. Which of the following voting methods will we discuss in this course. Select all that apply!

(a) Borda count method

(b) Plurality with elimination method

(c) Pairwise comparison method

(d) Jefferson's method

(e) Hamilton's method

(f) Plurality method

Answer.

- Plurality method
  Plurality with elimination method
  Borda count method
  Pairwise comparison method

3. Four candidates are running for president of the Worcester State Math Club: Alan (A), Brenda (B), Craig (C), and Diane (D). The club members are asked to submit preference ballots and the results are shown in the preference table below.

	19	15	11	7	2
First	B	C	D	A	C
Second	A	A	C	D	D
Third	C	D	A	C	A
Fourth	D	B	B	B	B

Note: All parts of the questions must be correct to earn credit!

Who would win the election under the plurality method?

(a) Alan

(b) Brenda

(c) Craig

(d) Diane

How many first place votes did this candidate receive?

____

Does the winning candidate receive the majority of first place votes?

(a) Yes

(b) No

Answer.

- B: Brenda
- 19
- B: No

4. Four candidates are running for president of the Worcester State Math Club: Alan (A), Brenda (B), Craig (C), and Diane (D). The club members are asked to submit preference ballots and the results are shown in the preference table below.

	19	15	11	7	2
First	B	C	D	A	C
Second	A	A	C	D	D
Third	C	D	A	C	A
Fourth	D	B	B	B	B

Note: All parts of the questions must be correct to earn credit!

Who would be the first candidate eliminated under the plurality with elimination method?

(a) Alan

(b) Brenda

(c) Craig

(d) Diane

After this candidate is eliminated and the votes are redistributed, how many first-choice votes does Diane have? (If Diane was the first eliminated, enter 0 as your answer here.)

____

Who wins this election under plurality with elimination?

(a) Alan

(b) Brenda

(c) Craig

(d) Diane

Answer.

- A: Alan
- 18
- D: Diane

5. Four candidates are running for president of the Worcester State Math Club: Alan (A), Brenda (B), Craig (C), and Diane (D). The club members are asked to submit preference ballots and the results are shown in the preference table below.

	19	15	11	7	2
First	B	C	D	A	C
Second	A	A	C	D	D
Third	C	D	A	C	A
Fourth	D	B	B	B	B

Note: All parts of the questions must be correct to earn credit!

Who would win the election under the Borda count method?

(a) Alan

(b) Brenda

(c) Craig

(d) Diane

How many points would the winning candidate receive under the Borda count?

____

Answer.

- A: Alan
- 156

6. Four candidates are running for president of the Worcester State Math Club: Alan (A), Brenda (B), Craig (C), and Diane (D). The club members are asked to submit preference ballots and the results are shown in the preference table below.

	19	15	11	7	2
First	B	C	D	A	C
Second	A	A	C	D	D
Third	C	D	A	C	A
Fourth	D	B	B	B	B

Note: All parts of the questions must be correct to earn credit!

Who would win the election under the pairwise comparison method?

(a) Alan

(b) Brenda

(c) Craig

(d) Diane

How many points would the winning candidate receive under the pairwise comparison method?

____

Answer.

- C: Craig
- 3

7. We have seen four different methods for determining the winner of an election. Through our examples and problems, we have also seen that these different methods can lead to different winners! In particular, in the Math Club President election considered throughout the exercises, it turns out that each of the four voting methods leads to each of the four candidates being declared the winner!

02. Activity Apportionment Methods

Apportionment is the problem of dividing up a fixed number of things among groups of different sizes. In politics, this takes the form of allocating a limited number of representatives amongst voters. This problem, presumably, is older than the United States, but the best-known ways to solve it have their origins in the problem of assigning each state an appropriate number of representatives in the new Congress when the country was formed. Apportionment theory is also used in many non-political situations, such as staffing a hospital with the appropriate number of nurses per shift and allocating buses to cover a set of bus routes based on the number of passengers.

The goal with any apportionment problem is to find a way to fairly distribute the items (e.g., doctors, buses, instructors) amoung the groups (e.g., clinics, routes, courses).

We face several real-world restrictions in this distribution process:

1 The things being divided up can exist only in whole numbers.

2 We must use all of the things being divided up, and we cannot use any more.

3 Each group must get at least one of the things being divided up.

4 The number of things assigned to each group should be at least approximately proportional to the population of the group. (Exact proportionality isn't possible because of the whole number requirement, but we should try to be close, and in any case, if Group A is larger than Group B, then Group B shouldn't get more of the things than Group A does.)

In terms of the apportionment of the United States House of Representatives, these rules imply:

1 We can only have whole representatives. Massachusetts can't have 8.4 representatives.

2 We can only use the (currently) 435 representatives available. If one state gets another representative, another state has to lose one.

3 Every state gets at least one representative.

4 The number of representatives each state gets should be approximately proportional to the state population. This way, the number of constituents each representative has should be approximately equal.

There are many different methods for solving an apportionment problem. In this course, we will discuss three of them: Hamilton's method, Jefferson's method, and Adams method. The first step for each method will be to compute the standard divisor and standard quotas.See Section for the apportionment notes.

1. Which of the following scenarios illustrates an apportionment problem? Select ALL that apply!

(a) A daycare must distribute 100 toys among 5 rooms based on the number of children in each room.

(b) The WSU Math Club needs to elect a president and will do so by selecting the candidate with the most votes.

(c) Four people pool their money to buy 60 shares of a stock, and the shares are distributed among the four based on the amount each person contributes.

(d) A school system purchases 500 new computers and must distribute them among 8 elementary schools based on the number of students at each school.

Answer.

- A daycare must distribute 100 toys among 5 rooms based on the number of children in each room.

  Four people pool their money to buy 60 shares of a stock, and the shares are distributed among the four based on the amount each person contributes.

  A school system purchases 500 new computers and must distribute them among 8 elementary schools based on the number of students at each school.

2. Which of the following apportionment methods will we discuss in this course. Select all that apply!

(a) Adam's Method

(b) Plurality method

(c) Plurality with elimination method

(d) Jefferson's method

(e) Pairwise comparison method

(f) Borda count method

(g) Hamilton's method

Answer.

- Hamilton's method
  Jefferson's method
  Adam's Method

3. An HMO has 40 doctors to be apportioned among four clinics (A, B ,C , D). The HMO decides to apportion the doctors based on the average weekly patient load for each clinic, shown in the table below. Find the standard divisor and the standard quota for each clinic. Round all answers, including the standard divisor, to two decimal places!

Clinic	A	B	C	D	Total
Avg Weekly Load	275	392	611	724	
Standard quota					

Standard divisor = ____________________
Standard quota for Clinic A = ____________________
Standard quota for Clinic B = ____________________
Standard quota for Clinic C = ____________________
Standard quota for Clinic D = ____________________

Answer.

- 50.05
- 5.49
- 7.83
- 12.21
- 14.47

4. An HMO has 40 doctors to be apportioned among four clinics (A, B, C, D). The HMO decides to apportion the doctors based on the average weekly patient load for each clinic, shown in the table below. Use Hamilton's method to apportion the doctors to the clinics.

Recall the standard divisor for this problem is 50.05.

Clinic	A	B	C	D	Total
Avg Weekly Load	275	392	611	724	2002
Standard quota	5.49	7.83	12.21	14.47	

Total doctors for Clinic A = ____________________
Total doctors for Clinic B = ____________________
Total doctors for Clinic C = ____________________
Total doctors for Clinic D = ____________________

Answer.

- 6
- 8
- 12
- 14

5. An HMO has 40 doctors to be apportioned among four clinics (A, B, C, D). The HMO decides to apportion the doctors based on the average weekly patient load for each clinic, shown in the table below. Use Jefferson's method to apportion the doctors to the clinics.

Recall the standard divisor for this problem is 50.05.

Clinic	A	B	C	D	Total
Avg Weekly Load	275	392	611	724	2002
Standard quota	5.49	7.83	12.21	14.47	

Total doctors for Clinic A = ________________
Total doctors for Clinic B = ________________
Total doctors for Clinic C = ________________
Total doctors for Clinic D = ________________

Answer.

- 5
- 8
- 12
- 15

6. An HMO has 40 doctors to be apportioned among four clinics (A, B, C, D). The HMO decides to apportion the doctors based on the average weekly patient load for each clinic, shown in the table below. Use Adams's method to apportion the doctors to the clinics.

Recall the standard divisor for this problem is 50.05.

Clinic	A	B	C	D	Total
Avg Weekly Load	275	392	611	724	2002
Standard quota	5.49	7.83	12.21	14.47	

Total doctors for Clinic A = __________________
Total doctors for Clinic B = __________________
Total doctors for Clinic C = __________________
Total doctors for Clinic D = __________________

Answer.

- 6
- 8
- 12
- 14

03. Activity Simple Interest

We have to work with money every day. While balancing your checkbook or calculating your monthly expenditures on coffee requires only arithmetic, when we start saving, planning for retirement, or need a loan, we need more mathematics!

See Section for the simple interest notes.Discussing interest starts with the principal, which is the amount your account starts with. This could be a starting investment or the starting amount of a loan. Interest is the amount earned for depositing money or the amount paid for borrowing money. It is calculated as a percentage of the principal (called the interest rate) and is written as a decimal (for example, 5% is 0.05).

For a simple example, if you borrowed \$100 from a friend and agreed to repay it with 5% interest, then the amount of interest you would pay would just be 5% of \$100:

$$\$100 \times 0.05 = \$5.$$

The total amount you would repay would be \$105, which is the original principal plus the interest. This is an example of one-time simple interest, where the interest is calculated based on the principal only.

1. Alex borrows \$1200 from their parents for college expenses and agrees to repay them with 7% interest.

a) What is the principle in this situation?
\$ ____________________
b) Write the interest rate as a decimal.
____________________ %
c) How much interest will Alex pay their parents?
\$ ____________________
d) What is the total amount, principle plus interest, Alex will repay?
\$ ____________________

Answer.

- 1200
- 0.07
- 84
- 1284

One-time simple interest is only common for extremely short-term loans. For longer-term loans, it is common for interest to be paid on a daily, monthly, quarterly, or annual basis. In that case, interest would be earned regularly. For example, bonds are essentially a loan made to the bond issuer (a company or government) by you, the bondholder. In return for the loan, the issuer agrees to pay interest, often annually. Bonds have a maturity date, at which time the issuer pays back the original bond value.

Example 5.1 Suppose your city is building a new park, and issues bonds to raise the money to build it. You obtain a $\$1,000$ bond that pays 5% interest annually that matures in 5 years. How much interest will you earn?

Solution. Each year you would earn interest:

$$\$1,000 \times 0.05 = \$50.$$

So, over the course of five years, you would earn a total of

$$\$50 \times 5 = \$250.$$

When the bond matures, you would receive back the $\$1,000$ you originally paid, leaving you with a total of $\$1,250$. ▲

Notice how the amount of interest (I) was computed in this example: we multiplied the principal by the interest rate (represented as a decimal), then multiplied this by the number of years. We can generalize this idea of simple interest over time to obtain a formula that we can use to find the amount of simple interest earned:

$$I = Prt$$

where:

- I is the amount of simple interest
- P is the principal (starting amount)
- r is the interest rate represented as a decimal
- t is the time in years

Note 5.2 You do NOT need to memorize this formula! It will be provided for you on all quizzes and exams!

Let's look at a quick example demonstrating the simple interest formula.

Example 5.3 A student took out a simple interest loan for $\$5,000$ for 3 years at a rate of 4% to purchase a used car. What is the interest on the loan?

Solution. We are given the following: $P = \$5,000$, $t = 3$, and $r = 0.04$. Using the simple interest formula, the amount of interest on the loan is:

$$I = \$5,000 \times 0.04 \times 3 = \$600$$

▲

2. In the simple interest formula $I = Prt$, what unit must t be given in?

(a) Years

(b) Weeks

(c) Quarters

(d) Months

(e) Decimals

(f) Days

(g) Whatever unit is given in the problem

(h) There is no particular unit

Answer.

- A: Years

3. You take out a simple interest loan for \$500 for 3 years at an interest rate of 4%to pay for textbooks. What is the simple interest on the loan?

\$ ____________________

Answer.

- 60

While finding the amount of interest accrued for a loan is important, it is perhaps more important that we determine the total amount that must be repaid. We will call this amount — that is, the total amount required to be repaid in a loan or the total amount earned in an investment — the **future value**. We will denote this quantity with the variable A.

As we saw with the example on the previous page involving a city issuing bonds to build a new park, the future value of a loan can be found by adding together the principal and the interest. This is true for an investment as well. This will allow us to develop a formula for the future value involving simple interest:

$$A = P + I$$

Since $I = Prt$, we can substitute this into the equation to get:

$$A = P + Prt = P(1 + rt)$$

This short calculation provides us with a formula for computing the future value for a simple interest loan or investment:

$$A = P(1 + rt)$$

where:

- A is the future value of the loan or investment
- P is the principal (starting amount) of the loan or investment
- r is the interest rate represented as a decimal
- t is the time in years

Note 5.4 You do NOT need to memorize this formula! It will be provided for you on all quizzes and exams!

Let's look at a quick example demonstrating the simple interest formula.

Example 5.5 A loan of $\$2,500$ has been made at 6% for 9 months. Find the loan's future value.

Solution. We are given the following: $P = \$2,500$ and $r = 0.06$.

For t, note that we are given a time in months, not years! For the simple interest formula to be used correctly, we must give this value in years. Since there are 12 months in a year, we can always convert from months to years by dividing the number of months by 12. In this case, we have:

$$t = \frac{9}{12} = 0.75 \text{ years}$$

Now, using the formula above we find a future value of:

$$A = \$2,500(1 + 0.06 \times 0.75) = \$2,612.50$$

▲

4. Match each of the quantities on the left with the variable which represents it in the simple interest formula.
Note all answers must be correct to earn credit!

(a) Future value

(b) Interest rate

(c) Time

(d) Principle

(a) P

(b) A

(c) r

(d) t

Answer.

- b c d a

5. A simple interest loan is taken out for 6 months. What value of t should be used in the simple interest formula?
$t =$ ____________________

Answer.

- 0.5

6. You deposit \$4500 in an investment paying a simple interest rate of 8%. How much will the investment be worth in nine months?
\$ ____________________

Answer.

- 4770

In addition to determining the future value for a simple interest loan or investment, the simple interest formula can be used to solve other types of problems. We might want to know how much we should deposit into an account with a given simple interest rate to ensure we have a given amount in the future. Or, like in the example below, we might want to determine the simple interest rate based on the principal and future values.

Example 5.6 You borrow \$500 from a friend and promise to pay back \$560 in 18 months. What simple interest rate will you pay?

Solution. We are given the following: $P = \$500$, $A = \$560$, and $t = \frac{18}{12} = 1.5$ years.

We are looking to find the value of r (hence, the "?"). We substitute the given values into the simple interest formula and solve for r:

$$A = P(1 + rt)$$

$$560 = 500(1 + r \times 1.5)$$

Divide both sides by 500:

$$1.12 = 1 + 1.5r$$

Subtract 1 from both sides:

$$0.12 = 1.5r$$

Divide by 1.5:

$$r = \frac{0.12}{1.5} = 0.08$$

Since $r = 0.08$, we convert this to a percent to get a simple interest rate of 8%. ▲

7. What was the primary goal in the preceding example?

(a) To find the amount of interest accrued over the course of a loan.

(b) To find the simple interest rate for a loan.

(c) To find the amount of time for which a loan was given.

(d) To find the amount deposited in a loan.

(e) To find the future value of a loan.

(f) None of the above.

Answer.

- B: To find the simple interest rate for a loan.

04. Activity Compound Interest

With simple interest, we were assuming that we pocketed the interest when we received it. In a standard bank account, any interest we earn is automatically added to our balance, and we earn interest on that interest in future years. This reinvestment of interest is called ***compounding***.

See Section for the compound interest notes.Suppose that we deposit $\$1,000$ in a bank account offering 12% simple interest. After 1 year, the account would have a balance of $\$1,120$.

Now, suppose that we deposit our $\$1,000$ in a bank account offering 12% interest, compounded monthly. How will our money grow?

In the first month (note we use $t = \frac{1}{12}$ so that time is given in years!):

$$I = \$1,000 \times 0.12 \times \frac{1}{12} = \$10$$

In the first month, we will earn $10 in interest, raising our account balance to $\$1,010$.

In the second month:

$$I = \$1,010 \times 0.12 \times \frac{1}{12} = \$10.10$$

Notice that in the second month we earned more interest than we did in the first month. This is because we earned interest not only on the original $\$1,000$ we deposited, but we also earned interest on the $10 of interest we earned the first month. This is the key advantage that compounding of interest gives us.

Calculating out a few more months, we obtain the following:

Month	Starting Balance	Interest Earned	Ending Balance
1	$1,000.00	$10.00	$1,010.00
2	$1,010.00	$10.10	$1,020.10
3	$1,020.10	$10.20	$1,030.30
4	$1,030.30	$10.30	$1,040.60
5	$1,040.60	$10.41	$1,051.01
6	$1,051.01	$10.51	$1,061.52
7	$1,061.52	$10.62	$1,072.14
8	$1,072.14	$10.72	$1,082.86
9	$1,082.86	$10.83	$1,093.69
10	$1,093.69	$10.94	$1,104.63
11	$1,104.63	$11.05	$1,115.68
12	$1,115.68	$11.16	$1,126.84

Notice that we earned slightly more interest with compound interest than with simple interest.

1. Answer the following questions about the \$1000 loan in the example above.

a) How much total interest was earned with the account using simple interest?
\$ ____________________

b) When using compound interest, how much interest was earned in month 6?
\$ ____________________

c) When using compound interest, what was the ending account balance after 9 months?
\$ ____________________

Answer.

- 30
- 2.53
- 1022.73

The calculation on the previous page was quite tedious! Fortunately, there are formulas that can be used to help us determine the future value of a loan or investment that undergoes compound interest and solve other problems relating to compound interest!

There are two formulas we will consider in this course. Both are mathematically equivalent, but are solved in terms of different values for our convenience.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$P = \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}}$$

where:

- P is the principal (starting amount) of the loan or investment, sometimes called the present value.
- A is the future value of the loan or investment.
- r is the interest rate represented as a decimal.
- n is the number of compounding periods per year.
- t is the time in years.

Note 5.7 You do NOT need to memorize this formula! It will be provided for you on all quizzes and exams!

The first formula (the one with A on the left-hand side) is called the **future value formula** since it is solved for the future value. It should be used when we want to determine the future value of a loan or investment with a given present value.

The second formula (the one with P on the left-hand side) is called the **present value formula** since it is solved for the present value (principal). It should be used when we want to determine the present value of a loan or investment with a given future value.

The new quantity here that was not present before is n, the number of compounding periods per year. This will need to be given to us, and is often done so using keywords. Some examples of these are:

- If the compounding is done annually (once a year), then $n = 1$.
- If the compounding is done semi-annually, then $n = 2$.
- If the compounding is done quarterly, then $n = 4$.
- If the compounding is done monthly, then $n = 12$.
- If the compounding is done weekly, then $n = 52$.
- If the compounding is done daily, then $n = 365$.

The most important thing to remember about using these formulas is that it assumes that we put money in the account once and let it sit there earning interest. An account where we make regular deposits is called an **annuity**, and will be discussed in our next unit! We look at some examples of using these formulas on the next page!

2. Which quantity is often called the present value of a loan or investment?

(a) t

(b) n

(c) A

(d) P

(e) r

(f) None of these

Answer.

- D: P

3. What would the value of n be in the compound interest formulas if interest is compounded

a) monthly?

____

b) daily?

____

c) semi-annually?

____

d) 7 times per year?

____

Answer.

- 12
- 365
- 2
- 7

Example 5.8 A certificate of deposit (CD) is a savings instrument that many banks offer. It usually gives a higher interest rate, but you cannot access your investment for a specified amount of time. Suppose you deposit $3,000 in a CD paying 6% interest compounded monthly. How much will you have in the account after 20 years? How much interest was earned?

Solution. We are given the following values:

- $P = \$3,000$ (the initial deposit)
- $r = 0.06$ (the interest rate as a decimal)
- $n = 12$ (interest is compounded monthly)
- $t = 20$ (we're looking for the amount after 20 years)

We are asked to find the value in the account in the future (i.e., the future value), so we use the future value formula for compound interest:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$A = 3,000\left(1 + \frac{0.06}{12}\right)^{12\times 20}$$

$$A = 3,000(1.005)^{240} \approx \$9,930.61$$

To find the amount of interest earned, we observe that everything in the account other than the initial deposit must be interest! Since we deposited $3,000 and ended up with $9,930.61, the amount of interest earned must be:

$$\text{Interest} = \$9,930.61 - \$3,000 = \$6,930.61$$

Note that we did NOT use the formula $I = Prt$ here to compute the amount of interest earned. This formula applies only to situations which use simple interest. If we used it here, we would not have calculated the correct amount of interest. ▲

4. You deposit $1000 in a savings account with an interest rate of 2.95%. The interest is compounded quarterly.
How much money will be in the account after 10 years? Round your answer to the nearest cent.
$ ____________________
How much of this is due to interest?
$ ____________________

Answer.

- 1341.67
- 341.67

Example 5.9 A new mother knows she will need $50,000 for her child's education in 18 years. If the account earns 6% compounded quarterly, how much should the mother deposit now to reach her goal? How much interest is earned?

Solution. We are given the following values:

- $A = \$50,000$ (the amount we want in 18 years)
- $r = 0.06$ (the interest rate as a decimal)
- $n = 4$ (interest is compounded quarterly)
- $t = 18$ (we're given the timeframe of 18 years)

We are asked to find the amount that needs to be deposited today (i.e., the present value), so we use the present value formula for compound interest:

$$P = \frac{A}{\left(1+\frac{r}{n}\right)^{nt}}$$

$$P = \frac{50,000}{\left(1+\frac{0.06}{4}\right)^{4\times 18}}$$

$$P = \frac{50,000}{(1.015)^{72}} \approx \$17,116.50$$

To find the amount of interest earned, we observe that everything in the account other than the initial deposit must be interest! Since we deposited $17,116.50 and ended up with $50,000, the amount of interest earned must be:

$$\text{Interest} = \$50,000 - \$17,116.50 = \$32,883.50$$

Again, note we did NOT use the formula $I = Prt$ to compute the amount of interest earned. This formula should be used only in situations in which simple interest is being used. ▲

5. How much money should you deposit in a CD today paying 8% interest compounded monthly so it will accumulate to $20,000 in 5 years?

$ ____________________

Answer.

- 13424.21

The effective annual yield (or effective rate) of an account is the simple interest rate that would produce the same account balance after one year as the compound interest at a stated rate and compounding period. The effective annual yield is determined by the following formula:

$$Y = \left(1 + \frac{r}{n}\right)^n - 1$$

where:

- Y is the effective annual yield
- r is the compound interest rate represented as a decimal
- n is the number of compounding periods per year

Note 5.10 You do NOT need to memorize this formula! It will be provided for you on all quizzes and exams!

Let's look at an example!

Example 5.11 Sue deposits $\$1,000$ in an account that pays 5% interest compounded monthly. Find the account's effective annual yield.

Solution. We are given $r = 0.05$ and $n = 12$ (monthly compounding). Using the formula we see that:

$$Y = \left(1 + \frac{0.05}{12}\right)^{12} - 1 \approx 0.0512$$

This means that if we were to deposit $\$1,000$ in an account using a simple interest rate of 5.12%, we would have the same balance after one year as the account with monthly compounding at 5%. ▲

6. A passbook savings account has an interest rate of 5%. The interest is compounded daily. Find the account's effective annual yield.

Round your answer to the nearest tenth of a percent (e.g., 7.3).

____________________ %

Answer.

- 5.1

05. Activity Annuities

Accounts with simple/compound interest require a one-time deposit. If we want to ensure the account balance reaches a certain amount by a particular point in the future, then we usually need to make a substantial deposit. For example, in a previous example we saw that to ensure a future value of $\$9,930.61$ in 20 years at 6% interest compounded monthly, we would need to deposit $\$3,000$ today!

For most of us, we aren't able to put a large sum of money like this in the bank today. Instead, we save for the future by depositing a smaller amount of money from each paycheck into the bank. This idea is called an **annuity**. Most retirement plans like 401(k) plans or IRA plans are examples of annuities.

In simplistic terms:

- For *compound interest*, we make a single deposit and this money sits in the account and earns interest.
- For an *annuity*, we make regular deposits (every month, week, year, etc.) and the money sits in the account and earns interest.

One technical thing: for the annuity formulas we will use, interest must be compounded at the same rate as the regular deposits are made. So if an annuity receives monthly deposits, interest will be compounded monthly.

See Section for the annuity notes.

1. Suppose you want to make weekly deposits into an annuity. How often should interest be compounded for this account?

(a) Daily

(b) Semi-annually

(c) Quarterly

(d) Monthly

(e) Annually

(f) It doesn't matter

(g) Weekly

Answer.

- G: Weekly

2. For each of the following scenarios, identify whether it is an account using compound interest or an annuity. Note that all must be correct to receive credit.

(a) Alex deposits \$150 every month into an account earning 7% interest compounded monthly.

(b) Alex deposits \$1000 in an account and earns 5% interest compounding monthly.

(c) Alex is saving for retirement and deposits \$50 into an account every week. The account has an interest rate of 3% and is compounded weekly.\$

(d) Alex deposits \$5 a day into an account earning 9.25% interest compounded daily.

(e) Alex deposits \$1000 every 3 months into an IRA that pays 6% compounded quarterly.

(f) Alex makes a one-time deposit of \$250 into an account with 4% interest compounded annually.

Answer.

- Alex deposits \$150 every month into an account earning 7% interest compounded monthly.

  Alex is saving for retirement and deposits \$50 into an account every week. The account has an interest rate of 3% and is compounded weekly.\$

  Alex deposits \$5 a day into an account earning 9.25% interest compounded daily.

  Alex deposits \$1000 every 3 months into an IRA that pays 6% compounded quarterly.

Like with compound interest, there are two formulas we will consider in this course. Both are mathematically equivalent, but are solved in terms of different values for our convenience.

$$A = \frac{P\left[\left(1+\frac{r}{n}\right)^{nt}-1\right]}{\left(\frac{r}{n}\right)}$$

$$P = \frac{A\cdot\left(\frac{r}{n}\right)}{\left(1+\frac{r}{n}\right)^{nt}-1}$$

where:

- P is the deposit amount made during each compounding period.
- A is the amount in the annuity after t years.
- r is the interest rate represented as a decimal.
- n is the number of compounding periods per year.
- t is the time in years.

Note 5.12 You do NOT need to memorize these formulas! They will be provided for you on all quizzes and exams! The first formula (the one with A on the left-hand side) should be used when we want to determine the future value of an annuity with a given deposit amount.

The second formula (the one with P on the left-hand side) should be used when we want to determine the deposit amount for an annuity to ensure a given future value. We look at some examples of using these formulas on the next page!

3. In which of the following situations would the formula $P = \dfrac{A\left(\frac{r}{n}\right)}{\left(1+\frac{r}{n}\right)^{nt}-1}$ be best used?

(a) You want to know how much would be in an annuity after 5 years with regular monthly deposits of \$250.

(b) You want to determine the amount of a one-time deposit into an account with 6% interest compounded weekly.

(c) You want to determine the amount to regularly deposit into an annuity so that there is \$10,000 in the account in 10 years.

(d) You want to know the future value of an account in 10 years which was started with a one-time deposit of \$5000.

Answer.

- C: You want to determine the amount to regularly deposit into an annuity so that there is \$10,000 in the account in 10 years.

The calculation on the previous page was quite tedious! Fortunately, there are formulas that can be used to help us determine the future value of a loan or investment that undergoes compound interest and solve other problems relating to compound interest!

There are two formulas we will consider in this course. Both are mathematically equivalent, but are solved in terms of different values for our convenience.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$P = \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}}$$

where:

- P is the principal (starting amount) of the loan or investment, sometimes called the present value.
- A is the future value of the loan or investment.
- r is the interest rate represented as a decimal.
- n is the number of compounding periods per year.
- t is the time in years.

Note 5.13 You do NOT need to memorize this formula! It will be provided for you on all quizzes and exams!

The first formula (the one with A on the left-hand side) is called the **future value formula** since it is solved for the future value. It should be used when we want to determine the future value of a loan or investment with a given present value.

The second formula (the one with P on the left-hand side) is called the **present value formula** since it is solved for the present value (principal). It should be used when we want to determine the present value of a loan or investment with a given future value.

The new quantity here that was not present before is n, the number of compounding periods per year. This will need to be given to us, and is often done so using keywords. Some examples of these are:

- If the compounding is done annually (once a year), then $n = 1$.
- If the compounding is done semi-annually, then $n = 2$.
- If the compounding is done quarterly, then $n = 4$.
- If the compounding is done monthly, then $n = 12$.
- If the compounding is done weekly, then $n = 52$.
- If the compounding is done daily, then $n = 365$.

The most important thing to remember about using these formulas is that it assumes that we put money in the account once and let it sit there earning interest. An account where we make regular deposits is called an **annuity**, and will be discussed in our next unit! We look at some examples of using these formulas on the next page!

4. At age 25, Joe decides to start saving for retirement. He decides to deposit \$200 at the end of each month into an IRA that pays 7.5% compounded monthly. How much money will be in the account when Joe retires at age 65? How much interest is earned?

Round your answer to the nearest dollar!

____

Answer.

- 604765

Example 5.14 You want to have $\$1,000,000$ in your account when you retire in 40 years. Your retirement account earns 7% interest compounded weekly. How much do you need to deposit each week to meet your retirement goal? How much of the $\$1,000,000$ is from interest?

Solution. We are given the following values:

- $A = \$1,000,000$ (the amount we want in 40 years)
- $r = 0.07$ (the interest rate as a decimal)
- $n = 52$ (interest is compounded weekly)
- $t = 40$ (we're given the timeframe of 40 years)

We are asked to find the amount of the regular deposits, so we use the deposit formula for an annuity:

$$d = \frac{A \cdot \frac{r}{n}}{\left(1 + \frac{r}{n}\right)^{nt} - 1}$$

$$d = \frac{1,000,000 \cdot \frac{0.07}{52}}{\left(1 + \frac{0.07}{52}\right)^{52\times 40} - 1}$$

$$d \approx \$87.47$$

To find the amount of interest earned, we observe that everything in the account other than the regular deposits must be interest! We first find the total amount deposited over the course of the annuity. Since we deposited $\$87.47$ each week (52 per year) for 40 years, the total amount deposited is:

$$\text{Total Deposited} = \$87.47 \times 52 \times 40 = \$181,937.60$$

Since we deposited $\$181,937.60$ and ended up with $\$1,000,000$, the amount of interest earned must be:

$$\text{Interest} = \$1,000,000 - \$181,937.60 = \$818,062.40$$

Note we did NOT use the formula $I = Prt$ here to compute the amount of interest earned. This formula applies only to situations which use simple interest. If we used it here, we would not have calculated the correct amount of interest. ▲

5. How much should you deposit at the end of each month into an annuity that pays 6.5% interest compounded monthly if you would like to have \$2 million when you retire in 45 years?

____

Answer.

- 619.48

06. Activity Loans

An **installment loan** is a loan which you pay off with regular weekly or monthly payments. Examples include car loans, student loans, and mortgages; we will discuss mortgages in the next unit.

Sometimes a loan requires a **down payment**, which is the amount paid at the time of purchase. The **amount of the loan** is the actual amount borrowed. It does ***not*** include the down payment (when there is one) because this money is not part of the debt.

The loan amount must be repaid, with interest, in regular payments (monthly, weekly, quarterly, etc.). The compounding frequency for interest is not always explicitly given, but is determined by how often you make payments. So, if you are required to make monthly payments on an installment loan, the interest will be compounded monthly.

For our purposes in this course, we will assume that you make loan payments on a regular schedule (every month, year, quarter, etc.) and are paying interest on the loan.

See Section for the Mortgages notes.

1. The amount paid at the time of purchase is called what?

(a) Down payment

(b) Loan amount

(c) Interest

(d) Principal

(e) Compounding amount

(f) Compounding period

(g) None of the above

Answer.

- A: Down payment

2. In this course, what assumptions will we be making about installment loans? Select ALL that apply!

(a) We will have a down payment.

(b) We will not have a down payment.

(c) We will be paying interest on the loan.

(d) We will make regular payments on the loan.

Answer.

- We will make regular payments on the loan.
  We will be paying interest on the loan.

Like with compound interest and annuities, there are two formulas we will consider in this course. Both are mathematically equivalent, but are solved in terms of different values for our convenience.

$$PMT = \frac{P \cdot \left(\frac{r}{n}\right)}{1 - \left(1 + \frac{r}{n}\right)^{-nt}}$$

$$P = \frac{PMT\left[1 - \left(1 + \frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$$

where:

- P is the amount of the loan (borrowed amount).
- PMT is the regular payment amount.
- r is the interest rate represented as a decimal.
- n is the number of compounding periods per year.
- t is the time in years.

Note 5.15 You do NOT need to memorize these formulas! They will be provided for you on all quizzes and exams! The first formula (the one with PMT on the left-hand side) should be used when we want to determine the payment amount for a given loan, and the second formula (the one with P on the left-hand side) should be used when we want to determine how much we can afford with a given payment amount.

We look at some examples of using these formulas on the next page!

3. Identfy the correct formula to use in each situation.

You borrow $\$25,000$ for a new car. You take out an installment loan for 3 years at 5% interest which requires regular monthly payments. How much will the monthly payments be?

(a) $PMT = \dfrac{P\left(\frac{r}{n}\right)}{\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}$

(b) $P = \dfrac{PMT\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$

(c) Neither of these

You can afford to pay $200 per month as a car payment. If you can get a loan at 3% interest for 5 years, how expensive of a car can you afford?

(a) $PMT = \dfrac{P\left(\frac{r}{n}\right)}{\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}$

(b) $P = \dfrac{PMT\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$

(c) Neither of these

Answer.

- A: $PMT = \dfrac{P\left(\frac{r}{n}\right)}{\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}$
- B: $P = \dfrac{PMT\left[1-\left(1+\frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$

Example 5.16 You decide to borrow $\$25,000$ for a new car. You take out an installment loan for 3 years at 5% interest which requires regular monthly payments. How much will the monthly payment be? How much interest will you pay in total?

Solution. We are given the following values:

- $P = \$25,000$ (the amount borrowed)
- $r = 0.05$ (the interest rate as a decimal)
- $n = 12$ (interest is compounded monthly)
- $t = 3$ (the loan is taken out for 3 years)

We are asked to find the amount of the monthly payment, so we use the formula for an installment loan:

$$PMT = \frac{P \cdot \frac{r}{n}}{1 - \left(1 + \frac{r}{n}\right)^{-nt}}$$

$$PMT = \frac{25,000 \cdot \frac{0.05}{12}}{1 - \left(1 + \frac{0.05}{12}\right)^{-12\times 3}}$$

$$PMT \approx \$749.38$$

To find the amount of interest paid, we first calculate the total amount paid over the course of the loan. We know that $\$25,000$ of this went to pay off the amount borrowed, and the rest must be interest. Since we paid $749.38 each month (12 per year) for 3 years, the total amount paid is:

$$\text{Total Paid} = \$749.38 \times 12 \times 3 = \$26,977.68$$

Since $\$25,000$ of this is the amount borrowed, the amount of interest you will pay will be:

$$\text{Interest} = \$26,977.68 - \$25,000 = \$1,977.68$$

▲

4. You borrow $\$40,000$ to purchase a new car. To pay for this, you take out a car loan for 5 years at 7.5% interest. How much will your monthly payments be?

____

Answer.

- 801.52

07. Activity Mortgages

A **mortgage** is a long-term loan, usually 15 or 30 years, for purposes of buying a home. Like other loans, they must be repaid on regular intervals (usually monthly) with regular payments and accrue interest over time compounded at the same rate as the payments.

See Section for the Mortgages notes.We begin by noting that we will be considering a very simplistic view of mortgages. In reality, they are extraordinarily complicated and technical, and the complete mathematics behind them is too advanced for us in this course. However, our goal is to understand the basics of how mortgages work and how math can be used to analyze them.

Mortgages have some special features that distinguish them from a typical loan. These include:

- **Down Payment**: This is a percent of the sale price of the home that is paid at the time of closing (purchase). Because it is paid at closing, money is ***not*** borrowed to cover the down payment. In other words, the amount of the mortgage is the sale price of the home minus the down payment amount.
- **Points**: This is a fee paid to a lender at the time of closing. Each point represents one percent of the mortgage amount, i.e.:
  - 1 point is 1% of the mortgage amount
  - 2 points is 2% of the mortgage amount
  - 3 points is 3% of the mortgage amount

Points are not included in the mortgage amount since they must be paid at closing, and is not considered interest. One important point (ha!) here: points are taken on the mortgage amount, not the sale price!

There are MANY other fees associated with mortgages that we will not discuss here; this is an example of the simplistic view mentioned above. One nice thing about mortgages is that once we take the down payment into consideration, the basic mathematics of mortgages is the same as with installment loans! For example, we use the same two formulas to determine the payment amount for a given mortgage and the maximum mortgage amount we can afford based on a given payment amount, respectively:

$$\text{Payment } PMT = \frac{P \cdot \left(\frac{r}{n}\right)}{1 - \left(1 + \frac{r}{n}\right)^{-nt}}, \text{ Mortgage } P = \frac{PMT\left[1 - \left(1 + \frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$$

where:

- P is the amount of the mortgage (borrowed amount).
- PMT is the regular payment amount.
- r is the interest rate represented as a decimal.
- n is the number of compounding periods per year.
- t is the time in years.

Note 5.17 You do NOT need to memorize these formulas! They will be provided for you on all quizzes and exams!

1. Suppose you are taking out a mortgage and you must pay two points on it. How much must be paid?

(a) 3% of the monthly payment

(b) 2% of the down payment

(c) 3% of the sale price

(d) 2% of the monthly payment

(e) 3% of the down payment

(f) 1% of the down payment

(g) 1% of the mortgage amount

(h) 2% of the sale price

(i) 3% of the mortgage amount

(j) 2% of the mortgage amount

(k) 1% of the monthly payment

(l) 1% of the sale price

Answer.

- J: 2% of the mortgage amount

2. Suppose you purchase a home which costs $\$500,000$ and requires a $\$50,000$ down payment. What is the amount of the mortgage required for this home?

$________________

Answer.

- 450000

3. Which of the following is true about the formulas used for calculations with mortgages?

(a) They are the same as for installment loans.

(b) They are the same as for compound interest.

(c) They are the same as for annuities.

(d) They are completely different from all other formulas we have encountered in the course so far.

(e) There are no formulas for mortgages.

Answer.

- A: They are the same as for installment loans.

See Example 2.31 and Example 2.32. The price of a home is \$195,000. The bank requires a 10% down payment and two points at the time of closing. The cost of the home is financed with a 30-year fixed-rate mortgage at 7.5%.

If we take a closer look at the example above, we notice something that is likely surprising to anyone who has not encountered a mortgage before: more than half of the amount paid covers the interest for the mortgage! In fact, over 60% of the total paid covers the interest. This is a consequence of the long length of the mortgage.

4. The price of a condo is \$280,000. The bank requires a 5% down payment and one point at the time of closing. The cost of the condo is financed with a 15-year fixed-rate mortgage at 8%.

a) Find the required down payment.
\$____________________
b) Find the mortgage amount.
\$____________________
c) Find the amount paid for the one point at closing.
\$____________________
d) Find the monthly payment amount. Round to the nearest dollar.
\$____________________
e) Find the total amount paid over the 15 years (not including the down payment or points).
\$____________________
f) Find the total amount of interest accrued over the 15 years.
\$____________________

Answer.

- 14000
- 266000
- 2660
- 2542
- 457560
- 191560

Example 5.18 Amanda and Fred are buying a house on a 30-year mortgage. They can only afford to pay \$800 per month for a mortgage. If they have an interest rate of 3.75%, what is the maximum price of a mortgage they can afford?

Solution. We are given the following values:

- $PMT = \$800$ (their max payment amount)
- $r = 0.0375$ (the interest rate as a decimal)
- $n = 12$ (interest is compounded monthly)
- $t = 30$ (the mortgage will be for 30 years)

We are asked to find the maximum amount of a loan we can afford, so we use the installment loan formula solved for P:

$$P = \frac{PMT\left[1 - \left(1 + \frac{r}{n}\right)^{-nt}\right]}{\left(\frac{r}{n}\right)}$$

$$P = \frac{800\left[1 - \left(1 + \frac{0.0375}{12}\right)^{-12\times 30}\right]}{\left(\frac{0.0375}{12}\right)}$$

$$P \approx \$172,743.05$$

Based on this calculation, if Amanda and Fred can only afford an \$800 monthly payment for a mortgage, then they should be sure not to take out a mortgage larger than around \$172,743.05. ▲

5. You want to purchase a home with a 15-year mortgage with 6.5% interest. You can afford to pay \$2000 each month. What is the maximum amount of a mortgage you can afford?

Round to the nearest dollar.

\$____________________

Answer.

- 229592

08. Activity Introduction to Probability

The probability of a specified event is the chance or likelihood that it will occur. There are several ways of viewing probability. One would be experimental in nature, where we repeatedly conduct an experiment. Suppose we flipped a coin over and over and over again and it came up heads about half of the time; The probability of a specified event is the chance or likelihood that it will occur. There are several ways of viewing probability. One would be experimental in nature, where we repeatedly conduct an experiment. Suppose we flipped a coin over and over and over again and it came up heads about half of the time; we would expect that in the future whenever we flipped the coin it would turn up heads about half of the time. When a weather reporter says "there is a 10% chance of rain tomorrow," they are basing that on prior evidence; that out of all days with similar weather patterns, it has rained on 1 out of 10 of those days.

In this course we will mostly be concerned with theoretical probability, which is defined as follows: Suppose there is a situation with equally likely possible outcomes and that m of those outcomes correspond to a particular event; then the probability of that event is defined as the fraction

$$\frac{m}{n}$$

where n is the total number of outcomes. We discuss the terminology a bit more below.

1. In this course we will be primarily concerned with which of the following?

(a) Descriptive probability

(b) Theoretical probability

(c) Distributive probability

(d) Experimental probability

(e) Inferential probability

(f) Subjective probability

(g) None of the above

Answer.

- B: Theoretical probability

2. Suppose an experiment has 21 equally-likely outcomes and 8 of them correspond to some event. According to the formula above, what is the probability of this event? Enter your answer as a fraction.

____

Answer.

- $\frac{8}{21}$

3. Suppose you roll a 12-sided die with sides labeled with the numbers 1 through 12. List the outcomes in the following events.

Rolling a number divisible by 3: ____________________
Rolling a number greater than 10: ____________________
Rolling a number which is odd and less than 6: ____________________

Answer.

- 3,6,9,12
- 11,12
- 1,3,5

If you roll a die, pick a card from a deck of playing cards, or randomly select a person and observe their hair color, we are executing an experiment or procedure in which the outcome is uncertain. In probability, we look at the likelihood of different outcomes. We begin with some terminology.

Definition 5.19 An **experiment** is an occurrence in which the outcome is uncertain.
The result of an experiment is called an **outcome**.
An **event** is any particular outcome or group of outcomes.
The **sample space** is the set of all possible outcomes from an experiment. ◆

Here is a simple example of an experiment.

Example 5.20 Suppose we roll a standard 6-sided die. The sample space is the set of all possible outcomes:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Some events for this experiment include:

- Rolling a 3: $E = \{3\}$
- Rolling an even number: $E = \{2, 4, 6\}$
- Rolling a number greater than 4: $E = \{5, 6\}$
- Rolling a number greater than 6: $E = \emptyset$ (the empty set!)

▲

Given that all outcomes of an experiment are equally likely, we can compute the probability of an event using this formula:

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in } S}$$

Important note: this fraction for $P(E)$ will sometimes be a reduced fraction, but not always; if it is not reduced, then we always want to reduce it!

We return to our die-rolling example.

Example 5.21 If we roll a 6-sided die, find the probabilities of the following events:

a Rolling a 3

b Rolling an even number

c Rolling a number greater than 3

d Rolling a number greater than 6

Solution. Since the sample space is $S = \{1, 2, 3, 4, 5, 6\}$, we have $n = 6$ total possible outcomes. To find the probability of each event, we need to determine the number of outcomes in each event and form the fraction with this number over 6.

a The event $E = \{3\}$ has only one outcome, so $m = 1$. Thus,

$$P(\text{rolling a 3}) = \frac{1}{6}$$

b The event $E = \{2, 4, 6\}$ has 3 outcomes, so $m = 3$. Thus,

$$P(\text{rolling an even number}) = \frac{3}{6} = \frac{1}{2}$$

c The event $E = \{4, 5, 6\}$ has three outcomes, so $m = 3$. Thus,

$$P(\text{rolling a number } > 3) = \frac{3}{6} = \frac{1}{2}$$

d The event here has no elements since it is impossible to roll a number greater than 6. So $m = 0$ and

$$P(\text{rolling a number } > 6) = \frac{0}{6} = 0$$

▲

4. Suppose you roll a 12-sided die with sides labeled with the numbers 1 through 12. Find the probabilities of the following events as reduced fractions. If it helps, you were asked to write down the outcomes to these same events in a previous question.

Rolling a number divisible by 3
$P(E) =$ ____________________
Rolling a number greater than 10
$P(E) =$ ____________________
Rolling a number which is odd and less than 6
$P(E) =$ ____________________

Answer.

- $\frac{1}{3}$
- $\frac{1}{6}$
- $\frac{1}{4}$

We now consider a few additional examples of experiments and probabilities of events. Many probability problems are concerned with playing cards. In case you are not familiar, here is everything you need to know about this! A standard deck of playing cards consists of:

- 52 total cards
- 4 suits: hearts, diamonds, spades, and clubs. As the names suggest, hearts and diamonds are red, while spades and clubs are black.
- 13 cards of each suit: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King.

The Jack, Queen, and King are collectively called **face cards** since they each have a face on them.

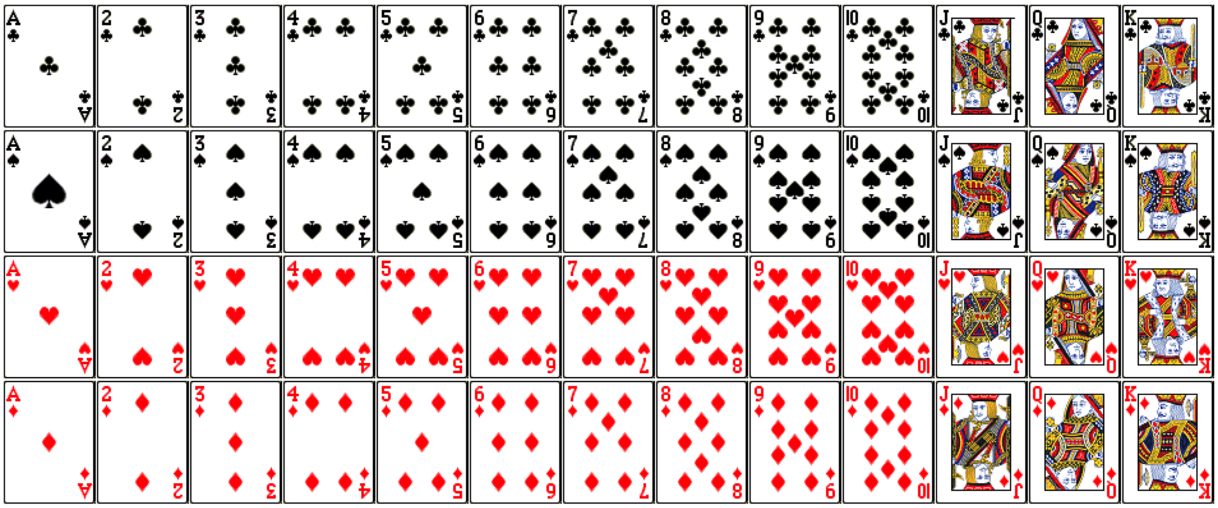


Figure 5.22 A standard deck of 52 playing cards.

5. You are dealt one card from a standard deck of playing cards. Find the probability of being dealt each of the following:

a An ace

b A spade

c An even heart

d A face card

Solution. Since there are 52 possible cards to be dealt, the total number of outcomes in the sample space is $n = 52$.

a There are four possible aces in the deck (one of each suit), so $m = 4$. Thus,

$$P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$$

b There are 13 spades in the deck, so $m = 13$. Thus,

$$P(\text{Spade}) = \frac{13}{52} = \frac{1}{4}$$

c There are 5 even hearts; namely, the 2, 4, 6, 8, and 10 of hearts. So $m = 5$ and

$$P(\text{Even Heart}) = \frac{5}{52}$$

d There are 12 face cards—the Jack, Queen, and King in each of the four suits—so $m = 12$. Thus,

$$P(\text{Face Card}) = \frac{12}{52} = \frac{3}{13}$$

6. You are dealt one card from a standard deck of 52 playing cards. Find the probability of being dealt the following.

(a) A black card

____

(b) An odd diamond (Note: We are only considering the number cards 2 through 10 as being odd; do NOT count the Ace.)

____

(c) The 5 of spades

____

(d) A red face card

____

Answer.

- $\frac{1}{2}$
- $\frac{1}{13}$
- $\frac{1}{52}$
- $\frac{3}{26}$

Tree diagram for three coin tosses as referred to in Figure 3.8.

7.

8. The sample space is shown in Table 3.4.

09. Activity Probability Events: Not, Or

Given an experiment with a sample space S, we know how to calculate the probability that an event E will happen:

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in } S}$$

For example, if we roll a standard 6-sided die, the probability of rolling a 3 is $\frac{1}{6}$ since there is only one way to roll a 3 and six total outcomes from the roll.

Now let us examine the probability that an event does **NOT** happen. Suppose we want to find the probability that we do **NOT** roll a 3. There are five outcomes in which we do not roll a 3 (namely, 1, 2, 4, 5, and 6), so the probability is $\frac{5}{6}$. Notice that:

$$\frac{1}{6} + \frac{5}{6} = 1$$

This is not a coincidence. Consider a generic situation with n possible outcomes and an event E that corresponds to m of these outcomes. Then the remaining $n - m$ outcomes correspond to E not happening, thus:

$$P(\text{not } E) = \frac{n-m}{n} = \frac{n}{n} - \frac{m}{n} = 1 - P(E)$$

Definition 5.23 The **complement** of an event E is the event "E does not happen." This is sometimes denoted by E^c (or E', $\overline{E}$).

The probability of the complement of E is calculated using the formula:

$$P(E^c) = 1 - P(E)$$

◆

Example 5.24 You draw a random card from a deck of playing cards. What is the probability that it is not a heart?

Solution. There are 13 hearts in a deck of 52 cards, so the probability of drawing a heart is:

$$P(\text{Heart}) = \frac{13}{52} = \frac{1}{4}$$

Using the complement formula above, the probability that the card is not a heart is:

$$P(\text{not Heart}) = 1 - P(\text{Heart}) = 1 - \frac{1}{4} = \frac{3}{4}$$

▲

1. Suppose E is an event and $P(E) = \frac{3}{10}$. What is $P(\text{not } E)$?

____

Answer.

- $\frac{7}{10}$

2. You draw a random card from a deck of playing cards.
What is the probability it is not a queen?

____

What is the probability that it is not a face card?

____

Answer.

- $\frac{12}{13}$
- $\frac{10}{13}$

Our previous examples have looked at a single event occurring. Now we will look at the probability of one of two events occurring; that is, we will consider the probability of A or B occurring, where A and B are two different events associated with the experiment being performed. Let us consider an example to demonstrate how to deal with such events.

Example 5.25 Suppose we flipped a coin and rolled a die, and wanted to know the probability of getting a head on the coin **or** a 6 on the die.

Solution. There are 12 possible outcomes for this experiment:

$$S = \{(H,1),(H,2),(H,3),(H,4),(H,5),(H,6),(T,1),(T,2),(T,3),(T,4),(T,5),(T,6)\}$$

By simply counting, we can see that 7 of the outcomes have a head on the coin or a 6 on the die or both; we use "or" inclusively here. These outcomes are $(H,1),(H,2),(H,3),(H,4),(H,5),(H,6)$, and $(T,6)$, so the probability is $\frac{7}{12}$.

How could we have found this from the individual probabilities? As we would expect, 6 of these outcomes have a head, and 2 of these outcomes have a 6 on the die. If we add these individual counts, we get $\frac{6}{12}+\frac{2}{12}=\frac{8}{12}$, which is not the correct probability.

Looking at the outcomes we can see why: the outcome $(H,6)$ would have been counted twice, since it contains both a head and a 6. The probability of getting both a head and rolling a 6 is $\frac{1}{12}$. If we subtract out this double count, we have the correct probability:

$$\frac{6}{12}+\frac{2}{12}-\frac{1}{12}=\frac{7}{12}$$

▲

The above example demonstrates how to compute the probability of an "or" event. We state the formula formally now.

Definition 5.26 If A and B are two events, then the probability of A **or** B (also written as $A \cup B$) is given by:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

◆

We consider another example involving cards.

Example 5.27 Suppose we draw one card from a standard deck.

a What is the probability that we get a Queen or a King?

b What is the probability that we get a red card or a King?

Solution. We know there are 52 possible cards, so $n = 52$.

a There are 4 Queens and 4 Kings in the deck, so $P(\text{Queen}) = \frac{4}{52}$ and $P(\text{King}) = \frac{4}{52}$. There are no cards which are both a Queen and a King, so $P(\text{Queen and King}) = \frac{0}{52} = 0$. Thus,

$$P(\text{Queen or King}) = \frac{4}{52}+\frac{4}{52}-0=\frac{8}{52}=\frac{2}{13}$$

b There are 26 red cards and 4 Kings in the deck, so $P(\text{Red}) = \frac{26}{52}$ and $P(\text{King}) = \frac{4}{52}$. There are 2 cards which are both red and a King (the King of Hearts and King of Diamonds), so $P(\text{Red and King}) = \frac{2}{52}$. Thus,

$$P(\text{Red or King}) = \frac{26}{52}+\frac{4}{52}-\frac{2}{52}=\frac{28}{52}=\frac{7}{13}$$

▲

Note 5.28 Note that it was advantageous in the previous example to wait until the final calculation to reduce the fractions. This allows us to easily add fractions with a common denominator—in this case, 52—and reduce the answer once.

3. Suppose we flipped a coin and rolled a die. What is the probability of getting a tail on the coin or an even number on the die?

____

Answer.

- $\frac{3}{4}$

4. Suppose we draw one card from a standard deck.

(a) What is the probability that we get a spade or a heart?

____

(b) What is the probability that we get a spade or a face card?

____

Answer.

- $\frac{1}{2}$
- $\frac{11}{26}$

We consider one additional example with information presented using a table. The table below shows the number of survey subjects who have received and not received a speeding ticket in the last year, and the color of their car.

Table 5.29

	Speeding ticket	**No speeding ticket**	**Total**
Red car	15	135	150
Not red car	45	470	515
Total	60	605	665

Example 5.30 Find the probability that a randomly chosen person:

a) has a red car **and** got a speeding ticket

b) has a red car **or** got a speeding ticket

Solution. The total number of people surveyed is $n = 665$.

a) From the table, we can see that 15 of the people surveyed had a red car and got a speeding ticket, so $m = 15$. Thus,

$$P(\text{Red Car and Ticket}) = \frac{15}{665} = \frac{3}{133}$$

b) From the table, we can see that 150 people have a red car, 60 people got a speeding ticket, and 15 people have a red car and got a speeding ticket. Using the addition rule to account for the double count, we find the number of favorable outcomes is $150 + 60 - 15 = 195$. Thus,

$$P(\text{Red Car or Ticket}) = \frac{150}{665} + \frac{60}{665} - \frac{15}{665} = \frac{195}{665} = \frac{39}{133}$$

▲

5. Using the table above, find the probability that a randomly choosen person:
(a) does not have a red car and did not get a speeding ticket.

____

(b) does not have a red can or did not get a speeding ticket.

____

Answer.

- $\frac{470}{665}$
- $\frac{650}{665}$

10. Activity The Fundamental Counting Principle

All of the probability problems we encountered in the previous unit had a relatively small sample space. Many situations involve very large sample spaces which cannot be written down, e.g., the number of 8-character passwords for a computer. For these types of situations, we need an efficient way to count certain types of items. We will ultimately use three primary counting tools: the Fundamental Counting Principle, Permutations, and Combinations. We discuss the first two of these in this activity.

Let's begin with a real-life situation. Suppose you are at a restaurant, and the menu has three choices for an appetizer (soup, salad, or breadsticks) and five choices for a main course (hamburger, sandwich, quiche, fajita, or pizza). You are allowed to choose exactly one appetizer and one main course for your meal. How many different options do you have?

There are different approaches to solving this problem. One possible approach is to list all possible meals. Here is an attempt to do so:

- Soup + hamburger
- Soup + fajita
- Salad + pizza
- Salad + hamburger
- Breadsticks + sandwich
- Breadsticks + quiche
- Salad + fajita
- Salad + quiche
- Salad + sandwich
- Soup + sandwich
- Soup + pizza
- Soup + quiche
- Breadsticks + pizza
- Breadsticks + hamburger

This list suggests there are 14 different options, but does this list include all of them? This is an issue with this particular approach; in order to know the number of meals possible, we must ensure that all options have been accounted for in our list. In this case, I intentionally left one out; the astute reader will notice Breadsticks + fajita does not appear in the above list.

Another approach is to make a table with rows for each appetizer and columns for each main course.

Table 5.31

	Hamburger	**Sandwich**	**Quiche**	**Fajita**	**Pizza**
Soup	1	2	3	4	5
Salad	6	7	8	9	10
Breadsticks	11	12	13	14	15

This tabular approach ensures we have all the options included! Counting the number of cells in the table shows there are 15 options at the restaurant. Moreover, from this approach we can see exactly how to calculate the total number of options based on the number of appetizers and the number of main courses: since there are 3 rows (corresponding to the appetizers) and 5 columns (corresponding to the main courses), we can see the number of options is $3 \times 5 = 15$. This demonstrates what we will refer to as the Fundamental Counting Principle.

Definition 5.32 If we are asked to choose one item from each of two separate categories where there are m items in the first category and n items in the second category, then the total number of available choices is:

$$m \times n$$

◆

Example 5.33 There are 8 novels and 5 volumes of poetry on a reading list for a college English course. How many different ways can a student select one novel and one volume of poetry to read for the course?

Solution. There are two categories here: novels and volumes of poetry. By the Fundamental Counting Principle, the number of ways to choose one item from each of the two categories is:

$$8 \times 5 = 40$$

▲

1. Which of the following counting techniques will we discuss in class. Select ALL that apply!

(a) Pigeonhole Principle

(b) Fundamental Counting Principle

(c) Permutations

(d) Combinations

(e) Conditional Counting Principle

(f) Inclusion-Exclusion Principle

Answer.

- Fundamental Counting Principle
  Permutations
  Combinations

2. Use the Fundamental Counting Principle to answer the following questions.

A restaurant offers 15 appetizers and 10 main courses. In how many different ways can a person order a two-course meal?

____

A college student needs to take a psychology course and an economics course during the summer session. There are 12 options for a psychology course and 9 options for an economics course. How many possible summer schedules does this student have to choose from?

____

You are buying a new car. There are 17 possible colors to choose from and you have the option of either manual or automatic transmission. How many different car options do you have?

____

Answer.

- 150
- 108
- 34

The Fundamental Counting Principle can be extended when there are more than two categories by applying it repeatedly. Here are some examples of this.

Example 5.34 A pizza shop has a special promotion which allows you to buy a one-topping pizza at half price! They offer three different sizes, four different crusts, and twelve different topping options. How many different pizzas can be purchased under this promotion?

Solution. There are three different categories here (size, crust, topping) from which we need to choose one option each. By the Fundamental Counting Principle, the total number of pizzas which can be purchased is:

$$3 \times 4 \times 12 = 144$$

▲

Example 5.35 A quiz consists of five true-or-false questions. In how many ways can a student answer the quiz?

Solution. In this scenario the categories are the questions on the quiz; we need to choose an answer for each of the five questions. Each question has 2 possible answers, either "True" or "False". This means we have 2 possible choices in each of the five categories, so the total number of ways to answer the quiz is:

$$2 \times 2 \times 2 \times 2 \times 2 = 32$$

Note this can be written as $2^5 = 32$. This can be a useful way to write this answer! ▲

Example 5.36 A telephone number consists of a 3-digit area code followed by a 7-digit local phone number. Area codes and local phone numbers cannot begin with a 0 or 1. How many different phone numbers are possible?

Solution. One way to think about a problem like this is to use what can be called a "blank diagram". A blank diagram will show a blank space for each item that needs to be chosen, and then we can fill in each blank with the number of choices we have for that item. For this particular scenario, a blank diagram might look like this, modeled after how we traditionally write phone numbers:

$$(\_\_ \; \_\_ \; \_\_) \; \_\_ \; \_\_ \; \_\_ - \_\_ \; \_\_ \; \_\_ \; \_\_$$

We start on the left side and fill in one blank at a time with the number of options we have for that position. The left-most number is the first digit in the area code. We are told this cannot be 0 or 1, which means it can be any of the digits $2, 3, 4, 5, 6, 7, 8$, or 9. There are 8 possible options here, so we fill in the first blank with an 8:

$$(\underline{8} \; \_\_ \; \_\_) \; \_\_ \; \_\_ \; \_\_ - \_\_ \; \_\_ \; \_\_ \; \_\_$$

The next number can be any digit; we have no restriction on the second digit of the area code. So we can fill in the second blank with the total number of options for that digit (the digits 0 through 9, which gives 10 choices):

$$(\underline{8} \; \underline{10} \; \_\_) \; \_\_ \; \_\_ \; \_\_ - \_\_ \; \_\_ \; \_\_ \; \_\_$$

Similarly, there are 10 options for the third digit of the area code:

$$(\underline{8} \; \underline{10} \; \underline{10}) \; \_\_ \; \_\_ \; \_\_ - \_\_ \; \_\_ \; \_\_ \; \_\_$$

Like with the area code, the first digit of the local phone number cannot be 0 or 1, so we have 8 possible choices. The remaining digits have no restriction, so they can be any of the 10 possible digits:

$$(\underline{8} \; \underline{10} \; \underline{10}) \; \underline{8} \; \underline{10} \; \underline{10} - \underline{10} \; \underline{10} \; \underline{10} \; \underline{10}$$

Once we have filled in all the blanks, we can multiply the number of options together to form the total number of possible phone numbers:

$$8 \times 10 \times 10 \times 8 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 6,400,000,000$$

▲

3. Answer the following questions using the Fundamental Couting Principle.

At a particular restaurant you have eight choices for an appetizer, eleven choices for a main course, and five choices for a desert. If you are allowed to choose exactly one item from each category, how many different meal options do you have?

____

You are purchasing a new car. You can choose from 9 different colors, with or without air conditioning, electric or gas powered, and with or without a satellite radio. How many different car selections are possible?

____

Answer.

- 440
- 72

4. A company puts a code on each product they sell. The code is made up of three digits followed by two letters (a through z, not case specific). For example, $123az$, $788cd$, and $010jk$ are all possible product codes. How many different codes are possible?

____

Answer.

- 676000

11. Activity Permutations

Let's now consider a slightly different counting problem.

Example 5.37 How many different ways can the letters of the word MATH be rearranged to form a four-letter code word? (E.g., MATH, AMTH, HTAM, etc.)

Solution. We can use a blank diagram to help us answer this question! We start with one with four blanks, one for each of the positions in the code word:

$$\_\_\ \_\_\ \_\_\ \_\_$$

There are 4 choices for the first letter of the code word; it can be M A, T, or H. Thus, we have:

$$\underline{4}\ \_\_\ \_\_\ \_\_$$

Once the first letter is chosen, there are only 3 choices for the second letter of the code word. For example, if M is chosen as the first letter, then the options for the second are A, T, and H, since M cannot be used a second time. Or, if A is chosen as the first letter, then the options for the second are M, T, and H. The key is that regardless of the choice for the first letter, there will always be 3 options to choose from for the second letter. So:

$$\underline{4}\ \underline{3}\ \_\_\ \_\_$$

Using a similar logic, once the first two letters are chosen there are only 2 options remaining for the third letter. For example, if M is chosen first and A chosen second, then the options for the third letter are only T and H. Thus:

$$\underline{4}\ \underline{3}\ \underline{2}\ \_\_$$

Finally, there is only 1 choice for the final letter, since all of the other letters have been used at this point. This means our completed blank diagram looks like:

$$\underline{4}\ \underline{3}\ \underline{2}\ \underline{1}$$

So by the Fundamental Counting Principle, the number of code words is:

$$4 \times 3 \times 2 \times 1 = 24$$

▲

This is a simple example of what is called a **permutation**. A permutation is an ordered arrangement of items that occurs when:

- No item is used more than once, and
- The order of arrangement makes a difference (e.g., MA is different than AM).

The product that we saw in the preceding example is called a **factorial** and is denoted $n!$. In general, $n!$ is read as "n factorial." For example:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

Definition 5.38 When a situation involves rearranging all of the objects in a given set, then the number of permutations is given by:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$$

where n is the total number of objects. ◆

Note 5.39 Scientific calculators have a button which will compute factorials automatically. If you have the TI-30XIIS calculator, you can access this by hitting the PRB button and arrowing over to the ! symbol.

Example 5.40 How many ways can five different door prizes be distributed among five people?

Solution. Here we want to rearrange the five people in order so that the first gets the first prize, the second gets the second prize, and so on. We must rearrange all five of the people since there are five prizes, so the number of ways to do this is:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

▲

1. Answer the following questions. Feel free to use the factorial key on your calculator to find the answers.

In how many different ways can a police department arrange 8 suspects in a police lineup if each lineup contains all eight people?

____

Twelve friends want to sit together at a movie theater. Conveniently, one of the rows has exactly 12 seats. How many different seating arrangements are possible?

____

Answer.

- 40320
- 479001600

Sometimes there are scenarios where we want to determine the number of ways to rearrange only a subset of a set of objects. The next two examples present two such scenarios.

Example 5.41 A charity benefit is attended by 25 people and three gift certificates are given away as door prizes: one gift certificate is in the amount of \$100, the second is worth \$25, and the third is worth \$10. Assuming that no person receives more than one prize, how many different ways can the three gift certificates be awarded?

Solution. Since the three gift certificates are worth different amounts, the order in which they are awarded makes a difference. There are 25 possible winners for the \$100 gift certificate. Once that has been awarded, there are 24 possible winners for the \$25 gift certificate since the first winner is not eligible. Finally, after the first two have been awarded there are 23 possible winners for the \$10 gift certificate since the first two winners are ineligible. The total number of ways the three gift certificates can be awarded is:

$$25 \times 24 \times 23 = 13,800$$

▲

Example 5.42 Eight sprinters have made it to the Olympic finals in the 100-meter race. In how many different ways can the gold, silver, and bronze medals be awarded?

Solution. Like the previous example, the order in which the medals are awarded makes a difference since they are different. There are 8 possible gold medalists, then 7 possible silver medalists (the runner who wins gold cannot win silver), and finally 6 possible bronze medalists (the gold and silver medalists cannot win bronze). This means the number of ways the medals can be awarded is:

$$8 \times 7 \times 6 = 336$$

▲

The two examples above are similar in that:

- They had a total group of people (25 and 8, respectively) and wanted to order a strictly smaller number of them (in both cases, 3).
- The order of selection was important (e.g., first place, second place, third place).
- Awards were done without replacement; whoever was chosen first could not be chosen again.

These examples demonstrate permutations where only a select number of items are chosen to be rearranged.

Here is the general scenario. Suppose we have n total objects and want to choose r of them without replacement when order matters (this is what was done in the preceding examples!). The number of ways this can be done is denoted ${}_nP_r$ and is given by the formula:

$${}_nP_r = \frac{n!}{(n-r)!}$$

Note 5.43 Your calculator should have a key to calculate this for you! If you are using the TI-30XIIS, the key is located in the PRB menu along with the factorial key!

Example 5.44 Bo has nine paintings and has room to display only four of them at a time on his wall. How many different ways could Bo do this?

Solution. In this example, we have a total of 9 paintings to choose from and need to choose 4 of them. The order of selection matters since they are being arranged on a wall, and none can be used more than once. This is a permutation! Using a calculator to find ${}_9P_4$, the number of ways for Bo to arrange four paintings is:

$${}_9P_4 = \frac{9!}{(9-4)!} = \frac{9!}{5!} = 9 \times 8 \times 7 \times 6 = 3,024$$

▲

Example 5.45 How many ways can a four-person executive committee (president, vice-president, secretary, treasurer) be selected from a 16-member board of directors of a non-profit organization?

Solution. In this example, we have a total of 16 people to choose from and need to choose 4 of them. The order of selection matters (e.g., being president is different than being secretary), and no one can serve in multiple roles. This is a permutation! Using a calculator to find ${}_{16}P_4$, the number of ways to form such a committee is:

$${}_{16}P_4 = \frac{16!}{(16-4)!} = \frac{16!}{12!} = 16 \times 15 \times 14 \times 13 = 43,680$$

▲

2. Answer the following questions. Use the ${}_nP_r$ button on your calculator!

You are the coach for a little league baseball team. There are 13 players on the team. You need to choose a batting order having 9 players. How many different batting orders can you make?

____

A TV network needs to fill up five different time slots and has 10 classic sitcoms available. In how many ways can they fill these available time slots?

____

Answer.

- 259459200
- 30240

3. You need to make a 5-character password using the letters A through Z.

How many passwords are possible if repeated letters are allowed?

____

How many passwords are possible if repeated letters are NOT allowed?

____

Answer.

- 11881376
- 7893600

12. Activity Combinations

Consider the following scenario, which is a slight modification of an example we considered in the previous section. A charity benefit is attended by 25 people and three gift certificates are given away as door prizes. Each gift certificate is \$100. Assuming that no person receives more than one prize, how many different ways can the three gift certificates be awarded?

In the previous example, the three gift certificates were for different amounts and therefore the order the three winners were selected in made a difference. Here, the order in which the three winners are chosen does **NOT** matter! We only care about which three individuals get the gift certificates since they are the same amount! Essentially, we want the number of ways to choose a group of 3 people from the 25 attendees.

This is an example of what is called a **combination**. Suppose we have n total objects and want to choose r of them without replacement when order does not matter. The number of ways this can be done is denoted ${}_nC_r$ and is given by the formula:

$$ {}_nC_r = \frac{n!}{r!(n-r)!} $$

Note 5.46 Your calculator will have a key to calculate the number of combinations directly. If you have the TI-30XIIS, it can be found under the PRB menu along with the factorial (!) and permutation (${}_nP_r$) functions.

To return to the example at the top of this section, we can find the number of ways to award the three gift certificates by using the combination function on a calculator:

$$ {}_{25}C_3 = \frac{25!}{3!(25-3)!} = \frac{25!}{3! \times 22!} = \frac{25 \times 24 \times 23}{3 \times 2 \times 1} = 2,300 $$

1. A charity benefit is attended by 20 people and five gift certificates are given away as door prizes. Each gift certificate is \$30. Assuming that no person receives more than one prize, how many different ways can the five gift certificates be awarded?

____

Answer.

- 15504

One key aspect of counting is to be able to distinguish between a permutation and a combination. The main point to consider is whether or not the order in which the items are selected matters. If the order matters, then we need to use permutations; if the order does not matter, we need to use combinations.

Example 5.47 A class has 30 total students.

a The class must choose a president, vice president, secretary, and treasurer for the year. In how many ways can this be done?

b The class must also choose four students to represent the class in the student council. In how many ways can this be done?

Solution.

a We need to choose four students, and the order makes a difference here because the roles of the students will be different (e.g., being president is different than being treasurer). This is a permutation, so the number of ways this can be done is:

$$_{30}P_4 = \frac{30!}{(30-4)!} = 30 \times 29 \times 28 \times 27 = 657,720$$

b In this case, the order in which the students are chosen does not matter; all four chosen students will serve the same role (namely, being a member of the student council). This is a combination, so the number of ways this can be done is:

$$_{30}C_4 = \frac{30!}{4!(30-4)!} = \frac{30 \times 29 \times 28 \times 27}{4 \times 3 \times 2 \times 1} = 27,405$$

▲

2. Answer the following questions. Note these are the same questions as in the previous exercise, where you determined whether they were permutations or combinations.

(a) In poker, a player is dealt five cards from a standard 52-card deck. How many different poker hands are possible?

____

(b) You volunteer to pet-sit for a friend who has 7 animals. How many different pet combinations are possible if you only have the room to pet-set three animals?

____

(c) Ten swimmers compete in a race. In how many ways can the first place, second place, and third place medals be awarded?

____

(d) An election ballot asks voters to select three city commissioners from a group of six candidates. In how many ways can this be done?

____

Answer.

- 2598960
- 35
- 720
- 20

3. For each scenario, determine whether the answer would be a permutation or combination.

(a) In poker, a player is dealt five cards from a standard 52-card deck. How many different poker hands are possible? `

(a) Permutation

(b) Combination

(b) You volunteer to pet-sit for a friend who has 7 animals. How many different pet combinations are possible if you only have the room to pet-set three animals?

(a) Permutation

(b) Combination

(c) Ten swimmers compete in a race. In how many ways can the first place, second place, and third place medals be awarded?

(a) Permutation

(b) Combination

(d) An election ballot asks voters to select three city commissioners from a group of six candidates. In how many ways can this be done?

(a) Permutation

(b) Combination

Answer.

- B: Combination
- B: Combination
- A: Permutation
- B: Combination

Sometimes combinations must be combined with the Fundamental Counting Principle to solve a problem! Here is an example of this!

Example 5.48 The US Senate of a particular Congress consisted of 48 Democrats, 50 Republicans, and 2 Independents. How many committees can be formed if each committee must have exactly 3 Democrats and 3 Republicans?

Solution. To answer this question, we need to break the problem down into three steps:

(a) Determine the number of ways to choose the Democrats from the possible Democrats.

(b) Determine the number of ways to choose the Republicans from the possible Republicans.

(c) Use the numbers determined in the first two steps to count the total number of possible committees.

For steps 1 and 2, note that the order in which we choose the Democrats and Republicans does not matter; all we care about is the set of people chosen from each party. Since the order does not matter, the number of ways to choose 3 Democrats from the total of 48 Democrats is:

$$_{48}C_3 = \frac{48!}{3!(48-3)!} = 17,296$$

and the number of ways to choose 3 Republicans from the possible 50 Republicans is:

$$_{50}C_3 = \frac{50!}{3!(50-3)!} = 19,600$$

Since we need to choose both the Democrats and the Republicans, we need to use the Fundamental Counting Principle to find the total number of possible committees. Thus, the number of committees is:

$$_{48}C_3 \times_{50} C_3 = 17,296 \times 19,600 = 339,001,600$$

Therefore, there are $339,001,600$ possible committees that can be formed. ▲

4. The Senate Appropriations Committee consists of 29 total members, including 15 Republicans and 14 Democrats. The Defense Subcommittee must consist of 19 of these committee members, and it is decided that 10 should be Republicans and 9 should be Democrats.

In how many ways can the 10 Republicans be chosen?

____

In how many ways can the 9 Democrats be chosen?

____

How many different ways can the Defense Subcommittee members be chosen from among the Appropriations Committee members?

____

Answer.

- 3003
- 2002
- 6012006

13. Activity Probability with FCP, nPr, nCr

We will turn our attention to combining the ideas of the past several units. Let us begin by recalling some of the main topics we have discussed:

- The probability of an event E is given by the formula:

$$P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in } S}$$

- **Fundamental Counting Principle**: When asked to choose items from different categories, we multiply together the number of options for each choice to determine the total number of possibilities.
- **Permutations** are arrangements with no repetition and where the order matters. If all n items need to be rearranged, the number of ways to do this is $n!$. If only r of the n items need to be rearranged, the number of ways to do this is ${}_nP_r$.
- **Combinations** are selections with no repetition where the order does **NOT** matter. The number of combinations involving r out of n items is ${}_nC_r$.

We can use permutations and combinations to help us answer more complex probability questions.

Example 5.49 Seven performers (A, B, C, D, E, F, and G) are to appear at a fundraiser. The order of performance is determined by random selection. Find the probability that:

a E will perform second and B will perform last.

b F or G will perform last.

Solution. First, note that since we are concerned with the order the performers will appear in, this example involves permutations. There are 7 performers and all will appear, so the total number of performance orders in the sample space is:

$$n = 7! = 5,040$$

a We want to determine the number of performance orders in which E performs second and B performs last. This means there is only 1 choice for the second performer (E) and only 1 choice for the last performer (B):

$$\underline{\quad}\ \underline{\ 1\ }\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\quad}\ \underline{\ 1\ }$$

There are no restrictions on the remaining performers, so we can proceed by counting the number of possibilities for each remaining position. There are 5 choices for the first performer (A, C, D, F, or G), then 4 choices for the third performer, 3 choices for the fourth performer, 2 choices for the fifth performer, and finally only 1 choice for the sixth performer:

$$\underline{\ 5\ }\ \underline{\ 1\ }\ \underline{\ 4\ }\ \underline{\ 3\ }\ \underline{\ 2\ }\ \underline{\ 1\ }\ \underline{\ 1\ }$$

Thus, the number of favorable performance orders is:

$$m = 5 \times 1 \times 4 \times 3 \times 2 \times 1 \times 1 = 120$$

Putting this all together, the probability of a randomly selected performance order with E second and B last is:

$$P(\text{E second and B last}) = \frac{120}{5,040} = \frac{1}{42}$$

b We want to determine the number of performance orders in which F or G performs last. This means there are 2 choices (F or G) for the final performer and no restrictions on the remaining ones:

$$\underline{\ 6\ }\ \underline{\ 5\ }\ \underline{\ 4\ }\ \underline{\ 3\ }\ \underline{\ 2\ }\ \underline{\ 1\ }\ \underline{\ 2\ }$$

So, there would be 6 choices for the first performer (any of the remaining performers), 5 choices for the second performer, and so on. The number of such performance orders is:

$$m = 6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 2 = 1,440$$

Thus, the probability is:

$$P(\text{F or G last}) = \frac{1,440}{5,040} = \frac{2}{7}$$

▲

1. Five performers (A, B, C, D, and E) are to appear at a fundraiser. The order of performance is determined by random selection. Find the probability that A or B will perform first.

Enter your answer as a fraction, though it need not be reduced.

____

Answer.

- $\frac{48}{120}$

Here is an example involving permutations.

Example 5.50 A 4-digit PIN is selected at random. What is the probability that there are no repeated digits?

Solution. There are 10 possible values for each digit of the PIN (namely: $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$). Since repetition is allowed for a standard PIN, by the Fundamental Counting Principle there are total possible PINs of:

$$n = 10 \times 10 \times 10 \times 10 = 10^4 = 10,000$$

To have no repeated digits, all four digits must be different, which means selecting 4 digits out of 10 without replacement where order matters. The number of ways this can be done is a permutation:

$$m =_{10} P_4 = \frac{10!}{(10-4)!} = 10 \times 9 \times 8 \times 7 = 5,040$$

The probability of no repeated digits is the number of 4-digit PINs with no repeated digits divided by the total number of 4-digit PINs. Thus,

$$P(\text{no repeated digits}) = \frac{{}_{10}P_4}{10,000} = \frac{5,040}{10,000} = \frac{63}{125} = 0.504$$

▲

2. A 5-digit PIN number is selected at random. What is the probability that there are no repeated digits? Enter your answer as a fraction, though it need not be reduced.

____

Answer.

- $\frac{30240}{100000}$

Next, we look at some examples involving combinations.

Example 5.51 Find the probability of randomly drawing five cards from a deck and getting exactly two Aces.

Solution. We first find the total number of ways to draw 5 cards from a standard 52-card deck. Since the order in which the cards are drawn does not matter, this is a combination:

$$n =_{52} C_5 = \frac{52!}{5!(52-5)!} = 2,598,960$$

Next, we need to find the number of 5-card hands which have exactly two Aces. Note that such a hand also includes three cards which are not Aces. We need to select the Aces and non-Aces separately. There are 4 Aces in the deck, so the number of ways to choose the two Aces is:

$${}_4C_2 = 6$$

There are 48 non-Aces in the deck, so the number of ways to choose the three non-Aces is:

$${}_{48}C_3 = 17,296$$

Since we must choose one of the sets of Aces and one of the sets of non-Aces in succession, the Fundamental Counting Principle tells us the total number of hands with exactly two Aces is:

$$m =_4 C_2 \times_{48} C_3 = 6 \times 17,296 = 103,776$$

Putting all of this together, the probability of getting a hand with exactly two Aces is:

$$P(\text{exactly 2 Aces}) = \frac{{}_4C_2 \times_{48} C_3}{{}_{52}C_5} = \frac{103,776}{2,598,960} = \frac{216}{5,4145} \approx 0.0399$$

▲

Example 5.52 A parent-teacher committee consisting of four people is to be selected from fifteen parents and five teachers. Find the probability of selecting two parents and two teachers.

Solution. We first find the total number of ways to select a committee of four people from the total group of 20 people ($15 + 5 = 20$). Since the order in which the people are selected does not matter, this is a combination:

$$n =_{20} C_4 = \frac{20!}{4!(20-4)!} = 4,845$$

Next, we need to find the number of four-person committees consisting of exactly two parents and two teachers. We first select the two parents from among the 15 possible parents, and this can be done in:

$${}_{15}C_2 = 105 \text{ ways}$$

We then select the two teachers from among the 5 possible teachers, and this can be done in:

$${}_5C_2 = 10 \text{ ways}$$

We must choose both a pair of parents and a pair of teachers in succession, so by the Fundamental Counting Principle the number of committees consisting of two parents and two teachers is:

$$m =_{15} C_2 \times_5 C_2 = 105 \times 10 = 1,050$$

Therefore, the probability of selecting a committee with two parents and two teachers is:

$$P(\text{2 parents and 2 teachers}) = \frac{{}_{15}C_2 \times_5 C_2}{{}_{20}C_4} = \frac{1,050}{4,845} = \frac{70}{323} \approx 0.2167$$

▲

3. Find the probability of randomly drawing five cards from a deck and getting exactly three Aces and two Kings. Give your answer as a fraction, though it need not be reduced.

____

Answer.

- $\frac{24}{2598960}$

4. A congressional committee is to be formed consisting of 5 people to be selected from 12 Republicans and 10 Democrats. Find the probability of selecting 2 Republicans and 3 Democrats.

Give your answer as a fraction, though it need not be reduced.

____

Answer.

- $\frac{7920}{26334}$

14. Activity Data and Graphs

Categorical data (also called qualitative data) allows us to sort individual subjects or objects into distinct categories or groups. To organize and make sense of this information, we typically construct a two-column **frequency table**. The first column lists the available unique categories, while the parallel column records the corresponding **frequency**, which is the total count of observations within each classification.

Example 5.53 Shirt Color Frequency Table. A student organizes the colors of 12 shirts in a wardrobe into the following frequency table:

Color	Tally Method	Frequency
Black	I	1
Blue	IIIII	5
Pink	II	2
White	IIII	4

Use the table above to answer the following questions.

a. Which shirt color is the most common in the wardrobe?

b. How many more blue shirts are there than black shirts?

c. What is the combined total of white and pink shirts?

Solution.

a. Blue is the most common color, with a maximum frequency of 5.

b. There are 4 more blue shirts than black shirts, calculated as $5 - 1 = 4$.

c. The combined total is 6 shirts, calculated as $4 \text{ white} + 2 \text{ pink} = 6$.

▲

1. The following data represents the age of 30 lottery winners.

20	29	30	30	31	31
33	35	45	49	51	52
53	54	57	57	58	58
61	61	63	68	69	70
72	74	74	76	79	82

Complete the frequency distribution for the data.

Age	Frequency
20-29	____
30-39	____
40-49	____
50-59	____
60-69	____
70-79	____
80-89	____

Answer.

- 2
- 6
- 2
- 8
- 5
- 6
- 1

Next, we will cover how to use data to create bar graphs, histograms, line graphs, and pie charts—visual tools designed to compare counts across independent categories.

2. Think about the similarities and differences of pie charts, bar graphs, and histograms. Select all of the following statements that are true.

(a) Pie charts, bar graphs, and histograms can be used interchangeably

(b) Pie charts and bar graphs are used to compare variables while histograms are used to examine the distribution of data.

(c) Depending on how data are categorized, the percentages associated with the bars can sum to more than 100%

(d) In a graph, if the percentages add to more than 100% then a pie chart can't be used.

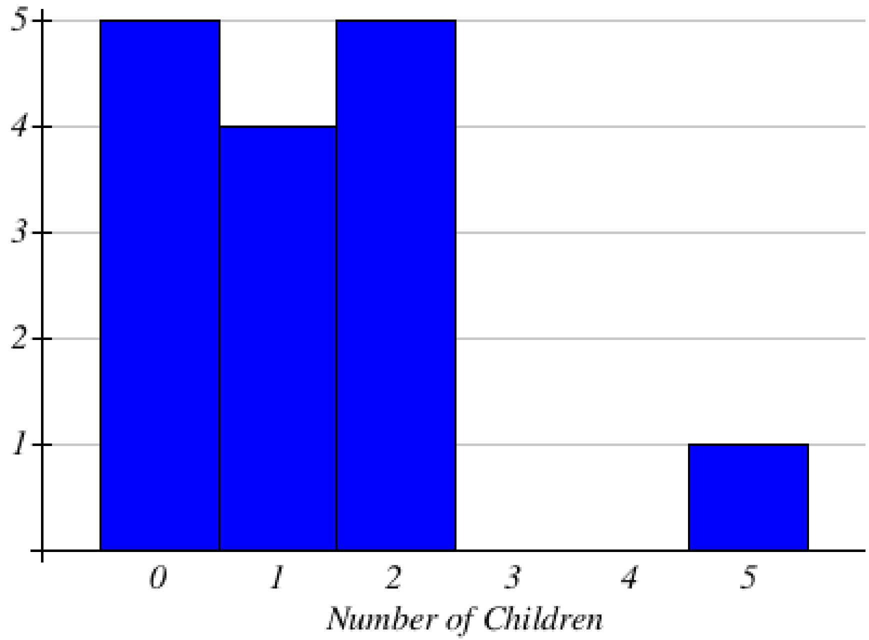
Answer.

- In a graph, if the percentages add to more than 100% then a pie chart can't be used.

  Depending on how data are categorized, the percentages associated with the bars can sum to more than 100%

  Pie charts and bar graphs are used to compare variables while histograms are used to examine the distribution of data.

3. A group of adults were asked how many children they have in their families. The bar graph below shows the number of adults who indicated each number of children.



How many adults were questioned?

____

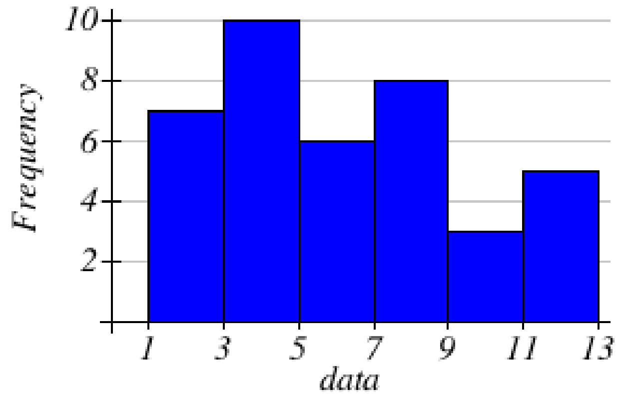
What percentage of the adults questioned had 5 children?

____________________ %

Answer.

- 15
- 6.6666666666667

4.



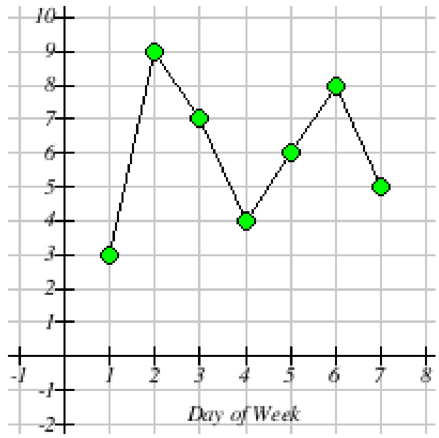
Based on the histogram above, what is the frequency of the class containing the value 10

____

Answer.

- 3

5. The line graph below shows the number of times Maria drove her car each day of the week.



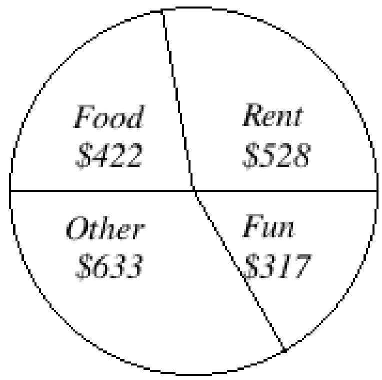
How many times did Maria drive her car on day 4?

____

Answer.

- 4

6. Kara categorized her spending for this month into four categories: Rent, Food, Fun, and Other. The amounts she spent in each category are pictured here.



What percent of her total spending did she spend on Food? Answer to the nearest whole percent.
____________________ %

Answer.

- 22

15. Activity Mean, Median, Mode and Standard Deviation

Mastering data distributions is key to analyzing statistics effectively. By utilizing measures of center and spread, you can instantly summarize and compare different datasets.

Measures of Center help identify a typical value within a dataset.

(a) *The Mean.*

The mathematical average and central balance point of a dataset. It is calculated by dividing the sum of all values by the total count of observations.

(b) *The Median.*

The exact middle value in an ordered dataset. If the dataset contains an even number of values, the median is computed as the average of the two central numbers.

(c) *The Mode.*

The value that appears most frequently in a dataset. Depending on the data, a set can be bimodal (two modes), multimodal (multiple modes), or have no mode at all.

1. The highway mileage (mpg) for a sample of 8 different models of a car company can be found below. Find the mean, median, and mode. Round to one decimal place as needed.

19, 23, 26, 27, 29, 31, 33, 33

Mean = ____________________

Median = ____________________

Mode = ____________________

Answer.

- 27.6
- 28
- 33
-

2.

Mini-Lesson: Mean, Median, and Mode

Work through the Lesson and record your work neatly in the Notebook. When you are finished, click the check box below a

Answer.

- I have completed this section of the Mini-Lesson and am ready to continue.

3. If you have the following test scores: 72, 78, 72, 78, 54, and 90, what is your mean test score?

____

Answer.

- 74

Moving beyond the range, the second primary measure of variation is the **standard deviation**. While the range is calculated from only two values, the standard deviation depends on every single data point within the dataset. Essentially, it summarizes how far each individual score typically deviates from the mathematical mean.

When comparing datasets, the standard deviation provides valuable insight into consistency:

List 5.54

A **smaller standard deviation** indicates that the data values are highly concentrated around the center.A **larger standard deviation** signals that the data points are highly dispersed and variable.

4.

Which of the following does NOT reveal the variation (spread) of the data? *Check all that apply.*

(a) Standard Deviation

(b) Mode

(c) Mean

(d) Median

(e) Range

Answer.

- Mean
  Mode
  Median

5. Two classes were given identical quizzes.
Class A had a mean score of 8.2 and a standard deviation of 0.3
Class B had a mean score of 7.7 and a standard deviation of 0.6
Which class scored better on average?

(a) Class A

(b) Class B

Which class had more consistent scores?

(a) Class A

(b) Class B

Answer.

- A: Class A
- A: Class A

6. Part 1 of 2

a) Suppose Nakala makes an 80 on the next quiz. Would the standard deviation of her 10 quiz scores be greater than, equal to, or less than 6? Justify your answer.

The standard deviation will be

(a) equal to

(b) less than

(c) greater than

The quiz score of 80 is

(a) less than

(b) greater than

(c) equal to

(a) 0

(b) 6

(c) 1

This will

(a) decrease

(b) increase

(c) not effect

Part 2 of 2

The standard deviation will be

(a) greater than

(b) equal to

(c) less than

The quiz score of 21 is

(a) very far from

(b) equal to

(c) very close to

(a) much more than 6

(b) much less than 6

This will

(a) not effect

(b) increase

(c) decrease

Answer.

- B: less than

- C: equal to
- A: 0
- A: decrease
- A: greater than
- A: very far from
- A: much more than 6
- B: increase

Chapter 6

Practical Application & Hands-On Sessions

These notes are intended strictly for a quick review of each chapter. They do not substitute for attending lectures or reading the assigned textbook.

1. Voting Methods Handout

This handout outlines the core procedural rules, calculation steps, and the four primary preference-ballot voting methods:

- *The Plurality Method.*

  Selecting the candidate who receives the most first-place votes, regardless of whether they achieve a strict majority.

- *The Plurality-with-Elimination Method.*

  An iterative process where the candidate with the fewest first-place votes is eliminated, and their ballots are redistributed, continuing until a candidate achieves a true majority.

- *The Borda Count Method.*

  A point-based system where points are assigned to each ranking position on a ballot, and the candidate with the highest total points across all ballots wins.

- *The Pairwise Comparison Method.*

  A head-to-head match-up approach where every candidate is compared one-on-one against every other candidate to see who wins the most direct duels, identifying a Condorcet candidate if one exists.

Exercises and Workspaces

1. Four students are running for president of their residence hall: Debra (D), Farah (F), Jorge (J), and Hillary (H). The votes of their fellow students are summarized in the following preferences table. Who is declared the president using the **plurality** method?

Number of Votes	**52**	**45**	**13**	**9**	**4**
First choice	D	F	J	F	H
Second choice	F	J	F	J	J
Third choice	H	H	H	D	D
Fourth choice	J	D	D	H	F

Solution. First-choice totals are D = 52, F = 54, J = 13, and H = 4. The winner by the plurality method is **F**.

2. Four students are running for president of their residence hall: Debra (D), Farah (F), Jorge (J), and Hillary (H). The votes of their fellow students are summarized in the following preferences table. Who is declared the president using the **plurality-with-elimination** method?

Number of Votes	**52**	**45**	**13**	**9**	**4**
First choice	D	F	J	F	H
Second choice	F	J	F	J	J
Third choice	H	H	H	D	D
Fourth choice	J	D	D	H	F

Solution. Eliminate H first (4 votes), which transfers to J, giving J 17 votes. Then eliminate J; all 17 J ballots transfer to F. Final totals are D = 52 and F = 71, so the plurality-with-elimination winner is **F**.

3. Four students are running for president of their residence hall: Debra (D), Farah (F), Jorge (J), and Hillary (H). The votes of their fellow students are summarized in the following preferences table. Who is declared the president using the **Borda count** method?

Number of Votes	**52**	**45**	**13**	**9**	**4**
First choice	D	F	J	F	H
Second choice	F	J	F	J	J
Third choice	H	H	H	D	D
Fourth choice	J	D	D	H	F

Solution. Using Borda scores 3-2-1-0, the totals are D = 169, F = 292, J = 155, and H = 122. The Borda count winner is **F**.

4. Four students are running for president of their residence hall: Debra (D), Farah (F), Jorge (J), and Hillary (H). The votes of their fellow students are summarized in the following preferences table. Who is declared the president using the **pairwise comparison** method? Is there a **Condorcet** winner?

Number of Votes	**52**	**45**	**13**	**9**	**4**
First choice	D	F	J	F	H
Second choice	F	J	F	J	J
Third choice	H	H	H	D	D
Fourth choice	J	D	D	H	F

Solution. In head-to-head matchups, F beats D, F beats J, and F beats H. Candidate **F** wins every pairwise comparison and is the Condorcet winner.

5. Four professors are running for president of the League of Innovation: Doug (D), Francis (F), Gallagher (G), and Smith (S). The votes of the members in the League of Innovation are summarized in the preference table below.

Number of Votes	**30**	**22**	**18**	**10**	**2**
First Choice	F	G	S	D	G
Second Choice	D	D	G	S	S
Third Choice	G	S	D	G	D
Fourth Choice	S	F	F	F	F

(a) How many members voted in this election?

(b) Using the plurality method, who becomes the new president?

(c) Using the plurality-with-elimination method, who becomes the new president?

(d) Using the Borda count method, who becomes the new president?

(e) Using the pairwise comparison method, who becomes the new president?

Solution. Total votes = 82. Plurality totals are F = 30, G = 24, S = 18, and D = 10, so **F** wins by plurality.

For plurality-with-elimination, D is eliminated first and its 10 votes transfer to S, then G is eliminated and its votes transfer to S, giving **S** the win.

Using Borda scores 3-2-1-0, the totals are D = 154, G = 148, S = 100, and F = 90. The Borda winner is **D**.

In pairwise comparison, G beats D, G beats F, and G beats S, so the pairwise winner is **G**.

2. Apportionment Handout

This handout covers the core metrics and sequential steps for the major historic apportionment methods:

- Calculating the Standard Divisor, Standard Quotas, Lower Quotas, and Upper Quotas.
- *Hamilton's Method.*

  Assigning initial seats using the Lower Quota, then distributing surplus seats based on the largest fractional remainders.
- *Jefferson's Method.*

  Finding a Modified Divisor that alters the quotas so that the sum of the Lower Quotas exactly matches the total seats.
- *Adams's Method.*

  Finding a Modified Divisor that alters the quotas so that the sum of the Upper Quotas exactly matches the total seats.

Exercises and Workspaces

1. A hospital has a nursing staff of 250 nurses working in four shifts:

- *Shift A: 7:00 A.M. to 1:00 P.M..*
- *Shift B: 1:00 P.M. to 7:00 P.M..*
- *Shift C: 7:00 P.M. to 1:00 A.M..*
- *Shift D: 1:00 A.M. to 7:00 A.M..*

The number of nurses apportioned to each shift is based on the average number of patients per shift, given in the table below. Use ***Hamilton's method*** to apportion the 250 nurses among the shifts at the hospital.

Shift	**A**	**B**	**C**	**D**
Average Patients	453	650	547	350

Solution. To complete the apportionment using Hamilton's method, we follow these sequential calculation steps:

(a) Find the Total Population: Sum the average number of patients across all shifts.

$$453 + 650 + 547 + 350 = 2000 \text{ total patients}$$

(b) Calculate the Standard Divisor (d): Divide the total population by the total number of items (250 nurses) to be allocated.

$$d = \frac{2000}{250} = 8$$

(c) Find Standard and Lower Quotas: Divide each shift's patient count by the Standard Divisor (8). The Lower Quota is found by rounding down to the nearest whole integer.

- Shift A: $\frac{453}{8} = 56.625$ (Lower Quota $= 56$)
- Shift B: $\frac{650}{8} = 81.250$ (Lower Quota $= 81$)
- Shift C: $\frac{547}{8} = 68.375$ (Lower Quota $= 68$)
- Shift D: $\frac{350}{8} = 43.750$ (Lower Quota $= 43$)

(d) Distribute Surplus Items: Summing the lower quotas gives $56 + 81 + 68 + 43 = 248$ nurses. Since we must allocate 250 nurses, there is a surplus of $250 - 248 = 2$ nurses. Hamilton's method gives these surplus spots to the shifts with the largest fractional remainder parts:

- Shift D has the largest remainder (0.750) and receives +1 nurse.
- Shift A has the second-largest remainder (0.625) and receives +1 nurse.

The complete apportionment layout is summarized below:

Metric / Step	**Shift A**	**Shift B**	**Shift C**	**Shift D**	**Total**
Patients	453	650	547	350	2000
Standard Quota	56.625	81.250	68.375	43.750	-
Lower Quota	56	81	68	43	248
Surplus Added	+1	0	0	+1	+2
Final Apportionment	57	81	68	44	250

2. A hospital has a nursing staff of 250 nurses working in four shifts. The number of nurses apportioned to each shift is based on the average number of patients per shift, given in the table below. Use ***Jefferson's method*** to apportion the 250 nurses among the shifts at the hospital.

Shift	**A**	**B**	**C**	**D**
Average Patients	453	650	547	350

Solution. To complete the apportionment using Jefferson's method, we apply a modified divisor so that the sum of the lower quotas exactly equals the required total.

(a) Find Total Population and Standard Divisor (d):

$$\text{Total Patients} = 453 + 650 + 547 + 350 = 2000$$

$$d = \frac{2000 \text{ patients}}{250 \text{ nurses}} = 8$$

(b) Test the Standard Divisor: Dividing each population by 8 and rounding down gives lower quotas of $56, 81, 68$, and 43. The sum of these initial lower quotas is $56 + 81 + 68 + 43 = 248$. This falls short of the 250 nurses needed.

(c) Determine a Modified Divisor (d_m): Because our initial sum is too small, Jefferson's method requires us to choose a smaller divisor to increase the individual quotas. Testing a modified divisor of $d_m = 7.93$ shifts the lower quotas upward to exactly equal 250.

The complete apportionment computations under Jefferson's method are detailed below:

Metric / Step	**Shift A**	**Shift B**	**Shift C**	**Shift D**	**Total**
Patients	453	650	547	350	2000
Standard Quota (d = 8)	56.625	81.250	68.375	43.750	—
Initial Lower Quota	56	81	68	43	248
Modified Quota (dm = 7.93)	57.125	81.967	68.979	44.136	—
Final Apportionment	57	81	68	44	250

3. A hospital has a nursing staff of 250 nurses working in four shifts. The number of nurses apportioned to each shift is based on the average number of patients per shift, given in the table below. Use ***Adams's method*** to apportion the 250 nurses among the shifts at the hospital.

Shift	**A**	**B**	**C**	**D**
Average Patients	453	650	547	350

Solution. To complete the apportionment using Adams's method, we apply a modified divisor so that the sum of the upper quotas (rounded up) exactly equals the required total.

(a) Find Total Population and Standard Divisor (d):

$$\text{Total Patients} = 453 + 650 + 547 + 350 = 2000$$

$$d = \frac{2000 \text{ patients}}{250 \text{ nurses}} = 8$$

(b) Test the Standard Divisor: Dividing each population by 8 and rounding up to the next highest integer gives upper quotas of $57, 82, 69$, and 44. The sum of these initial upper quotas is $57 + 82 + 69 + 44 = 252$. This exceeds the target of 250 nurses.

(c) Determine a Modified Divisor (d_m): Because our initial sum is too large, Adams's method requires us to choose a larger divisor to decrease the individual quotas. Testing a modified divisor of $d_m = 8.07$ shifts the upper quotas downward to exactly equal 250.

The complete apportionment calculations under Adams's method are organized in the accessible data layout below:

Metric / Step	**Shift A**	**Shift B**	**Shift C**	**Shift D**	**Total**
Patients	453	650	547	350	2000
Standard Quota ($d = 8$)	56.625	81.250	68.375	43.750	—
Initial Upper Quota	57	82	69	44	252
Modified Quota ($d_m = 8.07$)	56.134	80.545	67.782	43.371	—
Final Apportionment	57	81	68	44	250

4. The police department in a large city has 180 new officers to be apportioned among 6 high-crime precincts. The total number of recorded crimes for each respective precinct is documented in the data profile below.

Precinct	**A**	**B**	**C**	**D**	**E**	**F**
Crimes	446	526	835	227	338	456

(a) Use ***Hamilton's method*** to apportion the new officers among the precincts.

Solution. 1. Identify Metrics: Total crimes population equals $446 + 526 + 835 + 227 + 338 + 456 = 2828$. The Standard Divisor is:

$$d = \frac{2828}{180} \approx 15.7111$$

2. Distribute Remainders: Summing the lower whole quotas gives $28 + 33 + 53 + 14 + 21 + 29 = 178$ allocations, leaving a surplus of 2 seats. Hamilton's method awards these seats to the largest fractional remainders: Precinct B (0.4795) and Precinct E (0.5134).

Metric / Step	**A**	**B**	**C**	**D**	**E**	**F**	**Total**
Crimes	446	526	835	227	338	456	2828
Standard Quota	28.388	33.479	53.147	14.448	21.513	29.024	—
Lower Quota	28	33	53	14	21	29	178
Surplus Added	0	+1	0	0	+1	0	+2
Final Apportionment	28	34	53	14	22	29	180

(b) Use ***Jefferson's method*** to apportion the new officers among the precincts.

Solution. Because the initial lower quota sum (178) is too small, Jefferson's method requires modifying the divisor downward to raise the quotas. Choosing a smaller modified divisor of $d_m = 15.46$ shifts the lower whole quotas up to hit the targeted total allocation sum of exactly 180.

Metric / Step	**A**	**B**	**C**	**D**	**E**	**F**	**Total**
Crimes	446	526	835	227	338	456	2828
Modified Quota	28.849	34.023	54.010	14.683	21.863	29.495	—
Final Apportionment	28	34	54	14	21	29	180

(c) Use ***Adams's method*** to apportion the new officers among the precincts.

Solution. Testing the standard upper quotas (rounding fractions up) gives a sum of $29 + 34 + 54 + 15 + 22 + 30 = 184$, which exceeds the 180 seats. Adams's method resolves this by increasing the divisor. Choosing a larger modified divisor of $d_m = 15.94$ pushes the quotas downward so that their rounded-up total sums exactly to 180.

Metric / Step	**A**	**B**	**C**	**D**	**E**	**F**	**Total**
Crimes	446	526	835	227	338	456	2828
Modified Quota	27.980	32.999	52.384	14.241	21.205	28.607	—
Final Apportionment	28	33	53	15	22	29	180

3. Personal Finance Handout

This handout covers essential mathematical principles for evaluating individual savings, annuities, and debt management strategies:

- *Basic Interest Calculations.*

  Computing and comparing simple interest and compound interest models over various compounding periods.

- *Savings and Annuities.*

  Analyzing regular deposit schedules, future value growth patterns, and long-term retirement accounts.

- *Loans and Spendings.*

  Evaluating fixed-rate installment loans, calculating monthly amortization values, and managing large debts including home mortgages.

Exercises and Workspaces

1. In order to pay for baseball uniforms, a school takes out a ***simple interest*** loan for \$20,000 for 7 months at a rate of 12%. Find the future value of the loan.

Solution. To find the future value of a loan using a simple interest model, we extract our known variables and evaluate using the linear simple interest equation.

Identify Given Variables:

- Principal (P): $\$20,000$
- Annual Interest Rate (r): $12\% = 0.12$
- Time (t): 7 months $= \frac{7}{12}$ of a year

Step-by-Step Calculation:

(a) Calculate Accumulated Interest (I): Multiply Principal by Rate by Time.

$$I = P \cdot r \cdot t$$

$$I = 20000 \cdot 0.12 \cdot \left(\frac{7}{12}\right)$$

$$I = 20000 \cdot 0.07 = 1400$$

The total accumulated interest generated over the 7 months is $\$1,400$.

(b) Calculate Future Value (A): Sum the initial principal amount and the newly accumulated interest.

$$A = P + I$$

$$A = 20000 + 1400 = 21400$$

The final future value of the loan at maturity is $\$21,400$.

2. What is the amount of the deposit needed if your financial goal is to save \$24,000 in 3 years with a rate of 7.5% ***compounded semiannually***?

Solution. To find the initial deposit (present value) required to meet a future financial goal with compound interest, we rearrange the compound interest formula to solve for the principal P.

Identify Given Variables:

- Future Value Goal (A): $\$24,000$
- Annual Interest Rate (r): $7.5\% = 0.075$
- Compounding Periods per Year (n): 2 (semiannually)
- Time (t): 3 years

Step-by-Step Calculation:

(a) State the Present Value Formula:

$$P = \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}}$$

(b) Substitute the Values: Find the interest rate per period ($\frac{r}{n}$) and total number of compounding periods (nt).

$$\frac{r}{n} = \frac{0.075}{2} = 0.0375$$

$$nt = 2 \cdot 3 = 6 \text{ total compounding periods}$$

$$P = \frac{24000}{(1 + 0.0375)^6}$$

(c) Evaluate the Denominator:

$$P = \frac{24000}{(1.0375)^6}$$

$$P \approx \frac{24000}{1.247179}$$

(d) Calculate the Final Principal Amount: Divide the future goal by the compounding factor and round to the nearest cent.

$$P \approx 19243.44$$

The initial deposit amount needed to meet your financial goal is $\$19,243.44$.

3. Your sister has her first child and you decide to open an annuity for your baby nephew. You deposit \$30 ***at the end of each month into an annuity*** that returns 2.5% annual interest.

(a) How much will your nephew have in his annuity when he goes off to college (18 years from now)?

Solution. To find the future value of regular monthly contributions, we evaluate using the future value formula for an ordinary annuity.

Identify Given Variables:

- Monthly Payment (PMT): \$30
- Annual Interest Rate (r): $2.5\% = 0.025$
- Compounding Periods per Year (n): 12 (monthly)
- Time (t): 18 years

Step-by-Step Calculation:

(a) State the Annuity Future Value Formula:

$$A = \frac{PMT \cdot \left(\left(1+\frac{r}{n}\right)^{nt} - 1\right)}{\left(\frac{r}{n}\right)}$$

(b) Calculate Intermediate Terms: Find the monthly interest rate and the total number of deposits.

$$\text{Periodic Rate } \left(\frac{r}{n}\right) = \frac{0.025}{12} \approx 0.0020833$$

$$\text{Total Periods } (nt) = 12 \cdot 18 = 216 \text{ monthly deposits}$$

(c) Substitute and Evaluate:

$$A = \frac{30 \cdot \left(\left(1+\frac{0.025}{12}\right)^{216} - 1\right)}{\left(\frac{0.025}{12}\right)}$$

$$A \approx \frac{30 \cdot ((1.0020833)^{216} - 1)}{0.0020833}$$

$$A \approx \frac{30 \cdot (1.567431 - 1)}{0.0020833}$$

$$A \approx 30 \cdot 272.4377$$

$$A \approx 8173.13$$

When your nephew leaves for college, the account will have accumulated $\$8,173.13$.

(b) How much interest was earned?

Solution. To determine total interest earned, we calculate your total out-of-pocket contributions over the 18 years and subtract that value from the accumulated future value.

Step-by-Step Calculation:

(a) Calculate Total Out-of-Pocket Deposits: Multiply the monthly payment by the total number of compound periods over 18 years.

$$\text{Total Contributions} = PMT \cdot nt$$

$$\text{Total Contributions} = 30 \cdot 216 = 6480$$

The sum of all out-of-pocket deposits is $\$6,480.00$.

(b) Calculate Total Interest Earned: Subtract total out-of-pocket deposits from the accumulated future value found in Task 1.

$$\text{Interest} = A - \text{Total Contributions}$$

$$\text{Interest} = 8173.13 - 6480.00 = 1693.13$$

The total interest earned by the annuity is $\$1,693.13$.

4. The purchase cost of a Camry is \$32,500. This vehicle can be financed by providing a \$3,000 down payment and committing to monthly installment payments of \$680.25 per month for 60 months.

(a) Determine the amount financed.

Solution. The amount financed is the remaining vehicle balance that must be covered by a loan after subtracting the initial down payment from the base purchase price.

$$\text{Amount Financed} = \text{Purchase Cost} - \text{Down Payment}$$

$$\text{Amount Financed} = 32500 - 3000 = 29500$$

The base amount financed through the loan is $\$29,500.00$.

(b) Determine the total installment charge (total installment price).

Solution. The total installment price represents the absolute total cost of buying the car over time, found by adding the down payment to the sum of all monthly payments.

Step-by-Step Calculation:

(a) Calculate Total Monthly Deferred Payments: Multiply the monthly commitment by the total number of payments.

$$\text{Total Payments} = 680.25 \times 60 = 40815$$

The sum of all monthly installments across 60 months is $\$40,815.00$.

(b) Calculate Total Out-of-Pocket Cost: Combine the monthly payments with the initial down payment.

$$\text{Total Installment Price} = \text{Total Payments} + \text{Down Payment}$$

$$\text{Total Installment Price} = 40815 + 3000 = 43815$$

The total installment cost for purchasing the vehicle is $\$43,815.00$.

(c) Determine the finance charge.

Solution. The finance charge represents the total interest paid for borrowing money, calculated by finding the difference between the total installment price and the original cash purchase cost.

$$\text{Finance Charge} = \text{Total Installment Price} - \text{Purchase Cost}$$

$$\text{Finance Charge} = 43815 - 32500 = 11315$$

The absolute cost of borrowing (total interest expense) is $\$11,315.00$.

5. A new fax machine costs Worcester State University \$2,670. They are to pay it off in 18 months at a 9% ***simple interest rate***.

(a) What will each monthly payment be?

Solution. To find the monthly installment value, we first compute the total future value of the loan (principal plus interest) and then divide it equally by the total number of payments.

Identify Given Variables:

- Principal (P): $\$2,670$
- Annual Interest Rate (r): $9\% = 0.09$
- Loan Term (t): 18 months $= \frac{18}{12} = 1.5$ years

Step-by-Step Calculation:

(a) Calculate Total Accumulated Interest (I):

$$I = P \cdot r \cdot t$$
$$I = 2670 \cdot 0.09 \cdot 1.5$$
$$I = 360.45$$

The total interest accrued on the equipment purchase is \$360.45.

(b) Calculate Total Future Value Balance (A): Combine the principal cost and interest.

$$A = P + I$$
$$A = 2670 + 360.45 = 3030.45$$

The total amount to be repaid over the loan lifespan is $\$3,030.45$.

(c) Calculate Periodic Installment Payment (PMT): Divide total future value by the 18 months.

$$PMT = \frac{A}{18}$$
$$PMT = \frac{3030.45}{18} = 168.3583$$

Rounding to the nearest cent, each sequential monthly payment will be \$168.36.

(b) How much interest will they pay on the loan?

Solution. The absolute cost of credit (total interest expense) was calculated directly in step 1 using the simple interest formula:

$$I = P \cdot r \cdot t$$
$$I = 2670 \cdot 0.09 \cdot 1.5 = 360.45$$

Worcester State University will pay a total interest amount of \$360.45 over the lifespan of the financing agreement.

6. Congratulations! You purchase a sprawling home for $1,000,000. You make a 15% down payment and agree to a 4.5% interest rate for a 30-year fixed-rate mortgage. You are also required to pay 3 points at closing.

(a) What is the amount of your loan?

Solution. The loan amount is determined by finding the cash value of the down payment and subtracting it from the total purchase price of the home.

Step-by-Step Calculation:

(a) Calculate the Down Payment: Multiply the purchase price by the down payment percentage.

$$\text{Down Payment} = 1000000 \cdot 0.15 = 150000$$

The upfront cash down payment is $\$150,000$.

(b) Calculate the Principal Mortgage Amount (P): Subtract the down payment from the total home cost.

$$P = 1000000 - 150000 = 850000$$

The total principal amount of your mortgage loan is $\$850,000$.

(b) You paid 3 points at the closing. How much did you pay?

Solution. In real estate finance, 1 point is equal to 1% of the total amount borrowed (the mortgage loan principal), not the purchase price of the home.

$$\text{Cost of Points} = \text{Loan Amount} \cdot 3\%$$

$$\text{Cost of Points} = 850000 \cdot 0.03 = 25500$$

The total amount paid for 3 points at closing is $\$25,500$.

(c) What is your monthly mortgage payment?

Solution. To determine the regular monthly installment cost, we apply the standard fixed-rate loan amortization formula.

Identify Given Variables:

- Mortgage Loan Principal (P): $\$850,000$
- Annual Interest Rate (r): $4.5\% = 0.045$
- Compounding Periods per Year (n): 12 (monthly)
- Total Time (t): 30 years

Step-by-Step Calculation:

(a) State the Installment Formula:

$$PMT = \frac{P \cdot \frac{r}{n}}{\left(1 - (1 + \frac{r}{n})^{nt}\right)}$$

(b) Evaluate the Numerator: Find the periodic monthly rate.

$$\text{Numerator} = 850000 \cdot \frac{0.045}{12} = 850000 \cdot 0.00375 = 3187.5$$

(c) Evaluate the Denominator: Calculate the monthly compounding base over the lifetime total of 360 payments $(12 \cdot 30)$.

$$\text{Denominator} = 1 - (1 + 0.00375)^{-360}$$

$$\text{Denominator} = 1 - (1.00375)^{-360}$$

$$\text{Denominator} \approx 1 - 0.261596 = 0.738404$$

(d) Solve for PMT: Divide the numerator by the denominator.

$$PMT = \frac{3187.5}{0.738404} \approx 4306.85$$

Your base monthly mortgage payment is $\$4,306.85$.

(d) How much are you charged over the course of the mortgage loan, in interest?

Solution. The total interest expense is determined by adding up all monthly payments made over 30 years and subtracting the original amount borrowed.

Step-by-Step Calculation:

(a) Calculate Total Lifetime Payments: Multiply the monthly payment by 360 total months.

$$\text{Total Lifetime Outlay} = 4306.83 \cdot 360 = 1550458.80$$

The aggregate amount paid back to the lender over 30 years is $\$1,550,458.80$.

(b) Subtract the Initial Principal Balance:

$$\text{Total Interest} = \text{Total Lifetime Outlay} - \text{Loan Principal}$$

$$\text{Total Interest} = 1550458.80 - 850000 = 700458.80$$

The total interest charged over the standard lifespan of the loan is $\$700,458.80$.

4. Probability Handout

This handout bridges fundamental counting formulas with theoretical probability to help you solve real-world problems involving large sample spaces. By mastering factorials, permutations, and combinations, you will build an efficient toolkit for evaluating complex, multi-stage probability scenarios.

The core objectives are organized into the following conceptual blocks:

- *Fundamentals of Probability.*

  An introduction to theoretical probability, defining sample spaces, tracking experiments, and computing basic event fractions.

- *Events Involving NOT and OR.*

  Analyzing compound events using the complement rule for negations and the general addition rule to account for overlapping outcomes.

- *Counting Principles, Permutations, and Combinations.*

  Using the Fundamental Counting Principle and factorials ($n!$) to calculate expansive outcome paths without manual listing.

  Applying permutations ($_nP_r$) for ordered subsets and combinations ($_nC_r$) for unordered groupings or committees selected without replacement.

- *Probability with the FCP, nPr, and nCr.*

  Combining counting models with classical probability structures to solve high-density problems like lotteries, secure passwords, and multi-stage selections.

Exercises and Workspaces: Real-World Applications

1. A class is collecting data on eye color and gender. They organize the data they collected into the table shown. Numbers in the table represent the number of students in the class that belong to each of the categories.

Table 6.1

	Brown	**Blue**	**Green**	**Total**
Total	**40**	**38**	**22**	**100**
Male	22	18	10	50
Female	18	20	12	50

Find the probability that a randomly selected student from this class:

a Does not have brown eyes.

b Is female or has blue eyes.

Solution. First, compute the totals: Total students = 100. Total Brown = 40. Total Female = 50. Total Blue = 38. Female and Blue = 20.

a Using the complement rule:

$$P(\text{not Brown}) = 1 - P(\text{Brown}) = 1 - \frac{40}{100} = \frac{60}{100} = \frac{3}{5}$$

b Using the addition rule:

$$P(\text{Female or Blue}) = P(\text{Female}) + P(\text{Blue}) - P(\text{Female and Blue}) = \frac{50}{100} + \frac{38}{100} - \frac{20}{100} = \frac{68}{100} = \frac{17}{25}$$

2. Two dice are rolled. What is the probability that the sum of the faces on the two dice is 7?

Solution. When rolling two standard six-sided dice, the total number of outcomes in the sample space is $6 \times 6 = 36$.

The outcomes that result in a sum of 7 are:

$$E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$$

There are 6 favorable outcomes, so the probability is:

$$P(\text{Sum of } 7) = \frac{6}{36} = \frac{1}{6}$$

3. There are fifteen balloons: 3 blue, 4 red, 2 green, 3 yellow, and 3 orange. How many **distinct ways** can the balloons be arranged?

Solution. This is a permutation problem involving non-distinct objects. The total number of balloons is $n = 15$. The repeating counts are 3 blue, 4 red, 2 green, 3 yellow, and 3 orange.

The number of distinct arrangements is given by:

$$\frac{15!}{3! \times 4! \times 2! \times 3! \times 3!} = \frac{1,307,674,368,000}{6 \times 24 \times 2 \times 6 \times 6} = \frac{1,307,674,368,000}{10,368} = 126,126,000$$

4. Find the number of distinct ways to arrange the letters in the word "Pinocchio".

Solution. Count the total number of letters in "Pinocchio", which is $n = 9$. Identify the repeating letters:

- i appears 2 times
- o appears 2 times
- c appears 2 times

The letters P, n, and h appear 1 time each.

The number of distinct arrangements is:

$$\frac{9!}{2! \times 2! \times 2!} = \frac{362,880}{2 \times 2 \times 2} = \frac{362,880}{8} = 45,360$$

5. How many ways are there to form a 3-member committee from the 18 members of the club?

Solution. Since a standard committee has no specific titles or ranking positions, the order of selection does not matter. This is a combination problem $({}_nC_r)$ where we choose $r = 3$ out of $n = 18$ members.

$${}_{18}C_3 = \frac{18!}{3!(18-3)!} = \frac{18 \times 17 \times 16}{3 \times 2 \times 1} = 816$$

6. How many ways are there to form a 3-officer committee (President, Vice-President, and Secretary) from the 18 members of the club?

Solution. Because each committee seat represents a distinct, ranked official position, the order of selection matters. This is a permutation problem $({}_nP_r)$ where we arrange $r = 3$ out of $n = 18$ members.

$${}_{18}P_3 = \frac{18!}{(18-3)!} = 18 \times 17 \times 16 = 4,896$$

7. A group consists of 8 men and 9 women. Three people are selected to attend a conference.

a In how many ways can the 3 people be selected from the group?

b In how many ways can the 3 people be selected from the men only?

c Find the probability that the selected group will consist of all men.

Solution. Order does not matter for selection to attend a conference, so we use combinations.

a The total group size is $8 + 9 = 17$ people. Choosing 3:

$$_{17}C_3 = \frac{17 \times 16 \times 15}{3 \times 2 \times 1} = 680$$

b Choosing 3 people out of the 8 men only:

$$_{8}C_3 = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

c The probability is the number of all-male selections over total possible selections:

$$P(\text{All Men}) = \frac{_{8}C_3}{_{17}C_3} = \frac{56}{680} = \frac{7}{85}$$

8. Refer back to the chocolate selection problem (16 chocolates total: 4 nuts, 7 caramel, 8 cream filled). Answer the following additional tracking questions based on your selections without replacement:

a If four chocolates are selected, what is the probability of not selecting a cream-filled or caramel chocolate?

b How many ways can one choose a nut-filled chocolate followed by a caramel chocolate?

Solution. Let us evaluate based on the structural subset pools:

a The total number of chocolates that are either cream-filled or caramel depends on potential overlaps. Assuming a disjoint selection setting where remaining pools are cleanly isolated: The number of chocolates that are **neither** cream nor caramel is $16 - (7 + 8) = 1$ chocolate. Since you cannot select 4 chocolates sequentially without replacement from a remaining pool of only 1 item, this specific probability path is impossible:

$$P(\text{Not Cream or Caramel for 4 draws}) = 0$$

b We use the Fundamental Counting Principle for sequential independent category slots. There are 4 choices for the nut-filled chocolate first. There are 7 choices for the caramel chocolate second.

$$\text{Total Ways} = 4 \times 7 = 28 \text{ ways}$$

5. Statistics Handout

This handout provides an overview of different data distributions through graphs, including z-scores and empirical methods.

The handout is designed to help students understand the strengths and weaknesses of the mean, median, and mode, and how they can lead to different outcomes in data analysis. By working through the exercises, students will gain a deeper understanding of bell curves, normal distributions, and linear regression.

- *Data through Graphs.*

  An introduction to interpreting and creating various types of graphs, such as bar graphs, histograms, pie charts, and scatter plots, to visualize data effectively.

- *Measures of Center and Spread.*

  mean, median, mode, range, variance, and standard deviation are all measures that help us understand the central tendency and variability of a dataset.

- *The Empirical Rule.*

  A rule that describes the distribution of data in a normal distribution, stating that approximately 68% of the data falls within one standard deviation of the mean, 95% within two standard deviations, and 99.7% within three standard deviations.

- *z-scores.*

  Standardized scores that indicate how many standard deviations a data point is from the mean.

- *Regression.*

  A statistical method for modeling the relationship between a dependent variable and one or more independent variables, often used for prediction and understanding the strength of relationships between variables.

Exercises and Workspaces: Real-World Applications

1. **Data Representation: Daily Text Messages Sent.** The following dataset represents the number of text messages sent by 25 high school students during a single day:
   40, 20, 20, 50, 30, 30, 60, 20, 30, 40, 30, 10, 40, 50, 40, 30, 30, 40, 40, 50, 60, 20, 50, 60, 10

   a Complete a frequency table for the text message counts.

   b Construct a bar graph of the data.

   c Construct a line graph of the data.

   d Construct a pie chart of the data.

   e Construct a histogram of the data using 10-point class intervals.

   Solution. ***(a) Frequency Table:*** Sorting and counting each distinct message count yields the following frequency distribution:

   Table 6.2 Frequency Distribution of Daily Text Messages

Texts Sent	Frequency (Students)
10	2
20	4
30	6
40	6
50	4
60	3

   (b) Bar Graph: We represent each distinct count value on the horizontal axis and its corresponding frequency on the vertical axis using separated columns:

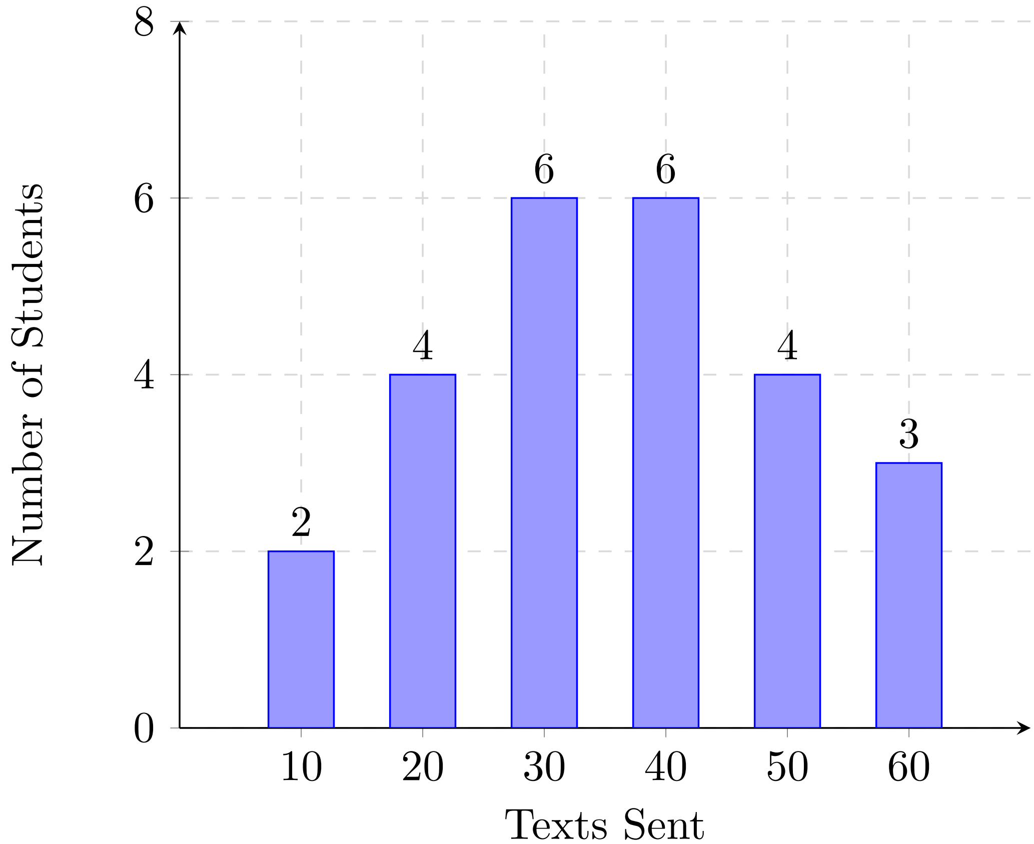


Figure 6.3

(c) Line Graph: Using coordinates, data markers are mapped out across consecutive value junctions and tied together using path lines:

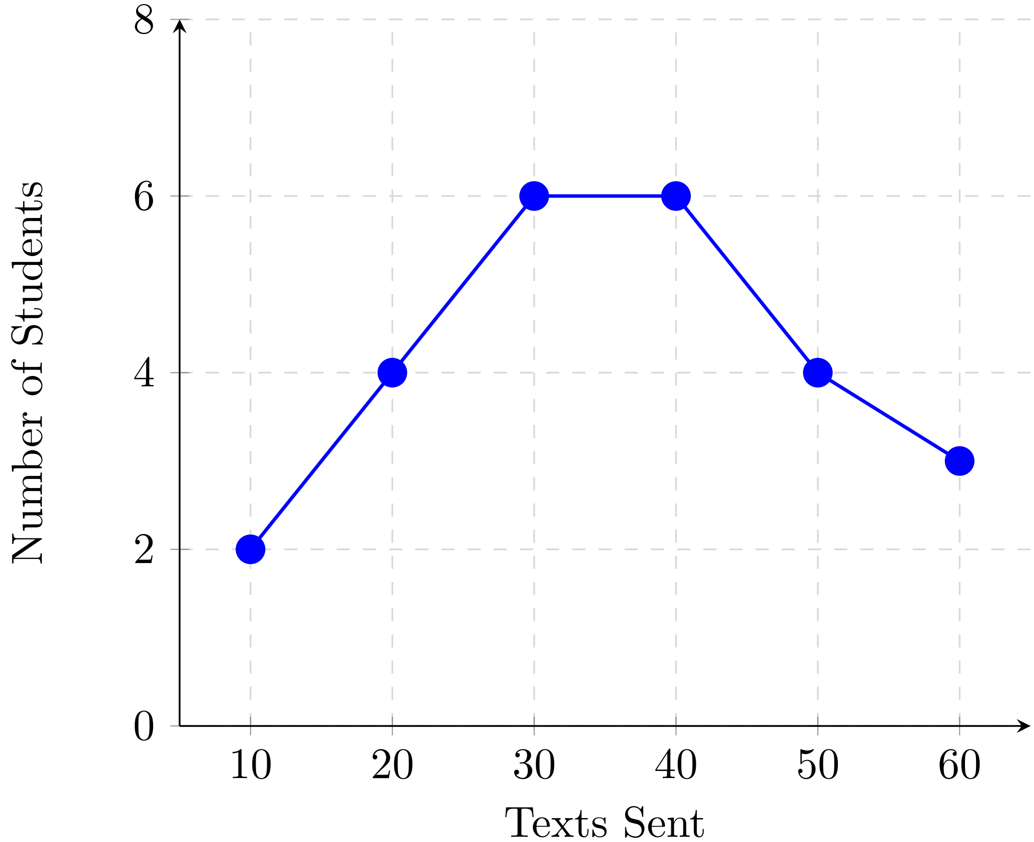


Figure 6.4

(d) Pie Chart: Proportional distributions are evaluated based on a complete baseline of $n = 25$ students using degree equations:

- ***10 Texts:*** 8.00 percent (28.8 degrees)
- ***20 Texts:*** 16.00 percent (57.6 degrees)
- ***30 Texts:*** 24.00 percent (86.4 degrees)
- ***40 Texts:*** 24.00 percent (86.4 degrees)
- ***50 Texts:*** 16.00 percent (57.6 degrees)
- ***60 Texts:*** 12.00 percent (43.2 degrees)

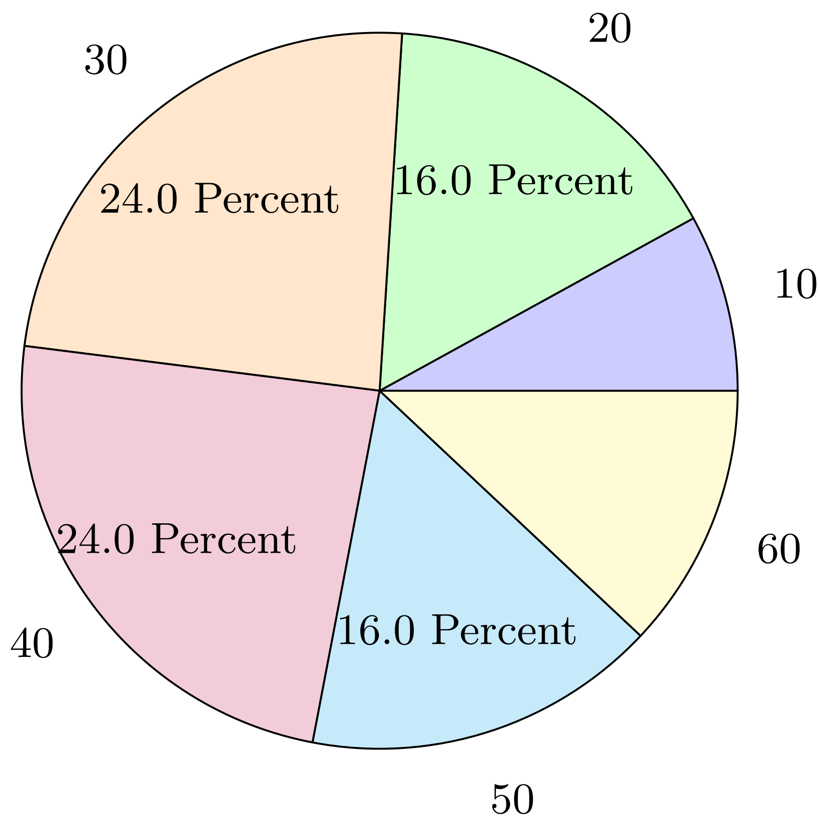


Figure 6.5

(e) Histogram: Grouping values into 10-point bins maps the dataset across continuous quantitative intervals, which forces columns to display directly side-by-side:

- ***[5, 15):*** 2 students (values of 10)
- ***[15, 25):*** 4 students (values of 20)
- ***[25, 35):*** 6 students (values of 30)
- ***[35, 45):*** 6 students (values of 40)
- ***[45, 55):*** 4 students (values of 50)
- ***[55, 65]:*** 3 students (values of 60)

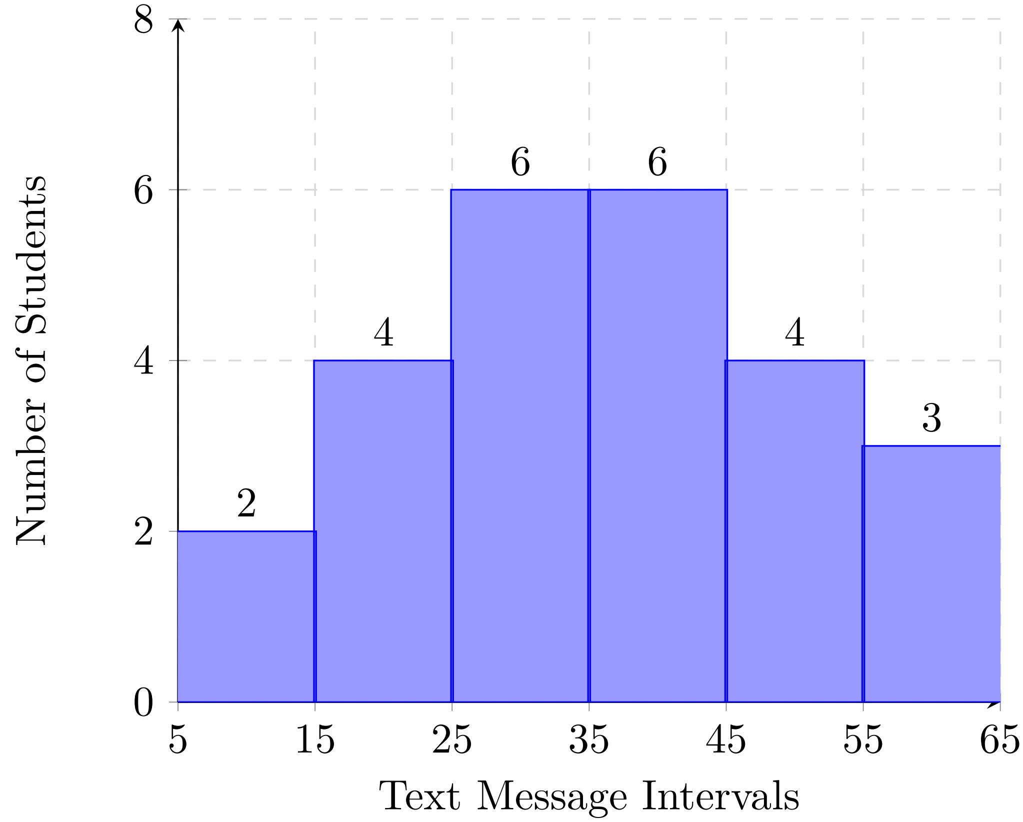

8
6
4
2
0
6
6
4
4
3
2
5
15
25
35
45
55
65
Number of Students
Text Message Intervals

Figure 6.6

2. **Employee Commute Times.** A small company tracks the daily one-way commute times (measured in minutes) for a random sample of 9 employees. The collected data values are listed below:
25, 40, 15, 30, 25, 20, 45, 25, 60
Calculate the following four summary metrics for this data set:

a The arithmetic mean commute time.

b The median commute time.

c The mode of the dataset.

d The total range of the commute times.

Solution. ***(a) Mean:*** To find the mean, sum all the data points together and divide by the total number of observations ($n = 9$):

$$\text{Mean} = \frac{25 + 40 + 15 + 30 + 25 + 20 + 45 + 25 + 60}{9} = \frac{285}{9} \approx 31.67 \text{ minutes}$$

(b) Median: To determine the median, first arrange the data points in ascending numerical order:
15, 20, 25, 25, ***25***, 30, 40, 45, 60
Since there is an odd number of scores (9), the median is the exact middle element located at position $\frac{9+1}{2} = 5$. The fifth value is 25 minutes.

(c) Mode: The mode is the specific data value that shows up most frequently in the sample. In this ordered list, the value 25 appears three times, while every other number appears only once. Therefore, the mode is 25 minutes.

(d) Range: The range is the absolute difference between the absolute highest value (maximum) and the lowest value (minimum) in the dataset:

$$\text{Range} = \text{Maximum} - \text{Minimum} = 60 - 15 = 45 \text{ minutes}$$

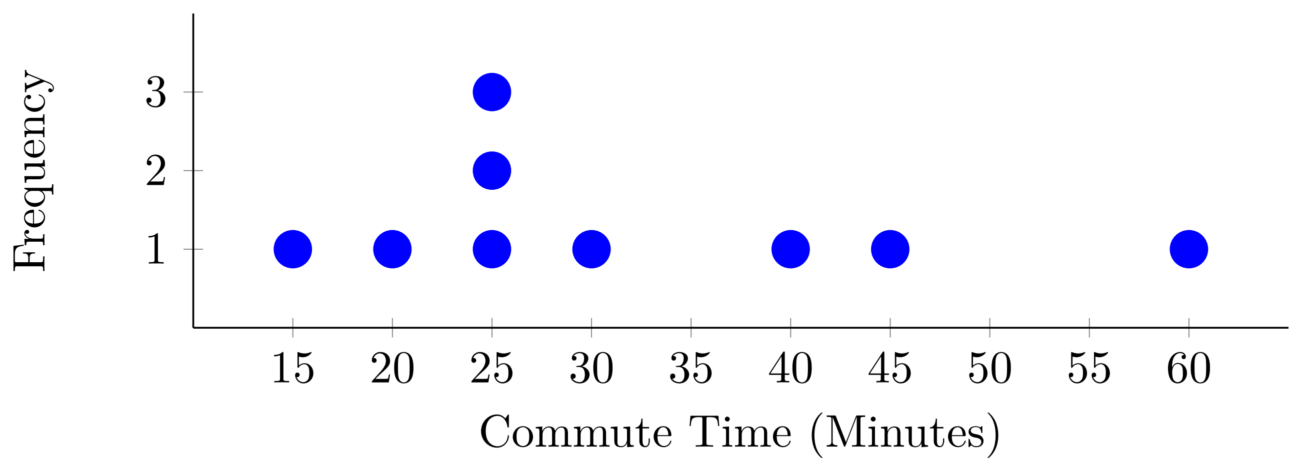


Figure 6.7 Dot Plot of Employee Commute Times

3. Female Height Distribution. Adult female heights in North America are approximately normally distributed with a mean of 65 inches and a standard deviation of 3.5 inches.

Use the Empirical Rule (68-95-99.7 Rule) to answer the following questions:

1 Approximately what percentage of women have heights between 61.5 inches and 68.5 inches?

2 Approximately 95 percent of women have heights between what two values?

3 Approximately what percentage of women have heights greater than 72 inches?

4 Approximately what percentage of women have heights between 58 inches and 65 inches?

Solution. ***1. Heights between 61.5 and 68.5 inches:*** The values 61.5 and 68.5 are found by subtracting and adding one standard deviation to the mean: $65-3.5 = 61.5$ and $65+3.5 = 68.5$. This range represents exactly 1 standard deviation from the mean (± 1) standard deviations. According to the Empirical Rule, approximately ***68 percent*** of the distribution falls within this range.

2. Range covering 95 percent of women: The Empirical Rule states that approximately 95 percent of data falls within 2 standard deviations of the (mean ± 2 sd). Calculating these boundaries:

$$\text{Lower Value} = 65 - 2(3.5) = 65 - 7 = 58 \text{ inches}$$

$$\text{Upper Value} = 65 + 2(3.5) = 65 + 7 = 72 \text{ inches}$$

Therefore, 95 percent of women have heights between ***58 inches and 72 inches***.

3. Percentage greater than 72 inches: A height of 72 inches sits exactly 2 standard deviations above the mean. Since 95 percent of women fall within 2 standard deviations (between 58 and 72 inches), the remaining 5 percent falls outside this span. Because the normal curve is perfectly symmetrical, half of that remainder is in the lower tail and half is in the upper tail. Dividing this gives $5 \div 2 =$ ***2.5 percent***.

4. Percentage between 58 and 65 inches: The value of 58 inches is exactly 2 standard deviations below the mean, while 65 inches is the mean itself. Since 95 percent of the data falls symmetrically within 2 standard deviations of the mean, the section from the lower boundary up to the center mean point represents exactly half of that cluster. Calculating this gives $95 \div 2 =$ ***47.5 percent***.

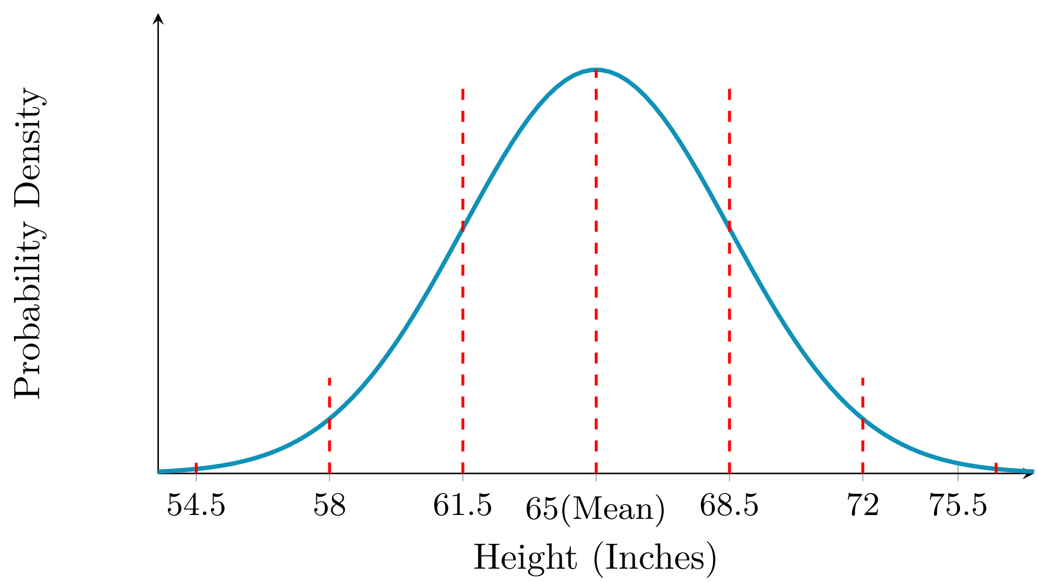


Figure 6.8 Normal Distribution Curve for Female Heights

4. **Calculating Z-Scores-1.** On a standardized certification exam, the mean score is 76 with a standard deviation of 5. Determine the z-score for an examinee who scored an 88.

   Solution.
$$z = \frac{88 - 76}{5} = 2.4$$

   This student scored 2.4 standard deviations above the population average.

5. **Calculating Z-Scores-2.** A track-and-field sprint group has a mean race completion time of 24.5 seconds with a standard deviation of 3.2 seconds. What is the z-score for a runner who completes the race in 19.1 seconds?

   Solution.
$$z = \frac{19.1 - 24.5}{3.2} \approx -1.69$$

   The runner's time is 1.69 standard deviations faster (below) than the group average.

6. **Comparing Z-Scores-3.** An industrial lighting brand has a mean runtime life of $15,000$ hours with a standard deviation of 850 hours. Find the specific operational lifespan x that corresponds to a z-score of -1.40.

   Solution.
$$x = mean + z(standard\ deviation)$$
$$x = 15000 + (-1.40)(850) = 13810 \text{ hours}$$

7. Negative Correlation: Car Mileage and Resale Value. A used car dealership wants to examine the relationship between a specific sedan's odometer mileage (X, measured in thousands of miles) and its current market resale value (Y, measured in thousands of dollars). The data collected from a sample of five vehicles is presented below.

Table 6.9 Vehicle Mileage vs. Resale Value

Vehicle	Mileage (X in thousands of miles)	Resale Value (Y in thousands of dollars)
1	10	35
2	20	29
3	30	26
4	40	20
5	50	14

Using the simple linear regression model calculated from the observed data table above, predict the total market resale value of a sedan if its odometer reaches 60 thousand miles (i.e., $X = 60$).

A. \$15.5 thousand dollars

B. \$9.5 thousand dollars

C. \$5.0 thousand dollars

D. \$2.1 thousand dollars

Answer. B.

Solution 1. To predict the resale value of the vehicle at 60 thousand miles, we first graph the data points from the table and calculate the line of best fit using linear regression.

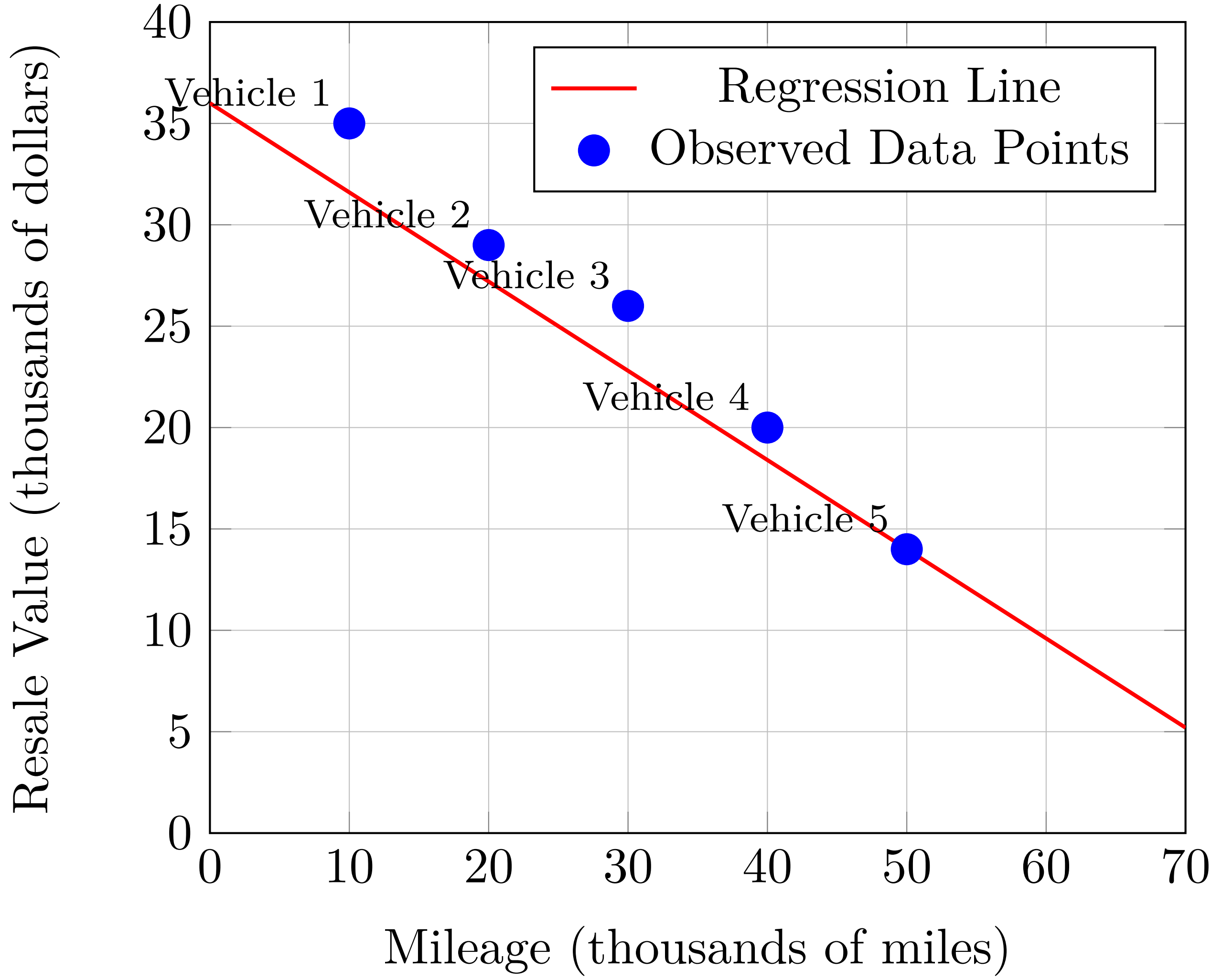


Figure 6.10 Scatter Plot of Mileage vs. Resale Value

Solution 2.

A. *Incorrect.*

Incorrect. This value fails to capture the continuous decline in value, overestimating how much the car is worth at 60 thousand miles.

B. *Correct.*

Correct! Observe the downward trend in the data points and plug the value directly into the line equation to find 9.5.

C. *Incorrect.*

Incorrect. While the vehicle loses value consistently, it has not depreciated this drastically according to the linear model.

D. *Incorrect.*

Incorrect. This underestimation assumes an exponential drop-off rather than a steady simple linear progression rate.

Chapter 7

Homework